ELS63, LE910Q1 and LE310Q1

AT Commands Reference Guide

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Applicability Table

Table 1: Applicability Table - Products

Products
ELS63-I
ELS63-IG
LE910Q1-WW
LE910Q1-SN
LE910Q1-WWG
LE910Q1-SNG
LE910Q1-SF
LE310Q1-WW
LE310Q1-SN
LE310Q1-EU

Table 2: Applicability Table - Software

Software
52.00.xx4

Introduction

Scope

This document is aimed to provide a detailed specification and a comprehensive listing for reference for the entire AT command set.

Audience

Readers of this document should be familiar with Telit modules and their ease of control by means of AT Commands.

Contact Information, Support

For technical support and general questions please e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use:

https://www.telit.com/contact-us/

Product information and technical documents are accessible 24/7 on our web site:

https://www.telit.com

Conventions

Note:

Provide advice and suggestions that may be useful when integrating the module.

Danger:

This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk:

Notifies the user to take proper grounding precautions before handling the product.

Warning:

Alerts the user on important steps about the module integration.

	Danger: This information MUST be followed or catastrophic equipment failure or personal injury may occur.
	Warning: Alerts the user on important steps about the module integration. If these points are not followed, the module and end user equipment may fail or malfunction.
	Note/Tip: This section provides all information related to the AT command involved. Each note can provide a different level of information: danger, caution/warning and tip/information
	Note/Tip: Provides advice and suggestions that may be useful when integrating the module.
	Set: This section provides all information related to SET functionality of the AT command involved. If there is strictly and relevant SET information, it can be found at the end of the section.

Read: This section provides all information related to READ functionality of the AT command involved. If there is strictly and relevant READ information, it can be found at the end of the section.

Test: This section provides all information related to TEST functionality of the AT command involved. If there is strictly and relevant TEST information, these can be found at the end of the section.

Additional info: This section provides any kind of additional and useful information related to the AT command section as well as command exceptions or special behavior cases.

Reference: This section provides useful references (standards or normative) related to the AT command involved.

Example: This section provides useful examples related to the AT command involved.

All dates are in ISO 8601 format, that is YYYY-MM-DD

Terms and Conditions

Refer to

https://www.telit.com/terms-conditions/

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Commands

The Telit family of wireless modules can be controlled using standard AT commands, via either the Serial UART or USB interfaces, where available.

The AT Command interface complies with:

- Hayes standard AT command set, to maintain the compatibility with existing software programs.
- 3GPP TS 27.007 specific AT command and GPRS specific commands.
- 3GPP TS 27.005 specific AT commands for SMS (Short Message Service) and CBS (Cell Broadcast Service)

In addition to the standards-based commands, Telit wireless modules also support our own proprietary AT commands, used for special purposes and to provide additional useful information.

The following is a description of how to use the AT commands with the modules mentioned in the Applicability Table.

Note:

The AT is an ATTENTION command and is used as a prefix to other parameters in a string.

AT commands, combined with their parameters, can be sent from your Host Application, a Script, or when using a Terminal emulator typed in manually as a command line instruction.

Definitions

The following syntactic definitions apply:

<CR> ASCII Carriage return character - is the command line and terminator character of the result code. The decimal value is between 0 and 255 and is specified within parameter S3. The default value is 13.

<LF> ASCII Linefeed character - is the character recognized as line feed character. The decimal value is between 0 and 255 and is specified within parameter S4. The default value is 10. The line feed character is output after the carriage return character if detailed result

codes are used (V1 option used). Otherwise, if numeric format result codes are used (V0 option used), it will not appear in the result codes.

<...> Name enclosed in angle brackets is a syntactic element. They are not displayed on the command line.

[...] The optional sub parameter of a command or an optional part of the TA (Terminal adapter) information response is enclosed in square brackets. These brackets are not displayed on the command line. If the sub parameter is not provided in AT commands that have a Read command, the new value remains the same as its previous value. Action type commands do not store the values of any of their sub parameters and therefore do not have a Read command. These commands are executed to invoke a function of the equipment, which generally involves more than the simple storage of a value for later use. When using action type commands, an action should be performed according to the recommended default setting of the sub parameter.

AT Commands Syntax

The syntax rules followed by Telit Cinterion implementation of either Hayes AT commands, GSM commands are very similar to those of standard basic and extended AT commands.

There are two types of extended command:

- Parameter type commands. This type of commands may be "set" (to store a value or values for later use), "read" (to determine the current value or stored values), or "tested" (to determine ranges of values supported). Each of them has a test command (trailing =?) to provide information about the type of its sub parameters; they also have a Read command (trailing?) to check the current values of sub parameters.

- Action type commands. This type of command may be "executed" or "tested".
  - "executed" to invoke a function of the equipment, which generally involves more than the simple storage of a value for later use
  - "tested" to determine:
    - if sub parameters are associated with the action, the ranges of sub parameters values that are supported; if the command has no sub parameters, issuing the correspondent Test command (trailing =?) raises the result code "ERROR".

      Note: issuing the Read command (trailing?) causes the command to be executed.
    - whether the equipment implements the Action Command or not (in this case issuing the correspondent Test command - trailing =? - returns the OK result code), and, if sub parameters are associated with the action, the ranges of sub parameters values that are supported.

Action commands do not store the values of any of their possible sub parameters.

Moreover:

The response to the Test Command (trailing =?) may be modified in the future by Telit Cinterion to allow for the description of new values/functionalities.

If all the sub parameters of a parameter type command +CMD are optional, issuing AT+CMD=<CR> causes the result code to be returned to OK and the previous values of the omitted sub parameters to be retained.

String Type Parameters

A string, either enclosed between quotes or not, is a valid string type parameter input. According to V25.ter space characters are ignored on the command line and may be used freely for formatting purposes, unless they are embedded in numeric or quoted string

constants; therefore a string containing a space character must be enclosed between quotes to be considered a valid string type parameter (e.g. typing AT+COPS=1,0,"A1" is the same as typing AT+COPS=1,0,A1; typing AT+COPS=1,0,"A BB" is different from typing AT+COPS=1,0,A BB).

A string is always case-sensitive.

A small set of commands always requires to write the input string parameters in quotes: this is explicitly reported in the specific descriptions.

Command Lines

A command line is made up of three elements: the prefix, the body and the termination character.

The command line prefix consists of the characters "AT" or "at", or, to repeat the execution of the previous command line, the characters "A/" or "a/" or AT#/ or at#/.

The termination character may be selected by a user option (parameter S3), the default being <CR>.

The basic structures of the command line are:

- ATCMD1<CR> where AT is the command line prefix, CMD1 is the body of a basic command (nb: the name of the command never begins with the character "+") and <CR> is the command line terminator character
- ATCMD2=10<CR> where 10 is a sub parameter
- AT+CMD1;+CMD2=, ,10<CR> These are two examples of extended commands (nb: the name of the command always begins with the character "+"2F2F). They are delimited by semicolon. In the second command the sub parameter is omitted.
- +CMD1?<CR> This is a Read command for checking current sub parameter values
- +CMD1=?<CR> This is a test command for checking possible sub parameter values

Note:

The set of proprietary AT commands differs from the standard one because the name of each of them begins with either "@", "#", "$" or "*". Proprietary AT commands follow the same syntax rules as extended commands.

These commands might be performed in a single command line as shown below:

ATCMD1 CMD2=10+CMD1;+CMD2=, ,10;+CMD1?;+CMD1=?<CR>

however, it is always preferable to separate basic and extended commands in different command lines; furthermore, it is recommended to avoid placing several action commands in the same command line, because if one of them fails, then an error message is received but it is not possible to claim which one of them has failed the execution.

If command V1 is enabled (detailed responses codes) and all commands in a command line have been performed successfully, the result code <CR><LF>OK<CR><LF> is sent from the TA to the TE, if sub parameter values of a command are not accepted by the TA or the command itself is invalid, or the command cannot be executed for some reason, result code <CR><LF>ERROR<CR><LF> is sent and no subsequent commands in the command line are processed.

If command V0 is enabled (numeric responses codes), and all commands in a command line has been executed successfully, the result code 0<CR> is sent from the TA to the TE, if sub-parameter values of a command are not accepted by the TA or the command itself is invalid, or command cannot be executed for some reason, result code 4<CR> and no subsequent commands in the command line are processed.

In case of errors depending on ME operation, ERROR (or 4) response may be replaced by +CME ERROR: <err> or +CMS ERROR: <err>.

Note:

The command line buffer accepts a maximum of 400 characters. If this number is exceeded, none of the commands will be executed and TA returns ERROR.

ME Error Result Code - +CME ERROR: <err>

This is NOT a command; it is the error response to +Cxxx 3GPP TS 27.007 commands.

Syntax: +CME ERROR: <err>

Parameter: <err> - error code can be either numeric or detailed (see +CMEE). The possible values of <err> are reported in the table:

Note:

"Numeric Format" Not all modules support the error codes shown in the table.

Note:

"Verbose Format" There could be small variations in the message depending on the module in use.

Table 3: ME Error Result Code

Numeric Format	Verbose Format
0	phone failure
1	no connection to phone
2	phone adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong
16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure

24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed - emergency calls only
34	numeric parameter instead of text parameter
35	text parameter instead of numeric parameter
36	numeric parameter out of bounds
37	text string too short
38	The GPIO Pin is already used
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required
49	EAP method not supported

50	Invalid EAP parameter
51	Parameter length error for all Auth commands
52	Temporary error for all Auth command
53	not verified hidden key
100	unknown
103	Illegal MESSAGE
106	Illegal ME
107	GPRS services not allowed
111	PLMN not allowed
112	Location area not allowed
113	Roaming not allowed in this location area
132	service option not supported
133	requested service option not subscribed
134	service option temporarily out of order
148	unspecified GPRS error
149	PDP authentication failure
150	invalid mobile class
257	network rejected request
258	retry operation
259	invalid deflected to number

260	deflected to own number
261	unknown subscriber
262	service not available
263	unknown class
264	unknown network message
273	Minimum TFT per PDP address error
274	Duplicate TFT eval prec index
275	Invalid TFT param combination
277	Invalid number of parameters
278	Invalid Parameter
320	Call index error
321	Call state error
322	Sys state error
323	Parameters error
550	generic undocumented error
551	wrong state
552	wrong mode
553	context already activated

554	stack already active
555	activation failed
556	context not opened
557	can not setup socket
558	can not resolve DN
559	time-out in opening socket
560	can not open socket
561	remote disconnected or time-out
562	connection failed
563	tx error
564	already listening
565	socket disconnection
566	can not resume socket
567	ip version type incompatible
568	ipv6 not enabled
569	
600	Generic undocumented error
601	wrong state
602	Can not activate
603	Can not resolve name
604	Can not allocate control socket

605	Can not connect control socket
606	Bad or no response from server
607	Not connected
608	Already connected
609	Context down
612	Resource used by other instance
613	Data socket yet opened in cmdmode
614	FTP CmdMode data socket closed
615	FTP not connected
616	FTP disconnected
617	FTP read command closed
618	FTP read command error
619	FTP write command closed
620	FTP write command error
621	FTP read data closed
622	FTP read data error
623	FTP write data closed
624	FTP write data error
625	FTP host not found
626	FTP accept failure
627	FTP listen failure
628	FTP bind failure

629	FTP file create failure
630	FTP file get failure
631	FTP file put failure
632	FTP file not found
633	FTP timed out
634	FTP login incorrect
635	FTP close error
636	FTP server not ready
637	FTP server shutdown
638	FTP unexpected reply
639	FTP user ID and password don't match
640	FTP user ID and password don't match
641	FTP user already logged in
642	FTP open channel timeout
643	FTP communication timeout
644	FTP unknown error
657	Network survey error (No Carrier)
658	Network survey error (Busy)
659	Network survey error (Wrong request)
660	Network survey error (Aborted)
680	LU processing

681	Network search aborted
682	PTM mode
683	Network search terminated
684	CSG Search processing
690	Active call state
691	RR connection established
770	SIM invalid
900	No Response for AT Command
1000	SSL not activated
1001	SSL certs and keys wrong or not stored
1002	SSL generic error
1003	SSL already activated
1004	SSL error during handshake
1005	SSL socket error
1006	SSL invalid state
1007	SSL cannot activate
1008	SSL not connected
1009	SSL already connected
1010	SSL error enc/dec data
1011	SSL disconnected
1012	SSL version mismatch

1100	Model not recognized
1101	Model information missing
1102	Unable to open the file
1103	Unable to close the file
1104	Unable to read the nv file
1105	Unable to write the nv file
1106	Input pattern is wrong
1113	Call establishment failed
1114	File name already exist
1150	GTP error timeout
1151	PDP is not active
1152	GTP is Disabled
1153	operation not supported
1154	Reboot failed. Reboot manually

ME Error Result Code for MQTT - +CME ERROR: <err>

If MQTT commands are used detailed error codes for possible values of <err> are reported in the following table:

Table 3.1: ME Error Result Code

Numeric Format	Verbose Format
1100	MQTT Initialization Fail
1101	MQTT De-initialization Fail
1102	MQTT Configuration Fail
1103	MQTT Connect Fail
1104	MQTT Disconnect Fail
1105	MQTT Publish Fail
1106	MQTT Subscribe Fail
1107	MQTT Unsubscribe Fail
1108	MQTT SSL Configuration Fail
1109	MQTT Command Timeout Fail
1110	MQTT PUBSEXT Fail
1111	MQTT Protocol Unsupported
1112	MQTT Identifier Rejected
1113	MQTT Server Unavailable
1114	MQTT Bad Username or Password
1115	MQTT Not Authorised

1200	MQTT-SN Initialization Fail
1201	MQTT-SN De-initialization Fail
1202	MQTT-SN Configuration Fail
1203	MQTT-SN Socket Fail
1204	MQTT-SN Connect Fail
1205	MQTT-SN Disconnect Fail
1206	MQTT-SN Register Fail
1207	MQTT-SN Publish Fail
1208	MQTT-SN Subscribe Fail
1209	MQTT-SN Unsubscribe Fail
1210	MQTT-SN Will Topic Update Fail
1211	MQTT-SN Will Message Update Fail
1212	MQTT-SN State Change Fail
1213	MQTT-SN Invalid Client Status

Message Service Failure Result Code - +CMS ERROR: <err>

This is NOT a command; it is the error response to +Cxxx 3GPP TS 27.005 commands.

Syntax: +CMS ERROR: <err>

Parameter: <err> - numeric error code.

The <err> values are reported in the table:

Table 4: Message Service Failure Result Code

Numeric Format	Meaning

According to 3GPP TS 24.011 section 8.2.5.4	
0...127	
According to 3GPP TS 23.040 sub clause 9.2.3.22 values	
128...255	
According to 3GPP TS 27.005 section 3.2.5 - Message Service Failure Result Code +CMS ERROR	
300	ME failure
301	SMS service of ME reserved
302	operation not allowed
303	operation not supported
304	invalid PDU mode parameter
305	invalid text mode parameter
310	SIM not inserted
311	SIM PIN required
312	PH-SIM PIN required
313	SIM failure
314	SIM busy
315	SIM wrong
316	SIM PUK required
317	SIM PIN2 required
318	SIM PUK2 required
320	memory failure
321	invalid memory index

322	memory full
330	SMSC address unknown
331	no network service
332	network time-out
340	no +CNMA acknowledgement expected
500	unknown error
510	msg blocked
<err> 512 and on are manufacturer specific	
512	No SM resources
513	TR1M timeout
514	LL error
515	No response from network

Carriage Returns, Line Feeds and Log

Generally, the number of carriage returns <CR> and line feeds <LF> at the end of command responses may vary. This scenario may also vary from software version to software version. We do not have evidence of this behavior in URC lines.

Command responses examples:

AT#LWM2MMON?<CR>

<CR><LF>

#LWM2MMON: "4"<LF>#LWM2MMON: "3"

Warning:

missing <CR>

<CR><LF>

<CR><LF>

OK

<CR><LF>

AT#LWM2MFYACKURI=0,2<CR>

<CR><LF>

#LWM2MNFYACKURI: "/3/0/9"<LF>#LWM2MNFYACKURI: "/3/0/8"

Warning:

missing <CR>

<CR><LF>

<CR><LF>

OK

<CR><LF>

The user must be aware of this aspect before writing a script to parse the command lines and the relative response lines to generate a log.

Information Responses and Result Codes

The TA response, in case of verbose response format enabled, for the previous examples command line could be as shown below:

- information response to +CMD1?
  <CR><LF>+CMD1:2,1,10<CR><LF>
- information response to +CMD1=?
  <CR><LF>+CMD1(0-2),(0,1),(0-15)<CR><LF>
- result code
  <CR><LF>OK<CR><LF>

Moreover, there are two other types of result codes:

- result codes that inform about the progress of the TA operation (e.g. connection establishment CONNECT)
- result codes that indicate the occurrence of an event not directly associated with the issuance of a command from TE (e.g. ring indication RING).

Here the basic result codes according to ITU-T V25Ter recommendation

Table 5: Basic result codes

Numeric form	Verbose form
0	OK

1	CONNECT or CONNECT <text> (CONNECT <text> not supported)
2	RING (not supported)
3	NO CARRIER
4	ERROR
6	NO DIALTONE (not supported)
7	BUSY (not supported)
8	NO ANSWER (not supported)
10	CONNECT 2400 (not supported)
11	CONNECT 4800 (not supported)
12	CONNECT 9600 (not supported)
15	CONNECT 14400 (not supported)
23	CONNECT 1200/75 (not supported)

Note:

<text> can be"300", "1200", "2400", "4800", "9600", "14400" or "1200/75".

Command Response Time-Out

Every command issued to the Telit modules returns a result response, if response codes are enabled (default). The time required to process the given command and return the response varies, depending on the command type. Commands that do not interact with the SIM or the network, and only involve internal setups or readings, have an immediate response. Commands that interact with the SIM or the network may take several seconds to send a response, depending on the SIM configuration (for example, number of contacts stored in the phonebook, number of stored SMS), or on the network the command may interact with.

Command Issuing Timing

The chain Command -> Response must always be respected, and a new command must not be issued before the module has finished all the ending of the response result code (whatever it is).

This applies especially to applications that "sense" the OK text and therefore may send the next command before the complete code <CR><LF>OK<CR><LF> is sent by the module.

However, It is advisable to wait for at least 20ms between the end of the reception of the response and the issue of the next AT command.

If the response codes are disabled and therefore the module does not report any response to the command, then at least the 20ms pause time must be respected.

Module Reboot

Reboot of the module can be triggered by AT command. Specific AT commands, to apply the changes requested, do include an implicit reboot. These AT commands are identified by the "This AT command causes a reboot" or "This AT command could cause a reboot" warning box statements in this Guide.

Warning:

The boot-up time of the module, after issuing any of the AT commands that implicitly trigger a Reboot procedure, could be considerably longer than expected.

Please allow up to 20 seconds for boot up before executing an HW_SHUTDOWN

Warning:

After every module reboot or power on there is the following URC on UART

^boot.rom

Discard it.

Storage

Factory Profile and User Profiles

The Telit wireless modules store the values, set by several commands, in the internal non-volatile memory (NVM), allowing to remember this setting even after power off. In NVM, these values are set either as factory or user profiles. There is one customizable user profile and one factory profile in the NVM of the device: by default, the device will start with user profile 0 equal to the factory profile.

Each profile is not divided in sub sections.

The &W command is used to save the current values of profile into the NVM user profile.

Command &P is used to set the profile to be loaded at startup.

The &F command resets to factory profile values the commands of the profile, the same as &F1.

The values set by other commands are stored in NVM outside the profile: some of them are always stored, without issuing any &W, some others are stored by issuing specific commands (+CSAS); all these values are read at power-up.

In this document, each AT command description begins with an "AT Command short overview table" in the following format:

Table 6: AT Command short overview table

SIM Presence	Setting saved	Can be aborted
/	see below	/

This chapter focuses on the values that the saved Setting field can have and their meaning. The meaning of the other fields will be described in the next chapter. The saved Setting field can have one of the values listed below (for information about the AT instance introduced hereafter, see the reference section of the #PORTCFG command):

Specific profile: the parameters values set by the command are stored in the profile. The stored values set is associated to the specific AT instance used to enter the command. It is a profile used by the specific AT instances.

Examples of the AT commands: +IPR, E, Q, V, X, etc.

Common profile: the parameters values set by the command are stored in the profile section. The stored values set is not associated to the specific AT instance used to enter the command. It is a profile shared between the AT instances.

Examples of the AT commands: +CSCS.

Auto: the parameters values set by the command are automatically stored in NVM, without issuing any storing AT command, and regardless of the profile (unique values). The values are automatically restored at startup.

AT commands examples: +COPS, #SCFG, etc.

Other: the parameters values set by the command are stored in NVM by issuing a specific command and regardless of the profile.

Examples of the AT commands:

+CSCA, +CSMP +CSCB setting is saved by +CSAS

Note:

AT Commands listed in the Examples may not be available on some modules.

AT Command Short Overview Table

As stated before, each AT command description begins with an "AT Command short overview table" having the following format:

Table 7: AT Command short overview table

SIM Presence	Setting saved	Can be aborted
Not required	/	No

Here are the table fields meanings:

SIM Presence: indicates if the AT command to be executed requires the presence of a SIM card.

Can be aborted: indicates if the AT command can be aborted during its execution.

AT COMMANDS

General Control and Config

Command Line Prefixes

AT - Starting a Command Line

AT is the prefix used to start a command line.

ITU-T Recommendation V.25 ter

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT

The prefix AT is a two-character abbreviation ("ATtention"), always used to start a command line to be sent from TE to TA, with the only exception of AT#/ prefix. As a command, it can be issued just to test if the device is responding to AT commands.

Generic Modem Control

AT&F - Set to Factory-Defined Configuration

Set configuration parameters to default values.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&F[<value>]

Execution command sets the configuration parameters stored in NVM profile to default values specified by manufacturer.

Parameter:

Name	Type	Default	Description
<value>	integer	0	parameters to reset

Values:

0 : profile parameters

1 : profile parameters

If parameter <value> is omitted, the command has the same behavior as AT&F0.

Both values 0 and 1 have the same description and effect.

ATZ - Soft Reset

Soft Reset

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATZ

Execution command sets the configuration parameters stored in NVM profile to default values specified by manufacturer.

AT&P - Default Reset Full Profile Designation

Define which full profile is loaded at startup.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT&P[<n>]

Execution command defines which full profile will be loaded at startup.

Parameter:

Name	Type	Default	Description
<n>	integer	0	full profile will be loaded on startup.

Value:

0	:	profile number

Differently from command Z<n>, which loads just once the desired profile, the one chosen through command &P will be loaded at every startup.

If parameter is omitted, the command has the same behavior as AT&P0

AT&W - Store Current Configuration

Execution command stores on profile <n> the complete configuration of the device.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&W[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	profile identifier

Value:

0	:	profile identifier

If parameter is omitted, the command has the same behavior of AT&W0.

AT&V - Display some Configuration and Profile

The command displays some of the basic modem configuration settings and parameters

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&V

Execution command returns some of the basic modem configuration settings and parameters, one for each row, in the format:

setting/parameter : value

Example of returned values.

```
AT&V
DTE SPEED : 115200
DTE FORMAT : 8N1
COMMAND ECHO : E1=YES
RESULT MESSAGES : Q0=YES
VERBOSE MESSAGES : V1=YES
EXTENDED MESSAGES : X0=NO
FLOW CONTROL OPTIONS : &K3=HW bidirect.
DTR (C108) OPTIONS : &D0=ignored
DCD (C109) OPTIONS : &C1=follows carrier
DEFAULT PROFILE : &P0=user profile 1
OK
```

AT+GCAP - Capabilities List

This command returns the equipment supported command set list.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GCAP

Execution command returns the equipment supported command set list:

+GCAP: +CGSM,+DS,+MS

Additional info:

Supported Command Set:

+CGSM: 3GPP TS command set

+DS: Data Service common modem command set

+MS: Mobile Specific command set

AT+GCAP=?

Test command returns ERROR result code.

AT+GMI - Manufacturer Identification

This command returns the manufacturer identification.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GMI

Execution command returns the manufacturer identification followed by an <OK> at newline.

AT+GMI=?

Test command returns OK result code.

AT+GMM - Model Identification

The command returns the model identification.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GMM

The execution command returns the model identification followed by an <OK> at newline.

AT+GMM=?

Test command returns OK result code.

AT+GMR - Revision Identification

The command returns the software revision identification.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GMR

Execution command returns the software revision identification followed by an <OK> at newline.

AT+GMR=?

Test command returns OK result code.

AT+CGMI - Request Manufacturer Identification

The command returns device manufacturer identification code.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGMI

Execution command returns the device manufacturer identification code followed by an OK at newline.

AT+CGMI=?

Test command returns OK result code.

AT+CGMM - Request Model Identification

This command returns the device model identification.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGMM

Execution command returns the device model identification code followed by an OK at newline.

AT+CGMM=?

Test command returns OK result code.

AT+CGMR - Request Revision Identification

The command returns device software revision number.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGMR

Execution command returns device software revision number followed by an OK at newline.

AT+CGMR=?

Test command returns OK result code.

AT+CGSN - Request Product Serial Number Identification

This command allows to retrieve the product serial number in form of IMEI of the mobile.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGSN[=<snt1>]

Execution command returns the product serial number, identified as the IMEI of the mobile.

Parameter:

Name	Type	Default	Description
<snt>	string	N/A	the serial number type

Values:

0 : returns <sn>

1 : returns <imei>

2 : returns <imeisv>

3 : returns <svn>

Additional info:

<sn> - serial number, identified as the IMEI of the mobile, without command echo.

<imei> - International Mobile station Equipment Identity; IMEI is composed of Type Allocation Code (TAC) (8 digits), Serial Number (SNR) (6 digits) and the Check Digit (CD) (1 digit).

<imeisv> - International Mobile station Equipment Identity and Software Version number; The 16 digits of IMEISV are composed of Type Allocation Code (TAC) (8 digits), Serial Number (SNR) (6 digits) and the software version (SVN) (2 digits).

<svn> - Software Version Number.

AT+CGSN=?

Test command returns supported <snt> values.

AT#CGSSN - Production Serial Number Identification

This command returns the Telit production Serial Number Identification.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CGSSN

Execution command returns the Telit production Serial Number Identification. This command shall be used when the IMEIW feature is used to track the original serial number of the module. In case of the IMEIW feature is not activated on the module, the result reported by the command is the same of command +CGSN.

AT#CGSSN=?

Test command returns the OK result code.

```
AT#CGSSN
#CGSSN: 357023009002206
OK
```

AT#IMEIW - IMEI Write

This command writes the IMEI in the module in case of IMEI Blank.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#IMEIW=<imeiCustom>

Set command writes the IMEI in the module. IMEI has the same format of command +CGSN and include the check digit (CD).

The check digit will be checked by the module to validate the IMEI.

Parameter:

Name	Type	Default	Description
<imeiCustom>	string	-	imei

Additional info:

The command returns ERROR if:

- the IMEI is not blank
- the IMEI string is not valid
- the check digit is not valid

AT#IMEIW?

Read command returns the programmed IMEI.

AT#IMEIW=?

Test command returns the OK result code.

IMEI will not be set if IMEI Blank is not configured.

IMEI will not be set if IMEI Blank is configured and already set once (one shot AT#IMEIW).

IMEI will be available after module power cycle.

```
AT#IMEIW?
#IMEIW: 000000000000000
OK
AT#IMEIW=358485000083981
OK
AT+CGSN 000000000000000
OK
AT#IMEIW?
#IMEIW: 358485000083981
OK

Power OFF
Power On

AT+CGSN
358485000083981
OK
```

AT#SWPKGV - Request Software Package Version

This command shows the software, modem, production, application packages versions. It provides two syntax formats: set and execution.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SWPKGV[=<version>]

Here is the set format.

Parameter:

Name	Type	Default	Description
<version>	string	N/A	selects the type of version to be shown; it must be entered between double quotes

Values:

M	:	shows modem version
A	:	shows App version
P	:	shows Production version
PK	:	shows the full package version

Additional info:

- The execution format returns the packages versions without #SWPKGV: command echo. The response is as follows:

AT#SWPKGV
<Telit Software Package Version>-<Production Parameters Version>
<Modem Package Version>
<Production Parameters Version>
<Application Software Version>

OK

AT#SWPKGV=?

Test command returns OK result code.

AT#SWOPTIONS - Read SW Options

Execution command returns a list of strings showing the currently supported SW features, with #SWOPTIONS: command echo.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SWOPTIONS

Dummy AT command

Does nothing and returns OK.

AT#SWOPTIONS

OK

AT#SWOPTIONS?

Test command returns OK result code.

AT+CFUN - Set Phone Functionality

This command selects the level of functionality in the ME.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CFUN=[<fun>[,<rst>]]

Set command selects the level of functionality in the ME.

Parameters:

Name	Type	Default	Description
<fun>	integer	1	Mobile functionality and optional power saving mode.

Values:

0 : minimum functionality with power saving disabled, see Additional info section

1 : mobile full functionality with power saving disabled

4 : disable both TX and RX

5 : full functionality with power saving enabled, see Additional info section (suggested for ELS63 and LE310Q1 product variants)

9 : deep power saving activation and exit with WAKEUP or ON/OFF, see Additional info section (only available for LE310Q1 product variants)

14 : full functionality with power saving enabled for GPIO12, see Additional info section (only available for LE910Q1 product variants and suggested for LE910Q1 product variants)

15 : full functionality with deep power saving enabled for GPIO12, see Additional info section (only available for LE910Q1 product variants)

<rst>	integer	0	reset flag

Values:

0 : do not reset the ME before setting it to <fun> functionality level.

1 : reset the ME before setting it to <fun> functionality level, this option works only with <fun> =1, with other it will return an error

Additional info:

- <fun>=0

  The module is in minimum functionality with power saving disabled.

  Disable both TX and RX.

  Disable SIM.

- <fun>=5

  Suggested for ELS63 and LE310Q1 product variants.

After set once +CFUN to 5 for power saving enable, plug out the USB, set the DTR (RS232) line to OFF to activate power saving.

Once in power saving, the CTS and DSR (RS232) lines switch to the OFF status to signal that the module is in power saving condition.

During the power saving condition the AT interface is not accessible by UART.

During the power saving any URC temporary activates UART.

During the power saving to deactivate it, before sending any AT command, set the DTR line to ON and wait for the CTS line to go in ON status.

With the DTR line in ON status the module will keep the power saving mode enabled, and to activate power saving mode again set the DTR line to OFF again.

- <fun>=9

This setting is not allowed in case of AT+CPSMS=1 configuration.

Only available for LE310Q1 product variants.

This is for deep power saving with very low current consumption (no RAM retention).

Plug out the USB, set the +CFUN to 9 to activate deep power saving.

Once in deep power saving, the PWRMON line switches to the OFF status to signal that the module is in power saving condition. The CTS and DSR (RS232) lines status in power saving condition have no meaning.

During the deep power saving condition the AT interface is not accessible by UART.

During the deep power saving any URC wakes up the device that returns to +CFUN 1.

During the deep power saving to deactivate it, before sending any AT command, set WAKEUP or ON/OFF to ON and wait for PWRMON to go in ON status, the device wakes up and returns to +CFUN 1.

To activate deep power saving mode again set +CFUN to 9 again.

- <fun>=14

Only available for LE910Q1 product variants.

Suggested for LE910Q1 product variants.

Enable GPIO wakeup functionality through AT#GPIO ALT setting selecting active/inactive logic.

If Low level ACTIVE (falling edge to wake up) is set then a small additional current consumption is expected compared to High level ACTIVE (raising edge to wake up).

After set once +CFUN to 14 for power saving enable with GPIO12, plug out the USB, set the GPIO12 to INACTIVE to activate power saving.

Once in power saving, the PWRMON line switches to the OFF status to signal that the module is in power saving condition. The CTS and DSR (RS232) lines status in power saving condition have no meaning.

During the power saving condition the AT interface is not accessible by UART.

During the power saving any URC temporary activates UART.

During the power saving to deactivate it, before sending any AT command, set the GPIO12 to ACTIVE and wait for PWRMON to go in ON status.

With the GPIO12 in ACTIVE status the module will keep the power saving mode enabled, and to activate power saving mode again set the GPIO12 to INACTIVE again.

▶▶ <fun>=15

This setting is not allowed in case of AT+CPSMS=1 configuration.

Only available for LE910Q1 product variants.

This is for deep power saving with very low current consumption (no RAM retention).

Enable GPIO wakeup functionality through AT#GPIO ALT setting selecting active/inactive logic.

If Low level ACTIVE (falling edge to wake up) is set then a significant additional current consumption is expected compared to High level ACTIVE (raising edge to wake up).

After set +CFUN to 15 for deep power saving enable with GPIO12, plug out the USB, set the GPIO12 to INACTIVE to activate deep power saving.

Once in deep power saving, the PWRMON line switches to the OFF status to signal that the module is in power saving condition. The CTS and DSR (RS232) lines status in power saving condition have no meaning.

During the deep power saving condition the AT interface is not accessible by UART.

During the deep power saving any URC wakes up the device that returns to +CFUN 1.

During the deep power saving to deactivate it, before sending any AT command, set the GPIO12 to ACTIVE and wait for PWRMON to go in ON status, the device wakes up and returns to +CFUN 1.

To activate deep power saving mode again set +CFUN to 15 again and set the GPIO12 to INACTIVE again.

- Issuing AT+CFUN=4[,0] causes the module to perform a network de-registration.
- If power saving is enabled, it reduces the power consumption during the idle time, thus allowing a longer standby time with a given battery capacity.
- The power saving condition does not affect the network behavior of the module, even during the power save condition the module remains registered on the network and reachable for incoming SMS.

  In case of incoming SMS during the power saving, the module will wake up and proceed normally with the SMS URC code (the same for other network related URC).

AT+CFUN?

Read command reports the current setting of <fun> in the format

+CFUN: <fun>

AT+CFUN=?

Test command returns the list of supported values for <fun> and <rst>.

AT+CLAC - Available AT Commands

This command shows the available AT commands list.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CLAC

The execution command causes the ME to return one or more lines reporting the AT commands that are available to the user. The format is:

<ATcmd1>[<CR><LF><ATcmd2>[...]]

<ATcmdn> is the AT command.

AT+CLAC=?

Test command returns the OK result code.

AT+CMEE - Report Mobile Equipment Error

The command enables the use of result code.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+CMEE=[<n>]

Set command disables/enables the use of result code +CME ERROR: <err> as an indication of an error relating to the +Cxxx command issued. When enabled, device related errors cause the +CME ERROR: <err> final result code instead of the default ERROR final result code. ERROR is returned normally when the error message is related to syntax, invalid parameters or DTE functionality.

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables +CME ERROR: <err> result code and selects the format

Values:

0 : disable

1 : enable and use numeric<err> values

2 : enable and use verbose <err> values

The detailed description of <err> is available in section "ME Error Result Code - +CME ERROR: <err>".

+CMEE has no effect on the final result code +CMS.

AT+CMEE?

Read command returns the current value of parameter <n> in the format:

+CMEE: <n>

OK

AT+CMEE=?

Test command returns the supported values of parameter <n>.

AT+CEER - Extended Error Report

Reports extended error related to the last unsuccessful call.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CEER

Execution command returns one or more lines of information text <report> offering the TA user an extended error report, in the format:

+CEER: <report>

This report regards some error condition that may occur:

the failure in the last unsuccessful call setup (originating or answering)

the last call release

If no error condition has occurred since power up, then "Normal, unspecified" condition is reported

AT+CEER=?

Test command returns OK result code.

AT#HWREV - Hardware Identification

This command returns the device Hardware revision identification code.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#HWREV

Execution command returns the device Hardware revision identification code with command echo.

AT#HWREV=?

Test command returns OK.

AT#SYSSTART - AT commands ready

This command enables/disables the AT commands ready unsolicited result code (URC).

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SYSSTART=<enable>

Set enables/disables the AT commands ready unsolicited result code in the two formats:

#SYSSTART

URC indicates that the UE is running in Normal Functionality level

#SYSSTART AIRPLANE MODE

URC indicates that the UE is running in Airplane Mode .

Parameter:

Name	Type	Default	Description
<enable>	integer	0	enables/disables the AT commands ready unsolicited result code

Values:

0 : disable the AT commands ready unsolicited result code

1 : enable the AT commands ready unsolicited result code

If #SYSTART URC is enabled it will be present after power on, restart, change between Normal Functionality level and Airplane Mode indicating the AT commands ready and the Functionality level.

AT#SYSSTART?

Read command returns the current value of <enable>

#SYSTART: enable>

AT#SYSSTART=?

Test command returns supported values for parameter <enable>.

Power on.

#SYSSTART

AT#PSMRI - Power Saving Mode Ring Indicator

The command enables or disables the Ring Indicator pin response to an URC message while modem is in power saving mode.

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT#PSMRI=<n>

Set command enables/disables the Ring Indicator pin response to an URC message while modem is in power saving mode. If enabled, a negative going pulse is generated, when URC message for specific event is invoked. The duration of this pulse is determined by the value of <n>.

Parameter:

Name	Type	Default	Description
<n>	integer	0	disables, enables/sets duration of the generated pulse.

Values:

0	:	disables RI pin response for URC message
50÷1150	:	enables RI pin response for URC messages with a duration specified in ms

the behavior for #PSMRI is invoked only when modem is in sleep mode (AT+CFUN=5 and DTR Off on Main UART).

AT#PSMRI?

Read command reports the duration in ms of the pulse generated, in the format:

#PSMRI: <n>

AT#PSMRI=?

Test command reports the supported range of values for parameter <n>

When RING signal for incoming call/SMS/socket listen is enabled, the behavior for #PSMRI will be ignored.

AT+CSCS - Select TE Character Set

The command purpose is to set different character sets that are used by the device.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT+CSCS=[<chset>]

Set command sets the current character set used by the device.

Parameter:

Name	Type	Default	Description
<chset>	string	IRA	character set to be used by the device.

Values:

GSM	:	GSM default alphabet (3GPP TS 23.038).
IRA	:	international reference alphabet (ITU-T T.50).
UCS2	:	16-bit universal multiple-octet coded character set (ISO/IEC10646).

AT+CSCS?

Read command returns the current value of the active character set.

AT+CSCS=?

Test command returns the supported values for parameter <chset>.

AT+CMUX - Multiplexing Mode

This command is used to enable/disable the multiplexing protocol control channel.

3GPP TS 27.007
3GPP TS 27.010

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CMUX=<mode>[,<subset>[,<port_speed>[,<N1>]]]

Set command is used to enable/disable the multiplexing protocol control channel.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	basic option is currently the only supported mode. Value: 0 : basic option mode.
<subset>	integer	0	the way in which the multiplexer control channel is set up Value: 0 : UIH frame
<port_speed>	integer	N/A	transmission rate. The default value is implementation specific.

Values:

1	:	9600 bps
2	:	19200 bps
3	:	38400 bps
4	:	57600 bps
5	:	115200 bps
6	:	230400 bps

<N1> integer 127 maximum frame size.

Value:

8÷1500	:	expressed in bytes

AT+CMUX?

Read command returns the current value of the parameters, in the format:

+CMUX: <mode>,<subset>,<port_speed>,<N1>

AT+CMUX=?

Test command returns the range of supported values for all parameters.

Version 52.00.004 supports two CMUX channels.

AT#USBCFG - USB Configuration

This command sets USB configuration on the modem device.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#USBCFG=<mode>

Parameter:

Name	Type	Default	Description
<mode>	integer	0	USB configuration mode. See #PORTCFG command.

Values:

0 : USB Modem Ports, RNDIS, AT COM1, AT COM2(PPP), DIAG;

For ELS63 and LE910Q1 product variants: VID 0x1BC7 PID 0x7020

For LE310Q1 product variants: VID 0x1BC7 PID 0x7028

1 : USB Modem Ports, ECM, AT COM1, AT COM2(PPP), DIAG;

For ELS63 and LE910Q1 product variants: VID 0x1BC7 PID 0x7021

For LE310Q1 product variants: VID 0x1BC7 PID 0x7029

50 : No Ports available (security setting)

- To make active the configuration the module must be rebooted.
- Once the setting #USBCFG=50 is applied, it will no longer be possible to give other AT commands on the USB Modem ports.

  Therefore, if you want to make the setting reversible by giving the AT#USBCFG command again, you need to be sure that you have other AT instances available, such as an UART port (see #PORTCFG).

AT#USBCFG?

Read command returns the current <mode> in the following format:

#USBCFG: <mode>

AT#USBCFG=?

Test command returns the list of supported values.

AT#PORTCFG - Connect Physical Ports to Service Access Points

This command allows to connect Service Access Points (software anchorage points) to the external physical ports.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#PORTCFG=<Variant>

Set command allows to connect Service Access Points to the external physical ports giving a great flexibility. Examples of Service Access Points: AT Parser Instance #1, #2, #3, etc..

Parameter:

Name	Type	Default	Description
<Variant>	integer	0	set port configuration. A short description, for each <Variant> value, is reported in test command section.

Value:

0÷max	:	see test command section

To enable the set port configuration, the module must be rebooted.

For ELS63 family:

- to enable AT#PORTCFG USIF1 composition it is mandatory to first configure GPIOs 16, 17, 18, 19 as ALT8 (se AT#GPIO)

AT#GPIO=16,0,9

OK

AT#GPIO=17,0,9

OK

AT#GPIO=18,0,9

OK

AT#GPIO=19,0,9

OK

AT#PORTCFG=3

OK

AT+CFUN=1,1

OK

-to restore GPIOs 16, 17, 18, 19 as input/output it is mandatory to first configure AT#PORTCFG as non-USIF1 composition

For LE910Q1 family:

- there is no relation between AT#GPIO and AT#PORTCFG.

AT#PORTCFG?

Read command returns the requested and the active port configuration in the format:

#PORTCFG: <requested>,<active>

Additional info:

▶▶ Parameters returned by the read command

Name	Type	Default	Description
<requested>	integer	-	value showing the requested configuration that will be activated at the next power ON.
<active>	integer	-	value showing the current configuration.

AT#PORTCFG=?

Test command returns a brief description of the supported ports arrangement solutions.

For each <Variant> are reported, on one row, the logical connections between a physical port (USIF0, USIF1, etc.) and a Service Access Point (AT#1, AT#2, etc.). To have information about the physical ports, refer to document [1].

The test command returns, for example, the following message:

AT#PORTCFG=?

#PORTCFG: Variant=0: AT= USIF0 USB0 USB1

#PORTCFG: Variant=3: AT= USIF0 USIF1 USB0

#PORTCFG: Variant=4: AT= USIF0 USIF1

#PORTCFG: Variant=8: AT= USB0 USB1

#PORTCFG: Variant=13: AT= USIF0 USB0

#PORTCFG: Variant=14: AT= USIF0 USIF1 USB0 USB1

OK

AT#FWSWITCH - Set Active Firmware Image

Set command allows enabling a specific firmware image on products embedding 2 or more different firmware images.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FWSWITCH=<imageNumber>[,<reboot>]

Parameters:

Name	Type	Default	Description
<imageNumber>	integer	-	To know the <imageNumber> values range refer to the test command. For the relation between <imageNumber> value and the relative customization refer to the table in Additional info section where are also described the default values. Values: 0÷Max : image number
<reboot>	integer	0	Reboot mode Values:

0 : Reboot the module after the network selection

1 : Do not reboot the module

Additional info:

▶▶ <imageNumber> identifies the image (customization) as shown in the following table.

Product	Customization	<imageNumber> default value
ELS63-I ELS63-IG	2 = India	2
LE910Q1-WW LE910Q1-WWG LE910Q1-SN LE910Q1-SNG LE910Q1-SF LE310Q1-WW LE310Q1-SN	0 = AT&T, 1 = Verizon, 2 = ROW	2
LE310Q1-EU	2 = Europe	2

AT#FWSWITCH?

Read command reports the current active firmware image:

#FWSWITCH: <imageNumber>

AT#FWSWITCH=?

Test command reports the range of supported values for parameter <imageNumber>

ELS63-I, ELS63-IG, LE310Q1-EU product variants:

The <imageNumber> is 2 and it is fixed.

AT#FWSWITCH test command returns <imageNumber> range only value 2.

AT#FWSWITCH set command accepts only <imageNumber> value 2.

AT#FWSWITCH read commands returns <imageNumber> value 2.

LE910Q1-WW, LE910Q1-WWG, LE910Q1-SN, LE910Q1-SNG, LE910Q1-SF, LE310Q1-WW, LE310Q1-SN product variants:

The <imageNumber> is not fixed, it is automatically selected and it depends on SIM IMSI MCC/MNC values.

AT#FWSWITCH test command returns <imageNumber> range only value corresponding to currently inserted SIM IMSI MCC/MNC value, if SIM is not inserted returns ERROR.

AT#FWSWITCH set command accepts only <imageNumber> corresponding to currently inserted SIM IMSI MCC/MNC value, if SIM is not inserted returns ERROR.

AT#FWSWITCH read commands returns <imageNumber> value corresponding to currently inserted SIM IMSI MCC/MNC value, if SIM is not inserted returns ERROR.

Read image:

AT#FWSWITCH?

#FWSWITCH: 0

OK

Switch to image 0:

AT#FWSWITCH =0

OK

AT#SECBTON - Secure boot enable status

This command check if secure boot is enabled on module

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SECBTON

Execution command return secure boot status in the following format:

AT#SECBTON

<secure boot enable status>,<fingerprint>

OK

Additional info:

- <secure boot enable status> will be the below value depending on the module's fusing status

  0: Secure boot is disabled

  1: Secure boot is enabled; module fused with some key

- <fingerprint> root certificate hash (fingerprint) as hex string

  hex string empty: if secure boot is disabled

  hex string: root certificate hash (fingerprint) as hex string

AT#SECBTON=?

Test command returns OK

AT#SECBTON

0,

OK

S Parameters

ATS2 - Escape Character

The command manages the ASCII character used as escape character.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

ATS2=<char>

Set command sets the ASCII character to be used as escape character.

Parameter:

Name	Type	Default	Description
<char>	integer	43	escape character decimal ASCII

Value:

0÷255	:	factory default value is '+'

The escape sequence consists of three escape characters preceded and followed by n ms of idle.

ATS2?

Read command returns the current value of S2 parameter.

The format of the numbers in output is always 3 digits, left-filled with 0s.

ATS2=?

Test command returns OK result code.

DTE - Modem Interface Control

ATE - Command Echo

This command allows to enable or disable the command echo.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

ATE[<n>]

The execution command allows to enable/disable the command echo.

Parameter:

Name	Type	Default	Description
<n>	integer	1	Configuration value

Values:

0 : disables command echo

1 : enables command echo, hence command sent to the device are echoed back to the DTE before the response is given.

If parameter is omitted, the command has the same behavior of ATE0

ATQ - Quiet Result Codes

This command allows to enable or disable the result code.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

ATQ[<n>]

Set command enables or disables the result codes.

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables result codes

Values:

0	:	enables result codes

If parameter is omitted, the command has the same behavior of ATQ0.

After issuing ATQ0 the OK result code is returned

```
AT+CGACT=?
+CGACT: (0-1)
OK
```

ATV - Response Format

Set command determines the contents of the header and trailer transmitted with result codes and information responses.

It also determines if result codes are transmitted in a numeric form or an alphanumeric form (according to [1]).

[1] ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

ATV[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	1	format of information responses and result codes. See Additional info section.

Values:

0 : limited headers and trailers and numeric format of result codes

1 : full headers and trailers and verbose format of result codes

Additional info:

▶▶

<n>=0

information responses	<text><CR><LF>
result codes	<numericCode><CR>

▶▶

<n>=1

information responses	<CR><LF> <text><CR><LF>
result codes	<CR><LF> <verboseCode><CR><LF>

The <text> portion of information responses is not affected by this setting.

If parameter is omitted, the command has the same behavior of ATV0.

ATI - Identification Information

This command returns identification information.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATI[<n>]

Execution command returns one or more lines of information text followed by a result code.

Parameter:

Name	Type	Default	Description
<n>	integer	0	information request

Values:

0 : numerical identifier

1 : module checksum

2 : checksum check result

3 : manufacturer

4 : product name

5 : DOB version

If parameter is omitted, the command has the same behavior of ATI0

AT&C - Data Carrier Detect (DCD) Control

This set command controls the DCD output behavior of the AT commands serial port.

ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&C[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	1	DCD output behavior

Values:

0 : DCD remains always High

1 : DCD follows the Carrier detect status: if carrier is detected DCD goes High, otherwise DCD is Low

2 : Not supported

3 : DCD line is High if one or more Internet services are in active state, otherwise DCD is Low. See Additional info section

Additional info:

Active state means:

- Online mode TCP or UDP client, SOCKET , HTTP: when state is "Connecting" or "Up".

- Online mode TCP Listener: when state is "Up", "Alerting", "Connected" and "Released".

- FTP: when data channel is connected, state is "Up".

If parameter is omitted, the command has the same behavior of AT&C0

AT&D - Data Terminal Ready (DTR) Control

This set command configures the behavior of the module according to the DTR control line transitions (RS232).

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT&D[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	defines the module behavior according to the DTR control line transitions

Values:

0 : module ignores DTR control line transitions

1 : when the module is connected, the high to low transition of DTR line sets the module in command mode, the current connection is not closed

2 : when the module is connected, the high to low transition of DTR line sets the module in command mode and the current connection is closed

3 : C108/1 operation is enabled. Not supported

4 : C108/1 operation is disabled. Not supported

- If AT&D2 has been issued, and the DTR line has been tied Low, auto-answering is inhibited and it is possible to answer only issuing command ATA.
- If parameter is omitted, the command has the same behavior of AT&D0.

AT&K - Flow Control

Flow Control settings.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT&K[<n>]

Set command sets the UART (AT channel) (AT instance) from which the command is sent

Parameter:

Name	Type	Default	Description
<n>	integer	3	flow control behavior

Values:

0 : no flow control

3 : hardware bi-directional flow control (both RTS/CTS active)

If parameter is omitted, the command has the same behavior as AT&K0

AT&K must be saved in profile through AT&W and there is a different setting for each UART (AT channel) (AT instance).

AT&K on USB returns OK and does nothing.

The flow control in command mode is affected by this command.
See AT#CFLO command for the flow control in command mode.

AT&S - Data Set Ready (DSR) Control

Set DSR pin behavior

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT&S[<n>]

Set command controls the RS232 DSR pin behavior.

Parameter:

Name	Type	Default	Description
<n>	integer	3	Configuration parameter

Values:

0 : DSR always High

1 : DSR High when online data mode connected (IPEasy and dialup)

ME in command mode: DSR is OFF/Low

ME in online data mode: DSR is ON/High

2 : DSR High when online data mode connected (IPEasy only)

ME in command mode: DSR is OFF/Low

ME in online data mode: DSR is ON/High

3 : High when device is ready to receive AT commands (including when device is ready to receive data in online data mode)

- If parameter is omitted, the command has the same behavior of AT&S0.
- In power saving mode the DSR pin always tied Low.

AT+IPR - UART DCE Interface Data Rate Speed

The command sets the speed of the USIF0 serial port, see document [1].

[1] Hardware Design Guide of the used module

[2] ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+IPR=<rate>

The command sets the UART speed of the USIF0 port during command mode operations.

Parameter:

Name	Type	Default	Description
<rate>	integer	115200	speed of the serial USIF0 port expressed in bit per second.

Values:

300	:	bps
600	:	bps
1200	:	bps
2400	:	bps
4800	:	bps

9600	:	bps
19200	:	bps
38400	:	bps
57600	:	bps
115200	:	bps
230400	:	bps
460800	:	bps
921600	:	bps

AT+IPR?

Read command returns the current value of <rate> parameter.

AT+IPR=?

Test command returns the list of <rate> values in the format:

+IPR: (list of <rate> values)

This command has no effect if it is sent on USB interface or CMUX instances: the DCE sends the OK result but the settings are ignored.

It is recommended to wait for at least 50 ms between the reception of the OK response and the issue of the next AT command in the new baudrate configuration.

Test command example

```
AT+IPR=?
+IPR:
(300,600,1200,2400,4800,9600,19200,38400,57600,115200,230400,460800,921600)
OK
```

AT+IFC - DTE-Modem Local Flow Control

This set command selects the flow control of the serial port in both directions.

ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+IFC=<byDTE>,<byDCE>

Parameters:

Name	Type	Default	Description
<byDTE>	integer	2	specifies the method used by the DTE to control the flow of data received from the device (DCE) Values: 0 : no flow control 2 : flow control by RTS control line (C105, Request to Send)
<byDCE>	integer	2	specifies the method used by the device (DCE) to control the flow of data received from the DTE Values:

0	:	no flow control
2	:	flow control by CTS control line (C105, Clear to Send)

The only possible commands are AT+IFC=0,0 and AT+IFC=2,2.

AT+IFC?

Read command returns active flow control settings.

AT+IFC=?

Test command returns all supported values of the parameters <byDTE> and <byDCE>.

AT+ICF - DTE-Modem Character Framing

This set command defines the asynchronous character framing.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+ICF=<format>[,<parity>]

Parameters:

Name	Type	Default	Description
<format>	integer	3	sets the number of Data bits, the presence of a Parity bit and the number of Stop bits.

Value:

1 : 8 Data, 2 Stop

2 : 8 Data, 1 Parity, 1 Stop

3 : 8 Data, 1 Stop

4 : 7 Data, 2 Stop

5 : 7 Data, 1 Parity, 1 Stop

6 : 7 Data, 1 Stop

<parity>	integer	0	sets the type of parity. This setting is considered only if <format> is either 2 or 5 otherwise it is ignored.

Values:

0	:	odd
1	:	even

AT+ICF?

Read command returns current settings for parameters <format> and <parity>.

If current setting of parameter <format> is neither 2 nor 5, the current setting of parameter <parity> will always be reported as 0

AT+ICF=?

Test command returns the ranges of values for the parameters <format> and <parity>.

This command has effect only on AT UART port.

If there is a request for a second UART AT port, in this case AT+ICF depends on which UART port is set.

AT+ICF does not have any effect on AT USB port; it responds OK but does nothing.

```
AT+ICF=3
OK

AT+ICF?
+ICF: 3,0
OK

AT+ICF=?
+ICF: (1-6),(0-1)
OK
```

AT#CFLO - Command Flow Control

This set command enables/disables the hardware flow control in command mode.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT#CFLO=<enable>

Parameter:

Name	Type	Default	Description
<enable>	integer	0	enable/disable hardware flow control

Values:

0 : disable flow control in command mode

1 : enable flow control in command mode

The flow control in online mode is not affected by this command.

See AT&K command for the flow control in online mode.

The flow control enabled in command mode (AT#CFLO=1) will be active only if the flow control in online mode is enabled (AT&K3).

See AT&K command for the flow control in online mode

AT#CFLO?

Read command returns current setting value in the format:

#CFLO: <enable>

AT#CFLO=?

Test command returns the range of supported values for parameter <enable>

AT#SKIPESC - Skip Escape Sequence

This command enables/disables skipping the escape sequence.

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

AT#SKIPESC=[<mode>]

Set command enables/disables skipping the escape sequence (+++) while transmitting during a data connection.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enable/disable skipping the escape sequence (+++)

Values:

0 : does not skip the escape sequence; its transmission is enabled.

1 : skips the escape sequence; its transmission is not enabled.

In FTP connection the escape sequence is not transmitted, regardless of the command setting.

AT#SKIPESC?

Read command returns the current value of the parameter <mode> in the format:

#SKIPESC: <mode>

AT#SKIPESC=?

Test command returns the supported values of parameter <mode>.

ATX - Extended Result Codes

This command allows to select the subset of result code messages.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

ATX[<n>]

Set command selects the subset of result code messages the modem uses to reply to the DTE upon AT commands execution.

Parameter:

Name	Type	Default	Description
<n>	integer	0	configuration value

Values:

0 : when entering in dial mode a CONNECT result code is relayed; see Additional info.

Additional info:

<n>=0
OK, CONNECT, RING, NO CARRIER, ERROR, NO ANSWER result codes are enabled. Dial tone and busy detection (NO DIALTONE and BUSY result codes) are disabled.

When the <n> parameter is omitted, the command acts like ATX0.

Current value is returned by AT&V.

Parameter:

<n>

0 - EXTENDED MESSAGES : X0=NO

1..4 - EXTENDED MESSAGES : X1=YES

For complete control on CONNECT response message see also +DR command.

Call (Voice and Data) Control

ATD - Dialup Connection

This command establishes a GPRS communication between the TE and the external PDN.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

ATD

The ATD execution command assumes the format shown in Additional info section. It causes the MT to perform whatever actions are necessary to establish communication between the TE and the external PDN.

Additional info:

- ATD*<GPRS_SC>[*[<called_address>][*[<L2P>][*[<cid>]]]]#

Name	Type	Default	Description
<GPRS_SC>	integer	N/A	is the GPRS Service Code, which identifies a request to use the GPRS communication

Value:

99 : GPRS Service Code

<called_address>	string	-	identifies the called party in the address space applicable to the PDP. Not supported
<L2P>	string	-	indicates the layer 2 protocol to be used. Only 1 supported.
<cid>	integer	-	cid range

On UART, DialUp (ATD*99#) +++ timings differs from AT#SD (and similar AT commands) +++ timings.

DialUp (ATD*99#) +++ timings:

+++ escape sequence must be preceded and followed by a pause of 1000 ms

+ characters must be entered in very quick succession, the time between each + must be less than (1000 / BAUDRATE) * 32 ms, e.g. for a BAUDRATE of 115200 bps it must be less than 0.27 ms

AT#SD (and similar AT commands) +++ timings:

+++ escape sequence must be preceded and followed by a pause of 1000 ms

+ characters must be entered in quick succession, the time between each + must be less then 1000 ms

ATH - Disconnect the Current Dial-up

This execution command hangs up/disconnects the current dial-up.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATH

When a dial-up is active the device is in on-line mode hence, to execute ATH command the device must be previously turned in command mode using the escape sequence or, if &D1 option is active, tying Low the DTR pin.

ATO - Return to ON-Line Mode

This execution command is used, during a suspended data conversation, to return in on-line mode from command mode. If there is no suspended conversation, it returns NO CARRIER.

ITU-T Recommendation V. 25 ter

SIM Presence	Setting saved	Can be aborted
Required	No	No

ATO

After issuing ATO command, the device returns in on-line mode and returns CONNECT.

To enter again command-mode you must issue the escape sequence, see register S2.

Network

AT+CNUM - Subscriber Number

Execution command returns the MSISDN (if the phone number of the device has been stored in the SIM card)

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CNUM

Execution command returns the MSISDN (if the phone number of the device has been stored in the SIM card) using the following format:

+CNUM: [<alpha>,]<number>,<type>[<CR><LF>
+CNUM: [<alpha>,]<number>,<type>[...]]

The parameters are described in the Additional info section.

Additional info:

List of the parameters meaning.

Name	Type	Default	Description
<alpha>	string	-	alphanumeric string associated to <number>; the character set depends on the value set with +CSCS.

<number>	string	-	numeric string containing the phone number in the format <type>.
<type>	integer	N/A	type of number.

Values:

129 : Unknown numbering scheme

145 : International numbering scheme (contains the character "+")

161 : National numbering scheme

The maximum number of reported MSISDN is 4, regardless the total number of phone numbers written in the SIM card.

AT+CNUM=?

Test command returns the OK result code.

AT+COPN - Read Operator Names

This command read operator names.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+COPN

Execution command returns the list of operator names from the ME in the format:

+COPN: <numeric1>,<alpha1>[<CR>

+COPN: <numeric2>,<alpha2>[...]]

The parameters are described in the Additional info section.

Additional info:

List of the parameters meaning.

Name	Type	Default	Description
<numericn>	string	-	operator in numeric format, see +COPS.
<alphan>	string	-	operator in long alphanumeric format, see +COPS.

Each operator code <numericn> that has an alphanumeric equivalent <alphan> in the ME memory is returned.

AT+COPN=?

Test command returns the OK result code.

AT+CREG - Network Registration Status

The command enables/disables the network registration unsolicited result code (URC) and selects its presentation format.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+CREG=[<mode>]

Set command enables/disables the network registration unsolicited result code and selects one of the three available formats:

short format: +CREG: <stat>
long format: +CREG: <stat>[,<lac>,<ci>[,<AcT>]]
long format 3: +CREG: <stat>[,[<lac>],[<ci>],[<AcT>][,<cause_type>,<reject_cause>]]

The parameter meanings are shown in Unsolicited code value section.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enables/disables the network registration unsolicited result code and selects one of the three formats: short, long format or long format 3. URC short format is displayed every time there is a change in the network registration status. URC long format and long format 3 are displayed every time there is a change of

the network cell and reject cause is available.

Values:

0 : disable the network registration unsolicited result code

1 : enable the network registration unsolicited result code, and selects the short format

2 : enable the network registration unsolicited result code, and selects the long format (includes the network cell identification data)

3 : enable the network registration unsolicited result code, and select the long format 3 (includes the cause value)

Unsolicited fields:

Name	Type	Description
<stat>	integer	network registration status of the module Values: 0 : not registered, terminal is not currently searching a new operator to register to 1 : registered, home network

		2 : not registered, but terminal is currently searching a new operator to register to 3 : registration denied 4 : unknown 5 : registered, roaming
<lac>	string	the parameter reports: Tracking Area Code when <AcT>=7
<ci>	string	Cell Id for the currently registered on cell
<AcT>	integer	access technology of the registered network Values: 7 : E-UTRAN
<cause_type>	integer	reject cause type Values: 0 : indicates that <reject_cause> contains an MM cause value 1 : indicates that <reject_cause> contains a manufacturer-specific cause
<reject_cause>	integer	cause of the failed registration

<lac>, <ci> and <AcT> network information is reported by URC only if <mode>=2 or <mode>=3 and the module is registered on some network cell.

AT+CREG?

Read command returns the current value of <mode>, the registration status <stat>, and the network information (<lac>, <ci> and <AcT>) according to the used <mode> parameter value.

+CREG: <mode>,<stat>[,<lac>,<ci>[,<AcT>][,<cause_type>,reject_cause>]]

<lac>, <ci>, and <AcT> network information is reported only if <mode>=2 or <mode>=3 and the module is registered on some network cell.

<cause_type>,<reject_cause> information is reported only if <mode>=3 and reject cause is available.

Please note that using +CREG to check the registration in LTE is not the best approach.

+CEREG, which is the specific command for LTE, should be used.

AT+CREG=?

Test command returns supported values for parameter <mode>.

Check the registration status of the module.

AT+CREG?

+CREG: 0,2

OK

The module is in network searching state

...

...

Check again module status
AT+CREG?

+CREG: 0,1

OK

The module is registered

+CREG refers to CS domain, so if the UE got a COMBINED ATTACH in LTE, +CREG will show the device is registered (the UE is registered for PS and CS domain, so +CEREG and +CREG show the same value 1 or 5).

If UE requested a COMBINED ATTACH, but it gets EPS only ATTACH accepted, while the CS domain is rejected, this is the case where +CEREG returns 1 or 5, while +CREG returns 0 or 3 (depending on the reject cause code for CS domain).

A good example is in packet only LTE networks: a COMBINED ATTACH attempt here is accepted for EPS only, while the CS domain is rejected with cause #18 (CS domain not available), which is a permanent reject cause. In this case the device is registered for PS only domain, so +CEREG shows 1, while +CREG shows 0 (because cause #18 is a permanent cause).

In such a condition, you can exchange data but you cannot send SMS over NAS.

AT+COPS - Operator Selection

The command selects a network operator and registers the module.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+COPS=[<mode>[,<format>[,<oper>[,<act>]]]]

The set command attempts to select a network operator and registers the module on the just chosen operator; the selection can be automatic or manual.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	defines the operator selection: automatic or manual.

Values:

0 : automatic selection, the parameter <oper> is ignored

1 : manual selection, the parameter <oper> must be present

2 : deregister from network. The module is unregistered until a +COPS with <mode>=0, 1 or 4 is issued

3 : set only <format> parameter, the parameter <oper> is ignored

4 : manual/automatic, <oper> parameter must be present. If manual selection fails, the module will try automatic mode (<mode>=0)

<format>	integer	0	specifies the operator name format, see <oper> parameter

Values:

0 : alphanumeric long form (max length 16 digits)

1 : alphanumeric short form

2 : numeric 5 or 6 digits [country code (3) + network code (2 or 3)]

<oper>	mixed	-	network operator in format defined by <format> parameter
<act>	integer	N/A	selects access technology.

Values:

7 : E-UTRAN

<mode> parameter setting is stored in NVM and available at next reboot. <mode>=3 is not saved.

If <mode>=1 or 4, the selected network is stored in NVM too and it is available at next reboot (this will happen also after inserting another SIM).

- <format> parameter setting is never stored in NVM.

- If AT+COPS=0 is issued after the switch-on, it causes a new attempt to select a network operator and registers the module on the selected operator.

AT+COPS?

Read command returns current value of <mode>, <format>, <oper> and <AcT> in format <format>. If no operator is selected, <format>, <oper> and <AcT> are omitted.

+COPS: <mode>[, <format>, <oper>,< act>]

If the module is deregistered, <format>, <oper>, and <act> parameters are omitted.

AT+COPS=?

Test command returns a list of quintuplets, each representing an operator present in the network. The quintuplets list is ended with the range values of the <mode> and <formats> parameters.

The quintuplets in the list are closed between round brackets, separated by commas, the <oper> parameter is returned in both formats.

+COPS: [quintuplets list (<stat>,long alphanumeric <oper>,shortalphanumeric <oper>,numeric <oper>,<act>),

(<stat>,long alphanumeric <oper>,short alphanumeric <oper>,numeric <oper>,<act>), ..]

[,,(range of <mode>),(range of <format>)]

<stat> parameter is described in the Additional info section.

Additional info:

▶▶ Meaning of the <stat> parameter.

Name	Type	Default	Description
<stat>	integer	N/A	operator availability

Values:

0	:	unknown
1	:	available
2	:	current
3	:	forbidden

Since with this command a network scan is done, this command may require some seconds before the output is given.

AT+CLCK - Facility Lock/Unlock

This command is used to lock or unlock a ME on a network facility.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CLCK=<fac>,<mode>[,<password>]

Set command is used to lock, unlock or interrogate a modem or a network lock facility. Password is normally needed to do such actions

Parameters:

Name	Type	Default	Description
<fac>	string	N/A	facility to lock, unlock or interrogate

Values:

"SC" : SIM (PIN request) (device asks SIM password at power-up and when this lock command issued)

"PN" : Network Personalization

Not supported

"PU" : network sUbset Personalization

Not supported

"PS" : PS lock

Not supported

<mode>	integer	N/A	defines the operation to be done on the facility
	Values:		
	0	:	unlock facility
	1	:	lock facility
	2	:	query status
<password>	string	-	shall be the same as password specified for the facility from the DTE user interface or with command Change Password +CPWD

Additional info:

▶▶ When <mode>=2 and command is successful, it returns:

+CLCK: <status>

Name	Type	Default	Description
<status>	integer	N/A	the current status of the facility
	Values:		
	0	:	not active
	1	:	active

AT+CLCK=?

Test command reports all the facilities supported by the device.

AT+CPWD - Change Facility Password

This command changes the password for the facility lock function defined by command Facility Lock +CLCK command.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Other	Yes

AT+CPWD=<fac>,<oldpwd>,<newpwd>

Execution command changes the password for the facility lock function defined by command Facility Lock +CLCK command.

Parameters:

Name	Type	Default	Description
<fac>	string	N/A	facility lock function. Values: "SC" : SIM (PIN request)
<oldpwd>	string	it shall be the same as password specified for the facility from the ME user interface or with command +CPWD.	

<newpwd>	string	-	new password.

Parameter <oldpwd> is the old password while <newpwd> is the new one.

The password can be changed only if PIN request is enabled (see +CLCK command).

AT+CPWD=?

Test command returns a list of pairs (<fac>,<pwdlength>) which represents the available facilities and the maximum length of their password (<pwdlength>).

MAX timeout indicates the time within which the command must be executed. MAX timeout for this command is 180 s.

AT+CPOL - Preferred Operator List

The command is used to edit or update the UICC preferred list of networks. The list is read in the UICC file selected by the command +CPLS.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CPOL=[<index>] [,<format>[,<oper>[,<GSM_AcT>,<GSM_Compact_AcT>,<UTRAN_AcT>,<E_UTRAN_AcTn>]]]

Execution command writes an entry in the UICC list of preferred operators.

Parameters:

Name	Type	Default	Description
<index>	integer	N/A	the order number of operator in the UICC preferred operator list.
	Value:		
	1÷n	:	order number in the list
<format>	integer	2	format for <oper> parameter.
	Value:		

	0 : long format alphanumeric <oper>. Not supported 1 : short format alphanumeric <oper>. Not supported 2 : numeric. Only 2 is allowed up to now		
<oper>	string	-	Operator Identifier.
<GSM_AcT>	integer	N/A	GSM access technology. Values: 0 : access technology not selected 1 : access technology selected
<GSM_Compact_AcT>	integer	N/A	GSM compact access technology. Currently the parameter is not supported but set value is accepted. Values: 0 : access technology not selected 1 : access technology selected
<UTRAN_AcT>	integer	N/A	UTRAN access technology. Currently the parameter is

			not supported but set value is accepted.
	Values: 0 : access technology not selected 1 : access technology selected		
<E_UTRAN_AcTn>	integer	N/A	E-UTRAN access technology
	Values: 0 : access technology not selected 1 : access technology selected		

- If <index> is used, and <oper> is not entered, the entry is deleted from the list of preferred operators.
- If <oper> is used, and <index> is not used, <oper> is put in the next free location.
- If only <format> is entered, the format of the <oper> in the read command is changed.
- If <index> is used, and <oper> is used, <oper> is put in the <index> location, replacing the existing entry.

AT+CPOL?

Read command returns all used entries from the UICC list of preferred operators.

AT+CPOL=?

Test command returns the <index> range supported by the UICC and the range for the <format> parameter.

Entry 3 in the preferred list of the operators is deleted.

AT+CPOL=3
OK

Operator identifier 22603 is inserted in the next free location of the list.
AT+CPOL=,2,22603
OK

Format of <oper> in the read command is changed (only 2 is allowed up to now).
AT+CPOL=,2
OK

Operator Identifier 22603 is inserted in the 4th position of the list.
AT+CPOL=4,2,22603
OK

Available range for <index> is 1 to 20, for <format>= 2.
AT+CPOL=?
+CPOL: (1-20),(2)

AT+CPLS - Selection of Preferred PLMN List

The command is used to select a list of preferred PLMNs in the SIM/USIM card.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CPLS=[<list>]

Set command select one PLMN selector with Access Technology list in the SIM card or active application in the UICC (GSM or USIM), that is used by +CPOL command.

Parameter:

Name	Type	Default	Description
<list>	integer	0	PLMNs list selector

Values:

0 : User controlled PLMN selector with Access Technology EFPLMNwAcT, if not found in the SIM/UICC then PLMN preferred list EFPLMNsel (this file is only available in SIM card or GSM application selected in UICC)

1 : Operator controlled PLMN selector with Access Technology EFOPLMNwAcT

2 : HPLMN selector with Access Technology EFHPLMNwAcT

The value set by command is directly stored in NVM and does not depend on the specific CMUX instance.

AT+CPLS?

Read command returns the selected PLMN selector <list> from the SIM/USIM.

AT+CPLS=?

Test command returns the whole index range supported <list>s by the SIM/USIM.

AT+CSQ - Signal Quality

Execution command returns received signal strength indication <rssi> and channel bit error rate <ber> from the MT.

[1] 3GPP TS 27.007
[2] 3GPP TS 07.07
[3] 3GPP TS 25.133

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSQ

See Additional info sections.

Additional info:

2G Networks

The execution command returns the following message:

+CSQ: <rssi>,<ber>

Name	Type	Default	Description
<rssi>	integer	N/A	measurements of the radio signal power, expressed in dBm, are mapped to <rssi> as shown below

Values:

0 : -113 dBm or less

1 : -111 dBm

2÷30	:	-109 dBm ... -53 dBm; 2 dBm per step
31	:	-51 dBm or greater
99	:	not known or not detectable

<ber>	integer	N/A	measurements of the channel bit error rate, expressed in %, are mapped to <ber> as shown below

Values:

0	:	less than 0.2%
1	:	0.2% to 0.4%
2	:	0.4% to 0.8%
3	:	0.8% to 1.6%
4	:	1.6% to 3.2%
5	:	3.2% to 6.4%
6	:	6.4% to 12.8%
7	:	more than 12.8%
99	:	not known or not detectable

▶▶ 4G Networks

The execution command returns the following message:

+CSQ: <rssi>,<rsrq>

Name	Type	Default	Description
<rssi>	integer	N/A	Received Signal Strength Indication. For <rssi> to be compliant with 3GPP TS27.007 specification, levels are mapped to range 0...31.

Values:

0	:	-113 dBm or less
1	:	-111 dBm
2÷30	:	-109...-53 dBm
31	:	- 51 dBm or greater
99	:	not known or not detectable

<rsrq>	integer	N/A	Reference Signal Received Quality. For <rsrq> levels are mapped to range 0...7.

Values:

0	:	-4...-3 dB
1	:	-6...-5 dB

2	:	-8...-7 dB
3	:	-10...-9 dB
4	:	-13...-11 dB
5	:	-15...-14 dB
6	:	-17...-16 dB
7	:	-19...-18 dB
99	:	not known or not detectable

AT+CSQ=?

Test command returns values supported as compound values.

AT#FPLMN - Periodical FPLMN Cleaning

Periodically delete the Forbidden PLMN list stored inside the SIM card, clear it or list it.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#FPLMN=<action>[,<period>]

The set command is used to manage the Forbidden PLMN List file (FPLMN) stored in the SIM card.

Parameters:

Name	Type	Default	Description
<action>	integer	0	kind of action for FPLMN file

Values:

0 : disable periodic FPLMN cleaning

1 : enable periodic FPLMN cleaning with period <period> (not supported)

2 : clear FPLMN file contents (one shot)

3 : list contents of FPLMN file (not supported)

<period>	integer	60	interval in minutes for FPLMN clearing

Value:

1÷60 : interval in minutes

AT#FPLMN?

Read command reports whether the periodic deletion is currently enabled or not, and the deletion period, in the format:

#FPLMN: <action>,<period>

AT#FPLMN=?

Test command reports available values for parameters <action> and <period>

AT#BND - Select Band

This command selects RF bands

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#BND=<band>[,<UMTS_band>[,<LTE_band>[,<TDSCDMA_band>[,<LTE_band_over_64>]]]]

Set command selects the current LTE bands.

Parameters:

Name	Type	Default	Description
<band>	integer	0	this parameter is used for backward compatibility Value: 0 : dummy parameter
<UMTS_band>	integer	0	this parameter is used for backward compatibility. Value: 0 : dummy parameter
<LTE_band>	integer	N/A	indicates the LTE supported bands expressed as the sum of Band number (1+2+4+8+16+ ...) calculated

as shown in the table (mask of 64 bits):

Band number	Band i
1	B1
2	B2
4	B3
8	B4
16	B5
128	B8
...	...
2147483648	B32
4294967296	B33
8589934592	B34
...	...
549755813888	B40
1099511627776	B41

Value:

1÷max	:	sum of Band number (1+2+4+8+16+ ...) max value depends on product variant

<TDSCDMA_band> integer 0 this parameter is used for backward compatibility

Value:

0	:	dummy parameter

<LTE_band_over_64> integer 0 indicates the LTE supported bands over 64 expressed as the sum of Band number (1+2+4+8+16+ ...) calculated as shown in the table (mask of 32 bits):

Band number	Band i
1	B65
2	B66

Values:

1÷max	:	0 if no band over 64 supported or sum of Band number (1+2+4+8+16+ ...)

max value depends on product variant

The command needs a reboot to be effective.

AT#BND?

Read command returns the current selected bands in the format:

#BND: <band>,<UMTS_band>,<LTE_band>,< TDSCDMA_band>,< LTE_band_over_64>

?

AT#BND=?

Test command returns the supported range of values of parameters <band>, <UMTS_band>,<LTE_band>,< TDSCDMA_band> and < LTE_band_over_64>.

```
Test command
AT#BND=?
#BND: (0),(0),(0,1-1649267441813),(0),(0)
OK

Read command
AT#BND?
#BND: 0,0,1649267441813,0,0
OK
```

AT#CEERNET - Extended Numeric Error Report for Network Reject Cause

The command is related to extended numeric error report.

3GPP TS 24.008
3GPP TS 24.301

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CEERNET

Execution command causes the TA to return a numeric code in the intermediate response format:

#CEERNET: <code>

which should offer the user of the TA a report for the last mobility management (MMGMM/EMM) or session management (SM/ESM) procedure not accepted by the network.

Additional info:

The following error codes are valid for mobility management (MM/GMM) or session management (SM), i.e. for 2G and 3G networks.

In 4G network the <code>s meanings are included in tables 9.9.4.4.1 (for ESM causes) and 9.9.3.9.1 (for EMM cause) of 3GPP TS 24.301 Release 9.

Name	Type	Default	Description
<code>	integer	N/A	error code

Values:

2	:	IMSI unknown in HLR
3	:	illegal MS
4	:	IMSI unknown in VISITOR LR
5	:	IMEI not accepted
6	:	illegal ME
7	:	GPRS not allowed
8	:	Operator determined barring (SM cause failure)/ GPRS and not GPRS not allowed (GMM cause failure)
9	:	MS identity cannot be derived by network
10	:	implicitly detached
11	:	PLMN not allowed
12	:	LA not allowed
13	:	roaming not allowed
14	:	GPRS not allowed in this PLMN

15 : no suitable cells in LA

16 : MSC TEMP not reachable

17 : network failure

20 : MAC failure

21 : SYNCH failure

22 : congestion

23 : GSM authentication unacceptable

24 : MBMS bearer capabilities insufficient for the service

25 : LLC or SNDCP failure

26 : insufficient resources

27 : missing or unknown APN

28 : unknown PDP address or PDP type

29 : user authentication failed

30 : activation rejected by GGSN

31 : activation rejected unspecified

32 : service option not supported

33 : req. service option not subscribed

34 : serv. option temporarily out of order

35 : NSAPI already used

36 : regular deactivation

37 : QOS not accepted

38 : call cannot be identified (MM cause failure) / SMN network failure (SM cause failure)

39 : reactivation required

40 : no PDP context activated (GMM cause failure) / feature not supported (SM cause failure)

41 : semantic error in TFT operation

42 : syntactical error in TFT operation

43 : unknown PDP context

44 : semantic err in PKT filter

45 : syntactical err in PKT filter

46 : PDP context without TFT activated

47 : multicast group membership timeout

48 : retry on new cell begin (if MM cause failure) / activation rejected BCM violation (if SM cause failure)

50 : PDP type IPV4 only allowed

51 : PDP type IPV6 only allowed

52 : single address bearers only allowed

63 : retry on new cell end

81 : invalid transaction identifier

95 : semantically incorrect message

96 : invalid mandatory information

97 : MSG type non-existent or not implemented

98 : MSG type not compatible with protocol state

99 : IE non-existent or not implemented

100 : conditional IE error

101 : MSG not compatible with protocol state

111 : protocol error unspecified

112 : APN restriction value incompatible with active PDP context

Telit Cinterion recommends that the host controlling the modem defines the proper retry/reboot scheme for reject causes 2, 7, 11, 14, 30, 33, 34, 38.

AT#CEERNET=?

Test command returns OK result code.

AT#CEERNETEXT - Extended Error Report for Network Reject Cause

This command is both a set and an execution command.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CEERNETEXT[=[<func>]]

Set command enables/disables the URC presentation or delates the last network information. The execution command (AT#CEERNETEXT<CR><LF>) gets the last reject error information from the network and returns the following message:

#CEERNETEXT: <code>,<AcT>,<MCC>,<MNC>

If no error information is present, the execution command returns OK

When URC is enabled, it will occur every time a mobility management (MM/GMM/EMM) or session management (SM/ESM) procedure is not accepted by the network.

The URC message is equal to the message returned by the execution command.

Parameter:

Name	Type	Default	Description
<func>	integer	0	enable/disable the URC or delete the last network info

Values:

0 : disable the #CEERNETEXT URC

1 : enable the #CEERNETEXT URC

2 : delete last info of <code>, <AcT>, <MCC> and <MNC>

Unsolicited fields:

Name	Type	Description
<code>	integer	last numeric Network Reject Cause from network, see <code> in #CEERNET
<AcT>	integer	access technology of the registered network Values: 7 : LTE
<MCC>	string	Mobile Country Code of the used network when last numeric code was received
<MNC>	string	Mobile Network Code of the used network when last numeric code was received

AT#CEERNETEXT?

The read command returns the current value of parameter <func> in the format:

#CEERNETEXT: <func>

Additional info:

Parameters returned by the read command.

Name	Type	Default	Description
<func>	integer	N/A	can assume the following values:

Values:

0 : #CEERNETEXT URC is disabled

1 : #CEERNETEXT URC is enabled

AT#CEERNETEXT=?

Test command reports the supported range of values for the <func> parameter.

AT+CEMODE - Set Mode of Operation for EPS

This set command configures the mode of operation for EPS.

[1] 3GPP TS 24.301

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CEMODE=<mode>

Parameter:

Name	Type	Default	Description
<mode>	integer	N/A	mode of operation. The default value depends on product and the support of VoLTE. UE modes of operation can be found in standard [1]. Other values are reserved and will result in an ERROR response to the set command.

Value:

0 : PS mode 2 of operation

2 : CS/PS mode 2 of operation

AT+CEMODE?

Read command returns the current value of parameter <mode> in the format:

+CEMODE: < mode >

AT+CEMODE=?

Test command returns the supported range of values of parameters <mode>.

AT+CESQ - Extended Signal Quality

Execution command returns received signal quality parameters according to the network on which the module is registered.

[1] 3GPP TS 27.007
[2] 3GPP TS 45.008
[3] 3GPP TS 25.133
[4] 3GPP TS 36.133

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CESQ

See Additional info section for networks on which the module can be registered.

Additional info:

The execution command returns the following message. Its format depends on the network on which the module is registered.

2G Networks

+CESQ: <rxlev>,<ber>,255,255,255,255

LTE Networks

+CESQ: 99,99,255,255,<rsrq>,<rsrp>

Name	Type	Default	Description
<rxlev>	integer	N/A	received signal strength level, see 3GPP TS 45.008 subclause 8.1.4.

Values:

0	:	rssi < -110 dBm
1	:	-110 dBm ≤ rssi < -109 dBm
2	:	-109 dBm ≤ rssi < -108 dBm
...	:	...
61	:	- 50 dBm ≤ rssi < - 49 dBm
62	:	- 49 dBm ≤ rssi < -48 dBm
63	:	- 48 dBm ≤ rssi
99	:	not known or not detectable or if the current serving cell is not a GERAN cell

<ber>	integer	N/A	channel bit error rate.

Values:

0÷7	:	as RXQUAL values, see 3GPP TS 45.008 subclause 8.2.4
99	:	not known or not detectable or if the current serving cell is not a GERAN cell

<rsrq>	integer	N/A	reference signal received quality, see 3GPP TS 36.133 subclause 9.1.7.

Values:

0	:	rsrq < -19.5 dB
1	:	-19.5 dB ≤ rsrq < -19.0 dB
2	:	-19.0 dB ≤ rsrq < -18.5 dB
...	:	...
32	:	-4 dB ≤ rsrq < -3.5 dB
33	:	-3.5 dB ≤ rsrq < -3 dB
34	:	-3 dB ≤ rsrq
255	:	not known or not detectable or if the current serving cell is not a E-UTRA cell

<rsrp>	integer	N/A	reference signal received power, see 3GPP TS 36.133 subclause 9.1.4.

Values:

0	:	rsrp < -140 dBm
1	:	-140 dBm ≤ rsrp < -139 dBm
2	:	-139 dBm ≤ rsrp < -138 dBm
...	:	...
95	:	-46 dBm ≤ rsrp < -45 dBm

96	:	-45 dBm ≤ rsrp < -44 dBm
97	:	-44 dBm ≤ rsrp
255	:	not known or not detectable or if the current serving cell is not a E-UTRA cell

AT+CESQ=?

Test command returns values supported as compound values.

AT+WS46 - PCCA STD-101 Select Wireless Network

This command selects the cellular network (Wireless Data Service, WDS).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+WS46=[<n>]

Set command selects the cellular network (Wireless Data Service, WDS) to operate with the TA (WDS-Side Stack Selection).

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	WDS-Side Stack to be used by the TA. The factory default value depends on each variant

Values:

28	:	E-UTRAN only

<n> parameter setting is stored in NVM, and available at next reboot.

AT+WS46?

Read command reports the currently selected cellular network, in the format:

+ WS46: <n>

AT+WS46=?

Test command reports the range for the parameter <n>.

AT+CEDRXS - eDRX Setting

This command controls the setting of the UEs eDRX parameters.

3GPP TS 27.007
3GPP TS 24.008

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CEDRXS=[<mode>[,<AcTtype>[,<Req_eDRX>]]]

Set command controls the setting of the UEs eDRX parameters. The command controls whether the UE wants to apply eDRX or not, as well as the requested eDRX value for each specified type of access technology.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	disable or enable the use of eDRX in the UE. This parameter is applicable to all specified types of access technology, i.e. the most recent setting of <mode> will take effect for all specified values of <AcTtype>.

Values:

0 : disable the use of eDRX

1 : enable the use of eDRX

2 : enable the use of eDRX and enable the unsolicited result code, see Additional info.

	3 :	disable the use of eDRX and discard all parameters for eDRX or, if available, reset to the manufacturer specific default values	
<AcTtype>	integer	N/A	type of access technology. Values: 0 : Access technology is not using eDRX. This parameter value is only used in the unsolicited result code, it cannot be used in the set command. 4 : E-UTRAN (WB-S1 mode)
<Req_eDRX>	string	-	half a byte in a 4-bit format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see subclause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008, Table 10.5.5.32/3GPP TS 24.008.

Additional info:

▶▶ If <mode>=2 and there is a change in the eDRX parameters provided by the network, the unsolicited result code reports:

+CEDRXS: <AcTtype>[,<Req_eDRX>[,<NW_prove_DRX>[,<PagTimeWindow>]]]

Name	Type	Default	Description
<NW_prove_DRX>	string	-	half a byte in a 4-bit format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see subclause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32/3GPP TS 24.008.

<PagTimeWindow>	string	-	half a byte in a 4-bit format. The paging time window refers to bit 8 to 5 of octet 3 of the Extended DRX parameters information element (see subclause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see the Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32/3GPP TS 24.008.

AT+CEDRXS?

Read command returns the current settings for each defined value of <AcT-type> in the format:

+CEDRXS: <AcTtype>,<Req_eDRX>

AT+CEDRXS=?

Test command returns the supported <mode>s and the value ranges for the access technology and the requested eDRX value as compound values.

Read command

AT+CEDRXS?

+CEDRXS: 4,"0011"

OK

Test command

AT+CEDRXS=?

+CEDRXS: (0-3),(0,4),("0001"-"1111")

OK

AT+CEDRXRDP - eDRX Read Dynamic Parameters

This command returns a message related to Extended Discontinuous Reception (eDRX).

3GPP TS 27.007
3GPP TS 24.008

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CEDRXRDP

Execution command returns the following message if eDRX is used for the cell that the MS is currently registered to, in the format:

+CEDRXRDP:<AcTtype>[,<Req_eDRX>[,<NW_prov_eDRX>[,<PagTimeWindow>]]]

If the cell that the MS is currently registered to is not using eDRX, <AcTtype>=0 is returned.

Additional info:

Here is the list of the meanings of the parameter returned by the +CEDRXRDP command.

Name	Type	Default	Description
<AcTtype>	integer	0	type of access technology.

Values:

0	:	access technology is not using eDRX
4	:	see +CEDRXS.

<Req_eDRX>	string	-	see +CEDRXS.
<NW_prov_eDRX>	string	-	see +CEDRXS.
<PagTimeWindow>	string	-	see +CEDRXS.

AT+CEDRXRDP=?

Test command returns OK result code.

AT+CEREG - EPS Network Registration Status

This command monitors the Evolved Packet System (EPS) network registration status in LTE.

[1] 3GPP TS 24.008
[2] 3GPP TS 24.301
[3] 3GPP TS 25.331

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+CEREG=[<mode>]

Set command enables/disables the EPS network registration unsolicited result code (URC) in LTE, and selects one of the available formats (see Parameter description); example:

short format: +CEREG: <stat>
long format: +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>]]
long format 3: +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>] [,<cause_type>,<reject_cause>]]
long format 4: +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>][,<Active-Time>,<Periodic-TAU>]]
long format 5: +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>][,[<cause_type>], [<reject_cause>][,[<Active-Time>],[<Periodic-TAU>]]]]

<tac>, <ci>, and <AcT> are reported by the command only if available.

In case of error, possible response(s): +CME ERROR: <err>

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enables/disables the network registration unsolicited result code (URC), and selects one of the available formats. The following events triggers the URC: URC short format is displayed every time there is a change in the EPS network registration status URC long format is displayed every time there is a change of network cell in LTE

Values:

0 : disable the network registration unsolicited result code

1 : enable network registration unsolicited result code +CEREG: <stat>

2 : enable network registration and location information unsolicited result code

+CEREG: <stat>[,[<tac>],[<ci>],[<AcT>]]

3 : enable network registration, location information and EMM cause value information unsolicited result code

+CEREG: <stat>[,[<tac>],[<ci>],[<AcT>] [,<cause_type>,<reject_cause>]]

4 : For a UE that wants to apply PSM, enable network registration and location information unsolicited result code

+CEREG: <stat>[,[<tac>],[<ci>],[<AcT>][,,[,[<Active-Time>],[<Periodic-TAU>]]]]

5 : For a UE that wants to apply PSM, enable network registration, location information and EMM cause value information unsolicited result code

+CEREG: <stat>[,[<tac>],[<ci>],[<AcT>][,[<cause_type>],[<reject_cause>][,[<Active-Time>],[<Periodic-TAU>]]]]

Unsolicited fields:

Name	Type	Description
<stat>	integer	EPS registration status Values: 0 : not registered, terminal is not currently searching a new operator to register to 1 : registered, home network 2 : not registered, but terminal is currently searching a new operator to register to 3 : registration denied 4 : out of coverage 5 : registered, roaming

<tac>	string	tracking area code (two bytes) in hexadecimal format (e.g. "00C3" equals 195 in decimal)
<ci>	string	LTE cell ID (four bytes) in hexadecimal format
<AcT>	integer	indicates the access technology of the serving cell. Values: 7 : E-UTRAN
<cause_type>	integer	reject cause type Values: 0 : indicates that <reject_cause> contains an EMM cause value 1 : indicates that <reject_cause> contains a manufacturer-specific cause
<reject_cause>	integer	cause of the failed registration
<Active-Time>	byte	one byte in an 8 bit format, indicates the Active Time value (T3324) allocated to the UE in E-UTRAN. See 24.008
<Periodic-TAU>	byte	one byte in an 8 bit format, indicates the extended periodic TAU value (T3412) allocated to the UE in E-UTRAN. See 24.008

AT+CEREG?

Read command returns the current value of <mode>, the registration status <stat>, and the information <tac>, <ci>, <AcT>, <cause_type>, <reject_cause>, <Active-Time> and <Periodic-TAU> according to the current <mode> parameter value.

+CEREG: <mode>,<stat>[,[<tac>],[<ci>],[<AcT>] [,[<cause_type>],[<reject_cause>][,[<Active-Time>],[<Periodic-TAU>]]]]

Please note that +CEREG is the specific command to use to check for LTE registration.

See also +CREG for a more detailed explanation of the difference between the two commands

AT+CEREG=?

Test command returns supported values for parameter <mode>.

AT#RFSTS - Read Current Network Status

Command reads current network status.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#RFSTS

Execution command returns the current network status. The format of the returned message is according to the network on which the module is registered.

LTE network

#RFSTS:<PLMN>,<EARFCN>,<RSRP>,<RSSI>,<RSRQ>,<TAC>,<RAC>,
[<TXPWR>],<DRX>, <MM>,<RRC>,<CID>,<IMSI>,
[<NetNameAsc>],<SD>,<ABND>,<T3402>,<T3412>,<SINR>

Parameters	Description
<PLMN>	Country code and operator code (MCC, MNC)
<EARFCN>	E-UTRA Assigned Radio Channel
<RSRP>	Reference Signal Received Power
<RSSI>	Received Signal Strength Indication
<RSRQ>	Reference Signal Received Quality
<TAC>	Tracking Area Code
<RAC>	Routing Area Code (dummy)

<TXPWR>	Tx Power (In traffic only)
<DRX>	Discontinuous reception cycle Length (cycle length in ms)

Parameter/values	Description
<MM>	Mobility Management state (dummy)

Parameters	Description
<RRC>	Radio Resource state
<CID>	Cell ID

Parameter/values	Description
<IMSI>	International Mobile Station ID
<NetNameAsc>	Operator name, quoted string type
<SD>	Service Domain
0	No Service
1	CS only
2	PS only
3	CS+PS

Parameters/values	Description
<ABND>	Active Band
1..63	According to 3GPP TS 36.101

Parameters	Description
<T3402>	Timer T3402 in seconds (dummy)
<T3412>	Timer T3412 in seconds (dummy)
<SINR>	Signal-to-Interface plus Noise Ratio

AT#RFSTS=?

Test command tests for command existence.

AT#MONI - Cell Monitor

This command is both a set and an execution command.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MONI[=[<number>]]

Set command sets the cells from which extract LTE related information.

After entering the set command, enter the execution command AT#MONI<CR> to get the LTE related information for the selected cell or cells in the format shown in the Additional info section.

Parameter:

Name	Type	Default	Description
<number>	integer	-	the parameter meaning depends on the network, see Additional info section.

Additional info:

▶▶ LTE network

Name	Type	Default	Description
<number>	integer	0	LTE network
	Value:		

0	:	it is the serving cell

Execution command AT#MONI<CR> reports LTE related information for selected cell and dedicated channel (if exists) in the following formats:

a) When extracting data for the serving cell and the network name is known the format is:

LTE network

#MONI: <netmame> RSRP:<rsrp> RSRQ:<rsrq> TAC:<tac> Id:<id> EARFCN:<earfcn> PWR:<dBm> DRX:<drx> PCI:< physicalCellId >

Name	Type	Default	Description
<netname>	string	-	name of network operator
<rsrp>	integer	-	Reference Signal Received Power
<rsrq>	integer	-	Reference Signal Received Quality
<tac>	integer	-	Tracking Area Code
<id>	integer	-	cell identifier
<earfcn>	integer	-	E-UTRA Assigned Radio Channel

<dBm>	integer	-	received signal strength in dBm.
<drx>	string	-	Discontinuous reception cycle length (dummy)
<physicalCellId>	integer	-	physical cell identifier

AT#MONI=?

Test command reports the maximum number of cells from which we can extract information, along with the ordinal number of the current selected cell, in the format:

#MONI: <MaxCellNo>

Additional info:

▶▶ Parameters meaning.

Name	Type	Default	Description
<MaxCellNo>	integer	-	maximum number of cells

Set command selects the cell 0 in the network

AT#MONI=0

OK

SIM

AT+CPIN - Enter the PIN

The command sends to the device a password which is necessary before it can be operated.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CPIN=<pin>[,<newpin>]

Set command sends to the device a password which is necessary before it can be operated (SIM PIN, SIM PUK, PH-SIM PIN, etc.).
If the PIN required is SIM PUK or SIM PUK2, the <newpin> is required. This <newpin>, will replace the old pin in the SIM.
The command may be used to change the SIM PIN by sending it with both parameters <pin> and <newpin>.

Parameters:

Name	Type	Default	Description
<pin>	string	-	PIN required or old PIN if the command is used to change the SIM PIN
<newpin>	string	-	new PIN that will replace old pin

AT+CPIN?

Read command reports the PIN/PUK/PUK2 request status of the device in the form:

+CPIN: <code>

Additional info:

▶▶ Parameters:

Name	Type	Default	Description
<code>	string	N/A	PIN/PUK/PUK2 request status code

Values:

READY	:	ME is not pending for any password
SIM PIN	:	ME is waiting SIM PIN to be given
SIM PUK	:	ME is waiting SIM PUK to be given
SIM PIN2	:	ME is waiting SIM PIN2 to be given; this <code> is returned only when the last executed command resulted in PIN2 authentication failure (i.e. +CME ERROR: 17)
SIM PUK2	:	ME is waiting SIM PUK2 to be given; this <code> is returned only when the last executed command resulted in PUK2 authentication failure (i.e. +CME ERROR: 18)

Pin pending status at startup depends on PIN facility setting; to change or query the default power up setting use the command AT+CLCK.

AT+CPIN=?

Test command returns OK result code.

```
AT+CMEE=1
OK

AT+CPIN?
+CME ERROR: 10
error: you have to insert the SIM

AT+CPIN?
+CPIN: READY
OK
you inserted the SIM and module is not waiting for PIN
```

AT+CCID - Read ICCID

Execution command reads on SIM the Integrated Circuit Card Identification (ICCID). It is the card identification number that provides a unique identification number for the SIM.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CCID

The command returns the following message:

+CCID: <ICCID>
OK

AT+CCID?

Read command returns the following message:

+CCID: <ICCID>
OK

AT+CCID=?

Test command returns the OK result code.

AT+CCID
+CCID: 89861109091740011006
OK

AT+CIMI - International Mobile Subscriber Identity (IMSI)

This command returns the International Mobile Subscriber Identity (IMSI number).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CIMI

Execution command returns the value of the International Mobile Subscriber Identity stored in the SIM, the returned message has the following format (with command no echoed):

<IMSI value>
OK

If the SIM is not inserted, the command returns an error. Return value depends on AT+CMEE configuration.

AT+CIMI=?

Test command returns OK result code

```
AT+CIMI
22201701202507
OK
```

AT#CCID - Read ICCID

Execution command reads on SIM the Integrated Circuit Card Identification (ICCID). It is the card identification number that provides a unique identification number for the SIM.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#CCID

The command returns the following message:

#CCID: <ICCID>
OK

AT#CCID=?

Test command returns the OK result code.

```
AT#CCID
#CCID: 89861109091740011006
OK
```

AT#QSS - Query SIM Status

Query SIM Status.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT#QSS=[<mode>]

Set command enables/disables the Query SIM Status unsolicited indication in the ME.

See Additional info section.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	Type of notification

Values:

0 : disabled. It is only possible to query the current SIM status through read command AT#QSS?

1 : enabled. The ME informs at every SIM status change through the unsolicited indication where <status> range is 0...1

2 : enabled. The ME informs at every SIM status change through the unsolicited indication where <status> range is 0...3

Additional info:

The format of the unsolicited indication is the following:

#QSS: <status>

The parameter is described in the Unsolicited field section.

Unsolicited field:

Name	Type	Description
<status>	integer	current SIM status Values: 0 : SIM not inserted 1 : SIM inserted 2 : SIM inserted and PIN unlocked 3 : SIM inserted and READY (SMS and Phonebook access are possible)

The command reports the SIM status change after the <mode> has been set to 2.

It is strongly suggested to set <mode>=2 and save the value in the user profile, then power off the module. The proper SIM status will be available at the next power on.

AT#QSS?

Read command reports whether the unsolicited indication #QSS is currently enabled or not, along with the SIM status, in the format:

#QSS: <mode>,<status>

The parameters are described in the previous sections.

AT#QSS=?

Test command returns the supported range of values for parameter <mode>.

AT#SIMSELECT - SIM Slot Switch

SIM Slot Switch

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SIMSELECT=<slot>

Set command selects to activate one of the two SIM slots to support Dual SIM Single Standby (DSSS).

Parameter:

Name	Type	Default	Description
<slot>	integer	1	SIM slot

Values:

1	:	SIM slot 1
2	:	SIM slot 2

To select any SIM slot, first set AT+CFUN=0, then select SIM slot with AT#SIMSELECT and restore original AT+CFUN value.

SIM hot swap function (only supported by ELS63 product variants) is only available for SIM slot 1.

SIM hot swap function (only supported by ELS63 product variants) is only available in idle/connected mode; it is not available in power saving modes (see AT+CFUN).

AT#SIMSELECT?

Read command reports the activated SIM slot, in the format:

#SIMSELECT: <slot>

AT#SIMSELECT=?

Test command reports the supported range of values for parameter <slot>.

Example of #SIMSELECT

```
AT#SIMSELECT?
#SIMSELECT: 1
OK
AT+CFUN?
+CFUN: 1
OK
AT+CFUN=0
OK
AT#SIMSELECT=2
OK
AT+CFUN=1
OK
AT#SIMSELECT?
#SIMSELECT: 2
OK
```

AT#SIMDET - SIM Detection Mode

This command manages the SIM detection mode.

[1] Telit Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	Common Profile	No

AT#SIMDET=<mode>

Set command forces the SIM status or selects the automatic SIM Detection status.

This command is used: by modules providing the SIMIN pin, dedicated or set through #SIMINCFG command; by modules without SIMIN pin, where the SIM presence or absence is managed through this command.

E.g.: if the SIM is removed, the user should advise the module by setting #SIMDET=0.

Parameter:

Name	Type	Default	Description
<mode>	integer	1	the <mode> parameter can be used as shown in Values section

Values:

0 : ignore dedicated SIMIN pin and simulate the status "SIM Not Inserted"

1 : ignore dedicated SIMIN pin and simulate the status "SIM Inserted"

2 : select automatic SIM detection using dedicated SIMIN Pin

- When #SIMDET=1 (that simulates "SIM Inserted") is issued, a query to detect the presence of the SIM is forced, regardless of SIMIN pin status.

  If SIM is correctly answering to query, then SIM status becomes SIM READY.

- For LE310Q1 family, default value of <mode> parameter is 1.

- If the SIMIN pin is not set through #SIMINCFG and not connected to the SIM holder, do not use #SIMDET=2.

AT#SIMDET?

Read command returns the currently selected SIM Detection Mode in the format:

#SIMDET: <mode>

AT#SIMDET=?

Test command reports the supported range of values for parameter <mode>.

This command is supported only in LE310Q1 family.

AT#SIMINCFG - SIMIN Pin Configuration

This command allows to configure a GPIO pin as SIMIN pin (detect switch of SIM holder) and the polarity of the SIMIN pin.

[1] Telit Hardware Design Guide of the used module

[2] SIM Integration Design Guide, 80000NT10001A

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SIMINCFG=<GPIOPin>,<siminDetMode>

Set command allows to configure either a General Purpose I/O pin (GPIO) as SIMIN pin to detect the SIM presence, and its status according to the used SIM holder.

Refer to document [1] to have information on the GPIO pins, see also document [2].

Parameters:

Name	Type	Default	Description
<GPIOPin>	integer	0	number of the GPIO pin used as SIMIN pin. The GPIO pin range starts from 1; 0 is used to reset #SIMINCFG settings, it is the default value. 0 is the only value allowed.

Values:

0 : reset #SIMINCFG settings

<siminDetMode>	integer	0	status of SIMIN pin for SIM detection

Values:

0 : SIMIN pin LOW means SIM inserted. HIGH means SIM not inserted (for normal SIM holder)

1 : SIMIN pin LOW means SIM not inserted, HIGH means SIM inserted (for micro and nano SIM holder)

SIMIN pin works only if #SIMDET is set to 2 (Automatic SIM Detection).

If SIMIN pin is not connected to the SIM Holder, do not use #SIMDET=2.

The chosen GPIO is configured as open drain with an internal pull up.

AT#SIMINCFG?

Read command reports the parameters values in the format:

#SIMINCFG: <GPIOPin>,<siminDetMode>

AT#SIMINCFG=?

Test command reports the supported range of value for parameters <GPIOPin> and <siminDetMode>.

This command is supported only in LE310Q1 family.

On LE310Q1 product family there is no dedicated SIMIN pin and AT#SIMDET=1 is the default value.

To use a GPIO as SIMIN pin #SIMDET=2 must be set.

AT+CRSM - Restricted SIM access

The command transmits to the UICC some specific commands and their required parameters.

3GPP TS 11.11
3GPP TS 31.101
3GPP TS 31.102

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CRSM=<command>[,<fileid>[,<P1>,<P2>,<P3>[,<data>[,<pathid>]]]]

Set command transmits to the UICC the specific command and its required parameters. The command response shows the status words and response data returned by the UICC in the format:

+CRSM: <sw1>,<sw2>[,<response>]

For parameters meanings see Additional info section.

Parameters:

Name	Type	Default	Description
<command>	integer	176	command passed on to the UICC

Values:

176	:	READ BINARY
178	:	READ RECORD
192	:	GET RESPONSE
214	:	UPDATE BINARY

220	:	UPDATE RECORD
242	:	STATUS

<fileId>	integer	-	identifier of an elementary data file on UICC; mandatory for every command except STATUS
<P1>	integer	0	parameter passed on to the UICC; it is mandatory for every command except GET RESPONSE and STATUS

Value:

0÷255	:	parameter P1 passed on to the UICC in a command APDU

<P2>	integer	0	parameter passed on to the UICC; it is mandatory for every command except GET RESPONSE and STATUS

Value:

0÷255	:	parameter P2 passed on to the UICC in a command APDU

<P3>	integer	0	parameter passed on to the UICC; it is mandatory for every command except GET RESPONSE and STATUS

Value:

	0÷255	:	parameter P3 passed on to the UICC in a command APDU

<data>	string	-	information to be read from UICC or written to the UICC (hexadecimal character format).
<pathid>	string	-	path of an elementary file on the SIM/ UICC in hexadecimal format as defined in ETSI TS 102 22 (for example "7F205F70" in SIM and UICC case). The <pathid> is only used in the mode "select by path from MF" as defined in ETSI TS 102 221

Additional info:

▶▶ Response data fields:

Name	Type	Default	Description
<sw1>	integer	-	information from the UICC about the execution of the actual command (successful or failed)
<sw2>	integer	-	information from the UICC about the execution of the actual command (successful or failed)

<response>	hex	-	on a successful completion of the previously issued command it shows the response data. It is not returned after a successful UPDATE BINARY or UPDATE RECORD command

AT+CRSM=?

Test command returns the OK result code

Read binary, ICCID(2FE2)

```
AT+CRSM=176,12258,0,0,10
+CRSM: 144,0,982850702001107686F4
OK
```

Read record, ADN(6F3A)

```
AT+CRSM=178,28474,1,4,40
+CRSM: 144,0,42434A554EFFFFFFFFFFFFFFFFFFFFFF06811056789282FFFFFFFFFFFFFF
OK
```

Update Binary, KcGPRS(6F52)

```
AT+CRSM=214,28539,0,0,8,C69018C7958C87
+CRSM: 144,0
OK
```

Update Record, ADN(6F3A)

```
AT+CRSM=220,28474,9,4,30,657469FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFF
+CRSM: 144,0
OK
```

Status, FPLMN(6F7B)

```
AT+CRSM=242,28539
+CRSM:144,0,623C820238218410A0000000871002FFFFFFFF89040300FFA51180013181030
```

10A3282011E8304000030E08A01058B032F0609C6099001C0830101830181
OK

AT+CSIM - Generic SIM Access

This command sends a generic command to the UICC.

[1] 3GPP TS 11.11
[2] 3GPP TS 31.101
[3] 3GPP TS 31.102

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSIM=<length>,<command>

Set command transmits to the MT the <command>, it then shall send as it is to the UICC. As response to the command, MT sends back the actual UICC <response> to the TA as it is. The response message of the command is in the format:

+CSIM: <length>,<response>

Error case:
+CME ERROR: <err>

The response messages parameters are described in the Additional info section.

Parameters:

Name	Type	Default	Description
<length>	integer	-	characters number of the <command> sent to UICC (two times the actual length of the command)
<command>	string	-	command passed on by the MT to the UICC in the format as described in standard [1] or [2] (hexadecimal character format, refer to +CSCS)

Additional info:

List of the meaning of the response messages parameters.

Name	Type	Default	Description
<length>	integer	-	characters number of the <response> sent to TE (two times the actual length of the response)
<response>	string	-	response to the command passed on by the UICC to the TE in the format as described in standard [1] or [2] (hexadecimal character format)
<err>	string	-	error values (numeric format followed by verbose format)

AT+CSIM=?

Test command returns the OK result code.

* 2G SIM, see standard [1]:

STATUS
AT+CSIM=10,A0F2000016
+CSIM:48,"000002A87F2002000000000009930022080083 8A838A9000"
OK

SELECT EF 6F07
AT+CSIM=14,A0A40000026F07
+CSIM: 4,"9F0F"
OK

GET RESPONSE
AT+CSIM=10,A0C000000F
+CSIM: 34,"000000096F0704001A001A010200009000"
OK

SELECT EF 6F30
AT+CSIM=14,A0A40000026F30
+CSIM: 4,"9F0F"
OK

READ BINARY
AT+CSIM=10,A0B00000FC
+CSIM:508,"FFFFFF1300831300901300541300301300651300381300801301801
3000113110913013013009813007713005913004313008113009513014013002313
0016330420130041FFFFFFFFFFFF21436542F41922F28822F201FFFFFFFFFFFF
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFFFFFFFFF9000"
OK

AT+CCHO - Open Logical Channel

Open Logical Channel

3GPP TS 31.101

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CCHO=<dfname>

Execution of the command causes the MT to return <sessionId> to allow the TE to identify a channel that is being allocated by the currently selected UICC, which is attached to ME.

The currently selected UICC will open a new logical channel; select the application identified by the <dfname> received with this command and return a <sessionId> as the response.

The ME shall restrict the communication between the TE and the UICC to this logical channel.

The response message of the command is in the format:
+CCHO: < sessionId >

The <sessionId> is described in the Additional info section.

Error case:
+CME ERROR: <err>

Parameter:

Name	Type	Default	Description
<dfname>	string	-	all selectable applications in the UICC are referenced by a DF name coded on 1 to 16 bytes (hexadecimal character format; refer +CSCS).

Additional info:

<sessionId> returned by the +CCHO command.

Name	Type	Default	Description
<sessionId>	integer	-	session Id to be used to target a specific application on the smart card (e.g. (U)SIM, WIM, ISIM) using logical channels mechanism. Session Id is used when sending commands with Restricted UICC Logical Channel access +CRLA, or Generic UICC Logical Channel access +CGLA commands.

The logical channel number is contained in the CLASS byte of an APDU command, thus implicitly contained in all APDU commands sent to a UICC.

In this case it will be up to the MT to manage the logical channel part of the APDU CLASS byte and to ensure that the chosen logical channel is relevant to the <sessionId> indicated in the AT command. Refer to 3GPP TS 31.101.

AT+CCHO=?

Returns the OK result code.

AT+CCHC - Close Logical Channel

Close a communication session.

3GPP TS 31.101

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CCHC=<sessionId>

Set command asks the ME to close a communication session with the active UICC. The ME shall close the previously opened logical channel. The TE will no longer be able to send commands on this logical channel. The UICC will close the logical channel when receiving this command.

Error case:

+CME ERROR: <err>

Parameter:

Name	Type	Default	Description
<sessionId>	integer	-	a session Id to be used to target a specific application on the smart card (e.g. (U)SIM, WIM, ISIM) using logical channels mechanism. <sessionId> is returned by the +CCHO command.

AT+CCHC=?

Test command returns the OK result code.

```
AT+CCHC=1
+CCHC

OK
```

AT+CGLA - Generic UICC Logical Channel Access

This command is used to control the currently selected UICC on the TE.

[1] 3GPP TS 11.11
[2] 3GPP TS 31.101
[3] 3GPP TS 31.102

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGLA=<sessionId>,<length>,<command>

Set command transmits to the MT the <command> it then shall send as it is to the selected UICC. In the same manner the UICC <response> shall be sent back by the MT to the TA as it is. This command allows a direct control of the currently selected UICC by a distant application on the TE. The TE shall then take care of processing the received UICC information. The response of the command is in the format:

+CGLA: <length>,<response>

Error case:

+CME ERROR: <err>

The response messages parameters are described in the Additional info section.

Parameters:

Name	Type	Default	Description
<sessionId>	integer	-	this parameter is the identifier of the session to be used in order to send the APDU commands to the UICC. It is

			mandatory to send commands to the UICC when targeting applications on the smart card using a logical channel other than the default channel (channel "0"). <sessionId> is returned by the +CCHO command.
<length>	integer	-	characters number of the <command> sent to UICC (two times the actual length of the command)
<command>	string	-	command passed on by the MT to the UICC in the format as described in standard [1] or [2] (hexadecimal character format; refer to +CSCS).

Additional info:

▶▶ List of the meaning of the response messages parameters.

Name	Type	Default	Description
<length>	integer	-	characters number of the <response> sent to TE (two times the actual length of the response)
<response>	string	-	response to the command passed on by the UICC to the TE in the format as described in standard [1] or [2] (hexadecimal character format)

<err>	string	-	error values (numeric format followed by verbose format)

AT+CGLA=?

Test command returns the OK result code.

AT+ICCID - Read ICCID

Execution command reads on SIM the Integrated Circuit Card Identification (ICCID). It is the card identification number that provides a unique identification number for the SIM.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+ICCID

The command returns the following message:

+ICCID: <ICCID>

AT+ICCID=?

Test command returns the OK result code.

```
AT+ICCID
+ICCID: 89861109091740011006
OK
```

SIM Toolkit

AT#STIA - SIM/USIM Toolkit Interface Action

The SIM/USIM Application Toolkit (SAT/USAT) provides an interface to the applications existing in the SIM/USIM device. The module must support the mechanisms required by the SIM/USIM applications.

[1] 3GPP TS 23.038

[2] 3GPP TS 31.111

[3] Telit SIM/USIM Application Toolkit Application Note

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#STIA=[<mode>[,<timeout>]]

Set command is used to enable/disable the SIM/USIM Application Toolkit (SAT/USAT). In addition, the command can enable the URCs sending.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	enables/disables SAT/USAT. In addition, <mode> parameter enables the: - #STN URCs notifying the user that the SIM/USIM application has issued a proactive command. Some proactive commands require a user response. - #STN URCs that are the SIM/USIM device responses concerning actions initiated by the user, refer to Additional info section.

If <mode>=2, the URC format depends on the <cmdType> as described in the Additional info sections. For <cmdType>, and all other URC parameters refer to #STGI command. The <mode> parameter values are listed below.

Values:

0 : disable SAT/USAT (partial SAT/USAT Terminal Profile without Setup menu)

1 : enable SAT/USAT without #STN URC

2 : enable SAT/USAT and extended #STN URC

if text is present

if the alphabet to be used is present

{

if the alphabet to be used for text indicated by the SIM is UCS2

the text will be displayed in hex format (example 0140)

else

the alphabet to be used for text will be the one indicated by the SIM

}

else

the alphabet to be used for text will be IRA

3 : enable SAT/USAT and reduced #STN URC

17 : enable SAT/USAT without #STN URC

18 : enable SAT/USAT and extended #STN URC

if text is present

if the alphabet to be used is present

{

if the alphabet to be used for text indicated by the SIM is UCS2

the text will be displayed in hex format (example 0140)

else

the alphabet to be used for text will be ASCII 8 bit

}

else

the alphabet to be used for text will be ASCII 8 bit

19 : enable SAT/USAT and reduced #STN URC

33 : enable SAT/USAT without #STN URC

34 : enable SAT/USAT and extended #STN URC

if text is present

if the alphabet to be used is present

{

the alphabet to be used for text will be UCS2 displayed in hex format (example 0140)

}

else

the alphabet to be used for text will be UCS2 displayed in hex format (example 0140)

35 : enable SAT/USAT and reduced #STN URC

<timeout>	integer	1	When an ongoing proactive command, requiring a user response, is not answered before <timeout> minutes, it is automatically aborted. In this case, the terminal response from the module is either "ME currently unable to process command" or, if applicable, "No response from user". In addition, the following URC is sent on the AT interface. For parameter meaning of the URC refer to Unsolicited fields section. #STN:<cmdTerminateValue>

Value:

1,2	:	timeout expressed in minutes

Additional info:

- <mode>=3, the URC has the following reduced format:

  #STN: <cmdType>

- <mode>=2, and <cmdType>=1 (REFRESH proactive command), the URC has the following extended format:

  #STN: 1,<refreshType>

- <mode>=2, and <cmdType>=33 (DISPLAY TEXT proactive command), the URC is sent if allowed by SIM/USIM, the extended format is:

  #STN: 33[,<cmdDetails>[,<textstring>]]

If bit 7 of <cmdDetails>=1, the response with the #STSR command is required.

- <mode>=2, and <cmdType>=34 (GET INKEY proactive command), the URC of the extended format is the same as the one of reduced format:

  #STN: 34

- <mode>=2, and <cmdType>=35 (GET INPUT proactive command), the URC of the extended format is the same as the one of reduced format:

  #STN: 35

- <mode>=2, and <cmdType>=36 (SELECT ITEM proactive command), the URC of the extended format is the same as the one of reduced format:

  #STN: 36

- <mode>=2, and <cmdType>=37 (SET UP MENU proactive command), the URC of the extended format is the same as the one of reduced format:

  #STN: 37

- <mode>=2, and <cmdType>=40 (SET UP IDLE MODE TEXT proactive command), the URC has the following extended format:

  #STN: 40[,<idleModeTextString>]

- <mode>=2 or <mode>=3, and <cmdType>=254 is a special URC <cmdType> to inform about the end of proactive session, the URC has the following extended format:

  #STN: 254

Unsolicited fields:

Name	Type	Description
<cmdTerminateValue>	integer	is defined as <cmdType> + terminate offset. Terminate offset = 100

The settings are saved on command execution and they are available on following reboot.

SIM/USIM Toolkit activation/deactivation is only performed at power on according to the saved setting.

AT#STIA parameters change is applied after restart.

A typical SAT/USAT session on AT interface starts after an #STN: 37 unsolicited code is received, if enabled (see above).

At that point usually an AT#STGI=37 command is issued (see #STGI), and after the SAT main menu has been displayed on TE an AT#STSR=37,0,x command is issued to select an item in the menu (see #STSR).

AT#STIA?

Read command can be used to get information about the SAT/USAT interface. The message format is:

#STIA: <state>,<mode>,<timeout>,<SatProfile>

Additional info:

Returned parameters:

Name	Type	Default	Description

<state>	integer	N/A	state of the sending of the SET UP MENU proactive command (37)
	Values:		
	0 :	SIM/USIM has not sent the SET UP MENU proactive command (37)	
	1 :	SIM/USIM has sent the SET UP MENU proactive command (37)	
<mode>	integer	-	refer to Set section
<timeout>	integer	-	refer to Set section
<SatProfile>	string	-	SAT/USAT Terminal Profile. Is the list of SIM/USIM Application Toolkit facilities supported by the ME. The profile cannot be changed by the TA.

In SAT/USAT applications which involve SMS, an SMS message is usually sent to the network provider containing service requests, e.g. to send the latest news.

The provider returns a message with the requested information. Before activating SAT/USAT, it is recommended to set the SMS text mode with the AT+CMGF=1 command and enable URC for incoming SMS messages with +CNMI command.

AT#STIA=?

Test command returns the range of available values for the parameters <mode> and <timeout>.

AT#STGI - SIM Toolkit Get Information

This command interfaces to the SIM/USIM Application Toolkit to get information on the ongoing proactive command.

3GPP TS 31.111 Telit SIM/USIM Application Toolkit Application Note

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#STGI=[<cmdType>]

Set command gets parameters of the ongoing proactive command. The command can be used after the reception of the #STN: <cmdType> URC. If no proactive command is ongoing, it returns an ERROR message.

Parameter:

Name	Type	Default	Description
<cmdType>	integer	N/A	proactive command code. For each proactive command listed below, the response format is described in the Additional info sections.

Values:

1	:	REFRESH
33	:	DISPLAY TEXT
34	:	GET INKEY

35 : GET INPUT

36 : SELECT ITEM

37 : SET UP MENU

40 : SET UP IDLE MODE TEXT

Additional info:

- <cmdType>=1 (REFRESH proactive command)

  #STGI response format:

  #STGI: 1,<refreshType>

Name	Type	Default	Description
<refreshType>	integer	N/A	identifies the refresh type

Values:

0 : SIM Initialization and Full File Change Notification

1 : File Change Notification

2 : SIM Initialization and File Change Notification

3 : SIM Initialization

4 : SIM Reset

5 : NAA Application Reset

6 : NAA Session Reset

7 : Steering of Roaming

8 : Steering of Roaming WLAN

▸▸ <cmdType>=33 (DISPLAY TEXT proactive command)

#STGI response format:

#STGI: 33,<cmdDetails>[,<text>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - normal priority 1 - high priority bits 1-6: reserved for future use bit 7: 0 - clear message after a delay

			1 - wait for user to clear message

Value:

0x00÷0xFF	:	mask

<text>	string	-	text provided by the SIM/ USIM device in ASCII format

▶▶ <cmdType>=34 (GET INKEY proactive command)

#STGI response format:

#STGI: 34,<cmdDetails>,<text>

Name	Type	Default	Description

<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - digits only (0-9, *, # and +) 1 - alphabet set bit 1: 0 - SMS default alphabet (GSM character set) 1 - UCS2 alphabet bit 2: 0 - character sets defined by bit 0 and bit 1 are enabled 1 - character sets defined by bit 0 and bit 1 are disabled and the "Yes/No" response is requested bits 3-6: 0 bit 7: 0 - no help information available 1 - help information available

Value:

0x00÷0x87	:	mask

<text>	string	-	string as prompt for test

▶▶ <cmdType>=35 (GET INPUT proactive command)

#STGI response format:

#STGI:
35,<cmdDetails>,<text>,<responseMin>,<responseMax>[,<defaultText>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - digits only (0-9, *, #, and +) 1 - alphabet set bit 1: 0 - SMS default alphabet (GSM character set) 1 - UCS2 alphabet bit 2: 0 - ME may echo user input on the display 1 - user input shall not be revealed in any way. Hidden entry mode is only available when using digit input. In hidden entry mode only characters ('0'-'9', '*' and '#') are allowed. bit 3: 0 - user input to be in unpacked format 1 - user input to be in SMS packed format

			bits 4-6: 0 bit 7: 0 - no help information available 1 - help information available

Value:

0x00÷0x8F	:	mask

<text>	string	-	string as prompt for text
<responseMin>	integer	N/A	minimum number of characters of the user input

Value:

0÷255	:	minimum length of user input.

<responseMax>	integer	N/A	maximum number of characters of the user input.

Value:

		0÷255	: maximum length of user input
<defaultText>	string	-	string supplied as default response text

▶▶ <cmdType>=36 (SELECT ITEM proactive command)

#STGI response format:

the first line of output is:

#STGI: 36,<cmdDetails>,<numOfItem>[,<titleText>]<CR><LF>

one line follows for every item, repeated <numOfItems> times:

#STGI: 36,<itemId>,<itemText>[,<nextActionId>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - presentation type is not specified 1 - presentation type is specified in bit 1 bit 1: 0 - presentation as a choice of data values if bit 0 = 1 1 - presentation as a

			choice of navigation options if bit 0 is 1 bit 2: 0 - no selection preference 1 - selection using soft key preferred bits 3-6: 0 bit 7: 0 - no help information available 1 - help information available

Value:

0x00÷0x87	:	mask

<numOfItems>	integer	-	number of items in the list
<titleText>	string	-	menu title string
<itemId>	integer	N/A	item identifier

Value:

1÷numOfItems	:	item identifier range

<itemText>	string	-	item title string

<nextActionId>	integer	-	is the code of next proactive command to be issued upon execution of the menu item. If <nextActionId>=0, no next action information available.

▶▶ <cmdType>=37 (SET UP MENU proactive command)

#STGI response format:

the first line of output is:

#STGI: 37,<cmdDetails>,<numOfItem>,<titleText><CR><LF>

one line follows for every item, repeated for <numOfItems>:

#STGI: 37,<itemId>,<itemText>[,<nextActionId>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - no selection preference 1 - selection using soft key preferred bit 1-6: 0 bit 7: 0 - no help information available

			1 - help information available
	Value:		
	0x00÷0x81	:	mask
<numOfItems>	integer	-	number of items in the list
<titleText>	string	-	menu title string
<itemId>	integer	N/A	item identifier
	Value:		
	1÷numOfItems	:	item identifier range
<itemText>	string	-	item title
<nextActionId>	integer	-	numerical code of next proactive command type to be issued upon execution of the menu item. If <nextActionId>=0, no next action information available.

▶▶ <cmdType>=40 (SET UP IDLE MODE TEXT proactive command)

#STGI response format:

#STGI: 40,<idleModeTextString>

Name	Type	Default	Description
<idleModeTextString>	string	-	text provided by the SIM/USIM device in ASCII format

The proactive commands are only those command types that use the AT interface.

SAT/USAT commands which are not using the AT interface (not MMI related SAT commands, e.g. PROVIDE LOCAL INFORMATION) are executed without sending any indication to the user.

AT#STGI?

Read command returns the ongoing proactive command and the SAT/USAT state. The message format is:

#STGI: <state>,<cmdType>

Additional info:

Returned parameters:

Name	Type	Default	Description

<state>	integer	-	state of the sending of the SET UP MENU proactive command (37), refer to AT#STIA? command
<cmdType>	integer	-	ongoing proactive command code

AT#STGI=?

Test command returns the supported values of parameters <state> and <cmdType>.

A typical SAT/USAT session, running on AT interface, starts when the user receives the #STN: 37 URC. The unsolicited result code must be previously enabled by the #STIA command. The #STN: 37 notifies the user that the main menu of the SIM/USIM Application has been sent to TA, and TA has stored the just received menu. Later, at any time, you can type in the AT#STGI=37 command to display the main menu of the SIM/USIM Application on TE.

Upon receiving the #STGI response, you must enter the #STSR command to confirm the execution of the proactive command and provide any required user response. In this case, you must enter the AT#STSR=37,0,x command to select the x item of the SIM/USIM Application menu.

The #STN: 237 URC indicates that the main menu of the SIM/USIM Application has been removed from TA, and it is no longer available. In this case, AT#STGI=37 command returns ERROR message.

AT#STSR - SIM Toolkit Send Response

This command allows the user to provide a response to confirm the execution of the ongoing proactive command.

3GPP TS 31.111

Telit SIM/USIM Toolkit Application Note

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#STSR=[<cmdType>[,<userAction>[,<data>]]]

Set command allows the user to provide a response action to the ongoing proactive command when the action is required by the command itself.

Parameters:

Name	Type	Default	Description
<cmdType>	integer	-	proactive command code, refer to #STGI command to have information on the <cmdType>
<userAction>	integer	0	identify the user action

Values:

0 : the user accepts the ongoing proactive command

16	:	proactive SIM/USIM session terminated by user
17	:	backward move in the proactive SIM/USIM session requested by the user
18	:	no response from user
19	:	help information required by the user
32	:	TA currently unable to process command

<data>	string	-	data entered by user, see Additional info section

Additional info:

- <data> parameter is used according to <cmdType>, and when <userAction>=0:

- <cmdType>=34 (GET INKEY proactive command)

  <data> contains the key pressed by the user. The character set is selected by +CSCS command.

  If the ongoing proactive command requires to the user a binary choice (yes/no), the valid content of <data> is:

  - "Y" or "y" (positive answer) and "N" or "n" (negative answer) for "IRA", "8859-1", "PCCP437" character sets
  - "0079" or "0059" (positive answer) and "006E" or "004E" (negative answer) for UCS2 alphabet

  The ongoing proactive command to require a binary choice sets bit 2 of the <cmdDetails> parameter to 1, see #STGI command.

- <cmdType>=35 (GET INPUT proactive command).

  <data> contains the string of characters entered by the user.

- <cmdType>=36 (SELECT ITEM proactive command).
  <data> contains the item identifier selected by the user.

- <cmdType>=37 (SET UP MENU proactive command).
  <data> contains the item identifier selected by the user.

Use of icons is not supported. All icon related actions will respond with no icon available.

AT#STSR?

Read command returns the ongoing proactive command and the SAT/USAT interface state. The format message is:

#STSR: <state>,<cmdType>

If there is no ongoing proactive command, an ERROR message is returned.

Additional info:

- Returned parameters.

Name	Type	Default	Description
<state>	integer	-	state of the sending of the SET UP MENU proactive command (37), refer to AT#STIA? command
<cmdType>	integer	-	proactive command code, refer to #STGI command to have information on the <cmdType>

?

AT#STSR=?

Test command returns the range for the parameters <state> and <cmdType>.

SMS & CB

AT+CSMS - Select Message Service

Set command selects messaging service <service>

3GPP TS 27.005
3GPP TS 23.040
3GPP TS 23.041

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CSMS=[<service>]

Set command selects messaging service <service>. It returns the types of messages supported by the ME:

+CSMS: <mt>,<mo>,<bm>

For parameters meaning see Additional info section.

Parameter:

Name	Type	Default	Description
<service>	integer	0	Select Message Service

Values:

0 : 3GPP TS 23.040 and 3GPP TS 23.041. The syntax of SMS AT commands is compatible with 3GPP TS 27.005

1 : 3GPP TS 23.040 and 3GPP TS 23.041. The syntax of SMS AT commands is compatible with 3GPP TS 27.005. The requirement of <service> setting 1 is mentioned under corresponding command descriptions

Additional info:

▶▶ Parameters meaning of the returned message.

Name	Type	Default	Description
<mt>	integer	1	mobile terminated messages support: Values: 0 : type not supported 1 : type supported
<mo>	integer	1	mobile originated messages support Values: 0 : type not supported 1 : type supported

<bm>	integer	1	broadcast type messages support

Values:

0 : type not supported

1 : type supported

AT+CSMS?

Read command reports current service setting along with supported message types in the format:

+CSMS: <service>, <mt>, <mo>, <bm>

AT+CSMS=?

Test command reports the supported value of the parameter <service>

AT+CPMS - Preferred Message Storage

The command selects the memory storage used by SMs (Short Messages).

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CPMS=<memr>[,<memw>[,<mems>]]

Set command selects memory storages <memr>, <memw> and <mems> to be used for reading, writing, sending and storing SMs

The command returns the memory storage status in the format:

+CPMS: <usedr>,<totalr>,<usedw>,<totalw>,<useds>,<totals>

The returned parameters are described in the Additional info section.

Parameters:

Name	Type	Default	Description
<memr>	string	ME	memory from which messages are read and deleted

Values:

ME	:	SMS memory storage in Flash
SM	:	SIM SMS memory storage

<memw>	string	ME	memory to which writing and sending operations are made

Values:

ME	:	SMS memory storage in Flash
SM	:	SIM SMS memory storage

<mems>	string	ME	memory to which received SMs are preferred to be stored

Values:

ME	:	SMS memory storage in Flash
SM	:	SIM SMS memory storage

Additional info:

▶▶ Here is the meaning of the parameters returned by the command.

Name	Type	Default	Description
<usedr>	integer	-	number of SMs stored in <memr>
<totalr>	integer	-	max number of SMs that <memr> can contain
<usedw>	integer	-	number of SMs stored in <memw>

<totalw>	integer	-	max number of SMs that <memw> can contain
<useds>	integer	-	number of SMs stored in <mems>
<totals>	integer	-	max number of SMs that <memw> can contain

AT+CPMS?

Read command reports the message storage status.

+CPMS:<memr>,<usedr>,<totalr>,<memw>,<usedw>,<totalw>,<mems>,<useds>,<totals>

The parameters are described in previous sections.

AT+CPMS=?

Test command reports the supported values for parameters <memr>, <memw> and <mems>.

```
AT+CPMS="SM","ME","SM"
+CPMS: 1,20,27, 50,1,20
OK

AT+CPMS?
+CPMS: "SM",1,20,"ME",27, 50,"SM",1,20
OK
```

You have 1 out of 20 SMS SIM positions occupied

When upgrading firmware using XFP all the SMS stored in ME (SMS memory storage in Flash) will be deleted

AT+CMGF - Message Format

Selects the format of SMS messages to be used in following SMS commands.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

AT+CMGF=[<mode>]

Set command selects the format of SMS messages used with send, list, read and write commands.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	format to use for SMS operations

Values:

0	:	PDU mode
1	:	text mode

AT+CMGF?

Read command reports the current value of the parameter <mode> in the format:

+CMGF: <mode>

AT+CMGF=?

Test command returns the supported values of parameter <mode>.

AT+CSCA - Service Center Address

This command allows to set the Service Center Address for SMS transmissions.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CSCA=<number>[,<type>]

Set command sets the Service Center Address to be used for mobile originated SMS transmissions

Parameters:

Name	Type	Default	Description
<number>	string	-	String type phone number of forwarding address in format specified by <type> parameter
<type>	integer	145	The type of number

Values:

129 : National numbering scheme

145 : International numbering scheme (contains the character "+")

AT+CSCA?

Read command reports the current value of the SCA in the format:

+CSCA: <number>,<type>

If SCA is not present the device reports an error message.

AT+CSCA=?

Test command returns the OK result code.

To use the SM service, is mandatory to set a Service Center Address at which service requests will be directed.

In Text mode, this setting is used by send and write commands; in PDU mode, setting is used by the same commands, but only when the length of the SMSC address coded into the <pdu> parameter equals zero.

At every power on or after every restart the Service Center Address is reset to SIM EF SMSP (Short Message Service Parameters) record 1 values.

The current settings are stored in NVM through +CSAS.

```
AT+CSCA="821029190903",145
OK
```

```
AT+CSCA?
+CSCA: "+821029190903",145
OK
```

AT+CSMP - Set Text Mode Parameters

This command is used to select values for additional parameters for storing and sending SMS when the text mode is used (AT+CMGF=1).

3GPP TS 27.005
3GPP TS 03.40/23.040
3GPP TS 03.38/23.038

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CSMP=[<fo>[,<vp>[,<pid>[,<dcs>]]]]

Set command is used to select values for additional parameters for storing and sending SMS when the text mode is used (AT+CMGF=1).

Parameters:

Name	Type	Default	Description
<fo>	integer	-	first octet of 3GPP TS 03.40/23.040 SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT (not supported), or SMS-COMMAND (not supported) in integer format.
<vp>	mixed	-	3GPP TS 03.40/23.040 TP-Validity-Period either in integer format (default 167) or in quoted time-string format.
<pid>	integer	-	3GPP TS 03.40/23.040 TP-Protocol-Identifier in integer format.

<dcs>	integer	-	3GPP TS 03.38/23.038 SMS Data Coding Scheme (default 0), or Cell Broadcast Data Coding Scheme.

AT+CSMP?

Read command returns the current setting in the format:

+CSMP: <fo>,<vp>,<pid>,<dcs>

AT+CSMP=?

Test command returns the OK result code.

- At every power on or after every restart the settings are reset to SIM EF SMSP (Short Message Service Parameters) record 1 values.
- The current settings are stored in NVM through +CSAS.
- <fo> First Octet parameter:

  bit 0/1 = Message Type Indicator

  bit 2 = Reject Duplicates

  bit 3/4 = Validity Period Format

  bit 5 = Status Report Request

  bit 6 = User Data Header Indicator

  bit 7 = Reply Path
- <fo> First Octet parameter bit 0/1 Message Type Indicator supports only the following values:

00 - SMS-DELIVER

01 - SMS-SUBMIT

<vp> Validity Period value depends on First Octet (<fo>) bit 3/4 Validity Period Format configuration.

If the First Octet (<fo>) and the Validity Period Format (<vp>) are not compatible with each other, the command response is still OK but the specifics parameters are discarded and the last valid configuration is kept:

```
AT+CSMP=40,"21/11/22,17:45:52+04",0,0
OK
AT+CSMP?
+CSMP: 17,167,0,0
```

The same, if the First Octet (<fo>) Validity Period Format changes and the Validity Period (<vp>) is not present, the command response is still OK but the specifics parameters are discarded and the last valid configuration is kept.

Set the parameters for an outgoing message with 24 hours of validity period and default properties:

```
AT+CSMP=17,167,0,0
OK
```

AT+CSDH - Show Text Mode Parameters

This command controls whether detailed header information is shown in text mode.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

AT+CSDH=[<show>]

Set command controls whether detailed header information is shown in text mode (AT+CMGF=1) result codes.

Parameter:

Name	Type	Default	Description
<show>	integer	0	control the display of the result codes.

Values:

0 : see Additional info section

1 : show the values in result codes

Additional info:

If <show>=0

do not show header values defined in commands +CSCA and +CSMP (<sca>,<tosca>, <fo>, <vp>, <pid> and <dcs>) nor <length>, <toda> or <tooa> in +CMT, +CMGL, +CMGR result codes for SMS-DELIVERs and SMS-SUBMITs in text mode.

For SMS-COMMANDs in +CMGR result code do not show <pid>, <mn>, <da>, <toda>, <length> or <cdata>

AT+CSDH?

Read command reports the current setting in the format:

+CSDH: <show>

AT+CSDH=?

Test command reports the supported range of values for parameter <show>.

AT+CSAS - Save Settings

Execution command saves settings which have been made by the +CSCA, and +CSMP commands in local non-volatile memory.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CSAS[=<profile>]

Parameter:

Name	Type	Default	Description
<profile>	integer	0	Index of the profile where the settings are saved

Values:

0 : it saves the settings to NVM

1÷n : SIM profile number; the value of <n> depends on the SIM (not supported)

If parameter is omitted (AT+CSAS<CR> is given) the settings are saved in the non-volatile memory.

AT+CSAS=?

Test command returns the possible range of values for the parameter <profile>.

AT+CRES - Restore Settings

Execution command restores message service settings saved by +CSAS command from local non-volatile memory.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CRES[=<profile>]

Parameter:

Name	Type	Default	Description
<profile>	integer	N/A	Defines which message service profiles to restore.

Values:

0 : restores message service settings from NVM

1÷n : restores message service settings from SIM. The n value depends on the SIM (not supported)

If parameter is omitted the command restores message service settings from NVM.

AT+CRES=?

Test command returns the possible range of values for the parameter <profile>.

AT+CMMS - More Message to Send

Set command controls the continuity of SMS relay protocol link. When feature is enabled (and supported by network) multiple messages can be sent much faster as link is kept open.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMMS=[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables the relay protocol link continuity.

Values:

0 : disable

1 : keep enabled until the time between the response of the latest message send command (+CMGS, +CMSS, etc.) and the next send command exceeds 5 seconds, then the link is closed and the parameter <n> is automatically reset to 0

2 : enable (if the time between the response of the latest message send command and the next send command exceeds 5 seconds, the link is closed but the parameter <n> remains set to 2)

Entering AT+CMMS= returns OK but has no effect.

AT+CMMS?

Read command reports the current value of the parameter <n> in the format:

+CMMS: <n>

AT+CMMS=?

Test command returns the range of supported <n>.

AT+CNMI - New Message Indications to Terminal Equipment

This command sets the parameters for receiving SMS messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CNMI=[<mode>[,<mt>[,<bm>[,<ds>[,<bfr>]]]]]

Set command selects the behavior of the device on how the receiving of new messages from the network is indicated to the DTE.

Omitting all parameters returns OK and does nothing.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	unsolicited result codes buffering option.

Values:

0 : Buffer unsolicited result codes in the TA. If TA result code buffer is full, indications can be buffered in some other place or the oldest indications may be discarded and replaced with the new received indications.

1 : Discard indication and reject new received message unsolicited result codes when TA-TE link is reserved, otherwise forward them directly to the TE.

2 : Buffer unsolicited result codes in the TA in case the DTE is busy and flush them to the TE after reservation. Otherwise forward them directly to the TE.

3 : if <mt> is set to 1, it enables the hardware ring line for 1 sec.

<mt>	integer	0	result code indication reporting for SMS-DELIVER.

Values:

0 : No SMS-DELIVER indications are routed to the TE and message is stored.

1 : If SMS-DELIVER is stored into ME/TA, indication of the memory location is routed to the TE using the URC described in Additional info section.

2 : SMS-DELIVERs (except class 2 messages and messages in the message waiting indication group) are routed directly to the TE using the URC described in the Additional info section.

3 : Class 3 SMS-DELIVERs are routed directly to TE using unsolicited result codes defined in <mt>=2. Messages of other data coding schemes result in indication as defined in <mt>=1.

<bm>	integer	0	broadcast reporting option

Configuration not supported

Values:

0 : Cell Broadcast Messages are not sent to the DTE

1 : If new Cell Broadcast Message is stored it will be sent to DTE with the URC described in Additional info section (not supported)

2 : New Cell Broadcast Messages are sent to the DTE with the URC described in Additional info section.

<ds>	integer	0	SMS-STATUS-REPORTs reporting option Configuration not supported

Values:

0 : status report receiving is not reported to the DTE and is not stored

1 : the status report is sent to the DTE with the URC described in the Additional info section.

2 : if a status report is stored, then the unsolicited result code, described in Additional info section, is sent.

<bfr>	integer	0	buffered result codes handling method Configuration not supported

Values:

0 : TA buffer of unsolicited result codes defined within this command is flushed to the TE when <mode>=1..3 is entered (OK response shall be given before flushing the codes)

1 : TA buffer of unsolicited result codes defined within this command is cleared when <mode>=1..3 is entered.

Additional info:

<mt>=1:

+CMTI: <mems>,<index>

<mt>=2:

PDU mode

+CMT: [<alpha>,]<length><CR><LF><PDU>

TEXT mode

+CMT:<oa>,
[<alpha>,]<scts>[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF>
<data>

The parameters written in italics will be present depending on +CSDH last setting.

Class 2 messages and messages in the message waiting indication group (stored message) result in indication as defined in <mt>=1.

Acknowledge for the received SMS-DELIVER SM is sent to network immediately when +CSMS <service> is set to '0' or when +CSMS <service> is set to '1', acknowledge is sent via +CNMA command during predefine time-out, an error is sent to network in case timeout expire, next +CMT response is depend on acknowledge of current received +CMT response in case +CSMS <service> parameter set to '1'.

<bm>=1 (not supported):

PDU mode and Text mode

+CBMI: <mems>,<index><CR><LF>

<bm>=2:

PDU mode

+CBM: <length><CR><LF><PDU>

Text mode

+CBM:<sn>,<mid>,<dcs>,<pag>,<pags><CR><LF><data>

<ds>=1:

PDU mode

+CDS: <length><CR><LF><PDU>

TEXT mode

+CDS: <fo>,<mr>,<ra>,<tora>,<scts>,<dt>,<st>

Acknowledge for the received SMS-STATUS-REPORT SM is sent to network immediately when +CSMS <service> is set to '0' or when +CSMS <service> is set to '1', acknowledge is sent via +CNMA command during pre-defined timeout, an error is sent to network in case timeout expire, next +CDS response is depend on acknowledge of current received +CDS response in case +CSMS <service> parameter set to '1'.

<ds>=2:

+CDSI: <mems>,<index>

Unsolicited fields:

Name	Type	Description
<mems>	string	memory storage where the new message is stored: "SM", "ME".
<index>	integer	location on the memory where SMS is stored.
<alpha>	string	alphanumeric representation of originator/destination number corresponding to the entry found in MT phonebook; used character set should be the one selected with command +CSCS.

<lenght>	integer	PDU length
<PDU>	string	PDU message
<oa>	string	originating address, string type converted in the currently selected character set (see +CSCS)
<alpha>	string	alphanumeric representation of <oa>; used character set should be the one selected with command +CSCS.
<scts>	string	arrival time of the message to the SC
<tooa>	integer	type of number <oa>: 129 - number in national format 145 - number in international format (contains the "+")
<fo>	string	first octet of message PDU, see 3GPP TS 03.40/23.040
<pid>	string	Protocol Identifier
<dcs>	string	Data Coding Scheme
<sca>	string	Service Centre address, string type, converted in the currently selected character set (see +CSCS)
<tosca>	integer	type of number <sca>: 129 - number in national format 145 - number in international format (contains the "+")
<length>	integer	text length
<data>	string	TP-User-Data If <dcs> indicates that GSM 03.38 default alphabet is used and <fo> indicates that GSM 03.40 TP-User-Data-Header-Indication is not set (bit 6 of <fo> is 0), each character of GSM alphabet will be converted into current TE character set (see +CSCS).

		If <dcs> indicates that 8-bit or UCS2 data coding scheme is used or <fo> indicates that GSM 03.40 TP-User-Data-Header-Indication is set (bit 6 of <fo> is 1), each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41). Class 2 messages and messages in the "store" message waiting indication group result in indication as defined in <mt>=1.
<sn>	integer	message serial number
<mid>	integer	message ID
<dcs>	string	Data Coding Scheme
<pag>	integer	page number
<pags>	integer	total number of pages of the message
<data>	string	CBM Content of Message If <dcs> indicates that GSM 03.38 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS) If <dcs> indicates that 8-bit or UCS2 data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41)
<mr>	integer	message reference number; 3GPP TS 23.040 TP-Message-Reference in integer format

<ra>	string	recipient address, string type, represented in the currently selected character set (see +CSCS)
<tora>	integer	type of number <ra>: 129 - number in national format 145 - number in international format (contains the "+")
<scts>	string	arrival time of the message to the SC
<dt>	string	sending time of the message
<st>	string	message status as coded in the PDU

DTR signal is ignored, hence the indication is sent even if the DTE is inactive (DTR signal is Low). In this case the unsolicited result code may be lost so if MODULE remains active while DTE is not, at DTE startup is suggested to check whether new messages have reached the device meanwhile with command AT+CMGL=0 that lists the new messages received.

AT+CNMI?

Read command returns the current parameter settings for +CNMI command in the form:

+CNMI: <mode>,<mt>,<bm>,<ds>,<bfr>

AT+CNMI=?

Test command reports the supported range of values for the +CNMI command parameters

```
AT+CMGF=1
OK

AT+CNMI=1,2,0,1,0
OK


Received message from network

+CMT: "+821020955219",,"07/07/26,20:09:07+36"

TEST MESSAGE
```

AT+CNMA - New Message Acknowledgement

This command is used to confirm the correct reception of a new message.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CNMA

Execution command confirms correct reception of a new message (SMS-DELIVER or SMS-STATUS-REPORT) which is routed directly to the TE.

Acknowledge with +CNMA is possible only if the +CSMS parameter is set to 1 (+CSMS=1) when a +CMT or +CDS indication is shown.

If no acknowledgement is given within the network timeout (17 seconds), an RP-ERROR is sent to the network, the <mt> and <ds> parameters of the +CNMI command are then reset to zero (do not show new message indication).

If command is executed, but no acknowledgement is expected, or some other ME related error occurs, result code +CMS ERROR: <err> is returned.

The AT command syntax and functionalities are different between SMS PDU Mode and SMS Text Mode, as explained in Additional info sections.

Additional info:

PDU Mode

AT+CNMA[=<n>[,<length>[<CR>PDU is given<ctrl-Z/ESC>]]]

Either positive (RP-ACK) or negative (RP-ERROR) acknowledgement to the network is possible. Parameter <n> defines which one will be sent. Optionally (when <length> is greater than zero) an acknowledgement TPDU (SMS-DELIVER-REPORT for RP-ACK or RP-

ERROR) may be sent to the network. The entering of PDU is done similarly as specified in command Send Message +CMGS, except that the SMSC address field is not present.

Name	Type	Default	Description
<n>	integer	N/A	type of acknowledgement in PDU mode Values: 0 : send RP-ACK without PDU (same as TEXT mode) 1 : send RP-ACK with optional PDU message 2 : send RP-ERROR with optional PDU message
<length>	integer	-	length of the PDU message

▶▶ Text Mode

AT+CNMA

Only positive acknowledgement to network (RP-ACK) is possible.

AT+CNMA=?

Test command returned information are different between SMS PDU Mode and SMS Text Mode, as explained below.

Additional info:

- PDU Mode

  Test command returns the possible range of values for the parameter <n>.

- Text Mode

  Test command returns the OK result code.

- In case that a directly routed message must be buffered in ME/TA (possible when +CNMI parameter <mode> equals 0 or 2) or AT interpreter remains too long in a state where result codes cannot be sent to TE (e.g. user is entering a message using +CMGS), acknowledgement (RP-ACK) is sent to the network without waiting +CNMA command from TE.

- It has been necessary to take the following decision to get over any incoherence problem, due to the possibility to have contemporaneous different settings of parameter <mt> and <ds> of the +CNMI command in different sessions (see #PORTCFG and +CMUX): only the <mt> and <ds> setting for session "0" are considered as valid to decide if +CNMA acknowledgment is expected or not.

- If ME does not get acknowledgement within required time (network timeout), ME should respond as specified in 3GPP TS 24.011 to the network. ME/TA shall automatically disable routing to TE by setting both <mt> and <ds> values of +CNMI to zero.

* PDU Mode

AT+CSMS=1
+CSMS: 1,1,1
OK

Set PDU mode.
AT+CMGF=0
OK

AT+CNMI=2,2,0,0,0
OK

Message is received from network.
+CMT: "",70
06816000585426000480980600F170110370537284...

Send positive acknowledgement to the network.
AT+CNMA=0
OK

Message is received from network.
+CMT: "",70
06816000585426000480980600F170110370537284...

Send negative acknowledgment (Unspecified error) to the network.
AT+CNMA=2,3<CR>
> 00FF00 <Ctrl-Z>
OK

* Text Mode

AT+CSMS=1
+CSMS: 1,1,1
OK

Set Text mode.
AT+CMGF=1
OK

AT+CNMI=2,2,0,0,0
OK

Message is received from network.
+CMT: "+821020955219",,"07/07/26,20:09:07+36"
TEST MESSAGE

Send positive acknowledgement to the network.
AT+CNMA
OK

AT+CMGL - List Messages

This command is used to list the messages.

3GPP TS 27.005
3GPP TS 23.040
3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGL[=<stat>]

Execution command reports the list of all the messages with status value <stat> stored into <memr> message storage (<memr> is the message storage for read and delete SMs as last settings of command +CPMS).

Parameter:

Name	Type	Default	Description
<stat>	mixed	-	<stat> parameter type and the command output depend on the last settings of the +CMGF command (message format to be used). There are two modes: PDU mode Text mode See the following Additional info sections.

Additional info:

When message format is PDU mode, the <stat> parameter is:

Name	Type	Default	Description

<stat>	integer	N/A	status value

Values:

0 : new message

1 : read message

2 : stored message not sent yet

3 : stored message already sent

4 : all messages

▶▶ In case of PDU mode the representation format (see +CMGF) is:

+CMGL: <index>,<stat>,
[<alpha>,]<length><CR><LF><pdu>[<CR><LF>
+CMGL: <index>,<stat>,[<alpha>,]<length><CR><LF><pdu>[...]]

Name	Type	Default	Description
<index>	integer	-	message position in the storage list.
<stat>	integer	-	message status. See the above <stat> parameter description.
<alpha>	string	-	String type alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character

			set is the one selected with command +CSCS.
<length>	integer	-	PDU length in bytes
<pdu>	string	-	message in PDU format, according to 3GPP TS 23.040

▶▶ When message format is TEXT mode, the <stat> parameter is:

Name	Type	Default	Description
<stat>	string	N/A	status value

Values:

"REC UNREAD"	:	new message
"REC READ"	:	read message
"STO UNSENT"	:	stored message not sent yet
"STO SENT"	:	stored message already sent
"ALL"	:	all messages

▶▶ In case of TEXT mode, the representation format for stored messages (either sent or unsent) or received messages (either read or unread, not message delivery confirm) is:

+CMGL: <index>,<stat>,<oa/da>,[<alpha>,][<scts>][,<tooa/toda>,<length>]<CR><LF> <data>[<CR><LF>
+CMGL: <index>,<stat>,<oa/da>,[<alpha>,][<scts>][,<tooa/toda>,<length>]<CR><LF> <data>[...]]

The information written in italics will be present depending on +CSDH last setting.

Name	Type	Default	Description
<index>	integer	-	message position in the storage list.
<stat>	string	-	message status. See the above <stat> parameter description.
<oa/da>	string	-	originator/destination address, represented in the currently selected character set (see +CSCS).
<alpha>	string	-	The alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS.
<scts>	string	-	TP-Service Centre Time Stamp in Time String Format.

<tooa/ toda>	integer	N/A	type of number <oa/da>
	Values:		
	129 :	number in national format	
	145 :	number in international format (contains the "+")	
<length>	integer	-	text length
<data>	string	-	TP-User-Data If <dcs> indicates that 3GPP TS 23.038 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS) If <dcs> indicates that 8-bit or UCS2 data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41) If <fo> indicates that a UDH is present each 8-bit octet will be converted into two IRA character long hexadecimal number. The <length> indicates text

			length in characters without UDH length.

▶▶ In case of TEXT mode, the representation format for delivery confirm messages is:

+CMGL: <index>,<stat>,<fo>,<mr>,<ra>,<tora>,
[<scts>,]<dt>,<st>[<CR><LF>
+CMGL: <index>,<stat>,<fo>,<mr>,<ra>,<tora>,
[<scts>,]<dt>,<st>[...]]

Name	Type	Default	Description
<index>	integer	-	message position in the storage list.
<stat>	string	-	Message status. See the last <stat> parameter description.
<fo>	integer	-	first octet of the message PDU
<mr>	integer	-	message reference number; 3GPP TS 23.040 TP-Message-Reference in integer format
<ra>	string	-	recipient address, represented in the currently selected character set (see +CSCS)
<tora>	string	-	type of number <ra>
<scts>	string	-	arrival time of the message to the SC

<dt>	string	-	sending time of the message
<st>	integer	-	message status as coded in the PDU

If parameter is omitted the command returns the list of SMS with "REC UNREAD" status.

The order in which the messages are reported by +CMGL corresponds to their position in the memory storage

AT+CMGL=?

Test command returns a list of supported <stat>s

AT+CMGR - Read Message

This command is used to read a message.

3GPP TS 27.005
3GPP TS 23.040
3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGR=<index>

Execution command reports the message with location value <index> from <memr> message storage (<memr> is the message storage for read and delete SMs as last settings of command +CPMS).

Parameter:

Name	Type	Default	Description
<index>	integer	-	message index. The command output depends on the last settings of command +CMGF (message format to be used). There are two modes: PDU mode Text mode See the following Additional info sections.

Additional info:

▶▶ In case of PDU mode, if there is a message in location <index>, the output has the following format:

+CMGR: <stat>,[<alpha>,]<length><CR><LF><pdu>

Name	Type	Default	Description
<stat>	integer	N/A	status of the message
	Values:		
	0 : new message		
	1 : read message		
	2 : stored message not yet sent		
	3 : stored message already sent		
<alpha>	string	-	string type alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS
<length>	integer	-	PDU length in bytes
<pdu>	string	-	message in PDU format, according to 3GPP TS 23.040

- In case of Text mode, if there is a received message in location <index>, the output has the following format (the information written in italics will be present depending on +CSDH last setting):

  +CMGR:<stat>,<oa>,[<alpha>,]<scts>
  [,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data>

  If there is either a sent or an unsent message in location <index> the output format is:

  +CMGR:<stat>,<da>,[<alpha>] [,<toda>,<fo>,<pid>,<dcs>,
  [<vp>],<sca>,<tosca>,<length>]<CR><LF><data>

  If there is a Message Delivery Confirm message in location <index> the output format is:

  +CMGR: <stat>,<fo>,<mr>,,,<scts>,<dt>,<st>

Name	Type	Default	Description
<stat>	string	N/A	status of the message
	Values:		
	"REC UNREAD"	:	new received message
	"REC READ"	:	received message read
	"STO UNSENT"	:	message stored not yet sent

"STO SENT"	:	message stored already sent

<fo>	integer	-	first octet of the message PDU
<mr>	integer	-	message reference number; 3GPP TS 23.040 TP-Message-Reference in integer format
<scts>	string	-	arrival time of the message to the SC
<dt>	string	-	sending time of the message
<st>	integer	-	message status as coded in the PDU
<pid>	integer	-	Protocol Identifier
<dcs>	integer	-	Data Coding Scheme
<vp>	mixed	-	Validity Period; its format depends on SMS-SUBMIT <fo> setting (see +CPMS): - Not present: if <fo> tells that Validity Period Format is not present - Integer: if <fo> tells that Validity Period Format is relative

			- Quoted time-string type: if <fo> tells that Validity Period Format is absolute - Quoted hexadecimal representation of 7 octets: if <fo> tells that Validity Period Format is enhanced
<oa>	string	-	Originator address, represented in the currently selected character set (see +CSCS).
<da>	string	-	Destination address, represented in the currently selected character set (see +CSCS).
<alpha>	string	-	The alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS.
<sca>	string	-	Service Centre Address
<tooa>	integer	N/A	type of number of <oa>

Values:

129	:	number in national format
145	:	number in international format (contains the "+")

<toda>	integer	N/A	type of number of <da>

Values:

129	:	number in national format
145	:	number in international format (contains the "+")

<tosca>	integer	N/A	type of number of <sca>

Values:

129	:	number in national format
145	:	number in international format (contains the "+")

<length>	integer	-	text length
<data>	string	-	TP-User-Data If <dcs> indicates that 3GPP TS 23.038 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS) If <dcs> indicates that 8-bit data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will

be converted as two characters 0x32 0x41)

In both cases (in PDU or Text mode) if status of the message is 'received unread', status in the storage changes to 'received read'.

Concatenated SMS are supported but AT+CMGR automatic join to long SMS text is not supported.

Concatenated SMS reading is supported through AT#CCMGR .

AT+CMGR=?

Test command returns the OK result code

```
AT+CMGF=0
OK

AT+CMGR=1
+CMGR:2,,21
0791280192919030116 40A8110516529700000A709027A794E77B95C2E
OK

AT+CMGF=1
OK

AT+CMGR=3
+CMGR:"REC READ","+821020955219",,"07/07/19,10:06:34+36" test
```

message/.....
OK

AT+CMGS - Send Short Message

The command is related to sending short messages.

3GPP TS 27.005

3GPP TS 23.040

3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGS

Execution command sends a short message to the network.

It can have two syntax formats according to the SMS format: PDU or Text mode (see +CMGF command).

If short message is successfully sent to the network, the result is shown with the following URC:

+CMGS: <mr>[,<scts>]

Additional info:

- In PDU mode the +CMGS command has the following syntax:

  AT+CMGS=<length>

  After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

  <CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

  and waits for the specified number of bytes.

The PDU shall be hexadecimal format (each octet of the PDU is given as two IRA character long hexadecimal number) and given in one line.

To send the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<length>	integer	N/A	length in bytes of the PDU to be sent (excluding the SMSC address octets)
	Value:		
	7÷164	:	number of bytes

In Text mode the +CMGS command has the following syntax:

AT+CMGS=<da>[,<toda>]

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

After this prompt, you can enter text that should be formatted as follows:

if current <dcs> (see +CSMP) indicates that 8-bit or UCS2 data coding scheme is used or current <fo> (see +CSMP) indicates that 3GPP TS 23.040 TP-User-Data-Header-Indication is set, the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octet (e.g. the 'asterisk' will be entered as 2A (IRA50

and IRA65) and this will be converted to an octet with integer value 0x2A)

To send the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS).
<toda>	string	129	type of destination address

Values:

129	:	number in national format
145	:	number in international format (contains the "+")

Unsolicited fields:

Name	Type	Description
<mr>	integer	TP-Message-Reference number as per 3GPP TS 23.040
<scts>	string	TP-Service Centre Time Stamp in Time String Format. <scts> is returned when +CSMS <service> value is 1 and network supports.

- The DCD signal shall be in ON state while data is entered. The echoing of data is controlled by echo command E.
- in PDU mode: when the length of the SMSC address equals 0, then the SMSC address set with command +CSCA is used; in this case the SMSC Type of Address octet shall not be present in the data.
- To ensure that during the command execution, which may take several seconds, no other SIM interacting commands issued, care must be taken.
- Concatenated SMS are supported but AT+CMGS automatic split of long SMS text is not supported; the maximum number of chars depends on the <dcs>:

  GSM-7, 160 chars

  IRA 8-bit, 140chars

  UCS-2, 70 chars

  Concatenated SMS sending is supported through AT#CCMGS.
- If message sending fails for some reason, then an error code is reported.

AT+CMGS=?

Test command returns the OK result code

To avoid malfunctions it is suggested to wait for the +CMGS: <mr> or +CMS ERROR: <err> response before issuing further commands.

Set PDU mode
AT+CMGF=0

AT+CMGS=18
> 0881280100990102591155078110209055 12F90000A704F4F29C0E

+CMGS: 124

OK

Set text mode
AT+CMGF=1

AT+CSMP=17,167,0,0

AT+CMGS="01090255219",129

>TEST MESSAGE

+CMGS:125

OK

AT+CMSS - Send Message from Storage

This command sends to the network a message which is already stored in the <memw> storage.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMSS=<index>[,<da>[,<toda>]]

Execution command sends to the network a message which is already stored in the <memw> storage (see +CPMS) at the location <index>.

Parameters:

Name	Type	Default	Description
<index>	string	-	location value in the message storage <memw> of the message to send
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS); if it is given it shall be used instead of the one stored with the message.
<toda>	integer	N/A	type of destination address Values: 129 : number in national format 145 : number in international format (contains the "+")

Additional info:

If message is successfully sent to the network then the result is sent in the format:

+CMSS: <mr>

where:

<mr> - message reference number.

If message sending fails for some reason, an error code is reported:

+CMS ERROR:<err>

To store a message in the <memw> storage see command +CMGW.

Care must be taken to ensure that during the command execution, which may take several seconds, no other SIM interacting commands are issued.

AT+CMSS=?

Test command returns the OK result code.

To avoid malfunctions is suggested to wait for the +CMSS: <mr> or +CMS ERROR: <err> response before issuing further commands.

AT+CMGW - Write Short Message to Memory

The command is related to writing short messages.

3GPP TS 27.005
3GPP TS 23.040
3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CMGW

Execution command writes a new short message in the <memw> memory storage (see +CPMS). It can have two syntax formats according to the SMS format: PDU or Text mode (see +CMGF command). If short message is successfully written the following URC is displayed:

+CMGW: <index>

Additional info:

In PDU mode the +CMGW command has the following syntax:

AT+CMGW=<length>[,<stat>]

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

and waits for the specified number of bytes.

To write the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<length>	integer	N/A	length in bytes of the PDU to be written

	Value:		
	7÷164	:	number of bytes
<stat>	integer	N/A	message status
	Values:		
	0	:	new message new message (received unread message; default for DELIVER messages (3GPP TS 23.040 SMS-DELIVER messages))
	1	:	read message
	2	:	stored message not sent yet (default for SUBMIT messages (3GPP TS 23.040 SMS-SUBMIT messages))
	3	:	stored message already sent
<data>	hex	-	PDU bytes, given in online mode

▶▶ In Text mode the +CMGW command has the following syntax:

AT+CMGW[=<da>[,<toda>[,<stat>]]]

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

After this prompt, you can enter text that should be formatted as follows:

if current <dcs> (see +CSMP) indicates that 8-bit or UCS2 data coding scheme is used or current <fo> (see +CSMP) indicates that 3GPP TS 23.040 TP-User-Data-Header-Indication is set, the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octet (e.g. the 'asterisk' will be entered as 2A (IRA50 and IRA65) and this will be converted to an octet with integer value 0x2A)

The command waits for the specified number of bytes.

To write the message issue Ctrl-Z char (0x1A hex). To exit without writing the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS).
<toda>	integer	N/A	type of destination address
	Values:		
	129 :		number in national format
	145 :		number in international format (contains the character "+")

<stat>	string	N/A	message status

Values:

"REC UNREAD"	:	new received message unread
"REC READ"	:	received message read
"STO UNSENT"	:	message stored not yet sent (default)
"STO SENT"	:	message stored already sent

Unsolicited field:

Name	Type	Description
<index>	integer	message location index in the memory <memw> (see +CPMS). If message storing fails for some reason, an error code is reported.

- The DCD signal shall be in ON state while <data> is entered. The echoing of <data> is controlled by echo command E.
- In PDU mode, not only SUBMIT messages can be stored in SIM, but also DELIVER and STATUS REPORT messages (3GPP TS 23.040 SMS-STATUS-REPORT messages).
- Care must be taken to ensure that during the command execution, which may take several seconds, no other SIM interacting commands are issued.

- It is possible to save a concatenation of at most 10 SMs; the maximum number of chars depends on <dcs>:

  2448 chars

  2144 chars if 8-bit is used

  1072 chars if UCS2 is used

- In text mode, not only SUBMIT messages can be stored in SIM, but also DELIVER messages.

  The type of saved message depends upon the current <fo> parameter (see +CSMP). For a DELIVER message, current <vp> parameter (see +CSMP) is used to set the message Service Centre Time Stamp <scts>, so it must be an absolute time string, e.g. "09/01/12,11:15:00+04".

- If message writing fails for some reason, then an error code is reported.

AT+CMGW=?

Test command returns the OK result code.

To avoid malfunctions it is suggested to wait for the +CMGW: <index> or +CMS ERROR: <err> response before issuing further commands.

```
AT+CMGF=0          set PDU mode
OK

AT+CMGW=18

> 0881280100990102591155078110209055 12F90000A704F4F29C0E

+CMGW: 29

OK

AT+CMGF=1          set text mode
OK
```

AT+CSMP=17,167,0,0
OK

AT+CSCA="821029190903",145
OK

AT+CMGW="0165872928"

> test message...

+CMGW: 28

AT+CMGD - Delete Message

This command allows to delete from memory messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGD=<index>[,<delflag>]

Execution command deletes SMS message(s) from a selected memory storage. Storage is selected by command +CPMS.

Parameters:

Name	Type	Default	Description
<index>	integer	-	Message index in the selected storage; it can have values form 1 to N, where N depends on the available space in the selected storage (see +CPMS)
<delflag>	integer	0	Type of multiple message deletion

Values:

0 : delete message specified in <index>

1 : delete all read messages from selected storage, leaving unread messages and stored mobile originated messages (whether sent or not) untouched

2 : delete all read messages from selected storage and sent mobile originated messages, leaving

unread messages and unsent mobile originated messages untouched

3 : delete all read messages from selected storage, sent and unsent mobile originated messages, leaving unread messages untouched

4 : delete all messages from selected storage.

If <delflag> is present and not set to 0 then, if <index> is greater than 0, <index> is ignored and the command follows the rules for <delflag> shown above.

AT+CMGD=?

Test command shows the valid memory locations <index> and the supported values of <delflag>.

```
AT+CMGD=?
+CMGD: (1,2,3,6,7,17,18,19,20,37,38,39,47),(0-4)
OK
```

AT#CCMGR - Read Concatenated Message

This command is used to read concatenated short messages.

3GPP TS 27.005
3GPP TS 23.040
3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#CCMGR=<index>

Execution command reports the message with location value <index> from <memr> message storage (<memr> is the message storage for read and delete SMs as last settings of command +CPMS).

It is only supported in the SMS format: Text mode (see +CMGF command).

Parameter:

Name	Type	Default	Description
<index>	integer	-	message index. The command output depends on the last settings of command +CMGF (message format to be used). There is only one mode: Text mode See the following Additional info sections.

Additional info:

- In case of PDU mode the #CCMGR is not supported.

▶▶ In case of Text mode, if there is a received message in location <index>, the output has the following format (the information written in italics will be present depending on +CSDH last setting):

#CCMGR:<stat>,<oa>,[<alpha>,]<scts>
[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]
[,<msg_ref>,<msg_max>,<msg_index>]<CR><LF><data>

If there is either a sent or an unsent message in location <index> the output format is:

#CCMGR:<stat>,<da>,[<alpha>] [,<toda>,<fo>,<pid>,<dcs>,
[<vp>],<sca>,<tosca>,<length>]
[,<msg_ref>,<msg_max>,<msg_index>]<CR><LF><data>

Name	Type	Default	Description
<stat>	string	N/A	status of the message

Values:

"REC UNREAD"	:	new received message
"REC READ"	:	received message read
"STO UNSENT"	:	message stored not yet sent
"STO SENT"	:	message stored already sent

<fo>	integer	-	first octet of the message PDU
<scts>	string	-	arrival time of the message to the SC
<pid>	integer	-	Protocol Identifier
<dcs>	integer	-	Data Coding Scheme
<vp>	mixed	-	Validity Period; its format depends on SMS-SUBMIT <fo> setting (see +CPMS): - Not present: if <fo> tells that Validity Period Format is not present - Integer: if <fo> tells that Validity Period Format is relative - Quoted time-string type: if <fo> tells that Validity Period Format is absolute - Quoted hexadecimal representation of 7 octets: if <fo> tells that Validity Period Format is enhanced
<oa>	string	-	Originator address, represented in the currently selected character set (see +CSCS).

<da>	string	-	Destination address, represented in the currently selected character set (see +CSCS).
<alpha>	string	-	The alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS.
<sca>	string	-	Service Centre Address
<tooa>	integer	N/A	type of number of <oa>

Values:

129	:	number in national format
145	:	number in international format (contains the "+")

<toda>	integer	N/A	type of number of <da>

Values:

129	:	number in national format

	145 : number in international format (contains the "+")		
<tosca>	integer	N/A	type of number of <sca>
	Values:		
	129 : number in national format		
	145 : number in international format (contains the "+")		
<length>	integer	-	text length
<data>	string	-	TP-User-Data If <dcs> indicates that 3GPP TS 23.038 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS) If <dcs> indicates that 8-bit data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41)

<msg_ref>	integer	-	Concatenated short message reference number. This is the reference number for the concatenated short message and is constant in every single split AT#CCMGR message
<msg_max>	integer	-	Total number of single split AT#CCMGR messages for the concatenated short message. This value is constant in every single split AT#CCMGR message
<msg_index>	integer	-	Index of the single split AT#CCMGR messages for the concatenated short message. This value is 1 for the first single split AT#CCMGR messages and increased by 1 up to <msg_max> for the next ones.

If status of the message is 'received unread', status in the storage changes to 'received read'.

Concatenated SMS are supported but automatic join to long SMS text is not supported; the join of long SMS text must be done after having used AT#CCMGR.

AT#CCMGR=?

Test command returns the OK result code

```
AT+CMGF=1
OK
AT+CMGL=ALL
+CMGL: 0,"REC UNREAD","01090255219",,"25/01/09,14:23:40+04"
test split 1
+CMGL: 1,"REC UNREAD","01090255219",,"25/01/09,14:24:17+04"
tests split 2
OK
AT#CCMGR=0
#CCMGR: "REC
READ","01090255219",,"25/01/09,14:23:40+04",145,4,0,0,"
01090255010",145,12,,,
test split 1
OK
AT#CCMGR=1
#CCMGR: "REC
READ","01090255219",,"25/01/09,14:24:17+04",145,4,0,0,"
01090255020",145,14,,,
tests split 2
OK
```

AT#CCMGS - Send Concatenated Short Message

The command is related to sending concatenated short messages.

3GPP TS 27.005

3GPP TS 23.040

3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#CCMGS

Execution command sends a concatenated short message to the network.

It can have only one syntax format, it is only supported in the SMS format Text mode (see +CMGF command).

If short message is successfully sent to the network, the result is shown with the following URC:

#CCMGS: <mr>

Additional info:

- In PDU mode the #CCMGS command is not supported.

- In Text mode the #CCMGS command has the following syntax:

  AT#CCMGS=<da>[,[<toda>][,<msg_ref>,<msg_max>,<msg_index>]]

  After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

  <CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

After this prompt, you can enter text that should be formatted as follows:

if current <dcs> (see +CSMP) indicates that 8-bit or UCS2 data coding scheme is used or current <fo> (see +CSMP) indicates that 3GPP TS 23.040 TP-User-Data-Header-Indication is set, the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octet (e.g. the 'asterisk' will be entered as 2A (IRA50 and IRA65) and this will be converted to an octet with integer value 0x2A)

To send the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS).
<toda>	string	129	type of destination address

Values:

129	:	number in national format
145	:	number in international format (contains the "+")

<msg_ref>	integer	0	Concatenated short message reference number. This is the reference number for the concatenated short message and must be kept constant in every single split AT#CCMGS message
<msg_max>	integer	0	Total number of single split AT#CCMGS messages for the concatenated short message. This value must be kept constant in every single split AT#CCMGS message

Values:

0	:	the single AT#CCMGS message is sent as a non-concatenated short message
1÷255	:	total number of single split AT#CCMGS messages

<msg_index>	integer	0	Index of the single split AT#CCMGS messages for the concatenated short message. This value must be 1 for the first single split AT#CCMGS messages and increased by

		1 up to <msg_max> for the next ones

Values:

0	:	the single AT#CCMGS message is sent as a non-concatenated short message
1÷255	:	index of the single split AT#CCMGS messages

Unsolicited fields:

Name	Type	Description
<mr>	integer	TP-Message-Reference number as per 3GPP TS 23.040

- The DCD signal shall be in ON state while data is entered. The echoing of data is controlled by echo command E.
- To ensure that during the command execution, which may take several seconds, no other SIM interacting commands issued, care must be taken.
- Concatenated SMS are supported but automatic split of long SMS text is not supported.

  The split of long SMS text must be done before start using AT#CCMGS; the maximum number of chars of the single split AT#CCMGS message depends on the <dcs>:

  GSM-7, 160 chars

IRA 8-bit, 140chars

UCS-2, 70 chars.

If message sending fails for some reason, then an error code is reported.

AT#CCMGS=?

Test command returns the OK result code

To avoid malfunctions it is suggested to wait for the #CCMGS: <mr> or +CMS ERROR: <err> response before issuing further commands.

Set text mode

```
AT+CMGF=1

OK

AT#CCMGS="01090255219",10,2,1


>TEST MESSAGE SPLIT 1


#CCMGS:1


OK

AT#CCMGS="01090255219",,10,2,2

>TEST MESSAGE SPLIT 2

#CCMGS:2

OK
```

AT#E2SMSRI - SMS Ring Indicator

This set command enables/disables the Ring Indicator pin response to an incoming SMS message. If enabled, a negative going pulse is generated when receiving an incoming SMS message.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#E2SMSRI=[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	RI enabling

Values:

0	:	disables RI pin response for incoming SMS messages
50÷1150	:	enables RI pin response for incoming SMS. The value of <n> is the duration in ms of the pulse generated on receipt of an incoming SMS.

If +CNMI=3,1 command is issued, and the module is in a GPRS connection, a 100 ms break signal is sent and a 1 sec. pulse is generated on RI pin, no matter if the RI pin response is either enabled or not.

If +CNMI=0,0,0,0,0 command is issued, no ring pulse will be generated, regardless of #E2SMSRI value

AT#E2SMSRI?

Read command reports the duration in ms of the pulse generated on receipt of an SMS, in the format:

#E2SMSRI: <n>

<n>=0 means that the RI pin response to an incoming SMS is disabled

AT#E2SMSRI=?

Reports the range of supported values for parameter <n>

AT+CSCB - Select Cell Broadcast

The command selects which types of Cell Broadcast Messages are to be received by the device.

3GPP TS 27.005
3GPP TS 23.041
3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSCB=[<mode>[,<mids>[,<dcss>]]]

Set command selects which types of Cell Broadcast Messages are to be received by the device

Parameters:

Name	Type	Default	Description
<mode>	integer	0	select which types of Cell Broadcast messages are to be received

Values:

0 : the message types defined by <mids> and <dcss> are accepted

1 : the message types defined by <mids> and <dcss> are rejected

<mids>	string	-	message Identifiers: all different possible combinations of the CBM message identifiers;
<dcss>	string	-	Data Coding Schemes: all different possible combinations of CBM data coding schemes;

The current settings are stored also by +CSAS command

AT+CSCB?

Read command reports the current value of parameters <mode>, <mids> and <dcss>

AT+CSCB=?

Test command returns the range of values for parameter <mode>

AT+CSCB?

+CSCB: 1,"",""

OK (all CBMs are accepted, none is rejected, only in UMTS NW)

AT+CSCB=0,"0,1,300-315,450","0-3"

OK

Time & Alarm

AT+CCLK - Clock Management

The command is related to real time clock management.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CCLK=<time>

Set command sets the real-time clock of the module.

Parameter:

Name	Type	Default	Description
<time>	string	N/A	Current time as quoted string in the format: "yy/MM/dd,hh:mm:ss±zz"

Values:

yy : year (two last digits are mandatory), range is 00..99

MM : month (two digits are mandatory), range is 01..12

dd : day (two digits are mandatory) The range for dd(day) depends either on the month and on the year it refers to. Available ranges are: (01..28) (01..29) (01..30) (01..31). Trying to enter

an out of range value will raise an ERROR message.

hh : hour (two digits are mandatory), range is 00..23

mm : minute (two digits are mandatory), range is 00..59

ss : seconds (two digits are mandatory), range is 00..59

±zz : time zone (indicates the difference, expressed in quarter of an hour, between the local time and GMT; two digits are mandatory), range is -96...+96

any real-time clock setting will be lost after reboot or shutdown.

AT+CCLK?

Read command returns the current setting <time> of the real-time clock, in the format:

+CCLK: <time>

AT+CCLK=?

Test command returns the OK result code.

Set date and time:
AT+CCLK="02/09/07,22:30:00+00"
OK

Read date and time:

```
AT+CCLK?
+CCLK: "02/09/07,22:30:25+00"
OK
```

AT+CTZR - Time Zone Reporting

This command enables and disables the time zone change event reporting.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+CTZR=<onoff>

Set command permits to enable/disable the time zone change event reporting.

If the reporting is enabled and whenever the time zone is changed, the MT returns the unsolicited result code:

+CTZV: <tz>

Parameter:

Name	Type	Default	Description
<onoff>	string	0	enable/disable the time zone change event reporting.

Values:

0 : Disable time zone change event reporting

1 : Enable time zone change event reporting by unsolicited result code +CTZV: <tz>.

2 : Enable extended time zone and local time reporting by unsolicited result code +CTZE: <tz>,<dst>,[<time>]. Not supported

3 : Enable extended time zone and universal time reporting by unsolicited result code +CTZEU: <tz>,<dst>,[<utime>]

Unsolicited field:

Name	Type	Description
<tz>	string	New time zone.
<dst>	integer	indicating whether <tz> includes daylight savings adjustment
<time>	string	local time. The format is YYYY/MM/DD,hh:mm:ss
<utime>	string	universal time. The format is YYYY/MM/DD,hh:mm:ss

AT+CTZR?

Read command reports the currently selected <onoff> in the format:

+CTZR: <onoff>

AT+CTZR=?

Test command reports the supported range of values for parameter <onoff>

AT+CTZU - Automatic Time Zone Update

Set command enables/disables the automatic time zone update via NITZ.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT+CTZU=<onoff>

Parameter:

Name	Type	Default	Description
<onoff>	integer	1	enables/disables the automatic time zone update via NITZ

Values:

0	:	disable
1	:	enable

The command +CTZU is the ETSI standard equivalent of Telit Cinterion custom command #NITZ (for the date and time update functionality).

Despite of the name, the command +CTZU enables automatic update of the date and time set by +CCLK command (not only time zone). This happens when a Network Identity and Time Zone (NITZ) message is sent by the network.

If the automatic date and time update functionality has been enabled by +CTZU or #NITZ (or both), NITZ message will cause a date and time update.

AT+CTZU?

Read command reports the current setting of <onoff> in the format:

+CTZU: <onoff>

AT+CTZU=?

Test command returns the supported values of parameter <onoff>.

AT#NITZ - Network Identity and Time Zone

This command handles Network Identity and Time Zone.

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT#NITZ=[<val>[,<mode>]]

Set command enables/disables the automatic date/time updating and the Full Network Name applying. It enables also the #NITZ URC in the format:

#NITZ: <datetime>

and permits to change its format.

Parameters:

Name	Type	Default	Description
<val>	integer	7	identifies the functionalities to enable. The <val> parameter is a sum of integer values, where every value corresponds to a functionality: 1 - enables automatic date/time updating 2 - enables Full Network Name applying; the value is valid but is does not have any effect (not supported) 4 - sets the #NITZ URC 'extended' format (see <datetime> below) 8 - sets the #NITZ URC 'extended' format with Daylight Saving Time (DST) support (see <datetime> below)

Values:

0	:	disables every functionality
1÷15	:	sum of integer values

<mode>	integer	0	enables/disables the #NITZ URC

Values:

0	:	disables the URC
1	:	enables the URC

Unsolicited field:

Name	Type	Description
<datetime>	string	string format depends on parameter <val> "yy/MM/dd,hh:mm:ss" - 'basic' format, if <val> is in (0..3) "yy/MM/dd,hh:mm:ss±zz" - 'extended' format, if <val> is in (4..7) "yy/MM/dd,hh:mm:ss±zz,d" - 'extended' format with DST support, if <val> is in (8..15) For the meaning of the <datetime> subfields, please check +CCLK and #CCLK commands

- AT#NITZ enables automatic update of the date and time set by +CCLK command.

  This happens when a Network Identity and Time Zone (NITZ) message is sent by the network.

  If the automatic date and time update functionality has been enabled by +CTZU or #NITZ (or both), NITZ message will cause a date and time update .

- If the DST information is not sent by the network, then the <datetime> parameter will have the format "yy/MM/dd,hh:mm:ss±zz".

- Date and time information can be sent by the network after registration or after PS attach.

AT#NITZ?

Read command reports whether

- automatic date/time updating

- Full Network Name applying

- #NITZ URC (as well as its format)

are currently enabled or not in the format:

#NITZ: <val>,<mode>

AT#NITZ=?

Test command returns supported values of parameters <val> and <mode>.

AT#CCLK - Clock Management

The command is related to real time clock management.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CCLK=<time>

Set command sets the real-time clock of the module.

Parameter:

Name	Type	Default	Description
<time>	string	N/A	Current time as quoted string in the format: "yy/MM/dd,hh:mm:ss±zz,d"

Values:

yy : year (two last digits are mandatory), range is 00..99

MM : month (two digits are mandatory), range is 01..12

dd : day (two digits are mandatory) The range for dd(day) depends either on the month and on the year, it refers to. Available ranges are: (01..28) (01..29) (01..30) (01..31). Trying to enter an out of range value will raise an ERROR message.

hh	:	hour (two digits are mandatory), range is 00..23
mm	:	minute (two digits are mandatory), range is 00..59
ss	:	seconds (two digits are mandatory), range is 00..59
±zz	:	time zone (indicates the difference, expressed in quarter of an hour, between the local time and GMT; two digits are mandatory), range is: -96..+96
d	:	number of hours added to the local TZ because of Daylight Saving Time (summertime) adjustment; range is 0-2.

AT#CCLK?

Read command returns the current setting of the real-time clock, in the format <time>.

If the time is set by the network but the Daylight-Saving Time (DST) information is missing, or the time is set by +CCLK command, then the <time> format is:

"yy/MM/dd,hh:mm:ss±zz"

AT#CCLK=?

Test command returns the OK result code.

The way of writing the <time>:

"yy/mm/dd,hh:mm:ss±zz,d"

AT#CCLK="02/09/07,22:30:00+00,1"

* Set command:

  AT#CCLK="14/02/18,08:00:00+12,1"

  OK

* Read command:

  AT#CCLK?
  #CCLK: "14/02/18,08:00:02+12,1"

  OK

AT#CCLKMODE - Clock Mode

This command allows to enable the local time or the UTC time.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CCLKMODE=<mode>

Set command enables the local time or the UTC time in +CCLK and #CCLK commands

Parameter:

Name	Type	Default	Description
<mode>	integer	0	Time and date mode

Values:

0 : Local time + local time zone offset

1 : UTC time + local time zone offset

The setting is saved automatically in NVM.

AT#CCLKMODE?

Read command reports whether the local time or the UTC time is enabled, in the format:

#CCLKMODE: <mode>

AT#CCLKMODE=?

Test command reports the supported range of values for parameter <mode>

WiFi

AT#WIFISCAN - WiFi RSSI and BSSID (MAC address) list

Detect surrounding WiFi RSSI and BSSID (MAC address) list.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#WIFISCAN[=[[<maxRespTime>],[[<maxRounds>],[[<maxBSSID>],[[<maxRoundTime>],[[<priority>],[<maxChannelTime>]]]]]]

Set command sets WiFi search parameters and starts the detection of surrounding WiFi RSSI and BSSID (MAC address) list.

It triggers in response a list of intermediate messages.

The intermediate messages have the following format:

#WIFISCAN: <ecn>,<ssid>,<RSSI>,<macAddress>,<channelNumber>

See Additional info section

Parameters:

Name	Type	Default	Description
<maxRespTime>	integer	12000	maximum response time of WiFi scan in ms
<maxRounds>	integer	1	maximum number of WiFi scan rounds
<maxBSSID>	integer	5	maximum required WiFi SSID numbers

<maxRoundTime>	integer	5	maximum search time per WiFi scan round in seconds
<priority>	integer	0	priority of the WiFi scan

Values:

0 : Data preferred

1 : WiFi preferred (drops connection)

<maxChannelTime>	integer	280	maximum search time per WiFi channel in ms

Additional info:

▶▶ Intermediate message parameters:

Name	Type	Description
<ecn>	integer	Not supported (empty field)
<ssid>	integer	Not supported (empty field)
<RSSI>	integer	RSSI in dBm
<macAddress>	string	MAC address (BSSID)
<channelNumber>	integer	channel number

The recommended UE state to start the detection of surrounding WiFi RSSI and BSSID (MAC address) list is the UE idle state.

If the priority is set to 0 and the UE is in connected state, the UE does not start the detection of surrounding WiFi RSSI and BSSID (MAC address) list until UE returns to the idle state.

If the priority is set to 1 and the UE is in connected state, the UE releases connection immediately and starts the detection of surrounding WiFi RSSI and BSSID (MAC address) list.

The value of <maxRespTime> should not be less than <maxRounds> multiplied by <maxRoundTime>.

AT#WIFISCAN?

Read command reports WiFi search parameters, in the format:

#WIFISCAN: <maxRespTime>,<maxRounds>,<maxBSSID>,<maxRoundTime>,<priority>,<maxChannelTime>

The parameters are described in the previous sections

AT#WIFISCAN=?

Test command returns the supported range of values of parameters.

This command is supported only by following product variants

Modules supporting the command

LE910Q1-	WW
LE910Q1-	WWG
ELS63-	I
ELS63-	IG

HW and Radio Control

AT#CBC - Battery and Charger Status

This command returns the current Battery and Charger state.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CBC

Execution command returns the current Battery and Charger state. The response is in the format:

#CBC: <ChargerState>,<BatteryVoltage>

Additional info:

- The response has its fields described below.

Name	Type	Default	Description
<ChargerState>	integer	0	Battery charger state

Values:

0 : charger not connected

1 : charger connected and charging

2 : charger connected and charge completed

<BatteryVoltage>	integer	-	battery voltage in units of 10 mV: it is the real battery voltage only if charger is not connected; if the charger is connected this value depends on the charger voltage

AT#CBC=?

Test command returns the OK result code.

AT#CBCMON - Battery Monitor

This command enables/disables the Battery Monitor function and URC.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CBCMON=<enable>[,<vthresh>][,<PowerSupplyDomain>]

Set command enables/disables and configures the Battery Monitor function.

The response to the set command is in the format:

#CBCMON: <vthresh_BB_curr>,<vthresh_BB_request>

OK

If it is enabled the unsolicited result codes are in the format:

#CBCMON: Undervoltage Warning

URC indicates that the UE is running in undervoltage, supply voltage is close to the defined undervoltage threshold.

#CBCMON: Undervoltage Shutdown

URC indicates that the UE is running in undervoltage and the undervoltage threshold exceeded. Module switches off within 5 seconds after sending the URC.

Parameters:

Name	Type	Default	Description
<enable>	integer	0	enables/disables the Battery Monitor function and unsolicited result code.

Values:

0	:	disable the Battery Monitor function and unsolicited result code
1	:	enable the Battery Monitor function and unsolicited result code

<vthresh>	integer	0	Undervoltage threshold value

Values:

-4 ÷ 4	:	Undervoltage threshold, see table below

<PowerSupplyDomain>	integer	0	Power Supply Domain

Values:

0	:	BB domain

Additional info:

▶▶ The response has its fields described below.

Name	Type	Default	Description
<vthresh_BB_curr>	integer	0	Current BB undervoltage threshold Values: -4 ÷ 4 : Current BB undervoltage threshold, see table below
<vthresh_BB_request>	integer	0	BB undervoltage threshold after next restart Values: -4 ÷ 4 : BB undervoltage threshold after next restart, see table below

▶▶ Undervoltage threshold table

Value	Voltage (V)
-4	2.90
-3	2.95
-2	3.00

-1	3.05
0	3.10
1	3.15
2	3.20
3	3.25
4	3.30

AT#CBCMON?

Read command returns the current values of <enable>, <vthresh_BB_curr> and <vthresh_BB_request>

#CBCMON: <enable>,<vthresh_BB_curr>,<vthresh_BB_request>

AT#CBCMON=?

Test command returns supported values for parameters <enable>, <vthresh> and <PowerSupplyDomain>.

AT#TEMPCFG - Temperature Monitor Configuration

Set command sets the temperature zones shown in Note section. Use #TEMPMON command to get the current internal temperature of the module.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#TEMPCFG=<etlz_clr>,<etlz>,<etlz_act_in>,<otlz_clr>,<otlz>,<otlz_act_in >, <otnz_clr>,<otnz>,<otnz_act_in>,<otuz_clr>,<otuz>,<otuz_act_in>, <etuz_clr>,<etuz>,<etuz_act_in>

Parameters:

Name	Type	Default	Description
<etlz_clr>	integer	-273	extreme low zone temperature threshold clear. It has only one valid value: -273 Celsius degrees. See note.
	Value:		
	-273 : Celsius degrees. See note		
<etlz>	integer	-33	extreme low zone temperature threshold. Default value -33 Celsius degrees.
	Value:		
	-40 ÷ 105 : Celsius degrees		

<etlz_act_in>	integer	0	extreme low zone action info. Default value 0.

Value:

0÷5	:	action

<otlz_clr>	integer	-35	operate low zone temperature threshold clear. Default value -35 Celsius degrees.

Value:

-40 ÷ 105	:	Celsius degrees

<otlz>	integer	-28	operate low zone temperature threshold. Default value -28 Celsius degrees.

Value:

-40 ÷ 105	:	Celsius degrees

<otlz_act_in>	integer	0	operate low zone action info. Default value 0.

Value:

	0÷5	:	action

<otnz_clr>	integer	-30	operate normal zone temperature threshold clear. Default value -30 Celsius degrees.

Value:

-40 ÷ 105	:	Celsius degrees

<otnz>	integer	94	operate normal zone temperature threshold. Default value 94 Celsius degrees.

Value:

40÷105	:	Celsius degrees

<otnz_act_in>	integer	0	operate normal zone action info. Default value 0.

Value:

0÷5	:	action

<otuz_clr>	integer	80	operate up zone temperature threshold clear. Default value 80 Celsius degrees.

Value:

40÷105	: Celsius degrees

<otuz>	integer	99	operate up zone temperature threshold. Default value 99 Celsius degrees.

Value:

40÷105	: Celsius degrees

<otuz_act_in>	integer	0	operate up zone action info. Default value 0.

Value:

0÷5	: action

<etuz_clr>	integer	82	extreme up zone temperature threshold clear. Default value 82 Celsius degrees.

Value:

40÷105	: Celsius degrees

<etuz>	integer	105	extreme up zone temperature threshold. It has only one valid value: 105 Celsius degrees. See note.

Value:

105	:	Celsius degrees. See note

<etuz_act_in>	integer	0	extreme up zone action info. Default value 0.

Value:

0÷5	:	action

- <etlz_clr> extreme low zone threshold clear is enforced to have value of '-273'.

  Module does not operate in such temperature, but this value is logically set in order to define clearly 'thermal state' to temperatures below -40 Celsius degrees.

- <etuz> extreme up zone threshold is enforced to have value of '105'.

  Module does not operate in such temperature, but this value is logically set to define clearly 'thermal state' to temperatures above 105 Celsius degrees.

- The module is shutdown (<action> 5) at temperature above the Extreme up zone threshold (105 Celsius degrees).

- #TEMPMON set command, changes field "actions" to "mitigate" or "none" to all zones.
- The temperature correctly set are saved in a configuration file in the module file system.

AT#TEMPCFG?

Read command reports the current parameter setting for #TEMPCFG command in the format:

#TEMPCFG:<etlz_clr>,<etlz>,<etlz_act_in>,<otlz_clr>,<otlz>,<otlz_act_in>,<otnz_clr>,<otnz>,<otnz_act_in>,<otuz_clr>,<otuz>,<otuz_act_in>,<etuz_clr>,<etuz>,<etuz_act_in>

AT#TEMPCFG=?

Test command reports the supported range values for parameters <x_clr>, <x>, <x_act_in>, where "x" is substitute for etlz, otlz, otnz, otuz, etuz.

Values are:

#TEMPCFG: (-40-105),(-40-105),(0-5)

Thermal mitigation mechanism

After setting new values, the mitigation algorithm operates with them if a power cycle or a #REBOOT command is executed. Thermal mitigation mechanism works like this:

the whole temperature scale is divided into 5 states (zones).

each measured temperature should belong to a state called the current state.

Each state is defined by the following fields:

threshold	upper temperature boundary of the state. Values are in Celsius degrees.
threshold_clr	lower temperature boundary of the state. Values are in Celsius degrees.
action_in	indicator that indicates if an action should be taken or not in the current state.

action_in can be none or mitigate, their values are shown in the table below.

none	mitigate	mitigation action
0		no mitigation
	1	data throttling (reducing uplink baud rate) Not supported
	2	TX back off (reducing MTPL - Max Tx Power Limit) Not supported
	3	emergency calls only Not supported

4	RF OFF. RX and TX circuits automatically disabled (set AT+CFUN=4 and AT+CFUN? reports +CFUN: 4)
5	automatic shutdown. Module is powered OFF

There are five limitations on setting temperature and actions, in-order to keep module safety.

- User is prohibited to set an action of "automatic shutdown" to 'operate normal zone'.

- User is prohibited to set an action of "no mitigation" or "data throttling" to "operate up zone" (rule not valid for ELS63/LE910Q1 product variants).

- User is prohibited to set an action of "no mitigation" or "data throttling" or "tx backoff" to "extreme up zone" (rule not valid for ELS63/LE910Q1 product variants).

- User is prohibited to set "normal zone" above 97 Celsius degrees.

- If the module enters into a state of "emergency only" calls, registration again to a regular call, happens just when the module returns to "no mitigation" state only.

- A "+CME ERROR: operation not supported" error will be received as a response.

When temperature exceeds the current state threshold, the thermal mitigation algorithm searches the next state when this temperature is lower than threshold. After it finds it, the current state is updated to that state and then it checks whether action is mitigate. If yes, then it activates the mitigation according to the action_in of the current state.

When temperature decreases below threshold_clr then it does the same algorithm as above, but in the opposite direction. It searches the next state when this temperature is greater than threshold_clr, updates the current state to that state, and activates mitigation as described above.

In the state definitions there are the following two rules:

rule 1: overlap between 2 adjacent states of at least 2 Celsius degrees, i.e.:

thr of state(x) - thr_clr of state(x+1) >= 2

rule 2: every state shall have free temperature range which has no part in any overlap range. This range should be at least 2 Celsius degrees, i.e.:

thre_clr state(x+2) - thre state(x) >= 2

Rule 1 comes to ensure hysteresis in the transition between two states.

Rule 2 comes to ensure a minimum range for a stable state.

Referring to the figure below:

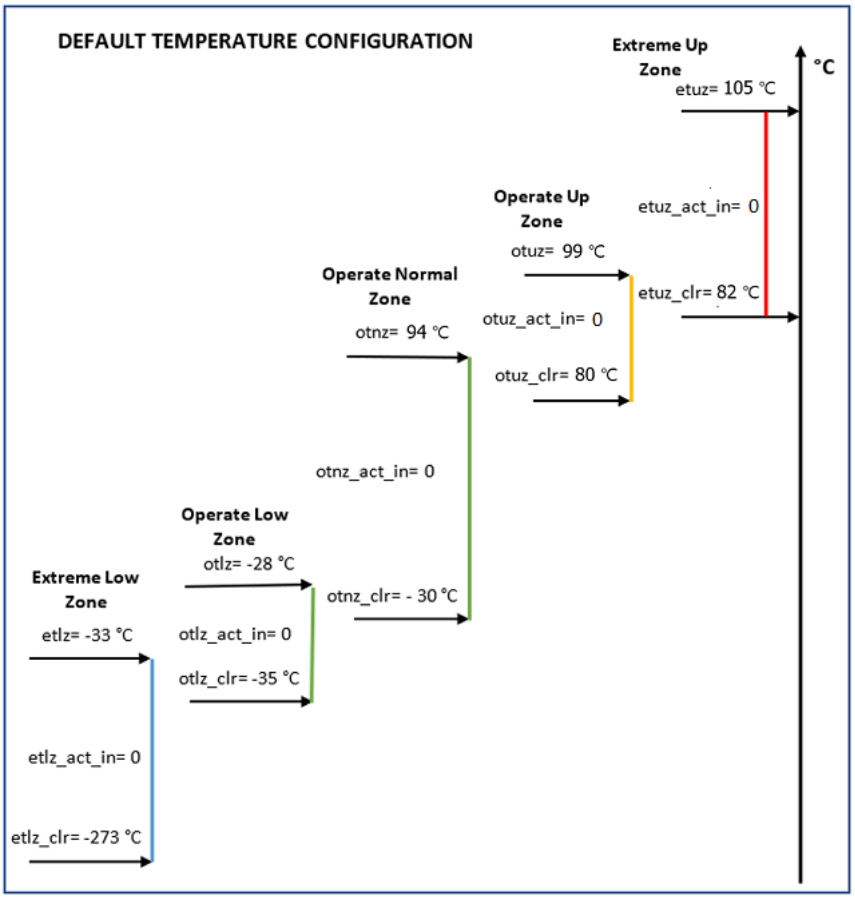
State 0 is Extreme low zone

State 1 is Operate low zone

State 2 is Operate normal zone

State 3 is Operate up zone

State 4 is Extreme up zone



After moving to zone with activity 3 (emergency call only), only when moving to zone with activity 0 (no mitigation) the device will register to the network.

<action> for high-zone cannot be 'no action' or 'data throttling'.

<action> for extreme high zone cannot be 'no action', 'data throttling' or 'tx backoff'.

Thermal management is active only when using a non-GCF test SIM. A SIM is a test SIM if it has an MCC of 001, 002, 003, 004, 005, 006, 007, 008, 009, 010, 011, 012.

</>

* AT#TEMPCFG=
-273,-33,3,-35,-28,2,-30,80,0,78,90,3,88,528,3
OK

Rules 1 and 2 are respected.

zone	thr_clr	thr	action_in
Extreme low zone	-273	-33	3: emergency call only
Operate low zone	-35	-28	2: TX back off
Operate normal zone	-30	80	0: no mitigation
Operate up zone	78	90	3: emergency call only
Extreme up zone	88	105	3: emergency call only

* AT#TEMPCFG=-273,-33,3,-35,-28,2,-30,80,0,79,90,3,88,105,3
+CME ERROR: operation not supported

Rule 1 is not respected: hysteresis less than 2 Celsius degrees

(thr of Operate normal zone) - (thr_clr of Operate up zone) = 1 < 2

zone	thr_clr	thr	action_in
Extreme low zone	-273	-33	3: emergency call only
Operate low zone	-35	-28	2: TX back off
Operate normal zone	-30	80	0: no mitigation
Operate up zone	79	90	3: emergency call only
Extreme up zone	88	105	3: emergency call only

* AT#TEMPCFG=-273,-33,3,-35,-28,2,-30,80,0,78,90,3,81,105,3

  +CME ERROR: operation not supported

Rule 2 is not respected: free temperature range less than 2 Celsius degrees

(thr_clr of Extreme up zone) - (thr of Operate normal zone) = 1 < 2

zone	thr_clr	thr	action_in
Extreme low zone	-273	-33	3: emergency call only

Operate low zone	-35	-28	2: TX back off
Operate normal zone	-30	80	0: no mitigation
Operate up zone	78	90	3: emergency call only
Extreme up zone	81	105	3: emergency call only

AT#GPIO - General Purpose Input/Output Pin Control

Configure the general-purpose input/output GPIO pins.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT#GPIO=[<pin>,<mode>[,<dir>[,<save>]]]

Execution command allows to configure the general purpose GPIO pin.

Parameters:

Name	Type	Default	Description
<pin>	integer	N/A	GPIO pin number. The supported range goes from Min to Max values that are hardware dependent. Use AT#GPIO=? test command to know Min and Max value. Value: Min÷Max : GPIO pin identifier
<mode>	integer	0	sets GPIO pin configuration, its action depends on <dir> value. Refer to Additional info sections. Value: 0÷4 : mode identifier
<dir>	integer	0	sets the GPIO pin in input, output, or alternate functions. Refer to Additional info sections.

Values:

0	:	pin set as input
1	:	pin set as output
2÷16	:	pin set in alternate functions (some values not supported).

<save>	integer	0	GPIO pin save configuration. If <save> is omitted, the configuration is stored in NVM only if user set ALTx function.

Values:

0	:	GPIO pin configuration is not saved
1	:	GPIO pin configuration is saved

Additional info:

▶▶ This table shows the GPIOs configurations set by <dir>=0 and <mode> values ranging from 0 to 4.

AT#GPIO=<pin>,<mode>,0

<mode>	Description
0	Set INPUT, any internal pull up/pull down removed.

1	Set INPUT, any internal pull up/pull down removed (same behavior as <mode>=0).
2	Read mode, <dir> can be omitted, see Additional info section below
3	Set INPUT, and internal pull up.
4	Set INPUT, and internal pull down.

▶▶ This table shows the GPIOs configurations set by <dir>=1 and <mode> values ranging from 0 to 4.

AT#GPIO=<pin>,<mode>,1

<mode>	Description
0	Set OUTPUT, and GPIO logical value to zero (Low).
1	Set OUTPUT, and GPIO logical value to one (High).
2	Read mode, <dir> can be omitted, see Additional info section below
3	<mode> has no meaning.
4	<mode> has no meaning.

▶▶ <mode>=2 selects the read mode. In read mode, <dir> can be omitted.

AT#GPIO=<pin>,2

Name	Type	Default	Description
<stat>	integer	-	the parameter can be: logic value read from pin GPIO<pin> in the case the pin <dir> is set to input. logic value present in output of the pin GPIO<pin> in the case the pin <dir> is currently set to output. no meaning value for the pin GPIO<pin> in the case the pin <dir> is set to alternate function or tristate pull down.

▶▶ <dir> values from 2 to 16 or 17 select an alternate function ranging respectively from ALT1 to ALT 15 or ALT16. <mode> must be set to 0 or 1 when an alternate function is selected. The table shows the relationship between ALTx, <dir> and the name of the associated function. Some some <dir> values are not supported.

AT#GPIO=<pin>,0,<dir>

For ELS63 family:

ALTx	Functions names (between quotes) associated to ALTx
ALT1 (<dir>=2)	#GPIO maps "STAT_LED" on GPIO5: AT#GPIO=5,0,2 see #SLED to configure the STAT_LED GPIO_x behavior.
ALT2 (<dir>=3)	Reserved
ALT3 (<dir>=4)	#GPIO maps "TEMPMON" on one of the available GPIOs. Example: AT#GPIO=<pin>,0,4 see #TEMPMON and #TEMPCFG to configure the TEMPMON GPIOx behavior.
ALT4 (<dir>=5)	Reserved
ALT5 (<dir>=6)	Reserved
ALT6 (<dir>=7)	Reserved #GPIO maps "FASTSHDN" on one of the available GPIOs. NOT SUPPORTED AS GPIO ALT BUT DEDICATED PIN 40 for ELS63 family See #FASTSHDN
ALT7 (<dir>=8)	#GPIO maps "SIM_SWITCH" on GPIO8: AT#GPIO=8,0,8

ALT8 (<dir>=9)	#GPIO maps "USIF1" on: GPIO16(RXD1), GPIO17(TXD1), GPIO18(RTS1), GPIO19(CTS1): AT#GPIO=16,0,9 AT#GPIO=17,0,9 AT#GPIO=18,0,9 AT#GPIO=19,0,9 Default is not ALT8 (<dir>=9) for GPIO16, GPIO17, GPIO18, GPIO19 See also AT#PORTCFG for USIF1 configuration
ALT9 (<dir>=10)	#GPIO maps "SPI" on: GPIO16(SPI_CLK), GPIO17(SPI_MISO), GPIO18(SPI_MOSI), GPIO19(SPI_CS): AT#GPIO=16,0,10 AT#GPIO=17,0,10 AT#GPIO=18,0,10 AT#GPIO=19,0,10
ALT10 (<dir>=11)	Reserved
ALT11 (<dir>=12)	#GPIO maps "GNSS_POWER_EN" on GPIO7 for GNSS variants: AT#GPIO=7,0,12 NOT SUPPORTED for non GNSS variants
ALT12 (<dir>=13)	Reserved
ALT13 (<dir>=14)	Reserved
ALT14 (<dir>=15)	#GPIO maps "GNSS_RESET" on GPIO23 for GNSS variants: AT#GPIO=23,0,15

	NOT SUPPORTED for non GNSS variants
ALT15 (<dir>=16)	#GPIO maps "GNSS_RTC_EINT" on GPIO22 for GNSS variants: AT#GPIO=22,0,16 NOT SUPPORTED for non GNSS variants

For LE910Q1 family:

ALTx	Functions names (between quotes) associated to ALTx
ALT1 (<dir>=2)	#GPIO maps "STAT_LED" on GPIO1: AT#GPIO=1,0,2 see #SLED to configure the STAT_LED GPIO_x behavior.
ALT2 (<dir>=3)	Reserved
ALT3 (<dir>=4)	#GPIO maps "TEMPMON" on one of the available GPIOs. Example: AT#GPIO=<pin>,0,4 see #TEMPMON and #TEMPCFG to configure the TEMPMON GPIOx behavior.
ALT4 (<dir>=5)	Reserved
ALT5 (<dir>=6)	Reserved
ALT6 (<dir>=7)	#GPIO maps "FASTSHDN" on one of the available GPIOs. Example: AT#GPIO=5,0,7 See #FASTSHDN to configure the FASTSHDN GPIO_x behavior
ALT7 (<dir>=8)	Reserved
ALT8 (<dir>=9)	Reserved
ALT9 (<dir>=10)	Reserved
ALT10 (<dir>=11)	#GPIO maps "SW_RDY" on GPIO8: AT#GPIO=8,0,11

	GPIO8 <dir>=11 is the default for LE910Q1 family.
ALT11 (<dir>=12)	#GPIO maps "GNSS_EN" on GPIO7 for GNSS variants: AT#GPIO=7,0,12 NOT SUPPORTED for non GNSS variants
ALT12 (<dir>=13)	#GPIO maps "WAKEUP" on GPIO12: AT#GPIO=12,0,13 Low level ACTIVE: Falling edge to wake up; default Pull-up.
ALT13 (<dir>=14)	#GPIO maps "WAKEUP" on GPIO12: AT#GPIO=12,0,14 High level ACTIVE: Raising edge to wake up; default Pull-down.
ALT14 (<dir>=15)	#GPIO maps "GNSS_RESET" on GPIO6 for GNSS variants: AT#GPIO=6,0,15 NOT SUPPORTED for non GNSS variants
ALT15 (<dir>=16)	#GPIO maps "GNSS_VCC_DIS" on GPIO4 for GNSS variants: AT#GPIO=4,0,16 NOT SUPPORTED for non GNSS variants

For LE310Q1 family:

ALTx	Functions names (between quotes) associated to ALTx
ALT1 (<dir>=2)	Reserved
ALT2 (<dir>=3)	Reserved
ALT3 (<dir>=4)	#GPIO maps "TEMPMON" on one of the available GPIOs. Example: AT#GPIO=<pin>,0,4 see #TEMPMON and #TEMPCFG to configure the TEMPMON GPIOx behavior.
ALT4 (<dir>=5)	Reserved
ALT5 (<dir>=6)	Reserved
ALT6 (<dir>=7)	#GPIO maps "FASTSHDN" on one of the available GPIOs. Example: AT#GPIO=5,0,7 See #FASTSHDN to configure the FASTSHDN GPIO_x behavior
ALT7 (<dir>=8)	Reserved
ALT8 (<dir>=9)	#GPIO maps "DTR" on GPIO1: AT#GPIO=1,0,9
ALT9 (<dir>=10)	#GPIO maps "RING" on: GPIO2: AT#GPIO=2,0,10 AT#GPIO=2,0,10
ALT10 (<dir>=11)	#GPIO is configured as "SIMIN" through #SIMINCFG command

ALT11 (<dir>=12)	Reserved
ALT12 (<dir>=13)	Reserved
ALT13 (<dir>=14)	Reserved
ALT14 (<dir>=15)	Reserved
ALT15 (<dir>=16)	#GPIO maps "DCD" on GPIO3: AT#GPIO=3,0,16
ALT16 (<dir>=17)	#GPIO maps "DSR" on GPIO4: AT#GPIO=4,0,17

- While using the pins in the alternate function, the GPIO read/write access to that pin is not accessible and must be avoided.
- GPIO <pin> values from 11 to 14 are available only in LE910Q1 product variants (they are not available in legacy form factors).
- A GPIO can be configured as "SIMIN", only through #SIMINCFG command.

AT#GPIO?

Read command reports, for any GPIO pin, a row showing the current parameters values. Row one shows GPIO pin one, row two shows GPIO pin two, and so on.

#GPIO: <dir>,<stat><CR><LF>
#GPIO: <dir>,<stat><CR><LF>
...

If GPIO was previously set to <mode>= 3 or 4, the format of the returned message is:

#GPIO: <dir>,<stat>,<mode><CR><LF>
#GPIO: <dir>,<stat>,<mode><CR><LF>
...

If GPIO is not supported, the format of the returned message is:

#GPIO: NA<CR><LF>
#GPIO: NA<CR><LF>
...

AT#GPIO=?

Test command reports the supported range of values of the command parameters <pin>, <mode>, <dir>, and <save>.

For ELS63 product variant family

Check the available values ranges (non GNSS product variant example)
AT#G PIO=?

#GPIO: (5-8,16-19,20-23,25),(0-4),(0-2,4,8-10),(0,1)

OK

Set GPIO_6 in output, and set logical value HIGH
AT#GPIO=6,1,1

OK

Set GPIO_7 as output, value HIGH and save setting
AT#GPIO=7,1,1,1

OK

Report GPIO_20 state
AT#GPIO=20,2
#GPIO: 1,1
OK

Read command (just a possible configuration scenario, not the default)
AT#GPIO?
#GPIO: NA // GPIO1

#GPIO: NA
#GPIO: NA
#GPIO: NA // GPIO4
#GPIO: 0,1 // GPIO5 is input and input value is HIGH
#GPIO: 1,1 // GPIO6 is output and output value is HIGH
#GPIO: 0,0
#GPIO: 0,0
#GPIO: NA // GPIO9
#GPIO: NA
#GPIO: NA
#GPIO: NA
#GPIO: NA
#GPIO: NA
#GPIO: NA // GPIO15
#GPIO: 9,0 // GPIO16 is ALT8
#GPIO: 9,0 // GPIO17 is ALT8
#GPIO: 9,0 // GPIO18 is ALT8
#GPIO: 9,0 // GPIO19 is ALT8
#GPIO: 0,0 // GPIO20
#GPIO: 0,0
#GPIO: 0,0
#GPIO: 0,0 // GPIO23
#GPIO: NA // GPIO24
#GPIO: 0,0 // GPIO25
OK

For LE910Q1 product variant family

Check the available values ranges (non GNSS product variant example)

AT#GPIO=?

#GPIO: (1-14),(0-4),(0-2,4,7,10-11,13-14),(0,1)

OK

..

Set GPIO_2 in output, and set logical value HIGH

AT#GPIO=2,1,1

OK

..

Set GPIO_3 as output, value HIGH and save setting

AT#GPIO=3,1,1,1

OK

..

Report GPIO_4 state

AT#GPIO=4,2

#GPIO: 1,1

OK

..

Read command (just a possible configuration scenario, not the default)

AT#GPIO?

#GPIO: 0,1 // GPIO1 is input and input value is HIGH

#GPIO: 1,1 // GPIO2 is output and output value is HIGH

#GPIO: 0,1

#GPIO: 0,0

#GPIO: 0,1 // GPIO5 is input and input value is HIGH

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0

#GPIO: 0,0 //GPIO14

OK

For LE310Q1 product variant family

Check the available values ranges

AT#GPIO=?

#GPIO: (1-6),(0-4),(0-1,4,7,9-11,16-17),(0,1)

OK

..

Set GPIO_2 in output, and set logical value HIGH

AT#GPIO=2,1,1

OK

..

Set GPIO_3 as output, value HIGH and save setting

AT#GPIO=3,1,1,1

OK

..

Report GPIO_4 state

AT#GPIO=4,2

#GPIO: 1,1

OK ..

Read command (just a possible configuration scenario, not the default)

AT#GPIO?

#GPIO: 0,1 // GPIO1 is input and input value is HIGH

#GPIO: 1,1 // GPIO2 is output and output value is HIGH

#GPIO: 0,1

#GPIO: 0,0

#GPIO: 0,1 // GPIO5 is input and input value is HIGH

#GPIO: 0,0 //GPIO6

OK

AT#SLED - STAT_LED GPIO Setting

The command configures the behavior of the STAT_LED GPIO.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT#SLED=<mode>[,<onDuration>[,<offDuration>]]

Set command sets the behavior of the STAT_LED GPIO.

To configure a GPIO pin as STAT_LED GPIO, the user must enter AT#GPIO=x,0,2 command to set the GPIO_x pin as ALT1 alternate function.

At the next power ON, the GPIO pin is low until the control reads the saved setting in NVM and configures the GPIO_x as STAT_LED GPIO.

Not all GPIOs can be configured as STAT_LED pin. See #GPIO description to know the available GPIOs for this function.

Parameters:

Name	Type	Default	Description
<mode>	integer	2	defines the STAT_LED GPIO behavior.

Values:

0 : GPIO tied low

1 : GPIO tied high

2 : GPIO is handled with specific timings. See Additional info section

3	:	GPIO is turned ON/OFF alternatively, with period defined by the sum <onDuration> + <offDuration>
4	:	GPIO is handled with specific timings. See Additional info section
5	:	status led disabled
11	:	GPIO is handled with specific timings. See Additional info section
12	:	GPIO is handled with specific timings. See Additional info section

<onDuration>	integer	10	duration of period in which STAT_LED GPIO is tied high while <mode>=3

Value:

1÷100	:	in tenth of seconds

<offDuration>	integer	10	duration of period in which STAT_LED GPIO is tied low while <mode>=3

Value:

1÷100	:	in tenth of seconds

Additional info:

- <mode>=2, the timings of STAT_LED GPIO are:

  - not registered: always ON

  - registered in idle: blinking 1 s ON and 2 s OFF

  - registered in idle with power saving: blinking time depends on network condition to minimize power consumption

- <mode>=4, the timings of STAT_LED GPIO are:

  - not registered: blinking 0.5 s ON and 0.5 s OFF

  - registered in idle: blinking 300 ms ON and 2.7 s OFF

  - registered in idle with power saving: blinking time depends on network condition to minimize power consumption

- <mode>=11, the timings of STAT_LED GPIO are:

  - not registered: blinking 0.5 s ON and 0.5 s OFF

  - registered in idle, no data transfer: always ON

  - data transfer: always ON

  - registered in idle with power saving: blinking time depends on network condition to minimize power consumption

- <mode>=12, the timings of STAT_LED GPIO are:

  - not registered: blinking 0.5 s ON and 0.5 s OFF

  - registered in idle, no data transfer: blinking 10 ms ON and 3.99 s OFF

  - data transfer: always blinking 10 ms ON and 1.99 s OFF

  - registered in idle with power saving: blinking time depends on network condition to minimize power consumption

SLED function of a GPIO corresponds to ALT1 function of the GPIO, so ALT1 function should be set through AT#GPIO command.

AT#SLED?

Read command returns the STAT_LED GPIO current setting, in the format:

#SLED: <mode>,<onDuration>,<offDuration>

<onDuration> and <offDuration> reported by AT#SLED? are always present and are the last value of the configured <onDuration> and <offDuration> parameters, regardless the <mode> value, regardless they are currently used in <mode> 3 configuration or not.

AT#SLED=?

Test command returns the range of available values for parameters <mode>, <onDuration> and <offDuration>.

The setting is saved using the #SLEDSAV command.

AT#SLEDSAV - Save STAT_LED GPIO Setting

This command allows to save the current STAT_LED GPIO setting.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SLEDSAV

Execution command saves the STAT_LED GPIO setting in NVM.

AT#SLEDSAV=?

Test command returns OK result code.

AT#ADC - Read Analog/Digital Converter Input

This command returns the current voltage value of the specified ADC inputs, expressed in mV.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#ADC=[<adc>,<mode>[,<dir>]]

Execution command reads selected <adc> pin voltage, converts it by baseband internal ADC, and prints out the result as shown in Additional info section.

Parameters:

Name	Type	Default	Description
<adc>	integer	N/A	index of input pin
	Values:		
	1÷n :		input pin index, where n is the maximum number of available ADCs: for ELS63 product family n is 1 for LE910Q1 product family n is 3
<mode>	integer	2	required action
	Value:		

2	:	query ADC value

<dir>	integer	0	direction. Its interpretation is currently not implemented.

Value:

0	:	no effect

Additional info:

Format of the message printed out by the execution command:

#ADC:<value>

Name	Type	Default	Description
<adc>	integer	-	pin <adc> voltage expressed in mV.

The command returns the last valid measure.

AT#ADC?

Read command reports all pins voltage, converted by ADC, in the format:

#ADC: <value>[<CR><LF>#ADC: <value>[...]]

AT#ADC=?

Test command reports the supported range of values of the command parameters <adc>, <mode> and <dir>.

AT#I2CWR - Write to I2C

This command is used to send data to an I2C peripheral connected to module.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#I2CWR=<sdaPin>,<sclPin>,<deviceId>,<registerId>,<len>

Execution command sends data to an I2C peripheral connected to module GPIOs. After the writing activity has been accomplished, the GPIOs will not be restored to the original setting. Use #GPIO command to see the status of the used GPIOs. To have information on GPIO pins refer to document [1].

Parameters:

Name	Type	Default	Description
<sdaPin>	integer	-	GPIO number for SDA. To know the range use #I2CWR test command.
<sclPin>	integer	-	GPIO number for SCL. To know the range use #I2CWR test command.
<deviceId>	hex	N/A	address of the I2C device (7 bits). The Least Significant Bit is used for read/ write command, but in this #I2CWR implementation, it does not matter if the LSB is set to 0 or 1. Address must be written in hexadecimal form without 0x. 10 bit address is also supported.

	Value:		
	0÷3FF	:	addressing range extended to 10 bit
<registerId>	hex	N/A	register to write data to
	Value:		
	0÷FF	:	value must be written in hexadecimal form without 0x
<len>	integer	N/A	number of data to send
	Value:		
	1÷254	:	number of data to send

Additional info:

▶▶ After entering the command, the module returns the prompt ">" and waits for the data to send. To complete the operation, send Ctrl-Z char (0x1A hex); to exit without writing the message send ESC char (0x1B hex). Data must be written in hexadecimal form.

If data are successfully sent, the response is OK, otherwise an error code is reported.

AT#I2CWR=?

Test command returns the range of available values for parameters <sdaPin>, <sclPin>, <deviceId>, <registerId>, <len>.

<sdaPin>, <sclPin> are fixed pins, they are not GPIOs.

<sdaPin>, <sclPin> only supported value is dummy value 0.

Test command reports only value 0 for <sdaPin>, <sclPin>.

Set GPIO_2 as SDA, and GPIO_3 as SCL. Device I2C address is 0x20; 0x10 is the address of the first register where to write I2C data; 14 data bytes will be written starting from register 0x10.

```
AT#I2CWR=2,3,20,10,14
> 00112233445566778899AABBCCDD<ctrl-z>
OK
```

AT#I2CRD - Read from I2C

This command is used to read data from an I2C peripheral connected to module.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#I2CRD=<sdaPin>,<sclPin>,<deviceId>,<registerId>,<len>

Execution command reads data from an I2C peripheral connected to module GPIOs. After the reading activity has been accomplished, the GPIOs will not be restored to the original setting. Use #GPIO command to see the status of the used GPIOs. To have information on GPIO pins refer to document [1].

Parameters:

Name	Type	Default	Description
<sdaPin>	integer	-	GPIO number for SDA. To know the range use #I2CRD test command.
<sclPin>	integer	-	GPIO number for SCL. To know the range use #I2CRD test command.
<deviceId>	hex	N/A	address of the I2C device (7 bits). The Least Significant Bit is used for read/ write command, but in this #I2CRD implementation, it does not matter if the LSB is set to 0 or 1. Address must be written in hexadecimal form without 0x. 10 bit address is also supported

	Value:		
	0÷3FF : addressing range extended to 10 bit		
<registerId>	hex	N/A	Register to read data from
	Value:		
	0÷FE : value must be written in hexadecimal form without 0x		
<len>	integer	N/A	Number of data to receive - Data Read from I2C will be dumped in hexadecimal format - If data requested are more than data available in the device, dummy data (normally 0x00 or 0xff) will be dumped
	Value:		
	1÷254 : number of data to receive		

AT#I2CRD=?

Test command returns the range of available values for parameters <sdaPin>, <sclPin>, <deviceId>, <registerId>, <len>.

<sdaPin>, <sclPin> are fixed pins, they are not GPIOs.

<sdaPin>, <sclPin> only supported value is dummy value 0.

Test command reports only value 0 for <sdaPin>, <sclPin>.

Read 12 bytes from I2C device with address 0x20, starting from register address 0x10. SDA is mapped on GPIO_02, SCL is mapped on GPIO_03.

```
AT#I2CRD=2,3,20,10,12
#I2CRD: 00112233445566778899AABBCC
OK
```

AT#I2CCF - Combined Format for I2C Writing and Reading

This command is used to write and read data to/from an I2C device using the I2C Combined Format. The module acts as an I2C master.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#I2CCF=<sdaPin>,<sclPin>,<deviceId>,<lenwr>,<lenrd>

The module, as master, transmits data to the slave and then, reads data from it through two GPIOs. Transfer direction is changed when writing section is ended. After the write/read activity has been accomplished, the GPIOs will not be restored to the original setting. Use #GPIO command to see the status of the used GPIOs. To have information on GPIO pins refer to document [1].

Parameters:

Name	Type	Default	Description
<sdaPin>	integer	-	GPIO number for SDA. To know the range use #I2CCF test command.
<sclPin>	integer	-	GPIO number for SCL. To know the range use #I2CCF test command.
<deviceId>	hex	N/A	address of the I2C device (7 bits). The Least Significant Bit is used for read/ write command, but in this #I2CCF implementation, it does not matter if the LSB is set to 0 or to 1. Address must be written in hexadecimal form without 0x. 10 bit address is also supported.

Value:

0÷3FF : addressing range extended to 10 bit

<lenwr> integer N/A number of data to write.

Value:

0÷254 : number of data to write.

<lenrd> integer N/A number of data to read.

Value:

0÷254 : number of data to read.

Additional info:

▶▶ After entering the command, and if <lenwr> > 0, the module returns the prompt ">" and waits for the data to send. To complete the operation enter Ctrl-Z char (0x1A hex); to exit without writing the message enter ESC char (0x1B hex).

Data must be written in hexadecimal form without 0x.

If data are successfully sent, the response is OK, otherwise an error code is reported.

AT#I2CCF=?

Test command returns the range of available values for parameters <sdaPin>, <sclPin>, <deviceId>, <lenwr>, <lenrd>.

* Set GPIO_2 as SDA, GPIO_3 as SCL; Device I2C address is 0x20;
  First is send data 0x0a; after a "RESTART", 4 data bytes are read.

  AT#I2CCF=2,3,20,1,4
  >0a<ctrl-z>
  #I2CCF: abcdef12

  OK

  The sequence is the following:
  START - 0x20- 0x0a -RESTART - 0X21 - data read 1 -...- data read 4 - STOP

* Set GPIO_2 as SDA, GPIO_3 as SCL; Device I2C address is 0x20;
  read data:

  AT#I2CCF=2,3,20,0,2
  #I2CCF: abcd

  OK

  The sequence is the following:
  START - 0x21- - data read 1 - data read 2 - STOP

AT#TEMPMON - Temperature Monitor

This command is used to retrieve internal temperature information of the module.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#TEMPMON=<mod>[,<urcMode>[,<action>[,<GPIO>]]]

Set command sets the internal temperature monitoring mode of the module, reads it, and enables/disables/triggers the presentation of the temperature monitoring message, that can be respectively an URC or an intermediate message according to the selected mode.

Both URC and intermediate message have the same format.

#TEMPMEAS: <level>,<value>

The temperature monitoring message parameters are described in the Unsolicited fields section.

Parameters:

Name	Type	Default	Description
<mod>	integer	N/A	select the temperature monitoring mode.

Values:

0 : sets the command parameters

1 : triggers the measurement of the module internal temperature, reporting the result using the format shown above

<urcMode>	integer	0	URC presentation mode

Values:

0 : disables the presentation of the temperature monitoring URC.

1 : enables the presentation of the temperature monitoring URC, whenever the module internal temperature reaches either operating or extreme levels and whenever the module internal temperature returns back to operating or normal levels, whenever the zone changes.

<action>	integer	0	sum of integers, each representing the action to be done whenever the module internal temperature reaches either operating or extreme levels.

Values:

0 : no action

1 : (01) Activating of thermal mitigation according to thermal configuration file

2 : (10) Output pin <GPIO> is tied HIGH when operating temperature bounds are reached; when the temperature is back to normal the output pin <GPIO> is tied LOW. If this <action> is required, it is mandatory to set the <GPIO> parameter too

		3 :	(11) This value contains <action=1> and <action=2> i.e. activate thermal mitigation and a GPIO indication. If this <action> is required, it is mandatory to set the <GPIO> parameter too
<GPIO>	integer	-	GPIO number. Valid range is any GPIO pin as described in #GPIO command. This parameter is needed and required only if <action> 2 or 3 is enabled.

Unsolicited fields:

Name	Type	Description
<level>	integer	threshold level (see Notes section) Values: -2 : Extreme temperature lower bound -1 : Operating temperature lower bound 0 : normal temperature 1 : Operating temperature upper bound 2 : Extreme temperature upper bound
<value>	integer	current temperature expressed in Celsius degrees.

AT#TEMPMON?

Read command reports the current parameter settings for the command in the format:

#TEMPMON: <urcMode>,<action>[,<GPIO>]

AT#TEMPMON=?

Test command reports the supported range of values for parameters <mod>, <urcMode>, <action> and <GPIO>.

- Thresholds levels are defined in #TEMPCFG command. See there for detailed description on thermal mitigation configuration.
- Last <GPIO> is saved in the NVM.
- Thermal management is active only when using a non-GCF test SIM. A SIM is a test SIM if it has an MCC of 001, 002, 003, 004, 005, 006, 007, 008, 009, 010, 011, 012.

AT#TESTMODE - Test Mode Configuration

Set module in Test Mode for configuring and testing the POWER level (not signaling mode).

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#TESTMODE=<cmd>

The Test Mode is entered using the following commands according to the network technology:

AT#TESTMODE="TM" followed by AT#TESTMODE="INIT4G" for 4G networks

Only after this set, AT#TESTMODE can be used with the other allowed commands. To exit from Test Mode and go back to Operative Mode, enter the following commands according to the network technology:

AT#TESTMODE="DEINIT4G" followed by AT#TESTMODE="OM"> for 4G

Parameter:

Name	Type	Default	Description
<cmd>	string	N/A	Identifies one of the commands listed in the Values: section. The command is a quoted string. PL4G command must be followed by the value of the <pwr> power transmission in dBm: AT#TESTMODE="PL4G -10" CH4G command must be followed by the <earfcn> channel and <bw> bandwidth: AT#TESTMODE="CHG4 20000 0"

Values:

"TM"	:	forces the module in Test Mode
"OM"	:	forces the module in Operative Mode and reboots module
"INIT4G"	:	4G command: initialize Radio for 4G transmission; reboots the module
"CH4G earfcn bw"	:	4G command: set the earfcn UL or DL channel for transmitting and receiving and set the bandwidth
"PL4G pwr"	:	4G command: change the transmission power. Range is: -40 ÷ 23 in dBm
"TX4G"	:	4G command: starts the non-stop module transmission
"RL4G"	:	4G command: reads Rx power level in dBm
"DEINIT4G"	:	4G command: de-initialize Radio for 4G transmission and reception
"ESC"	:	stop the transmission or reception

Additional info:

▶▶

<earfcn> UL	Band

18000 - 18599	1
18600 - 19199	2
19200 - 19949	3
19950 - 20399	4
20400 - 20649	5
21450 - 21799	8
23010 - 23179	12
23180 - 23279	13
23280 - 23379	14
23850 - 23999	18
24000 - 24149	19
24150 - 24449	20
26040 - 26689	25

26690 - 27039	26
27040 - 27209	27
27210 - 27659	28
131972 - 132671	66
133122 - 133471	71
134002 - 134181	85

<earfcn> DL	Band
0 - 599	1
600 - 1199	2
1200 - 1949	3
1950 - 2399	4
2400 - 2649	5
3450 - 3799	8
5010 - 5179	12
5180 - 5279	13

5280 - 5379	14
5850 - 5999	18
6000 - 6149	19
6150 - 6449	20
8040 - 8689	25
8690 - 9039	26
9040 - 9209	27
9210 - 9659	28
66436 - 67335	66
68586 - 68935	71
70366 - 70545	85

<bw>	Bandwidth (kHz)
0	180

- Bands support varies depending on the product
- 4G support varies depending on the product

When testing 4G transmission and reception, the minimum and maximum channels that can be used for a specific band have to be included in the following range:

the lowest channel + 2 ÷ the highest channel - 1.

E.g. if we consider B4 UL, the usable channels start from 19952 up to 20398.

If we consider B4 DL, the usable channels start from 1952 up to 2398

For RL4G power level test, use a CW (continuous wave) signal with a frequency offset of 500 kHz.

AT#TESTMODE is intended for RF functional verification or connection testing.

The reported output power level (dBm) is not calibrated and therefore should not be used as a reference for accurate power measurement.

AT#TESTMODE?

Read command reports the currently selected <command> in the format:

#TESTMODE: <testModeStatus>

Additional info:

Parameter meaning:

Name	Type	Default	Description
<testModeStatus>	integer	0	status

Values:

0 : module is in Operative Mode

1 : module is in Test Mode

AT#TESTMODE=?

Test command returns the OK result code

Set Test Mode

AT#TESTMODE="TM"

OK

Initialize Test Mode for 4G operations

AT#TESTMODE="INIT4G"

OK

Module reboot

Set B2 with 180 kHz bandwidth, download

AT#TESTMODE="CH4G 900 0"

OK

Read power level

AT#TESTMODE="RL4G"

TESTMODE: -9

OK

Stop reception

AT#TESTMODE="ESC"

OK

Set B4 with 180 kHz bandwidth, upload

AT#TESTMODE="CH4G 20000 0"

OK

Set power level

AT#TESTMODE="PL4G 20"

OK

Start transmission

AT#TESTMODE="TX4G"

OK

Stop transmission

AT#TESTMODE="ESC"

OK

De-initialize Test Mode for 4G operations

AT#TESTMODE="DEINIT4G"

OK

Exit from Test Mode

AT#TESTMODE="OM"

OK

Module reboot

Power Down

AT#REBOOT - Module Reboot

Immediate module reboot.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#REBOOT

Execution command reboots immediately the unit.
It can be used to reboot the system after a remote update of the script in order to have the new one running.

If #REBOOT follows an AT command that stores some parameters in NVM, it is recommended to insert a delay of at least 5 seconds before to issue #REBOOT, to permit the complete NVM storing.

AT#REBOOT=?

Test command returns OK result code.

```
* Reboot the module
AT#REBOOT
OK
(the module reboots)
```

AT#ENHRST - Periodic Reset

Enable or Disable the one shot or periodic unit reset

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#ENHRST=<mode>,<delay>

Set commands enables/disables the unit reset after the specified <delay> in minutes

Parameters:

Name	Type	Default	Description
<mode>	integer	0	Enable\Disable mode
	Values:		
	0 :		disables the unit reset
	1 :		enables the unit reset only one time (one shot reset)
	2 :		enables periodically unit reset
<delay>	integer	-	time interval in minutes after that the unit reboots

- The settings are saved automatically in NVM only if old or new <mode> value is 2, i.e. unit set in periodic reset mode. Therefore, any change from 0 to 1 or conversely is not stored in NVM.
- The command AT#ENHRST=1,0 causes the immediate module reboot. If it follows an AT command that stores some parameters in NVM, it is strongly recommended to insert a delay of at least 5 sec before issuing it, to permit the complete NVM storing process.

AT#ENHRST?

Read command reports the current parameter settings in the following format:

#EHNRST: <mode>[,<delay>,<remainingTime>]

Additional info:

- Read command parameter for <mode> =1 or 2.

Name	Type	Default	Description
<remainingTime>	integer	-	time in minutes remaining before next reset

AT#ENHRST=?

Test command reports supported range of values for parameters <mode> and <delay>.

- This AT command causes a reboot

Example of #ENHRST usage and expected unit behavior.

* AT#ENHRST=1,60

  ...

  Module reboots after 60 minutes

* AT#ENHRST=1,0

  Module reboots immediately

* AT#ENHRST=2,40

  ...

  Module reboots after 40 minutes, and after every following power on, it will continue to reboot always after 40 minutes.

AT#SHDN - Software Shutdown

This command turns the module OFF.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SHDN

Execution command causes device to detach from the network and shut down. Before definitive shut down an OK response is returned.

When issuing the command any previous activity terminated and the device will not respond to any further command.

To turn it on again hardware pin ON/OFF must be tied low.

The maximum time to shut down the device, completely is 25 seconds.

AT#SHDN=?

Test command returns OK result code.

AT#FASTSHDN - Fast Shutdown Configuration

This command can be used as a set command to configure a GPIO pin performing a fast shutdown when on it is forced a High to Low level transition; or it can be used as an execute command to force immediately a fast shutdown.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#FASTSHDN[=<enable>[,<GPIO>]]

Parameters:

Name	Type	Default	Description
<enable>	integer	0	enable/disable the fast shutdown execution via a GPIO pin. Values: 0 : disabled 1 : enabled
<GPIO>	integer	-	selects the GPIO to execute the fast shutdown. When the selected <GPIO> pin goes from High to Low level and the <enable> is set to 1, the module execute immediately the fast shutdown.

Additional info:

▶▶ The execution command #FASTSHDN<CR><LF> forces the module to execute immediately the fast shutdown.

For ELS63 product variant family:

the FASTSHDN pin is not a GPIO ALT function, it is a fixed pin.

<enable> default value is 1 and the supported values are 0 and 1.

<GPIO> default value is 0 and the only supported value is 0, meaning a fixed pin is used.

AT#FASTSHDN=1,0

AT#FATSTHDN=0,0

are the only allowed configurations.

For LE910Q1 and LE310Q1 product variant family:

the FASTSHDN is a GPIO ALT function for any GPIO pin.

<enable> default value is 0 and the supported values are 0 and 1.

<GPIO> must be in the range of supported GPIO pin values.

The GPIO pin selected by the #FASTSHDN command must be used for this purpose only.

If you need to use the selected GPIO pin for different activities, it must be set free with the following command:

AT#FASTSHDN=0,<GPIO>

For LE910Q1 in AT+CFUN=14 configuration and during power saving the FASTSHDN does not work as GPIO ALT function for the other GPIOs than GPIO12.

AT#FASTSHDN?

Read command reports the currently selected configuration in the format:

AT#FASTSHDN: <enable>,<GPIO>

AT#FASTSHDN=?

Test command returns the supported range of values for all the parameters.

For ELS63 product variant family

The FASTSHDN pin is not a GPIO ALT function, it is a fixed pin

AT#FASTSHDN=1,0

OK

AT#FASTSHDN?

#FASTSHDN: 1,0

OK

Force immediate fast shutdown

AT#FASTSHDN

OK

For LE910Q1 and LE310Q1 product variant family

The FASTSHDN is a GPIO ALT function for any GPIO pin.

Enable fast shutdown on GPIO_05

AT#FASTSHDN=1,5

OK

AT#FASTSHDN?

#FASTSHDN: 1,5

OK

Force immediate fast shutdown

AT#FASTSHDN

OK

Jamming Detection and Report

AT#JDR4GCFG - LTE Jamming Detection Threshold Configuration

The command configures the LTE Jamming Detection thresholds.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#JDR4GCFG=<P_RSRP_T4G>[,<P_RSRQ_T4G>,<UNUSED1>,<UNUSED2>,<UNUSED3>[,<P_RSSI_S4G>]]

Set command allows to configure the LTE Jamming Detection thresholds. After configuration, use the #JDRENH2 command to enable/disable LTE jamming detection and select reporting mode.

Parameters:

Name	Type	Default	Description
<P_RSRP_T4G>	integer	10	set the threshold of RSRP. The threshold (RSRP_Thr) is calculated as: RSRP_Thr = RSRP_Av*(1 - (<P_RSRP_T4G>/100)) where RSRP_Av is the average of the last 20 RSRP measures

Value:

0÷100	:	threshold of RSRP

<P_RSRQ_T4G>	integer	20	set the threshold of RSRQ. The threshold (RSRQ_Thr) is calculated as: RSRQ_Thr = RSRQ_Av*(1 + (<P_RSRQ_T4G>/100)) where RSRQ_Av is the average of th e last 20 RSRQ measures
	Value:		
	0÷100	:	threshold of RSRQ
<UNUSED1>	integer	-	Unused
<UNUSED2>	integer	-	Unused
<UNUSED3>	integer	-	Unused
<P_RSSI_S4G>	integer	-50	RSSI threshold level in dBm, above which the Jamming condition is detected
	Value:		
	-120÷-20	:	Value of RSSI Threshold

If you change the <P_RSRQ_T4G> parameter of the AT#JDR4GCFG command, it will automatically change the same parameter of the AT#JDRENH2 command and vice versa.

All the parameter settings are saved in NVM memory.

AT#JDR4GCFG?

Read command returns the current settings in the format:

#JDR4GCFG: <P_RSRP_T4G>,<P_RSRQ_T4G>,,,,<P_RSSI_S4G>

AT#JDR4GCFG=?

Test command returns the range of supported values for all the parameters.

AT#JDRENH2 - Enhanced Jamming Detection and Report

This command enables/disables jamming detection, and reports the relative result to the user.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#JDRENH2=<mode>,<UNUSED1>,<UNUSED2>,<UNUSED3>,<UNUSED4>,<UNUSED5>,<UNUSED6>,<UNUSED7> [,<P_RSSI_T4G>[,<P_RSRQ_T4G>]]

The set command performs the following activities:

- enables/disables the detection of the presence of jamming in the module coverage.

- selects one of the following two reports mode or both: on dedicated GPIO, by means of the URC, or GPIO + URC.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	enable/disable jamming detection and select reporting mode

Values:

0 : disable jamming detection

1 : enable jamming detection, and report its condition on a GPIO pin, see Additional info section

2 : enable jamming detection, and report its condition with an URC, see Additional info section

3 : enable jamming detection, and report its condition as <mode>=1 and <mode>=2

4 : enable jamming detection, and report its condition with an URC every 3 sec, see <mode>=2

5 : enable jamming detection, and report its condition as <mode>=1 and <mode>=4

6 : not used

<UNUSED1>	integer	-	unused
<UNUSED2>	integer	-	unused
<UNUSED3>	integer	-	unused
<UNUSED4>	integer	-	unused
<UNUSED5>	integer	-	unused
<UNUSED6>	integer	-	unused
<UNUSED7>	integer	-	unused

Value:

			0÷100 : dummy value
<P_RSSI_T4G>	integer	10	set the threshold of RSSI. The threshold (RSSI_Thr) is calculated as: RSSI_Thr = RSSI_Av*(1 - (<P_RSRP_T4G>/100)) where RSSI_Av is the average of the last RSSI measures Value: 0÷100 : threshold of RSSI
<P_RSRQ_T4G>	integer	20	Set the threshold of RSRQ. The threshold (RSRQ_Thr) is calculated as RSRQ_Thr = RSRQ_Avg*(1 + (<P_RSRQ_T4G>/100)) where RSRQ_Avg is the average of the last RSRQ measures Value: 0÷100 : RSRQ threshold values

Additional info:

▶▶ <mode>=1

The jamming condition is reported on pin GPIO_02 (JDR):

GPIO_02 (JDR) = Low, Normal Operating Condition

GPIO_02 (JDR) = High, Jammed Condition

To have information on GPIO_02 pin, refer to document [1]. GPIO_02 pin can be used also by other functionality, see #GPIO command.

▶▶ <mode>=2

the jamming condition is reported with a single URC on serial line, in the format:

#JDR: <status>

Unsolicited field:

Name	Type	Description
<status>	string	jamming condition status, <mode>=2 Values: JAMMED <RAT> : jamming condition detected, <RAT> is the Radio Access Technology for operates: GSM or LTE. OPERATIVE <RAT> : Normal Operating condition restored. Status shown only after a jamming condition has occurred.

AT#JDRENH2?

Read command reports the current parameters values, in the format:

#JDRENH2:<mode>,,,,,,,,,<P_RSSI_T4G>,<P_RSRQ_T4G>

AT#JDRENH2=?

Test command reports the supported range of the parameters values.

Packet Domain

AT+CGDCONT - Define PDP Context

Define PDP Context.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGDCONT=[<cid>[,<PDP_type>[,<APN>[,<PDP_addr>[,<d_comp>[,<h_comp>

[,<IPv4AddrAlloc>[,<request_type>[,<PCSCF_discovery>[,<IM_CN_Signalling_Flag_Ind>[,<NSLPI>

[,<securePCO> [,<IPv4_MTU_discovery>][,<Local_Addr_Ind>][,<Non-IP_MTU_discovery>]
[,<Reliable_Data_Service>]]]]]]]]]]]]]

Set command specifies PDP context parameter values for a PDP context identified by the (local) context identification parameter, <cid>.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	(PDP Context Identifier) numeric parameter which specifies a particular PDP context definition. 1..max - where the value of max is returned by the Test command.

<PDP_type>	string	N/A	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. Values: "IP" : Internet Protocol "IPV6" : Internet Protocol version 6 "IPV4V6" : Virtual introduced to handle dual IP stack UE capability. Non-IP : Transfer of Non-IP data to external packet data network (not supported)
<APN>	string	-	(Access Point Name) a string parameter which is a logical name that is used to select the GGSN or the external packet data network. If the value is empty ("") or omitted, then the subscription value will be requested.
<PDP_addr>	string	-	A string parameter that identifies the terminal in the address space applicable to the PDP (ignored with the set command). The allocated address may be read using the +CGPADDR command.
<d_comp>	integer	0	Numeric parameter that controls PDP data compression.

Values:

0	:	PDP data compression off (Not applicable)
1	:	PDP data compression on (Not applicable)
2	:	V.42bis (Not applicable)

<h_comp>	integer	0	Numeric parameter that controls PDP header compression.

Values:

0	:	PDP header compression off (Not applicable)
1	:	PDP header compression on (Not applicable)
2	:	RFC1144 (Not applicable)
3	:	RFC2507 (Not applicable)
4	:	RFC3095 (Not applicable)

<IPv4AddrAlloc>	integer	0	a numeric parameter that controls how the MT/TA requests to get the IPv4 address information.

Value:

	0 : IPv4 Address Allocation through NAS Signaling (default)		
<request_type>	integer	0	a numeric parameter used to indicate the type of PDP context activation request (for emergency bearer services or not).
	Value:		
	0 : PDP context is for new PDP context establishment and it is not for emergency bearer services (default)		
<P-CSCF_discovery>	integer	0	a numeric parameter that controls how the MT/TA requests to get the P-CSCF address.
	Value:		
	0 : Preference of P-CSCF address discovery not influenced by +CGDCONT (default)		
<IM_CN_Signalling_Flag_Ind>	integer	0	a numeric parameter that indicates to the network whether the PDP context is for IM CN subsystem-related signaling only or not.
	Value:		

	0 :	UE indicates that the PDP context is not for IM CN subsystem-related signaling only (default)	
<NSLPI>	integer	0	a numeric parameter that indicates the NAS signaling priority requested for this PDP context.
	Value:		
	0 :	Indicates that this PDP context is to be activated with the value for the low priority indicator configured in the MT (default)	
	1 :	Indicates that this PDP context is to be activated with the value for the low-priority-indicator set to "MS is not configured for NAS signaling low priority"	
<securePCO>	integer	0	a numeric parameter that specifies if security-protected transmission of PCO is requested or not.
	Value:		
	0 :	Security protected transmission of PCO is not requested (default)	

1 : Security protected transmission of PCO is requested

<IPv4_MTU_discovery>	integer	0	a numeric parameter that influences how the MT/TA requests to get the IPv4 MTU size.

Value:

0 : Preference of IPv4 MTU size discovery not influenced by +CGDCONT (default)

1 : Preference of IPv4 MTU size discovery through NAS signaling

<Local_Addr_Ind>	integer	0	a numeric parameter that indicates to the network whether the MS supports a local IP address in TFTs.

Value:

0 : Indicates that the MS does not support local IP address in TFTs (default)

<Non-IP_MTU_discovery>	integer	0	a numeric parameter that influences how the MT/TA requests to get the Non-IP MTU size.

Value:

		0 :	Preference of Non-IP MTU size discovery not influenced by +CGDCONT (default)
<Reliable_Data_Service>	integer	0	a numeric parameter that indicates whether the UE is using Reliable Data Service for a PDN connection or not.
	Value:		
		0 :	Reliable Data Service is not being used for the PDN connection (default)

To erase all parameters for a <cid> give as follows:

AT+CGDCONT=<cid>,<pdp_type>,""

AT+CGDCONT?

Read command returns the current settings for each defined context in the format:

+CGDCONT: <cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>

[,<IPv4AddrAlloc>[,<request_type>[,<PCSCF_discovery>[,<IM_CN_Signalling_Flag_Ind>[,<NSLPI>

[,<securePCO>[,<IPv4_MTU_discovery>[,<Local_Addr_Ind> [,<Non-IP_MTU_discovery>[,<Reliable_Data_Service>]]]]]]]]] <CR><LF>

+CGDCONT: <cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>

[,<IPv4AddrAlloc>[,<request_type>[,<PCSCF_discovery>[,<IM_CN_Signalling_Flag_Ind> [,<NSLPI>

[,<securePCO>[,<IPv4_MTU_discovery>[,<Local_Addr_Ind> [,<Non-IP_MTU_discovery>[,<Reliable_Data_Service>]]]]]]]]] [...]

AT+CGDCONT=?

Test command returns values supported as a compound value.

In Context Id 1 line:

- before 4G Attach, AT+CGDCONT? returns configured values for <PDP_type> and <APN>, no value for <PDP_addr>

- after 4G Attach, AT+CGDCONT? returns configured values for <PDP_type> and <APN>, no value for <PDP_addr> (real <APN> only if configured <APN> was empty)

Context Id 1 line exception:

- if SIM IMSI MCC/MNC values are for Verizon SIM, before and after 4G Attach, AT+CGDCONT? returns for <APN> the Verizon apn class 3 stored in NVM APN table

```
AT+CGDCONT=1,"IP","APN"
OK

AT+CGDCONT?
+CGDCONT: 1,"IP","APN"
OK

AT+CGDCONT=?
+CGDCONT: (1-6),"IP",,,,,(0),(0,2),(0),(0),(0,1),(0),(0,1),(0)
+CGDCONT: (1-6),"IPV6",,,,,(0),(0,2),(0),(0),(0,1),(0),(0),(0)
+CGDCONT: (1-6),"IPV4V6",,,,,(0),(0,2),(0),(0),(0,1),(0),(0,1),(0)
```

OK

AT+CGPADDR - Show PDP Address

This command returns a list of PDP addresses for the specified context identifiers.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGPADDR=[<cid>[,...]]

Execution command returns a list of PDP addresses for the specified context identifiers.

Parameter:

Name	Type	Default	Description
<cid>	integer	-	specifies a PDP context definition, see +CGDCONT command. If no <cid> specified, the addresses for all defined contexts are returned.

Additional info:

- The command returns a row of information for every <cid> whose context has been defined. No row is returned for a <cid> whose context has not been defined. Here is the response format:

  +CGPADDR: <cid>[,<PDP_addr_1>[,<PDP_addr_2>]]

Name	Type	Default	Description
<PDP_addr>	string	-	identifies the terminal in an address space applicable to the PDP. The

address may be static or dynamic:

for a static address, it will be the one set by the +CGDCONT command when the context was defined

for a dynamic address it will be the one assigned during the last PDP context activation that used the context definition referred to by <cid>; <PDP_addr> is omitted if none is available

AT+CGPADDR=?

Test command returns a list of defined <cid>.

```
AT#SGACT=3,1
#SGACT: xxx.yyy.zzz.www
OK

AT+CGPADDR=3
+CGPADDR: 3,"xxx.yyy.zzz.www"
OK

AT+CGPADDR=?
+CGPADDR: (3)
 OK
```

AT+CGAUTH - Define PDP Context Authentication Parameters

This command allows the TE to specify authentication parameters for a PDP context.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGAUTH=<cid>[,<auth_proto>[,<userid>[,<password>]]]

Set command allows the TE to specify authentication parameters for a PDP context identified by the (local) context identification parameter <cid>, used during the PDP context activation and the PDP context modification procedures.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	specifies a particular PDP context definition, see the +CGDCONT command.
<auth_proto>	string	NONE	selects the authentication protocol used for this PDP context.

Values:

NONE	:	no authentication
PAP	:	PAP authentication
CHAP	:	CHAP authentication

<username>	string	-	User name for access to the IP network
<password>	string	-	Password for access to the IP network

AT+CGAUTH?

It returns the current setting in the format:

+CGAUTH: <cid>,<auth_type>[,<username>]

AT+CGAUTH=?

Test command reports the supported range of values for parameters <cid>, <auth_type> and the strings <username> and <password>.

AT+CGCONTRDP - PDP Context Read Dynamic Parameters

The execution command returns the relevant information for a PDP Context established by the network.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGCONTRDP[=<cid>]

The execution command returns the relevant information on a PDP Context established by the network with the context identifier <cid>. If the parameter <cid> is omitted, the information for all established PDP contexts is returned. The response message has the following format:

+CGCONTRDP: <cid>,<bearer_id>,<apn>[,<local_addr and subnet_mask>

[,<gw_addr>[,<DNS_prim_addr>[,<DNS_sec_addr>[,<PCSCF_prim_addr>

[,<PCSCF_sec_addr>[,<IM_CN_Signalling_Flag>[,<LIPA_indication>

[,<IPv4_MTU>[,<WLAN_Offload>[,<Local_Addr_Ind>
[,<Serving_PLMN_rate_control_value>]]]]]]]]]]]]

If the context cannot be found an ERROR response is returned.

The response message parameters are described in the Additional info section.

Parameter:

Name	Type	Default	Description
<cid>	integer	-	identifies a non secondary PDP context definition. The parameter is local to the TE-

MT interface and is used in other PDP context-related commands.

Additional info:

▶▶ List of the meaning of the response message parameters.

Name	Type	Default	Description
<bearerId>	integer	-	identifies the EPS Bearer.
<apn>	string	-	logical name used to select the GGSN or the external packet data network.
<local_addr and subnet_mask>	string	-	IP address and subnet mask of the MT. For more information, see next Additional info section.
<gw_addr>	string	-	Gateway address of the MT.

<DNS_prim_addr>	string	-	IP address of the primary DNS Server.
<DNS_sec_addr>	string	-	IP address of the secondary DNS Server.
<P_CSCF_prim_addr>	string	-	IP address of the primary P-CSCF Server.
<P_CSCF_sec_addr>	string	-	IP address of the secondary P-CSCF Server.
< IM_CN_Signalling_Flag >	integer	-	integer type; shows whether the PDP context is for IM CN subsystem-related signalling only or not. 0PDP context is not for IM CN subsystem-related signalling only

			1PDP context is for IM CN subsystem-related signalling only.
< LIPA_indication >	integer	-	indicates that the PDP context provides connectivity using a LIPA PDN connection. This parameter cannot be set by the TE. 0indication not received that the PDP context provides connectivity using a LIPA PDN connection 1indication received that the PDP context provides connectivity using a LIPA

			PDN connection.
< IPv4_MTU >	integer	-	shows the IPv4 MTU size in octets.
<WLAN_Offload >	integer	-	indicates whether traffic can be offloaded using the specified PDN connection via a WLAN or not. This refers to bits 1 and 2 of the WLAN offload acceptability IE as specified in 3GPP TS 24.008 subclause 10.5.6.20. 0offloading the traffic of the PDN connection via a WLAN when in S1 mode or when in Iu

			mode is not acceptable. 1offloading the traffic of the PDN connection via a WLAN when in S1 mode is acceptable, but not acceptable in Iu mode. 2offloading the traffic of the PDN connection via a WLAN when in Iu mode is acceptable, but not acceptable in S1 mode. 3offloading the traffic of the PDN connection via a WLAN when in S1 mode or when in Iu mode is acceptable..
< Local_Addr_Ind >	integer	-	indicates whether or not the MS and the

network support local IP address in TFTs (see 3GPP TS 24.301 and 3GPP TS 24.008 subclause 10.5.6.3).

0indicates that the MS or the network or both do not support local IP address in TFTs

1indicates that the MS and the network support local IP address in TFTs.

< Serving_PLMN_rate_control_value >	integer	-	indicates the maximum number of uplink messages the UE is allowed to send in a 6 minute interval. This refers to octet 3 to 4 of the Serving PLMN rate control IE as specified in 3GPP TS 24.301 subclause 9.9.4.28.

▶▶ Referring to < local_addr> and <subnet_mask > parameter:

the string is given as dot-separated numeric (0-255) parameters. The format is:

for IPv4:

"a1.a2.a3.a4.m1.m2.m3.m4"

for IPv6:

"a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16. m1.m2.m3.m4.m5.m6.m7.m8.m9.m10.m11.m12. m13.m14.m15.m16"

The format of this parameter depends also on AT+CGPIAF setting for IPv6.

- Referring to all other IP address parameters than < local_addr> and <subnet_mask >:

  the string is given as dot-separated numeric (0-255) parameters. The format is:

  for IPv4:

  "a1.a2.a3.a4"

  for IPv6:

  "a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16"

  the format of this parameter depends also on AT+CGPIAF setting for IPv6

The dynamic part of the PDP context will only exist if established by the network. The test command returns a list of <cid>s associated with active contexts.

If the MT has dual stack capabilities, two lines of information are returned per <cid>. First one line with the IPv4 parameters followed by one line with the IPv6 parameters.

AT+CGCONTRDP=?

Return the list of <cid>s associated with active contexts.

AT+CGPIAF - Printing IP Address Format

This command selects the printout format of the IPv6 address.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGPIAF=[<IPv6_AddressFormat>[,<IPv6_SubnetNotation>[,<IPv6_LeadingZeros>[,<IPv6_CompressZeros>]]]]

Parameters:

Name	Type	Default	Description
<IPv6_AddressFormat>	integer	0	selects the IPv6 address format. Relevant for all AT command parameters that can hold an IPv6 address. Values: 0 : use IPv4-like dot-notation. IP addresses, and subnetwork mask if applicable, are dot-separated. 1 : use IPv6-like colon-notation. IP address, and subnetwork mask if applicable and when given explicitly, are separated by a space.
<IPv6_SubnetNotation>	integer	0	selects the subnet-notation for remote address and subnet mask. Setting does not apply if IPv6 address

			format <IPv6_AddressFormat>=0.
	Values:		
	0 :	both IP address, and subnet mask are started explicitly, separated by a space.	
	1 :	the printout format is applying / (forward slash) subnet-prefix Classless Inter-Domain Routing (CIDR) notation.	
<IPv6_LeadingZeros>	integer	0	selects whether leading zeros are omitted or not. Setting does not apply if IPv6 address format <IPv6_AddressFormat>=0.
	Values:		
	0 :	leading zeros are omitted.	
	1 :	leading zeros are included.	
<IPv6_CompressZeros>	integer	0	selects whether 1-n instances of 16-bit- zero values are replaced by only "::". This applies only once. Setting does not apply if IPv6 address format <IPv6_AddressFormat>=0.
	Values:		

0	:	no zero compression.
1	:	use zero compression.

AT+CGPIAF?

Read command returns the current parameter setting.

AT+CGPIAF=?

Test command returns values supported as compound values.

```
AT+CGPIAF=0,0,0,0
OK

AT#SGACT=1,1
#SGACT: 252.1.171.171.205.205.239.224.0.0.0.0.0.0.0.1
OK

AT+CGPIAF=1,0,0,0
OK

AT#SGACT=1,1
#SGACT: FC01:ABAB:CDCD:EFE0:0:0:0:1
OK
```

AT+CGACT - PDP Context Activate or Deactivate

This command activates or deactivates the specified PDP context(s).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGACT=<state>,<cid>

Parameters:

Name	Type	Default	Description
<state>	integer	N/A	activate/deactivate the PDP context
	Values:		
	0	:	deactivate
	1	:	activate
<cid>	integer	-	specifies a PDP context definition (see +CGDCONT command)

<cid> is not optional, it must be specified.

Only one <cid> is accepted in a single AT+CGACT command.

It is not supported multiple <cid>s activation or deactivation in a single AT+CGACT command.

AT+CGACT?

Read command returns the current activation state for all the defined PDP contexts in the format:

+CGACT: <cid>,<state>[<CR><LF>
+CGACT: <cid>,<state>[...]]

AT+CGACT=?

Test command reports information on the supported PDP context activation states <state>.

```
AT+CGACT=1,1
OK

AT+CGACT?
+CGACT: 1,1

OK
```

AT+CGEREP - Packet Domain Event Reporting

This command enables or disables the presentation of unsolicited result codes.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

AT+CGEREP=[<mode>[,<bfr>]]

Set command enables/disables sending of unsolicited result codes in case of certain events occurring in the module or in the network. The URC formats and related events are shown in the Additional info sections.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	controls the processing of URCs specified with this command.

Values:

0 : buffer unsolicited result codes in the TA. If TA result code buffer is full, the oldest one can be discarded. No codes are forwarded to the TE.

1 : discard unsolicited result codes when TA-TE link is reserved (e.g. in on-line data mode); otherwise forward them directly to the TE.

2 : buffer unsolicited result codes in the TA when TA-TE link is reserved (e.g. in on-line data mode) and flush them to the TE when TA-TE link becomes available; otherwise forward them directly to the TE.

Not supported

<bfr>	integer	0	controls the effect on buffered codes when <mode> 1 or 2 is entered.

Values:

0 : TA buffer of unsolicited result codes defined within this command is cleared when <mode> 1 or 2 is entered.

1 : TA buffer of unsolicited result codes defined within this command is flushed to the TE when <mode> 1 or 2 is entered (OK response shall be given before flushing the codes)

Not supported

Additional info:

- The network has deactivated a context. The associated <cid> for this context is provided to the TE.

  +CGEV: NW PDN DEACT <cid>

- The mobile equipment has deactivated a context. The associated <cid> for this context is provided to the TE. +CGEV: ME PDN DEACT <cid>

- The network has forced a PS detach. This implies that all active contexts have been deactivated. These are not reported separately.

  +CGEV: NW DETACH

- The mobile equipment has forced a PS detach. This implies that all active contexts have been deactivated. These are not reported separately.

  +CGEV: ME DETACH

Unsolicited fields:

Name	Type	Description
<PDP_type>	string	Packet Data Protocol type, which specifies the type of packet data protocol
<PDP_addr>	string	identifies the terminal in the address space applicable to the PDP
<cid>	integer	PDP Context Identifier

AT+CGEREP?

Read command returns the current <mode> and <bfr> settings, in the format:

+CGEREP: <mode>,<bfr>

AT+CGEREP=?

Test command reports the supported range of values for the +CGEREP command parameters.

AT+CGATT - PS Attach or Detach

This execution command is used to register (attach) the terminal to or deregister (detach) the terminal from the Packet Domain service.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGATT=<state>

Parameter:

Name	Type	Default	Description
<state>	integer	N/A	state of PS attachment

Values:

0 : detached

1 : attached

AT+CGATT?

Read command returns the current PS state in the format:

+CGATT: <state>

AT+CGATT=?

Test command returns the values range of the <state> parameter.

```
AT+CGATT?
+CGATT: 0
OK

AT+CGATT=?
+CGATT: (0,1)
OK

AT+CGATT=1
OK
```

AT+CGEQOS - Define EPS Quality of Service

The command specifies the EPS Quality of Service parameters.

[1] 3GPP TS 23.203

[2] 3GPP TS 24.301

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGEQOS=[<cid>[,<QCI> [,<DL_GBR>,<UL_GBR> [,<DL_MBR>,<UL_MBR>]]]]

Set command specifies the EPS Quality of Service parameters.

A special form of the set command, +CGEQOS= <cid> causes the values for context number <cid> to become undefined.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	it specifies a particular EPS Traffic Flows definition in EPS
<QCI>	integer	9	it specifies a class of EPS QoS, see standard [1].

Values:

0	:	QCI is selected by network
1÷4	:	value range for guaranteed bit rate Traffic Flows

5÷9	:	value range for non-guaranteed bit rate Traffic Flows
75	:	value range for guaranteed bit rate Traffic Flows
79	:	value range for non-guaranteed bit rate Traffic Flows

<DL_GBR>	integer	-	it indicates DL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].
<UL_GBR>	integer	-	indicates UL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].
<DL_MBR>	integer	-	it indicates DL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].
<UL_MBR>	integer	-	it indicates UL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].

Additional info:

▶▶ Possible Response(s):

+CME ERROR: <err>

AT+CGEQOS?

Read command returns the current settings for each defined QoS.

+CGEQOS: <cid>, <QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>]
[<CR>>LF>+CGEQOS: <cid>, <QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>]
[...]]

AT+CGEQOS=?

Test command returns the ranges of the supported parameters:

+CGEQOS: (list of supported <cid>s),(list of supported <QCI>s)

- AT+CGEQOS? returns only OK if no AT+CGEQOS valid setting set before.
- <DL_GBR>, <UL_GBR>, <DL_MBR>, <UL_MBR> are valid parameter but their values are not used (dedicated bearer not supported).

  AT+CGEQOS=? does not report <DL_GBR>, <UL_GBR>, <DL_MBR>, <UL_MBR> parameters range.

AT+CGEQOSRDP - EPS Quality of Service Read Dynamic Parameters

The command returns the parameters related to the Quality of Service.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGEQOSRDP=[<cid>[,...]]

The execution command returns the Quality of Service parameters of the established PDN connection associated to the provided context identifier <cid>.

The format of the returned message is:

+CGEQOSRDP:<cid>,<QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>]
[<CR>>LF>
+CGEQOSRDP:<cid>,<QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>] [...]]

See +CGEQOS command to have information on the meaning of the returned parameters.

Parameter:

Name	Type	Default	Description
<cid>	integer	-	It specifies a particular Traffic Flows definition in EPS and a PDN connection definition in UMTS/GPRS.

- If the context cannot be found an ERROR response is returned.
- If the parameter <cid> is omitted, the Quality of Service parameters for all established PDN connections are returned.
- <DL_GBR>, <UL_GBR>, <DL_MBR>, <UL_MBR> parameters are omitted for a non-GBR QCI.

AT+CGEQOSRDP=?

The test command returns a list of <cid>s associated with active contexts.

Parameters of both network and MT/TA initiated PDN connections will be returned.

IPEasy

AT#SGACT - Context Activation

This command enables/disables the PDP context activation.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SGACT=<cid>,<stat>[,<userId>[,<pwd>]]

Execution command is used to activate the specified PDP context, followed by binding data application to the PS network. Also, it is used to deactivate the PDP context and unbind data application from PS network.

Execution command returns a list of IP addresses for the specified context identifiers in the format:

for IP or IPV6 PDP context: #SGACT: <ipAddr>

for DUAL STACK IPV4V6 PDP context: #SGACT: [<ipAddrV4>],[<ipAddrV6>]

Parameters:

Name	Type	Default	Description
<cid>	integer	-	specifies a particular PDP context definition. To know the range see +CGDCONT command.
<stat>	integer	0	activates/disactivates the PDP context specified
	Values:		

0	:	deactivate the context
1	:	activate the context

<userId>	string	-	user identifier, used only if the context requires it
<pwd>	string	-	password, used only if the context requires it

Additional info:

Meaning of the parameters returned by the command.

Name	Type	Default	Description
<ipAddr>	string	-	ip address ipv4 or ipv6
<ipAddrV4>	string	-	ip address ipv4 (if v4 PDP context activated)
<ipAddrV6>	string	-	ip address ipv6 (if v6 PDP context activated)

Context activation/deactivation returns ERROR if there is not any socket associated to it, see #SCFG.

In LTE network, default PDP context (<cid>=1) is activated by piggybacking on LTE attach procedure and maintained until detached from network. This command with <cid>=1 is just binding or unbinding application to the default PDP context.

#SCFG=x,0 will close all unclosed AT#SD sockets on <cid>=x.

AT#SGACT?

Read command returns the state of all the contexts that have been defined in the format:

#SGACT: <cid$_1$>,<stat$_1$><CR><LF>

...

#SGACT: <cid$_n$>,<stat$_n$>

Each row in the read command's answer is optional.

AT#SGACT=?

Test command reports the range for the parameters <cid> and <stat>.

AT#SCFG - Socket Configuration

The command sets the configuration for the socket.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SCFG=<connId>,<cid>,<pktSz>,<maxTo>,<connTo>,<txTo>

Set command sets the socket configuration parameters.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier. Value: 1÷max : socket connection identifier value (max value is returned by Test command)
<cid>	integer	-	PDP context identifier. To know the range see +CGDCONT command.
<pktSz>	integer	300	Packet size in bytes to be used by the TCP/UDP/IP stack for data sending. Values: 0 : select automatically default value

1÷1500	:	packet size in bytes

<maxTo>	integer	90	Exchange timeout in seconds (or socket inactivity timeout); if there's no data exchange within this timeout period the connection is closed.

Values:

0	:	no timeout
1÷65535	:	timeout

<connTo>	integer	600	Overall connection timeout in tenths of seconds. If the module cannot establish a connection to the remote within this timeout period, an error is raised. An error can also be raised earlier than <connTo> expiries caused by lower layer TCP/IP retry/timeout mechanism.

Values:

10÷1200	:	timeout value in tenths of seconds

<txTo>	integer	50	data sending timeout; data are sent even if they are less than max packet size, after this period. Used for online data mode only.

Values:

0	:	no timeout
1÷255	:	timeout in tenths of seconds
256	:	timeout value of 10 ms
257	:	timeout value of 20 ms
258	:	timeout value of 30 ms
259	:	timeout value of 40 ms
260	:	timeout value of 50 ms
261	:	timeout value of 60 ms
262	:	timeout value of 70 ms
263	:	timeout value of 80 ms
264	:	timeout value of 90 ms

AT#SCFG?

Read command returns the current socket configuration parameters values for all the six sockets, in the format:

#SCFG: <connId1>,<cid1>,<pktsz1>,<maxTo1>,<connTo1>,<txTo1><CR><LF>
#SCFG: <connId2>,<cid2>,<pktsz2>,<maxTo2>,<connTo2>,<txTo2><CR><LF>
...
#SCFG: <connId10>,<cid10>,<pktsz10>,<maxTo10>,<connTo10>,<txTo10>

AT#SCFG=?

Test command returns the range of supported values for all the parameters.

Example of read command.

```
AT#SCFG?
#SCFG: 1,1,300,90,600,50
#SCFG: 2,1,300,90,600,50
#SCFG: 3,1,300,90,600,50
#SCFG: 4,1,300,90,600,50
#SCFG: 5,1,300,90,600,50

...
#SCFG: 10,1,300,90,600,50
OK
```

AT#SCFGEXT - Socket Configuration Extended

This command sets the socket configuration extended parameters.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SCFGEXT=<connId>,<srMode>,<recvDataMode>,<keepalive>[,<ListenAutoRsp>[,<sendDataMode>]]

Set command sets the socket configuration extended parameters.

Parameters:

Name	Type	Default	Description
<connId>	integer	1	socket connection identifier
	Value:		
	1÷max :		socket connection identifier value (max value is returned by Test command)
<srMode>	integer	0	SRING unsolicited mode, see Additional info section.
	Values:		
	0 :		Normal
	1 :		Data amount
	2 :		Data view

3 : Data view with UDP datagram information

<recvDataMode>	integer	0	Data view mode for received data in command mode (#SRECV or <srMode> = 2)

Values:

0 : Text mode

1 : Hexadecimal mode

<keepalive>	integer	0	Set the TCP Keepalive value in minutes

Values:

0 : deactivated

1÷240 : keepalive time in minutes

<ListenAutoRsp>	integer	0	set the listen auto-response mode, that affects the commands #SL and #SLUDP

Values:

0 : deactivated

1 : activated

<sendDataMode>	integer	0	Data mode for sending data in command mode (#SSEND)

Values:

0 : data represented as text

1 : 0ata represented as sequence of hexadecimal numbers (from 00 to FF). Each octet of the data is given as two IRA character long.

Additional info:

▶▶ These are the SRING formats, depending on <srMode> setting:

if <srMode> = 0 (Normal):

SRING: <connId>

if <srMode> = 1 (Data amount):

SRING: <connId>,<recData>

if <srMode> = 2 (Data view):

SRING: <connId>,<recData>,<data>

if <srMode> = 3 (Data view with UDP datagram information):

SRING: <sourceIP>,<sourcePort>,<connId>,<recData>,<dataLeft>,<data>

Name	Type	Default	Description

<recData>	integer	-	amount of data received on the socket connection number <connId>
<data>	mixed	-	data received displayed following <recvDataMode> value
<sourceIP>	string	-	IP address of the source of data
<sourcePort>	string	-	IP port of the source of data
<dataLeft>	integer	-	number of bytes left in the UDP datagram

Keepalive is available only on TCP connections.

For the behavior of #SL and #SLUDP in case of auto response mode or in case of no auto response mode, see the description of the two commands.

AT#SCFGEXT?

Read command returns the current socket extended configuration parameters values for all the six sockets, in the format:

#SCFGEXT: <connId$_1$>,
<srMode>,<dataMode>,<keepalive>,<ListenAutoRsp>,0<CR><LF>

. . .

#SCFGEXT:<connId$_{conMax}$>,<srMode>,<dataMode>,<keepalive>,<ListenAutoRsp>,0<CR><LF>

AT#SCFGEXT=?

Test command returns the range of supported values for all the sub parameters.

</>

* Socket 1 set with data view sring, text data mode, a keepalive time of 30 minutes and listen auto-response set.

  Socket 3 set with data amount sring, hex recv data mode, no keepalive and listen auto-response not set.

  Socket 4 set with hex recv and send data mode.

```
AT#SCFGEXT?
#SCFGEXT: 1,2,0,30,1,0
#SCFGEXT: 2,0,0,0,0,0
#SCFGEXT: 3,1,1,0,0,0
#SCFGEXT: 4,0,1,0,0,1
...
...
OK
```

AT#SCFGEXT2 - Socket Configuration Extended 2

Socket Configuration Extended.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SCFGEXT2=<connId>,<bufferStart>[,<abortConnAttempt>[,<unused_B>[,<unused_C>[,<noCarrierMode>]]]]

Set command sets the socket configuration extended parameters for features not included in #SCFGEXT command.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier Value: 1÷max : socket connection identifier value (max value is returned by Test command)
<bufferStart>	integer	0	select one of the two data sending timeout methods, the first one defined "old" the second one "new". The "old" data sending timeout method is set - by default - by #SCFG command, which sets also the <txTo> data sending timeout value. With #SCFGETXT2 command, you can set either the

"old" or the "new" data sending timeout method. If the "new" method is selected, the "old" one is automatically disabled.

The checking if new data have been received from serial port is done with a granularity that is directly related to #SCFG <txTo> setting with a maximum period of 1 sec.

Values:

0	:	select "old" method: start transmission timer only first time if new data are received from the serial port
1	:	select "new" method: restart transmission timer when new data is received from serial port

<abortConnAttempt>	integer	0	enable the abort of an ongoing connection attempt started by #SD command and before the reception of the CONNECT message (in online mode) or OK message (in command mode). Values automatically saved in NVM.

Values:

0	:	not possible to interrupt connection attempt

1 : it is possible to interrupt the connection attempt (<connTo> set by #SCFG or DNS resolution running if required) and give back control to AT interface by reception of a character. As soon as the control given to the AT interface, the ERROR message will be received on the interface itself.

<unused_B>	integer	-	reserved for future use
<unused_C>	integer	-	reserved for future use
<noCarrierMode>	integer	0	select the NO CARRIER message format received when the socket is closed.

Values:

0 : no additional information is attached to NO CARRIER message

1 : NO CARRIER: <connId> message

2 : NO CARRIER: <connId>, <cause> message. Refer to Additional info section

Additional info:

▶▶ <noCarrierMode>=2 selects the following NO CARRIER message format:

NO CARRIER: <connId>, <cause>

Name	Type	Default	Description
<cause>	integer	-	is the socket disconnection cause. Refer to #SLASTCLOSURE command to know its values and meanings.

- Is necessary to avoid overlapping of the two methods. Enabling new method, the old method for transmission timer (#SCFG) is automatically disabled to avoid overlapping.
- Check if new data have been received from serial port is done with a granularity directly related to <txTo> parameter which is set by #SCFG command. The maximum period is 1 sec.
- Like #SLASTCLOSURE, in case of subsequent consecutive closure causes received, the original disconnection cause indicated.
- In the case of command mode connection and remote closure with subsequent inactivity timeout closure without retrieval of all available data (#SRECV or SRING mode 2), it is indicated cause 1 for both possible FIN and RST from remote.

AT#SCFGEXT2?

Read command returns the current socket extended configuration of the six sockets. The format is:

#SCFGEXT2:<connId$_1$>,<bufferStart$_1$>,<abortConnAttempt$_1$>,0,0,<noCarrierMode$_1$><CR><LF>
. . .
#SCFGEXT2:<connId$_6$>,<bufferStart$_6$>,<abortConnAttempt$_6$>0,0,<noCarrierMode$_6$><CR><LF>

AT#SCFGEXT2=?

Test command returns the range of supported values for all parameters.

</>

* Set the new transmission timer behavior for <connId>=1 and <connId>=2 sockets.

```
AT#SCFGEXT2=1,1
OK

AT#SCFGEXT2=2,1
OK
```

Check the current extended configuration of the six sockets

```
AT#SCFGEXT2?
#SCFGEXT2: 1,1,0,0,0,0
#SCFGEXT2: 2,1,0,0,0,0
#SCFGEXT2: 3,0,0,0,0,0
#SCFGEXT2: 4,0,0,0,0,0
#SCFGEXT2: 5,0,0,0,0,0
#SCFGEXT2: 6,0,0,0,0,0
OK
```

Check the current configuration of the six sockets

```
AT#SCFG?
#SCFG: 1,1,300,90,600,50
#SCFG: 2,1,300,90,600,50
#SCFG: 3,1,300,90,600,50
#SCFG: 4,2,300,90,600,50
#SCFG: 5,2,300,90,600,50
```

#SCFG: 6,2,300,90,600,50
OK

Change the <txTo> data sending timeout of the <connId>=1 socket.

AT#SCFG=1,1,300,90,600,30
OK

AT#SCFGEXT3 - Socket Configuration Extended 3

This command sets the socket configuration extended parameters for features not included in #SCFGEXT command nor in #SCFGEXT2 command.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SCFGEXT3=<connId>,<immRsp>[,<closureTypeCmdModeEnabling>[,<fastSRING>[,<ssendTimeout>[,<unusedD>]]]]

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<immRsp>	integer	0	enables #SD command mode immediate response
	Values:		

		0 : #SD in command mode (see #SD) returns after the socket is connected 1 : #SD in command mode returns immediately; then the state of the connection can be read by the AT command #SS	
<closureTypeCmdModeEnabling>	integer	0	it has no effect and is included only for backward compatibility
	Value:		
		0 : factory default	
<fastSRING>	integer	0	enables the fast SRING (active only when #SCFGEXT parameter <srmode>=2) in TCP and UDP sockets
	Values:		
		0 : SRING unsolicited is received periodically if data are available every 200 ms	

		1 :	if data are available SRING unsolicited is received asynchronous as fast as possible
<ssendTimeout>	integer	0	timeout for #SSEND (dummy)
	Values:		
		0 :	no timeout
		1 :	immediate exit(error) in case of stack temporary suspended internally
		100÷600 :	timeout in 100 msec units
<unusedD>	mixed	0	unused parameter
	Value:		
		0 :	factory default

If <immRsp> value is 1 and #SD in command mode, in case of server not responding there will be a NO CARRIER indicating the end of connection attempts.

Monitor the state of the connection through the AT command #SS.

Parameter <ssendTimeout> is dummy.

The set value is only tested against range error according to range reported in AT#SCGFEXT=? test command

AT#SCFGEXT3?

Read command returns the current socket extended configuration parameters values for all the max sockets, in the format:

#SCFGEXT3:<connId$_1$>,<immRsp$_1$>,<closureTypeCmdModeEnabling>,<fastsring>,<ssendTimeout>,0<CR><LF>

. . .

#SCFGEXT3:<connId$_{max}$>,<immRsp$_{max}$>,<closureTypeCmdModeEnabling>,<fastsring>,<ssendTimeout>,0<CR><LF>

AT#SCFGEXT3=?

Test command returns the range of supported values for all the parameters.

AT#SCFGEXT4 - Socket Configuration Extended 4

This command sets additional socket configuration extended parameters.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SCFGEXT4=<connId>,<listenAutoReopen>[,<listenClientIP>[,<unusedA>[,<unusedB>[,<unusedC>]]]]

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷max :		socket connection identifier value (max value is returned by Test command)
<listenAutoReopen>	integer	0	enables automatic re-open of the socket listening for an incoming TCP or UDP connection on the socket closure after client connection
	Values:		
	0 :		disables automatic re-open of the socket listening

1 : enables automatic re-open of the socket listening

<listenClientIP>	integer	0	enables client IP reporting on client connection to socket listening for an incoming TCP or UDP connection

Values:

0 : disable client IP reporting

1 : enable client IP reporting

<unusedA>	mixed	0	unused parameter

Value:

0 : factory default

<unusedB>	mixed	0	unused parameter

Value:

0 : factory default

<unusedC>	mixed	0	unused parameter

Value:

0 : factory default

Additional info:

- If the <listenClientIP> parameter is set to value 1 (for the specific <connId>) and also <ListenAutoRsp> is set to 1 (for the same <connId>) when the client connection to TCP or UDP listener service is accepted in online mode the CONNECT indication will report the client IP address in the following format:

  CONNECT [<connId>,<clientIP>,<clientPort>]

  Parameter:

Name	Type	Default	Description
<clientIP>	string	-	IP address of the client
<clientPort>	integer	-	Client port number
	Values:		
	1÷65535	:	Client port number

- About <clientIP> parameter, the string is given as dot-separated numeric (0-255) parameters. The format is:

  for IPv4:

  "a1.a2.a3.a4"

  for IPv6:

  "a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16"

  the format of this parameter depends also on AT+CGPIAF setting for IPv6.

About <listenAutoReopen> parameter see also AT#SL and AT#SLUDP.

AT#SCFGEXT4?

Read command returns the current socket extended configuration parameters values for all the max sockets, in the format:

#SCFGEXT4:<connId$_1$>,<listenAutoReopen$_1$>,<listenClientIP$_1$>,0,0,0<CR><LF>
. . .
#SCFGEXT4:<connId$_{max}$>,<listenAutoReopen$_{max}$>,<listenClientIP$_{max}$>,0,0,0<CR><LF>

AT#SCFGEXT4=?

Test command returns the range of supported values for all the parameters.

AT#SKTRST - Socket Parameters Reset

Socket Parameters Reset

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SKTRST

Execution command resets the #SCFG socket parameters to the "factory default" configuration and stores them in the NVM of the device.

AT#SKTRST=?

Test command returns the OK result code.

Reset of the socket parameters

```
AT#SKTRST
OK
```

AT#SD - Socket Dial

Execution command opens a remote connection via socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SD=<connId>,<txProt>,<rPort>,<lPaddr>[,<closureType>[,<lPort>[,<connMode>[,<txTime>[,<userIpType>]]]]]

Execution command opens a remote connection via socket.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier.

Value:

1÷max	:	socket connection identifier value (max value is returned by Test command)

Name	Type	Default	Description
<txProt>	integer	N/A	Transmission protocol.

Values:

0	:	TCP
1	:	UDP

<rPort>	integer	N/A	Remote host port to contact.

Value:

1÷65535	:	remote host port number

<IPaddr>	string	-	IP address of the remote host: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any host name to be solved with a DNS query - any valid IPv6 address in the format: xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx or xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx. xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx
<closureType>	integer	0	Socket closure behavior only for TCP when remote host has closed. The parameter has no effects for UDP connections.

Values:

0	:	local host closes immediately when remote host has closed
255	:	local host closes after an escape sequence (+++)

<lPort>	integer	N/A	UDP connections local port only for UDP connections. The parameter has no effects for TCP connections.

Value:

1÷65535	:	UDP local port number

<connMode>	integer	0	Connection mode.

Values:

0	:	online mode connection
1	:	command mode connection

<txTime>	integer	0	adjusting a time interval for series of UDP data packets will be uploaded.

Values:

0	:	Time interval is not requested
1÷1000	:	Time interval in milliseconds

<userIpType>	integer	0	ip type for socket to open

Values:

0	:	no ip type chosen
1	:	ipv4
2	:	ipv6

.

- <userIpType> parameter is only valid when <IPaddr> is domain name and dual stack connection is open by #SGACT.

  When <userIpType> is "no ip type chosen" ipv6 will be requested firstly. When ipv6 DNS server does not support so ipv4 will be requested.

- <closureType> parameter is valid for TCP connections only and has no effect (if used) for UDP connections.

- <lPort> parameter is valid for UDP connections only and has no effect (if used) for TCP connections

- If we set <connMode> to online mode connection and the command is successful we enter in online data mode and we see the intermediate result code CONNECT. After the CONNECT we can suspend the direct interface to the socket connection (N.B. the socket stays open) using the escape sequence (+++): the module moves back to command mode and we receive the final result code OK after the suspension.

  After such a suspension, it is possible to resume it in every moment (unless the socket inactivity timer timeouts, see #SCFG) by using the #SO command with the corresponding <connId>.

- If we set <connMode> to command mode connection and the command is successful, the socket is opened, and we remain in command mode and we see the result code OK.

- If there are input data arrived through a connected socket and not yet read because the module entered command mode before reading them (after an escape sequence or after #SD has been issued with <connMode> set to command mode connection), these data are buffered and we receive the SRING URC (SRING presentation format depends on the last #SCFGEXT setting); it is possible to read these data afterwards issuing #SRECV. Under the same hypotheses it is possible to send data while in command mode issuing #SSEND.

- <txTime> parameter is valid for UDP connections only and has no effect (if used) for TCP connections. For slow servers it is recommended to adjust the time interval for uploading series of data packets in order to do not lose data. The following data packet will be sent after the previous data packet's time interval has been expired.

- AT#SD (and similar AT commands) +++ timings:

  +++ escape sequence must be preceded and followed by a pause of 1000 ms

  + characters must be entered in quick succession, the time between each + must be less then 1000 ms

- The maximum number of sockets connected at the same time is 6.

  It is possible to configure up to 10 <connid>, but it is only possible to have 6 <connid> connected at the same time.

  If FTP is used at the same time the number of <connid> connected at the same time is decreased by 2.

  If HTTP is used at the same time the number of <connid> connected at the same time is decreased by 1.

AT#SD=?

Test command reports the range of values for all the parameters.

Examples of socket dial in online and command mode.

* Open socket 1 in online mode

```
AT#SD=1,0,80,"www.google.com",0,0,0

CONNECT
...
...
```

Open socket 1 in command mode

```
AT#SD=1,0,80,"www.google.com",0,0,1
OK
```

AT#SO - Socket Restore

Execution command resumes the direct interface to a socket connection which has been suspended by the escape sequence.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SO=<connId>

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier

Value:

1÷max	:	socket connection identifier value (max value is returned by Test command)

AT#SO=?

Test command reports the range of values for <connId> parameter

AT#SH - Socket Shutdown

The set command closes a socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SH=<connId>

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier to be closed

Value:

1÷max : socket connection identifier value (max value is returned by Test command)

Socket cannot be closed in states "resolving DNS" and "connecting", see #SS command.

AT#SH=?

Test command reports the range for parameter <connId>

AT#SL - Socket Listen

The command opens/closes socket listening.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SL=<connId>,<listenState>,<listenPort>[,<lingerT>]

Set command opens/closes a socket listening for an incoming TCP connection on a specified port.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<listenState>	integer	N/A	listening action
	Values:		
	0	:	close socket listening
	1	:	start socket listening
<listenPort>	integer	N/A	local listening port

Value:

1÷65535	:	local listening port value

<lingerT>	integer	N/A	linger time

Values:

0	:	immediate closure after remote closure
255	:	local host closes only after an escape sequence (+++)

If successful, command returns a final result code OK.

If the ListenAutoRsp flag has not been set through the command #SCFGEXT (for the specific <connId>), then, when a TCP connection request comes on the input port, if the sender is not filtered out by internal firewall, an URC is received:

+SRING : <connId>

Afterwards we can use #SA to accept the connection or #SH to refuse it.

If the connection is accepted in online mode: the CONNECT indication is given and the modem goes into online data mode. If the ListenAutoRsp flag has been set, then, when a TCP connection request comes on the input port, if the sender is not filtered out by the internal firewall, the connection is automatically accepted: the CONNECT indication is given and the modem goes into online data mode.

If the listenClientIP flag was set through the command #SCFGEXT4 (for the specific <connId>) the CONNECT indication will report the client IP address, see AT#SCFGEXT4 for more details.

The connection can be refused by moving to command mode with +++ followed by AT#SH.

For both cases ListenAutoRsp flag not set and set, as soon as the client connection is accepted the listener service will be deactivated and the same socket/connId used for listen will be reused for data exchange.

On connection closure, and on connection refuse with AT#SH, there will be no listener service unless listenAutoReopen flag was set through the command #SCFGEXT4 (for the specific <connId>).

If the socket is closed by the network the following URC is received:

#SL: ABORTED

The maximum number of sockets connected at the same time is 6.

It is possible to configure up to 10 <connId>, but it is only possible to have 6 <connId> connected at the same time.

If FTP is used at the same time the number of <connId> connected at the same time is decreased by 2.

If HTTP is used at the same time the number of <connId> connected at the same time is decreased by 1.

AT#SL?

Read command returns all the actual listening TCP sockets.

AT#SL=?

Test command returns the range of supported values of the parameters.

Open a socket listening for TCP on port 3500.

```
AT#SL=1,1,3500
OK
```

AT#SLUDP - Socket Listen UDP

This command opens/closes a socket listening for an incoming UDP connection on a specified port.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#SLUDP=<connId>,<listenState>,<listenPort>

Execution command opens/closes a socket listening for an incoming UDP connection on a specified port.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<listenState>	integer	0	indicates the action that will be performed
	Values:		
	0	:	closes socket listening
	1	:	starts socket listening

<listenPort>	integer	1	local listening port

Value:

1÷65535	:	available port numbers

If successful, command returns a final result code OK.

If the ListenAutoRsp flag has not been set through the command #SCFGEXT (for the specific <connId>), then, when an UDP connection request comes on the input port, if the sender is not filtered out by internal firewall, an URC is received:

+SRING : <connId>

Afterwards we can use #SA to accept the connection or #SH to refuse it.

If the connection is accepted in online mode: the CONNECT indication is given and the modem goes into online data mode.

If the ListenAutoRsp flag has been set, then, when an UDP connection request comes on the input port, if the sender is not filtered out by the internal firewall, the connection is automatically accepted: the CONNECT indication is given and the modem goes into online data mode.

If the listenClientIP flag was set through the command #SCFGEXT4 (for the specific <connId>) the CONNECT indication will report the client IP address, see AT#SCFGEXT4 for more details.

The connection can be refused by moving to command mode with +++ followed by AT#SH.

For both cases ListenAutoRsp flag not set and set, as soon as the client connection is accepted the listener service will be

deactivated and the same socket/connId used for listen will be reused for data exchange.

On connection closure, and on connection refuse with AT#SH, there will be no listener service unless listenAutoReopen flag was set through the command #SCFGEXT4 (for the specific <connId>).

If the socket is closed by the network the following URC is received:

#SLUDP: ABORTED

When closing the listening socket <listenPort> is a do not care parameter

AT#SLUDP?

Read command returns all the actual listening UDP sockets.

AT#SLUDP=?

Test command returns the range of supported values for all the sub parameters.

Next command opens a socket listening for UDP on port 3500.

```
AT#SLUDP=1,1,3500
OK
```

AT#SA - Socket Accept

Execution command accepts an incoming socket connection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SA=<connId>[,<connMode>]

Execution command accepts an incoming socket connection after an URC

SRING: <connId>

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier.
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<connMode>	integer	0	Connection mode, as for command #SD.
	Values:		
	0	:	online mode connection
	1	:	command mode connection

- The SRING URC has to be a consequence of a #SL issue.
- Setting the command before to having received a SRING will result in an ERROR indication, giving the information that a connection request has not yet been received.

AT#SA=?

Test command reports the range of values for all the parameters.

AT#SSEND - Send Data in Command Mode

This command is used to send data through a connected socket.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSEND=<connId>[,<Rai>]

Execution command permits, while the module is in command mode, to send data through a connected socket.

After entering AT#SSEND=... command, and terminated the command line with <CR>, the module returns the following four characters sequence prompt, and waits for data to send:

<CR><LF><greater_than><space> (see IRA character set: 13, 10, 62, 32)

To send the entered data, enter Ctrl-Z char (0x1A hex); to abort the operation enter ESC char (0x1B hex).

If data are successfully sent, the command returns OK. If data sending fails for some reason, an error code is reported.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		

1÷max	:	socket connection identifier value (max value is returned by Test command)

<Rai>	integer	N/A	RAI (Release Assistance Indication) configuration. NOT supported.

Values:

1	:	Set RAI to: "No further uplink or downlink data transmission subsequent to this transmission"
2	:	Set RAI to: "Only a single downlink data transmission and no further uplink data transmissions subsequent to this transmission"

- The maximum number of bytes to send is 1500 bytes; trying to send more data will cause the surplus to be discarded and lost.
- It is possible to use #SSEND only if the connection was opened by #SD, else the ME is raising an error.
- A byte corresponding to BS char (0x08) is treated with its corresponding meaning; therefore previous byte will be cancelled (and BS char itself will not be sent).

  RAI (Release Assistance Indication) is not supported, so it is ignored.

AT#SSEND=?

Test command returns the range of supported values for parameter <connId> and <Rai>.

Send data through socket number 2

```
AT#SSEND=2
>Test<CTRL-Z>
OK
```

AT#SSENDEXT - Send Data in Command Mode extended

This command allows to send data through a connected socket including all possible octets (from 0x00 to 0xFF).

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSENDEXT=<connId>,<bytesToSend>[,<Rai>]

Execution command allows, while the module is in command mode, to send data through a connected socket including all possible octets (from 0x00 to 0xFF).

After entering AT#SSENDEXT=... command, and terminated the command line with <CR>, the module returns the following four characters sequence prompt, and waits for data to send:

<CR><LF><greater_than><space> (see IRA character set: 13, 10, 62, 32)

When <bytesToSend> bytes have been sent, operation is automatically completed.

If data are successfully sent, the command returns OK. If data sending fails for some reason, an error code is reported.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier

Value:

1÷max	:	socket connection identifier value (max value is returned by Test command)

<bytesToSend>	integer	N/A	number of bytes to be sent

Value:

1÷maxBytes	:	maxBytes is the maximum number of bytes that can be sent and it is reported by Test command

<Rai>	integer	N/A	RAI (Release Assistance Indication) configuration. NOT supported.

Values:

1	:	Set RAI to: "No further uplink or downlink data transmission subsequent to this transmission"
2	:	Set RAI to: "Only a single downlink data transmission and no further uplink data transmissions subsequent to this transmission"

It is possible to use #SSENDEXT only if the connection was opened by #SD, else the modem returns an error.

All special characters are sent like a generic byte. For example, 0x08 is not interpreted as a BS (BackSpace) but it is simply sent through the socket.

RAI (Release Assistance Indication) is not supported, so it is ignored.

AT#SSENDEXT=?

Test command returns the range of supported values for parameters <connId>, <bytesToSend> and <Rai>.

Open the socket in command mode:

AT#SD=1,0,<port>,"IP address",0,0,1
OK

Enter the command specifying total number of bytes as second parameter:

AT#SSENDEXT=1,256

> ..... ; Terminal echo of bytes sent is displayed here
OK

All possible bytes (from 0x00 to 0xFF) are sent on the socket as generic bytes.

AT#SRECV - Socket Receive Data in Command Mode

The command permits the user to read data arrived through a connected socket when the module is in command mode.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SRECV=<connId>,<maxByte>[,<UDPInfo>]

Execution command permits the user to read data arrived through a connected socket but buffered and not yet read because the module entered command mode before reading them; the module is notified of these data by a SRING: URC, whose presentation format depends on the last #SCFGEXT setting.

Parameters:

Name	Type	Default	Description
<connId>	integer	NA	socket connection identifier
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<maxByte>	integer	NA	max number of bytes to read
	Value:		
	1÷1500	:	max number of bytes to read
<UDPInfo>	integer	0	enables/disables the visualization of UDP datagram information.

Values:

0 : UDP information disabled

1 : UDP information enabled, see Additional info section.

Additional info:

If <UDPInfo> is set to 1 (AT#SRECV=<connId>,<maxBytes>,1), the command returns a message having the following format:

#SRECV: <remoteIP>,<remotePort><connId>,<recData>,<dataLeft>

Name	Type	Default	Description
<remoteIP>	string	-	remote ip address
<remotePort>	string	-	remote port address
<recData>	integer	-	received data
<dataLeft>	integer	-	remaining bytes in the datagram.

Issuing #SRECV when there is no buffered data raises an error.

AT#SRECV=?

Test command returns the range of supported values for parameters <connId> <maxByte> and <UDPInfo>.

AT#SSENDUDP - Send UDP Data to a Specific Remote Host

This command allows to send data over UDP to a specific remote host.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSENDUDP=<connId>,<remoteIP>,<remotePort>[,<Rai>]

This command allows, while the module is in command mode, to send data over UDP to a specific remote host. UDP connection has to be previously completed with a first remote host through #SLUDP / #SA. Then, if module receives data from this or another host, it is able to send data to it. Like command #SSEND, the device responds with ">" prompt and waits for the data to send.

Parameters:

Name	Type	Default	Description
<connId>	integer	1	socket connection identifier
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<remoteIP>	string	-	IP address of the remote host in dotted decimal notation, string type: "xxx.xxx.xxx.xxx"
<remotePort>	integer	1	remote host port
	Value:		

			1÷65535 : host port number
<Rai>	integer	N/A	RAI (Release Assistance Indication) configuration. NOT supported. Values: 1 : Set RAI to: "No further uplink or downlink data transmission subsequent to this transmission" 2 : Set RAI to: "Only a single downlink data transmission and no further uplink data transmissions subsequent to this transmission"

After SRING that indicates incoming UDP data and issuing #SRECV to receive data itself, through #SS is possible to check last remote host (IP/Port).

If successive resume of the socket to online mode is performed (#SO), connection with first remote host is restored as it was before.

RAI (Release Assistance Indication) is not supported, so it is ignored.

AT#SSENDUDP=?

Test command reports the supported range of values for parameters <connId>,<remoteIP>, <remotePort> and <Rai>.

</>

* Starts listening on <LocPort> (previous setting of firewall through #FRWL has to be done)

```
AT#SLUDP=1,1,<LocPort>
OK

SRING: 1            UDP data from a remote host available

AT#SA=1,1
OK

SRING: 1

AT#SI=1
#SI: 1,0,0,23,0       23 bytes to read
OK

AT#SRECV=1,23
#SRECV:1,23
message from first host
OK

AT#SS=1
#SS: 1,2,<LocIP>,<LocPort>,<RemIP1>,<RemPort1>
OK
```

AT#SSENDUDP=1,<RemIP1>,<RemPort1>
>response to first host
OK

SRING: 1 UDP data from a remote host available

AT#SI=1
#SI: 1,22,23,24,0 24 bytes to read
OK

AT#SRECV=1,24
#SRECV:1,24
message from second host
OK

AT#SS=1
#SS: 1,2,<LocIP>,<LocPort>,<RemIP2>,<RemPort2>
OK

Remote host has changed, we want to send a response:

AT#SSENDUDP=1,<RemIP2>,<RemPort2>
>response to second host
OK

AT#SSENDUDPEXT - Send UDP Data to a Specific Remote Host EXTENDED

This command allows, while the module is in command mode, to send data over UDP to a specific remote host including all possible octets (from 0x00 to 0xFF)

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT#SSENDUDPEXT=<connId>,<bytestosend>,<remoteIP>,<remotePort>[,<Rai>]

Set command allows, while the module is in command mode, to send data over UDP to a specific remote host including all possible octets (from 0x00 to 0xFF).

As indicated about #SSENDUDP, UDP socket has to be previously opened through #SLUDP / #SA, then we are able to send data to different remote hosts.

Like #SSENDEXT, the device responds with the prompt '> ' and waits for the data to send, operation is automatically completed when <bytestosend> have been sent.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷max	:	socket connection identifier value (max value is returned by Test command)
<bytestosend>	integer	N/A	bytes to be sent
	Value:		

	1÷1500 : number of bytes to be sent		
<remoteIP>	string	-	IP address of the remote host in dotted decimal notation ("xxx.xxx.xxx.xxx")
<remotePort>	integer	N/A	remote host port Value: 1÷65535 : host port number
<Rai>	integer	N/A	RAI (Release Assistance Indication) configuration. NOT supported. Values: 1 : Set RAI to: "No further uplink or downlink data transmission subsequent to this transmission" 2 : Set RAI to: "Only a single downlink data transmission and no further uplink data transmissions subsequent to this transmission"

RAI (Release Assistance Indication) is not supported, so it is ignored.

AT#SSENDUDPEXT=?

Test command reports the supported range of values for parameters <connId>, <bytestosend>, <remoteIP>, <remotePort> and <Rai>.

AT#SLASTCLOSURE - Detect the Cause of a Socket Disconnection

The command detects the cause of a socket disconnection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SLASTCLOSURE=<connId>

Execution command reports the socket disconnection cause.

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier

Value:

1÷max	:	socket connection identifier value (max value is returned by Test command)

Additional info:

The execution command reports the disconnection cause of the selected socket. The format of the returned message is:

#SLASTCLOSURE: <connId>,<cause>

Name	Type	Default	Description
<cause>	hex	0	socket disconnection cause.

Values:

0 : not available (socket has not yet been closed)

1 : remote host TCP connection close due to FIN/END: normal remote disconnection decided by the remote application

2 : remote host TCP connection close due to RST, all other cases in which the socket is aborted without indication from peer (for instance because peer doesn't send ack after maximum number of retransmissions/peer is no more alive). All these cases include all the "FATAL" errors after recv or send on the TCP socket (named as different from EWOULDBLOCK)

3 : socket inactivity timeout

4 : network deactivation (PDP context deactivation from network)

5 : DNS Error

- Any time socket is re-opened, last disconnection cause is reset. Command report 0 (not available).
- User closure cause (#SH) is not considered and if a user closure is performed after remote disconnection, remote disconnection cause remains saved and is not overwritten.
- If more consecutive closure causes are received, the original disconnection cause is saved.
  (For instance: if a TCP FIN is received from remote and later a TCP RST because we continue to send data, FIN cause is saved and not overwritten)
- Also in case of <closureType> (#SD) set to 255, if the socket has not yet been closed by user after the escape sequence, #SLASTCLOSURE indicates remote disconnection cause if it has been received.
- In case of UDP, cause 2 indicates abnormal (local) disconnection. Cause 3 and 4 are still possible.
  (Cause 1 is obviously never possible)
- In case of command mode connection and remote closure with subsequent inactivity timeout closure without retrieval of all available data (#SRECV or SRING mode 2), it is indicated cause 1 for both possible FIN and RST from remote.

AT#SLASTCLOSURE=?

Test command reports the supported range for parameter <connId>

AT#SS - Socket Status

Execution command reports the current sockets status.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SS[=<connId>]

Execution command with <connId> reports the current sockets status using the following message format:

#SS: <connId>,<state>,<locIP>,<locPort>,<remIP>,<remPort>

If no <connId> is given, it reports the status for all sockets in the format:

#SS: <connId>,<state>,<locIP>,<locPort>,<remIP>,<remPort><CR><LF>

[<connId>,<state>,<locIP>,<locPort>,<remIP>,<remPort><CR><LF>[...]]

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier

Value:

1÷max	:	socket connection identifier value (max value is returned by Test command)

Additional info:

▶▶ Parameters meaning.

Name	Type	Default	Description
<state>	integer	0	actual state of the socket Values: 0 : socket closed 1 : socket with an active data transfer connection 2 : socket suspended 3 : socket suspended with pending data 4 : socket listening 5 : socket with an incoming connection. Waiting for the user accept or shutdown command 6 : socket in DNS resolving process 7 : socket connecting
<locIP>	string	-	IP address associated by the context activation to the socket
<locPort>	integer	-	two meanings:

			the listening port if we put the socket in listen mode the local port for the connection if we use the socket to connect to a remote machine
<remIP>	string	-	when we are connected to a remote machine this is the remote IP address
<remPort>	string	-	it is the port we are connected to on the remote machine

AT#SS=?

Test command reports the range for <connId> parameter.

</>

* Get information about all sockets.

```
AT#SS
#SS: 1,3,91.80.90.162,61119,88.37.127.146,10510
#SS: 2,4,91.80.90.162,1000
#SS: 3,0
#SS: 4,0
#SS: 5,3,91.80.73.70,61120,88.37.127.146,10509
#SS: 6,0
OK
```

Socket 1: opened from local IP 91.80.90.162/local port 61119 to remote IP 88.37.127.146/remote port 10510 is suspended with pending data.

Socket 2: listening on local IP 91.80.90.162/local port 1000.

Socket 5: opened from local IP 91.80.73.70/local port 61120 to remote IP 88.37.127.146/remote port 10509 is suspended with pending data.

AT#SI - Socket Info

This command is used to get socket information.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SI[=<connId>]

Execution command returns information about all sockets data traffic.

Parameter:

Name	Type	Default	Description
<connId>	integer	NA	socket connection identifier. Refer to Additional info sections to have information about the use of the <connId> parameter

Value:

1÷max	:	socket connection identifier value (max value is returned by Test command)

Additional info:

- If the execution command is used with the <connId> socket identifier, it returns data traffic information on the selected socket. The format of the returned message is:

  #SI: <connId>,<sent>,<received>,<buff_in>,<ack_waiting>

Name	Type	Default	Description
<sent>	integer	-	total amount (in bytes) of data sent since the last time the socket connection identified by <connId> has been opened
<received>	integer	-	total amount (in bytes) of received data since the last time the socket connection identified by <connId> has been opened
<buff_in>	integer	-	total amount (in bytes) of data just arrived through the socket connection identified by <connId> and currently buffered, not yet read
<ack_waiting>	integer	-	total amount (in bytes) of sent and "not yet acknowledged data" since the last time the socket connection identified by <connId> has been opened. The data "not yet acknowledged" are available only for TCP connections. For UDP connections <ack_waiting> value is always 0.

If the AT#SI command is used without the <connId> socket identifier, it returns data traffic information on all sockets. For each socket, the format of the returned message is:

#SI: <connId$_n$>,<sent$_n$>,<received$_n$>,<buff_in$_n$>,<ack_waiting$_n$>

AT#SI=?

Test command reports the range of <connId> parameter.

```
* Get information about data traffic of all sockets.

  AT#SI
  #SI: 1,123,400,10,50
  #SI: 2,0,100,0,0
  #SI: 3,589,100,10,100
  #SI: 4,0,0,0,0
  #SI: 5,0,0,0,0
  ...
  OK

* Assume that sockets 1,2,3 are opened and having some
  data traffic. To get traffic information only for the socket
  <connId>=1 enter the following command:

  AT#SI=1
  #SI: 1,123,400,10,50
  OK
```

Socket <connId>=1 has 123 bytes sent, 400 bytes received, 10 bytes waiting to be read and 50 bytes waiting to be acknowledged from the remote side.

AT#ST - Socket Type

Socket Type

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#ST[=<connId>]

Set command reports the current type of the socket (TCP/UDP) and its direction (Dialer/Listener)

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷conMAX	:	socket connection identifier. conMax value is returned by test command

Additional info:

The response format is:

#ST: <connId>,<type>,<direction>

Name	Type	Default	Description
<type>	integer	N/A	socket type
	Values:		

0	:	No socket
1	:	TCP socket
2	:	UDP socket

<direction>	integer	N/A	direction of the socket

Values:

0	:	None
1	:	Dialer
2	:	Listener

Issuing #ST command without <connId> socket identifier, it returns information about type of all sockets. For each socket, the format of the returned message is:

#ST: <connId$_n$>,<type$_n$>,<direction$_n$><CR><LF>

AT#ST=?

Test command reports the range for parameter <connId>.

Examples for single socket and for all sockets

* For single socket

  AT#ST=3
  #ST: 3,2,1

  Socket 3 is an UDP dialer

* for all socket

AT#ST
#ST: 1,0,0
#ST: 2,0,0
#ST: 3,2,1
#ST: 4,2,2
#ST: 5,1,1

...

Socket 1 is closed.

Socket 2 is closed.

Socket 3 is an UDP dialer

Socket 4 is an UDP listener

Socket 5 is a TCP dialer

AT#E2SLRI - Socket Listen Ring Indicator

This command enables the Ring Indicator pin response to a Socket Listen connect.

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT#E2SLRI=[<n>]

Set command enables/disables the Ring Indicator pin response to a Socket Listen connect and, if enabled, the duration of the negative going pulse generated on receipt of connect.

Parameter:

Name	Type	Default	Description
<n>	integer	0	RI enabling

Values:

0	:	RI disabled for Socket Listen connect
50÷1150	:	RI enabled for Socket Listen connect; a negative going pulse is generated on receipt of connect and <n> is the duration in ms of this pulse

AT#E2SLRI?

Read command reports whether the Ring Indicator pin response to a Socket Listen connect is currently enabled or not, in the format:

#E2SLRI: <n>

AT#E2SLRI=?

Test command returns the allowed values for parameter <status>.

AT#FRWL - Firewall Setup

This command controls the internal firewall settings.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FRWL=[<action>[,<ip_addr>[,<net_mask>]]]

Set command controls the internal firewall settings

Parameters:

Name	Type	Default	Description
<action>	integer	0	command action

Values:

0 : remove selected chain

1 : add an ACCEPT chain

2 : remove all chains (DROP everything); <ip_addr> and <net_mask> have no meaning in this case

<ip_addr>	string	-	remote address to be added into the ACCEPT chain; it can be any valid IP address in the format: xxx.xxx.xxx.xxx
<net_mask>	string	-	mask to be applied on the <ip_addr>; it can be any valid IP address mask in the format: xxx.xxx.xxx.xxx

Additional info:

Firewall criterion

The firewall applies for incoming (listening) connections only. Its general policy is DROP, therefore all packets that are not included into an ACCEPT chain rule will be silently discarded.

When a packet comes from the IP address incoming_IP, the firewall chain rules will be scanned for matching with the following criteria:

incoming_IP & <net_mask> = <ip_addr> & <net_mask>

If criterion is matched, then the packet is accepted and the rule scan is finished; if criteria are not matched for any chain the packet is silently dropped

AT#FRWL?

Read command reports the list of all ACCEPT chain rules registered in the Firewall settings in the format:

#FRWL: <ip_addr>,<net_mask>
#FRWL: <ip_addr>,<net_mask>
...
OK

AT#FRWL=?

Test command returns the allowed values for parameter <action>.

Default AT#FRWL is:

AT#FRWL?

#FRWL: "000.000.000.000","000.000.000.000"

OK

Default white list is:

- all IP address allowed

If firewall need to be set than first delete default white list with

AT#FRWL=0,"000.000.000.000","000.000.000.000"

or with

AT#FRWL=2

and then set the desired white list with

AT#FRWL=1,<ip_addr>,<net_mask>

in order to have the desired white list

AT#FRWL?

#FRWL: "<ip_addr>","<net_mask>"

Only three zero format 000 is accepted for both <ip_addr> and <net_mask>.

Do not use single zero format 0 for both <ip_addr> and <net_mask>.

The firewall setting from AT#FRWL applies only to TCP (AT#SL scenario) not to UDP (AT#SLUDP).

</>

* Let assume we want to accept connections only from our devices which are on the IP addresses ranging from

197.158.1.1 to 197.158.255.255

We need to add the following chain to the firewall:

AT#FRWL=1,"197.158.1.1","255.255.0.0"
OK

AT#ICMP - Ping Support

Set command enables/disables the ICMP Ping support.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#ICMP=<mode>

Parameter:

Name	Type	Default	Description
<mode>	integer	1	ICMP mode selection.

Values:

0 : disable ICMP Ping support

1 : enable firewalled ICMP Ping support: the module is sending a proper ECHO_REPLY only to a subset of IP Addresses pinging it; not filtered by internal firewall.

2 : enable free ICMP Ping support; the module is sending ECHO_REPLY to every IP Address pinging it.

AT#ICMP?

Read command returns whether the ICMP Ping support is currently enabled or not, in the format:

#ICMP: <mode>

AT#ICMP=?

Test command reports the supported range of values for the <mode> parameter.

Currently internal firewall is fully open: all IP address are allowed.

AT#PING - Send PING Request

This command is used to send Ping Echo Request.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#PING=<IPaddr>[,<retryNum>[,<len>[,<timeout>[,<ttl>[,<pdpId>]]]]]

Execution command is used to send Ping Echo Request messages and to receive the corresponding Echo Reply. Before sending PING Request the PDP context must be activated by #SGACT command.

Once the single Echo Reply message is received, a string like that is displayed:

#PING: <replyId>,<Ip Address>,<replyTime>,<ttl>

Parameters:

Name	Type	Default	Description
<IPaddr>	string	-	address of the remote host, string type. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any host name to be solved with a DNS query
<retryNum>	integer	4	the number of Ping Echo Request to send
	Value:		

			1÷64 : Ping Echo Request number
<len>	integer	32	the length of Ping Echo Request message Value: 32÷1460 : Ping Echo Request length
<timeout>	integer	50	the timeout, in 100 ms units, waiting a single Echo Reply Value: 1÷600 : timeout, in 100 ms units
<ttl>	integer	128	time to live Value: 1÷255 : time to live
<pdpId>	integer	-	specifies a PDP context definition. Use the AT+CGDCONT=? test command to get the range of the supported values. Refer to additional info section for default value.

Additional info:

- <pdpId> default value.

The default value of this parameter depends on the software customization as shown in the following table.

Software customization	<pdpId> default value
Verizon	3
All other	1

Unsolicited fields:

Name	Type	Description
<replyId>	integer	Echo Reply number
<IpAddress>	string	IP address of the remote host
<replyTime>	integer	time, in 100 ms units, required to receive the response
<ttl>	integer	time to live of the Echo Reply message

When the Echo Request timeout expires (no reply received on time) the response will contain <replyTime> set to 600 and <ttl> set to 255.

To receive the corresponding Echo Reply is not required to enable separately #ICMP

AT#PING=?

Test command reports the supported range of values for the #PING command parameters.

```
AT#PING="www.telit.com"
#PING: 01,"81.201.117.177",6,50
#PING: 02,"81.201.117.177",5,50
#PING: 03,"81.201.117.177",6,50
#PING: 04,"81.201.117.177",5,50

OK
```

AT#QDNS - Query DNS

The command executes a DNS query

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#QDNS[=<hostName>]

Execution command executes a DNS query to solve the host name into an IP address. If the DNS query is successful, then the IP address will be reported in the result code as follows:

#QDNS: <hostName>,<IPaddress>

Parameter:

Name	Type	Default	Description
<hostName>	string	-	Host name string

Additional info:

- IP address in the result code

Name	Type	Default	Description
<IPaddress>	string	-	IP or IPv6 address or empty string if DNS query was unsuccessful.

- <IPaddress> address parameters can be:

  - any valid IP address in the format: "xxx.xxx.xxx.xxx"

  - any valid IPv6 address in one of the following formats depending on AT+CGPIAF setting:

  "xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx"

  or

  "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx".

- The command activates the PDP context if it was not previously activated. In this case the context is deactivated after the DNS query.

- This command requires that the authentication parameters are correctly set, and the PS network is present.

AT#QDNS=?

Test command returns the OK result code.

AT#DNS - Manual DNS Selection

This command manually set primary and secondary DNS servers.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#DNS=<cid>,<primary>,<secondary>

Set command allows to manually set primary and secondary DNS servers for a PDP context defined by +CGDCONT

Parameters:

Name	Type	Default	Description
<cid>	integer	N/A	context identifier
	Value:		
	1÷max	:	specifies a PDP context definition. The max value is returned by the test command
<primary>	string	-	manual primary DNS server address used for the specified cid; we are using this value instead of the primary DNS server coming from the network (default is "0.0.0.0")
<secondary>	string	-	manual secondary DNS server address used for the specified cid; we are using this value instead of the secondary DNS server coming from the network (default is "0.0.0.0")

- <primary> and <secondary> addresses parameters can be:

  - any valid IP address in the format: "xxx.xxx.xxx.xxx"

  - any valid IPv6 address in one of the following formats depending on AT+CGPIAF setting:

  "xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx"

  or

  "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx".

- If <primary> is "0.0.0.0" and <secondary> is not "0.0.0.0", then issuing AT#DNS=... returns an error.

- If <primary> is "0.0.0.0" we are using the primary DNS server coming from the network as consequence of a context activation.

- if <primary> is not "0.0.0.0" and <secondary> is "0.0.0.0", then we are using only the manual primary DNS server.

- The context identified by <cid> must be previously defined, otherwise issuing AT#DNS=... returns an error.

- Issuing AT#DNS=... returns an error if the context identified by <cid> has already been activated by AT commands.

AT#DNS?

Read command returns the manual DNS servers set for every defined PDP context, in the format:

[#DNS: <cid>,<primary>,<secondary>[<CR><LF>

#DNS: <cid>,<primary>,<secondary>]]

AT#DNS=?

Test command reports the supported range of values for the <cid> parameter only.

AT#NWDNS - DNS from Network

The command allows to get the primary and secondary DNS addresses for selected PDP context identifiers

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#NWDNS=[<cid>[,...]]

Execution command returns a list of primary and secondary DNS addresses for the specified PDP context identifiers. The command returns a row of information for every specified <cid> whose context has been already defined. No row is returned for a <cid> whose context has not been defined yet. The response is in the form:

#NWDNS: $<cid_n>$,$<PDNSaddress_n>$,$<SDNSaddress_n>$<CR><LF>
#NWDNS: $<cid_p>$,$<PDNSaddress_p>$,$<SDNSaddress_p>$<CR><LF>
..
#NWDNS: $<cid_q>$,$<PDNSaddress_q>$,$<SDNSaddress_q>$<CR><LF>

Parameter:

Name	Type	Default	Description
<cid>	integer	N/A	Generic context identifier. The value of max is returned by the test command

Value:

1÷max : specifies a particular PDP context definition (see +CGDCONT command)

Additional info:

Parameters response description:

Name	Type	Default	Description
<PDNSaddress>	string	-	Generic primary DNS address or assigned during PDP context activation.
<SDNSaddress>	string	-	Generic secondary DNS address or assigned during PDP context activation.

- Entering AT#NWDNS= (no <cid> specified), the DNS addresses for all defined contexts are returned.
- Issuing the command with more than 6 input parameters raises an error.
- The command returns only one row of information for every specified <cid>, even if the same <cid> is present more than once.

AT#NWDNS=?

Test command returns a list of defined <cid>s.

AT#NTP - Calculate and Update Date and Time with NTP

The command handles the date and time update using NTP protocol.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#NTP=<NTPAddr>,<NTPPort>[,<updModClock>[,<timeout>[,<timeZone>]]]

Execution command permits to calculate and update date and time through NTP protocol sending a request to a NTP server. The command returns an intermediate response having the following format:

#NTP: <ntpTime>[,<ntpLI>,<ntpStratum>]<CR><LF>OK<CR><LF>

or

<CR><LF>ERROR<CR><LF>

The #NTP intermediate response parameters are described in Additional info section.

Parameters:

Name	Type	Default	Description
<NTPAddr>	string	address of the NTP server. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" or "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx"	

		- any host name to be solved with a DNS query	
<NTPPort>	integer	N/A	NTP server port to contact

Value:

1÷65535	:	port

<updModClock>	integer	0	update mode

Values:

0	:	no update module clock
1	:	update module clock

<timeout>	integer	10	waiting timeout for server response in seconds

Value:

1÷10	:	waiting timeout for server response in seconds

<timeZone>	string	0	Time Zone: indicates the difference, expressed in quarter of an hour, between the local time and GMT.

Value:

- 47 ÷ + 48 : Time Zone

Additional info:

▶▶ Intermediate response parameter.

Name	Type	Default	Description
<ntpTime>	string	-	current time as quoted string in the format: "yy/MM/dd,hh:mm:ss±zz"
<ntpLI>	integer	-	Leap Indicator

Values:

0 : no warning

1	:	last minute of the day has 61 seconds
2	:	last minute of the day has 59 seconds
3	:	unknown

<ntpStratum> integer - Stratum

Values:

0	:	unspecified or invalid
1	:	primary server (e.g. equipped with GPS receiver)
2÷15	:	secondary server (via NTP)
6	:	unsynchronized
17÷255	:	reserved

In case of error, the command returns the following numeric or verbose error message:

+CME ERROR: <err>

Parameter <err> can be either numeric or verbose (see +CMEE).

The available values are reported in the table:

Numeric Format	Verbose Format
30001	sntp - not enough resources
30002	sntp - net timeout
30003	sntp - socket error
30005	sntp - host name resolve error
30006	sntp - invalid parameter
30008	sntp - send timeout
30009	sntp - send error
30011	sntp - response timeout
30015	sntp - context not opened
30020	sntp - generic error
30030	sntp - server authentication fail
30031	sntp - server keys file missing
30200	sntp - cannot set time

See AT#NTPCFG for context id, authentication and Stratum parameters.

By default, AT#NTP uses the PDP context id <cid>=1, see AT+CGDCONT command.

The Time Zone is applied directly in the Date and Time received by the NTP Server, that is, by definition, GMT+0.

AT#SGACT must be called to activate the PDP context before calling AT#NTP command.

AT#NTP=?

Test command returns the maximum length for <NTPAddr> string, and supported range of values of parameters: <NTPPort>, <updModClock>, <timeout> and <timeZone>.

```
AT#NTP="ntp1.inrim.it",123,1,2,4
#NTP: 12/01/27,14:42:38+04
OK
AT+CCLK?
+CCLK: "12/01/27,14:42:39+04"
OK
```

AT#NTPCFG - Configure NTP Parameters

This set command allows to configure additional parameters to be used for NTP operations.

[1] Standard RFC5905

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#NTPCFG=<cid>[,<authType>,<KeyId>,<keysFilePath>[,<stratumEnable>]]

Parameters:

Name	Type	Default	Description
<cid>	integer	-	context id used to perform NTP operations. Refer to test command to know the supported range of values.
<authType>	integer	0	authentication type to be used with NTP server

Values:

0 : do not authenticate server

1 : Not supported

use symmetric key to authenticate server (see standard [1]), using <KeyId> and <KeysFilePath> to retrieve the required parameters

<KeyId>	integer	-	if <authType> is 1, it provides the key Id to authenticate the server if <authType> is 0, use value 0
<keysFilePath>	string	-	if <authType> is 1, it provides the absolute path of the file where the server keys can be found. Only MD5 keys are supported. if <authType> is 0, use value 0
<stratumEnable>	integer	0	enable/disable <ntpLI> and <ntpStratum> parameters in AT#NTP response

Values:

0 : disable <ntpLI> and <ntpStratum> parameters in AT#NTP response

1 : enable <ntpLI> and <ntpStratum> parameters in AT#NTP response

AT#NTPCFG?

Read command reports the parameters current values in the format:

If <stratumEnable> is 0

#NTPCFG:<cid>,<authType>

else

#NTPCFG:<cid>,<authType>,<stratumEnable>

AT#NTPCFG=?

Test command reports the available range of <cid>, <authType>, <KeyId> and <stratumEnable> parameters values.

Set <cid>=3

```
AT#NTPCFG=3
OK
```

FTPEasy

AT#FTPAPP - FTP Append

This command is used to append data to an already existing file via FTP during an FTP session.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPAPP=<fileName>[,<connMode>]

Set command, issued during an FTP connection, opens a data connection and append data to existing <fileName> file.

If the data connection succeeds, a CONNECT indication is sent, afterward a NO CARRIER indication is sent when the socket is closed.

Parameters:

Name	Type	Default	Description
<fileName>	string	-	the file name
<connMode>	integer	0	the connection mode

Values:

0	:	online mode
1	:	command mode

- If <connMode> is set to 1, the data connection is opened, the device remains in command mode and the OK result code is displayed (instead of CONNECT).
- Use the escape sequence +++ to close the data connection.
- The command causes an ERROR result code if no FTP connection has been opened yet.

AT#FTPAPP=?

Test command reports the maximum length of <fileName> and the supported range of values of <connMode>. The format is:

#FTPAPP: <length>, (list of supported <connMode>s)

Additional info:

Parameter meaning.

Name	Type	Default	Description
<length>	integer	-	is the maximum length of <fileName>

AT#FTPAPPEXT - FTP Append Extended

The command sends data on a FTP data port while the module is in command mode.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPAPPEXT=<bytesToSend>[,<eof>]

Execution command sends data on a FTP data port while the module is in command mode.

FTP data port must be previously opened by #FTPPUT (or #FTPAPP) with <connMode> parameter set to command mode connection.

After command line is terminated with <CR>, the module responds sending a four characters sequence prompt, and waits for the specified number of bytes:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

When <bytesToSend> bytes have been sent, operation is automatically completed. If (all or part of the) data are successfully sent, then the response is:

#FTPAPPEXT: <sentBytes>
OK

If data sending fails for some reason, an error code is reported.

Parameters:

Name	Type	Default	Description
<bytesToSend>	integer	N/A	number of bytes to be sent
	Value:		
	1÷1500	:	number of bytes
<eof>	integer	0	data port closure
	Values:		

0 : normal sending of data chunk

1 : close data port after sending data chunk

Additional info:

Parameters:

Name	Type	Default	Description
<sentBytes>	integer	N/A	the number of sent bytes

Value:

1÷1500 : number of bytes

<sentBytes> could be less than <bytesToSend>.

AT#FTPAPPEXT=?

Test command reports the supported values of parameters <bytesToSend> and <eof>.

```
AT#FTPOPEN="IP",username,password
OK

AT#FTPPUT=<filename>,1
OK
```

the second param (1) means that we open the connection in command mode

Here data socket will stay opened, but interface will be available (command mode)

AT#FTPAPPEXT=Size
>binary data
#FTPAPPEXT: <sentBytes>
OK

write here the binary data. As soon Size bytes are written, data are sent and OK is returned

Last #FTPAPPEXT will close the data socket, because second (optional) parameter has this meaning:

AT#FTPAPPEXT=Size,1
>binary data
#FTPAPPEXT: <sentBytes>
OK

write here the binary data. As soon Size bytes are written, data are sent and OK is returned and the data socket is closed.

If the user has to reopen the data port to send another (or append to the same) file, they can restart with #FTPPUT (or #FTPAPP).
Then #FTPAPPEXT to send the data chunks on the reopened data port.

If, while sending the chunks, the data port is closed from remote, user will be aware of it because #FTPAPPEXT will indicate ERROR and cause (available if previously issued the command AT+CMEE=2) will indicate that socket has been closed.

Also in this case obviously, data port will have to be reopened with #FTPPUT and the related commands.

AT#FTPCLOSE - FTP Close Command

The command purpose is to close the previously open FTP connection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPCLOSE

Execution command closes an FTP connection.

AT#FTPCLOSE=?

Test command returns the OK result code.

AT#FTPCMD - FTP Send Commands

This command sends an FTP command or a combination of two commands to an FTP server.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#FTPCMD=<command1>,<respCode1>[,<command2>,<respCode2>]

Execution command, issued during an FTP connection, sends <command1> to an FTP server and expects <respCode1> reply. Optionally, after the execution of the first command, a second command can be sent (<command2> and <respCode2>).

Parameters:

Name	Type	Default	Description
<command1>	string	-	name of the 1st command (maximum length 200 characters)
<respCode1>	integer	N/A	Expected response code from FTP server after <command1> execution

Values:

1	:	Expected code 1XY
2	:	Expected code 2XY
3	:	Expected code 3XY
4	:	Expected code 4XY
5	:	Expected code 5XY

<command2>	string	-	name of the 2nd command (maximum length 200 characters)

<respCode2>	integer	N/A	Expected response code from FTP server after <command2> execution

Values:

1	:	Expected code 1XY
2	:	Expected code 2XY
3	:	Expected code 3XY
4	:	Expected code 4XY
5	:	Expected code 5XY

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

To read unexpected ERROR result codes returned from the FTP server, use AT#FTPMSG command.

AT#FTPCMD=?

Test command reports the maximum length of <command1> and <command2> and the supported range of values of <respCode1> and <respCode2>.

Additional info:

The format is:

#FTPCMD: <length>, (list of supported <respCode1>s), <length>, (list of supported <respCode2>s)

Name	Type	Default	Description

<length>	integer	-	maximum length of <command1> or <command2>

Examples of command executions.

* Rename file on remote server:

  at#ftpcmd="RNFR test.txt",3,"RNTO test2.txt",2

  350 File or directory exists, ready for destination name

  250 Rename successful

  OK

  ...

  ...

  Create directory on remote server:

  at#ftpcmd="MKD testDir",2

  257 "/testDir" - Directory successfully created

  OK

  ...

  ...

  Remove Directory on remote server:

  at#ftpcmd="RMD testDir",2

  250 RMD command successful

  OK

  ...

  ...

AT#FTPCWD - FTP Change Working Directory

Command to change the working directory on FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPCWD=[<dirname>]

Execution command, issued during an FTP connection, changes the working directory on FTP server.

Parameter:

Name	Type	Default	Description
<dirname>	string	-	Name of the new working directory.

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPCWD=?

Test command returns the OK result code.

AT#FTPDELE - FTP Delete

This command, issued during a FTP connection, allows to delete a file from the remote working directory.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPDELE=[<filename>]

Execution command, issued during a FTP connection, deletes a file from the remote working directory.

Parameter:

Name	Type	Default	Description
<filename>	string	-	Name of the file that must be deleted

This command returns an ERROR result code if no FTP connection has been opened yet.

This command returns an ERROR result code in case of delayed server response.

If this is the case, the #FTPMSG response is temporarily empty; a later check of the #FTPMSG response will show the server response.

AT#FTPDELE=?

Test command returns OK result code.

AT#FTPFSIZE - Get File Size from FTP Server

This command returns the size of a file located on a FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPFSIZE=<filename>

Execution command, issued during an FTP connection, permits to get the size of a file located on a FTP server. The response format is:

#FTPFSIZE: <size>

Parameter:

Name	Type	Default	Description
<filename>	string	-	the name of the file that you want to know the size

Additional info:

Parameter:

Name	Type	Default	Description
<size>	integer	-	dimension in bytes of the file located on the FTP server

AT#FTPTYPE=0 command must be issued before #FTPFSIZE command, to set file transfer type to binary mode.

AT#FTPFSIZE=?

Test command returns OK result code.

AT#FTPGET - FTP Get Command

This command executes the FTP Get function during an FTP connection.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPGET=[<filename>]

Execution command opens a data connection and starts getting a file from the FTP server. If the data connection succeeds a CONNECT indication is sent and the file is received on the serial port.

Parameter:

Name	Type	Default	Description
<filename>	string	-	file name to get from server.

The command causes an ERROR result code to be returned in case no FTP connection has been opened yet.

Command closure should always be handled by application. To avoid download stall situations a timeout should be implemented by the application.

AT#FTPGET=?

Test command returns the OK result code.

AT#FTPGETPKT - FTP Get in Command Mode

FTP gets in command mode.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPGETPKT=<fileName>[,<viewMode>]

Execution command, issued during an FTP connection, opens a data connection and starts getting a file from the FTP server while remaining in command mode.

The data port is opened, we remain in command mode and we see the result code OK.
Retrieval from FTP server of <fileName> is started, but data are only buffered in the module.
It is possible to read data afterwards issuing #FTPRECV command.

Parameters:

Name	Type	Default	Description
<fileName>	string	-	file name. Maximum length: 200 characters.
<viewMode>	integer	0	choose the view mode Values: 0 : text format 1 : hexadecimal format

The command causes an ERROR result code to be returned in case no FTP connection has been opened yet.

Command closure should always be handled by application. To avoid download stall situations a timeout should be implemented by the application.

AT#FTPGETPKT?

Read command reports current download state for <fileName> with <viewMode> chosen, in the format:

#FTPGETPKT: <remotefile>,<viewMode>,<eof>

Additional info:

The following parameter signals the state of the file transmission.

Name	Type	Default	Description
<eof>	integer	N/A	End of file

Values:

0 : file currently being transferred

1 : complete file has been transferred to FTP client

AT#FTPGETPKT=?

Test command returns OK result code.

AT#FTPLIST - FTP List

This command is used during a FTP connection.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPLIST[=[<name]]

Execution command, issued during an FTP connection, opens a data connection and starts getting from the server the list of contents of the specified directory or the properties of the specified file

Parameter:

Name	Type	Default	Description
<name>	string	-	is the name of the directory or file

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

Issuing AT#FTPLIST<CR> opens a data connection and starts getting from the server the list of contents of the working directory.

AT#FTPLIST=?

Test command returns the OK result code.

AT#FTPMSG - FTP Read Message

This command returns the last response received from the FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPMSG

Execution command returns the last response received from the server during an FTP connection.

AT#FTPMSG=?

Test command returns the OK result code.

AT#FTPOPEN - FTP Connection Opening

This execution command opens an FTP connection toward the FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPOPEN=[<server:port>,<username>,<password>,<mode>[,<cid>]]

Parameters:

Name	Type	Default	Description
<server:port>	string	-	address and port of FTP server (factory default port 21), in the format: "ipv4" / "ipv4:port" "ipv6" / "[ipv6]" / "[ipv6]:port" "dynamic_name" / "dynamic_name:port"
<username>	string	-	authentication user identification for FTP
<password>	string	-	authentication password for FTP
<mode>	integer	0	active or passive mode Values: 0 : active mode 1 : passive mode
<cid>	string	-	PDP context identifier

- In FTP Open case, the solution dependency limits the maximum time out to 1200 (120 seconds). The FTPTO value that exceed 1200 is considered as 1200.
- Before opening FTP connection the PDP context must been activated with #SGACT.

AT#FTPOPEN=?

Test command returns the OK result code

AT#FTPPUT - FTP Send File

This command sends a file to the FTP server.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#FTPPUT=[<filename>[,<connMode>]]

Execution command, issued during an FTP connection, opens a data connection and starts sending <filename> file to the FTP server.

Parameters:

Name	Type	Default	Description
<filename>	string	-	name of the file (maximum length 200 characters)
<connMode>	integer	0	select online or command mode: If online mode is selected (default) and the data connection succeeds, a CONNECT indication is sent; afterward a NO CARRIER indication is sent when the socket is closed. If command mode is selected and the data connection succeeds, we remain in command mode and we see the result code OK (instead of CONNECT).

Values:

0	:	online mode
1	:	command mode

- Use the escape sequence +++ to close the data connection.
- The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPPUT=?

Test command reports the maximum length of <filename> and the supported range of values of <connMode>.

Additional info:

- The format is:

  #FTPPUT: <length>, (list of supported <connMode>s)

Name	Type	Default	Description
<length>	integer	-	maximum length of <filename>

AT#FTPPWD - FTP Print Working Directory

This command, issued during an FTP connection, shows the current working directory on FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPPWD

Execution command, issued during an FTP connection, shows the current working directory on FTP server.

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPPWD=?

Test command returns the OK result code.

AT#FTPRECV - Receive Data in Command Mode

The command permits the user to read a given amount of data already transferred via FTP from a remote file.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#FTPRECV=<blockSize>

Execution command permits the user to transfer at most <blocksize> bytes of remote file, provided that retrieving from the FTP server has been started with a previous #FTPGETPKT command, onto the serial port.

This number is limited to the current number of bytes of the remote file which have been transferred from the FTP server.

Parameter:

Name	Type	Default	Description
<blockSize>	integer	N/A	maximum number of bytes to read

Value:

1÷3000	:	maximum number of bytes to read

It is necessary to have previously opened FTP data port and started download and buffering of remote file through #FTPGETPKT command.

- Issuing #FTPRECV when there is no FTP data port opened raises an error.

- Data port will stay opened if socket is temporary waiting to receive data (#FTPRECV returns 0 and #FTPGETPTK gives an EOF 0 indication).

AT#FTPRECV?

Read command reports the number of bytes currently transferred from FTP server in the format:

#FTPRECV: <available>

Additional info:

- Parameter:

Name	Type	Default	Description
<available>	integer	-	number of transferred bytes and available for reading

AT#FTPRECV=?

Test command returns the supported values for parameter <blocksize>.

```
AT#FTPRECV?
#FTPRECV: 3000
OK
```

Read required part of the buffered data:

```
AT#FTPRECV=400
#FTPRECV:400

Text row number 1 * 111111111111111111111111111 *
Text row number 2 * 222222222222222222222222222 *
Text row number 3 * 333333333333333333333333333 *
Text row number 4 * 444444444444444444444444444 *
Text row number 5 * 555555555555555555555555555 *
Text row number 6 * 666666666666666666666666666 *
Text row number 7 * 777777777777777777777777777 *
Text row number 8 * 88888888888888888888

OK
```

```
AT#FTPRECV=200
#FTPRECV:200

88888 *
Text row number 9 * 999999999999999999999999999 *
Text row number 10 * AAAAAAAAAAAAAAAAAAAAAAAAAAA*
Text row number 11 * BBBBBBBBBBBBBBBBBBBBBBBBBBB *
Text row number 12 * CCCCCCCCCCCCCCCC

OK
```

To check when you have received complete file it is possible to use #FTPGETPKT read command:

```
AT#FTPGETPKT?
#FTPGETPKT:sample.txt,0,1
OK
```

(you will get <eof> set to 1)

AT#FTPREST - Set Restart Position for FTP GET

Set command sets the restart position for successive #FTPGET (or #FTPGETPKT) command. It permits to restart a previously interrupted FTP download from the selected position in byte.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#FTPREST=[<restartPosition>]

Parameter:

Name	Type	Default	Description
<restartPosition>	integer	-	position in byte of restarting for successive #FTPGET (or #FTPGETPKT)

- It is necessary to issue #FTPTYPE=0 before successive #FTPGET (or #FTPGETPKT command) to set binary file transfer type.
- Setting <restartPosition> takes effect on successive FTP download. After successive successfully initiated #FTPGET (or #FTPGETPKT) command <restartPosition> is automatically reset.
- Value set for <restartPosition> takes effect on next data transfer (data port opened by #FTPGET or #FTPGETPKT). Then <restartPosition> value is automatically assigned to 0 for next download.

AT#FTPREST?

Read command returns the current <restartPosition>:

#FTPREST:<restartPosition>

AT#FTPREST=?

Test command returns the OK result code.

AT#FTPTO - FTP Time Out

Set the FTP time out.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPTO=[<tout>]

Set command sets the time out used when opening either the FTP control channel or the FTP traffic channel.

Parameter:

Name	Type	Default	Description
<tout>	integer	100	time out in 100 milliseconds units

Value:

100÷5000	:	hundreds of milliseconds

AT#FTPTO?

Read command returns the current FTP operations time out in the format:

#FTPTO: <tout>

AT#FTPTO=?

Test command returns the supported values of parameter <tout>.

AT#FTPTYPE - FTP Type

This command sets the FTP file transfer type.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPTYPE=[<type>]

Set command, issued during a FTP connection, sets the file transfer type.

Parameter:

Name	Type	Default	Description
<type>	integer	N/A	file transfer type

Values:

0	:	binary
1	:	ASCII

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPTYPE?

Read command returns the current file transfer type, in the format:

#FTPTYPE: <type>

AT#FTPTYPE=?

Test command returns the range of available values for parameter <type>:

#FTPTYPE: (0,1)

AT#FTPCFG - FTP Configuration

This command sets the time-out used when opening either the FTP control channel or the FTP traffic channel.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPCFG=<tout>,<IPPignoring>[,<FTPSEn>[,<FTPext>]]

Parameters:

Name	Type	Default	Description
<tout>	integer	100	time out in 100 milliseconds units
	Value:		
	100÷5000	:	hundreds of ms
<IPPignoring>	integer	0	enable or disable IP private ignoring
	Values:		
	0	:	No IP Private ignoring. During a FTP passive mode connection client uses the IP address received from server, even if it is a private IPV4 address.

1 : IP Private ignoring enabled. During a FTP passive mode connection if the server sends a private IPV4 address the client doesn't consider this and connects with server using the IP address used in #FTPOPEN.

<FTPSEn>	integer	0	disable FTPS security

Values:

0 : disable FTPS security: all FTP commands will perform plain FTP connections

1 : enable FTPS security (FTPS explicit)

2 : enable FTPS security (FTPS implicit)

<FTPext>	integer	1	Use PORT/PASV commands

Values:

0 : always use EPRT and EPSV commands

1 : if both module and server ipv4 use PORT and PASV commands. Option added to pass-through firewall that is unaware of the extended FTP commands for #FTPPUT, #FTPLIST, #FTPAPP, #FTPGET

If parameter <tout> is omitted the behavior of set command is the same as read command.

AT#FTPCFG?

Read command reports the currently selected parameters in the format:

#FTPCFG: <tout>,<IPPignoring>,<FTPSEn>,<FTPext>

AT#FTPCFG=?

Test command reports the supported range of values for parameter(s): <tout>, <IPPignoring>, <FTPSEn>, and <FTPext>.

HTTP

AT#HTTPCFG - Configure HTTP Parameters

This command sets the parameters needed to the HTTP connection.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#HTTPCFG=<prof_id>[,<server_address>[,<server_port>[,<auth_type>[,<username>[,<password>[,<ssl_enabled>[,<timeout>[,<cid>[,<pkt_size>[,<unused1>[,<unused2>]]]]]]]]]]]

Parameters:

Name	Type	Default	Description
<prof_id>	integer	N/A	select the profile identifier. Value: 0÷2 : Profile identifier
<server_address>	string	-	IP address of the HTTP server. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any valid IPv6 address in one of the following format: "xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx. xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx" "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx" - any host name to be solved with a DNS query. Default: "" for first and second profile "m2mlocate.telit.com" for third profile
<server_port>	integer	N/A	select TCP remote port.

for first and second profile 80, by default

for third profile 9978, by default

Value:

1÷65535	:	TCP remote port of the HTTP server to connect to

<auth_type>	integer	0	select HTTP authentication type.

Values:

0	:	no authentication
1	:	basic authentication

<username>	string	-	configure authentication user identification string for HTTP.
<password>	string	-	configure authentication password string for HTTP.
<ssl_enabled>	integer	0	enable/disable SSL encryption.

Values:

0	:	SSL encryption disabled
1	:	SSL encryption enabled

<timeout>	integer	120	timeout, data from HTTPS server.

Value:

	1÷65535	:	time interval in seconds to wait for receiving data from HTTP server.

<cid>	integer	-	PDP context identifier. See +CGDCONT command.
<pkt_size>	integer	0	size for data sending (#HTTPSND) or receiving (#HTTPRCV).

Values:

0	:	select automatically value 300
1÷1500	:	send or receive size in bytes

<unused1>	integer	-	parameter for future use. Must be set to 0.
<unused2>	integer	-	parameter for future use. Must be set to 0.

A special form of the set command, AT#HTTPCFG=<prof_id>, causes the values for profile number <prof_id> to reset to default values.

Only one profile can use the SSL encryption.

AT#HTTPCFG?

Read command returns the current settings for each defined profile in the format:

#HTTPCFG:<prof_id>,<server_address>,<server_port>,<auth_type>,<username>,<password>,<ssl_enabled>,<timeout>,<cid>,<pkt_size>,0,0<CR><LF>[<CR><LF>

#HTTPCFG:<prof_id>,<server_address>,<server_port>,<auth_type>,<username>,<password>,<ssl_enabled>,<timeout>,<cid>,<pkt_size>,0,0]<CR><LF>[...]]

AT#HTTPCFG=?

Test command returns the supported range of parameters <prof_id>, <server_port>, <auth_type>, <ssl_enabled>, <timeout>, <cid>,<pkt_size>,<unused1>,<unused2>, and the maximum length of <server_address>, <username> and <password> parameters in the format:

#HTTPCFG:(list of supported <prof_id>s),<s_length>,(list of supported <server_port>s), (list of supported <auth_type>s),<u_length>,<p_length>,(list of supported <ssl_enabled>s),(list of supported <timeout>s),(list of supported <cid>s),(list of supported <pkt_size>s),(0),(0)

Additional info:

▶▶ Meaning of the <..._length> parameters:

Name	Type	Default	Description
<s_length>	integer	-	maximum length of parameter <server_address>
<u_length>	integer	-	maximum length of parameter <username>.
<p_length>	integer	-	maximum length of parameter <password>.

AT#HTTPQRY - Send HTTP GET, HEAD or DELETE Request

This command performs a GET, HEAD or DELETE request to HTTP server.

Standard RFC 2616

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#HTTPQRY=<prof_id>,<command>,<resource>[,<extra_header_line>]

Parameters:

Name	Type	Default	Description
<prof_id>	integer	N/A	profile identifier
	Value:		
	0÷2	:	identifier values
<command>	integer	0	identifies command requested to HTTP server
	Values:		
	0	:	GET
	1	:	HEAD
	2	:	DELETE
<resource>	string	-	is the HTTP resource (URI), object of the request. It must

			always start with "/" (slash character)
<extra_header_line>	string	-	is the optional HTTP header line

Additional info:

- When the HTTP server answer is received, then the following URC is put on the serial port:

  #HTTPRING: <prof_id>,<http_status_code>,<content_type>,<data_size>

  If there are no data from server or the server does not answer within the time interval specified in <timeout> parameter of #HTTPCFG command, then the URC #HTTPRING <http_status_code> parameter has value 0.

Name	Type	Default	Description
<http_status_code>	string	-	is the status code, as received from the server, see RFC 2616
<content_type>	string	-	reports the "Content-Type" header line, as received from the server, see RFC 2616. This field is empty if <http_status_code> has a value out of range (200-206).

<data_size>	string	-	is the byte amount of data received from the server. If the server does not report the "Content-Length:" header line, or <http_status_code> has a value out of range (200-206), the parameter value is 0.

To set more than one HTTP header line in parameter <extra_header_line>, they have to be separated by ">>"

Example:
AT#HTTPQRY=0,0, "myURI","Content-Type: xyz>>Authorization: something"

Where:
"myURI" must always start with "/" (slash).

Here an example where #HTTPQRY requests for the homepage of www.myhost.com:

AT#HTTPCFG=0,"www.myhost.com",80
OK
AT#HTTPQRY=0,0,"/"
OK

If sending ends successfully, the response is OK; otherwise an error code is reported.

The HTTP request header sent with #HTTPQRY always contains the "Connection: close" line, and it cannot be removed.

AT#HTTPQRY=?

Test command reports the supported range of values for the parameters <prof_id> and <command> and the maximum length of <resource> parameter in the format:

#HTTPQRY:(list of supported <prof_id>s),(list of supported <command>s),<r_length>, <m_length>

Additional info:

▶▶ Meaning of <..._length> parameters:

Name	Type	Default	Description
<r_length>	integer	-	maximum length of parameter <resource>.
<m_length>	integer	-	maximum length of parameter <extra_header_line>.

AT#HTTPSND - Send HTTP POST or PUT request

This command performs a POST or PUT request to HTTP server and starts sending data to the server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#HTTPSND=<profId>,<command>,<resource>,<dataLen>[,<postParam>[,<extraHeaderLine>]]

After entering AT#HTTPSND=... command, and terminated the command line with <CR>, the module returns the following three characters sequence prompt, and waits for data to send:

<greater_than><greater_than><greater_than> (see IRA character set: 62, 62, 62)

Now, data can be entered from TE, sized <dataLen> bytes. If sending ends successfully the response will be OK, otherwise an error code will be reported.

When the HTTP server answer is received, the URC will be available on the serial port with the following format:

#HTTPRING: <profId>,<httpStatusCode>,<contentType>,<dataSize>

The URC parameters are described in Unsolicited Field section.

Parameters:

Name	Type	Default	Description
<profId>	integer	N/A	profile identifier
	Value:		
	0÷2	:	profile identifier
<command>	integer	N/A	command requested to HTTP server
	Values:		

	0 : POST command 1 : PUT command		
<resource>	string	-	HTTP resource (uri), object of the request
<dataLen>	integer	-	data length to send in bytes
<postParam>	string	N/A	HTTP Content-type identifier, used only for POST command, optionally followed by colon character (:) and a string that extends with sub-types the identifier
	Values: 0[:extension] : "application/x-www-form-urlencoded " with optional extension 1[:extension] : "text/plain" with optional extension 2[:extension] : "application/octet-stream" with optional extension 3[:extension] : "multipart/form-data" with optional extension other : free string corresponding to other content type and possible sub-types		
<extraHeaderLine>	string	-	optional HTTP header line

Unsolicited fields:

Name	Type	Description
<httpStatusCode>	integer	status code, as received from the server (see RFC 2616)
<contentType>	string	"Content-Type" header line, as received from the server (see RFC 2616). This field is empty if <httpStatusCode> has a value out of range (200-206)
<dataSize>	integer	byte amount of data received from the server. If the server does not report the "Content-Length:" header line or <httpStatusCode> has a value out of range (200-206), the parameter value is 0.

The HTTP request header sent with #HTTPSND always contains the "Connection: close" line, and it cannot be removed.

If there are no data from server or the server doesn't answer within the time interval specified in <timeout> parameter of #HTTPCFG command, then the URC #HTTPRING <httpStatusCode> parameter will have value 0.

AT#HTTPSND=?

Test command returns the supported range of parameters <profId>, <command> and <dataLen> and the maximum length of <resource>, <postParam> and <extraHeaderLine> string parameters in the format:

HTTPSND: (list of supported <profId>s),(list of supported <command>s), <rLength>, (list of supported <dataLen>s),<pLength>,<mLength>

Additional info:

<...Length> parameters meaning:

Name	Type	Default	Description
<rLength>	integer	-	maximum length of parameter <resource>
<pLength>	integer	-	maximum length of parameter <postParam>
<mLength>	integer	-	maximum length of parameter <extraHeaderLine>

POST commands examples.

* Post 100 byte without "Content-type" header

  AT#HTTPSND=0,0,"/",100

  >>>

* Post 100 byte with "application/x-www-form-urlencoded"

  AT#HTTPSND=0,0,"/",100,0

  >>>

* Post 100 byte with "multipart/form-data" and extension

  AT#HTTPSND=0,0,"/",100,"3:boundary=----FormBoundary"

  >>>

AT#HTTPRCV - Receive HTTP Server Data

This command permits the user to read data from HTTP server in response to a previous HTTP module request.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#HTTPRCV=<profId>[,<maxByte>]

The module is notified of these data by the #HTTPRING URC.

The device shall prompt a three-character sequence followed by the data:

<less_than><less_than><less_than> (see IRA character set: 60,60,60)

If reading ends successfully, the response is OK; otherwise an error code is reported.

Parameters:

Name	Type	Default	Description
<profId>	integer	N/A	profile identifier
	Value:		
	0÷2	:	profile identifier
<maxByte>	integer	0	max number of bytes to read at a time
	Value:		
	0, 64÷1500	:	0 means infinite size.

If <maxByte> is unspecified, server data will be transferred all in once.

If the data are not present or the #HTTPRING <httpStatusCode> parameter has value 0, an error code is reported.

AT#HTTPRCV=?

Test command reports the supported range of values for <profId> and <maxByte> parameters in the format:

HTTPRCV: (list of supported <profId>s,<maxByte>)

SSL

AT#SSLCFG - Configure General Parameters of a SSL Socket

This command configures SSL connection parameters.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SSLCFG=<SSId>,<cid>,<pktSz>,<maxTo>,<defTo>,<txTo>[,<SSLSRingMode>[,<noCarrierMode>[,<skipHostMismatch>[,<equalizeTX> [,<connTo>[,<Unused1>]]]]]]

Set command allows configuring SSL connection parameters.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<cid>	integer	-	PDP context identifier, see +CGDCONT command.
<pktSz>	integer	0	packet size to be used by the SSL/TCP/IP stack for data sending
	Values:		
	0	:	select automatically default value (300)

	1÷1500 : number of bytes		
<maxTo>	integer	90	exchange timeout or socket inactivity timeout. In online mode, if there is no data exchange within this timeout period, the connection is closed.
	Values:		
	0 : no timeout		
	1÷65535 : timeout in seconds		
<defTo>	integer	100	timeout that will be used by default whenever the corresponding parameter of each command is not set.
	Value:		
	10÷5000 : timeout in tenth of seconds		
<txTo>	integer	50	data sending timeout. In online mode, after this period data are sent also if they're less than max packet size.
	Values:		
	0 : no timeout		

1÷255 : timeout value in hundreds of milliseconds

<SSLSRingMode>	integer	0	SSLRING unsolicited mode

Values:

0 : disable

1 : enable decoded data format, see Additional info section

2 : enable chunk of data format, see Additional info section

<noCarrierMode>	integer	0	selects the NO CARRIER indication format when the secure socket is closed.

Values:

0 : NO CARRIER without additional information

1 : NO CARRIER:SSL,<SSId> See Additional info section

2 : NO CARRIER:SSL,<SSId>,<cause> See Additional info section

<skipHostMismatch>	integer	1	enables/disables Host Mismatch alert

Values:

0 : same behavior as value 1 (not supported "do not ignore")

1 : ignore

<equalizeTX>	integer	0	enable/disable equalized data transmission

Values:

0 : disable equalized data transmission

1 : enable equalized data transmission, i.e.: after successful data write for sending on SSL connection, next check for available <pktSz> data is triggered after <txTo>

<connTo>	integer	0	Overall connection timeout in tenths of seconds. If the module cannot establish a connection to the remote within this timeout period, an error is raised. An error can also be raised earlier than <connTo> expiry caused by lower layer TCP/IP retry/timeout mechanism

Values:

0 : no timeout

	10÷1200	:	Not supported (timeout in tenth of seconds)
<Unused1>	integer	-	parameter for future use. Must be set to 0.

Additional info:

- When <SSLSRingMode>= 1 a new unsolicited is sent whenever the amount of data ready to be read changes, the unsolicited format is:

  SSLSRING: <SSId>,<recData>

  <recData> is described in the Unsolicited fields section.

- When <SSLSRingMode>= 2 the unsolicited format is:

  SSLSRING: <SSId>,<dataLen>,<data>

  <dataLen> and <data> are described in the Unsolicited fields section.

- The NO CARRIER indication can be followed by additional information:

  NO CARRIER:SSL,<SSId>

  NO CARRIER:SSL,<SSId>,<cause>

  The fixed "SSL" string distinguishes secure sockets from TCP sockets.

Name	Type	Default	Description
<cause>	integer	N/A	cause of the secure socket closure

Values:

0 : not available (secure socket has not yet been closed)

1 : the remote TCP connection has been closed (RST, or any fatal error in send/recv are all included within this case)

2 : socket inactivity timeout

3 : network deactivation (PDP context deactivation from network)

4 : SSL "Close Notify Alert" message has been received

5 : the remote TCP connection has been closed (FIN) after all data have been retrieved from socket

6 : closure due to any other SSL alert different from the previous ones

Unsolicited fields:

Name	Type	Description
<recData>	integer	number of bytes available to be read from the TLS/SSL record that is currently being processed (obviously: already decoded from TLS/SSL record) plus the

		number of bytes available to be read in the TCP/IP stack.
<dataLen>	integer	length of the current chunk of data in bytes
<data>	hex	received data in ASCII format

If secure socket is not enabled using #SSLEN only test requests can be made.

This version of command reordered the parameters of the command, so this is not interchangeable with previous version

AT#SSLCFG?

Read command reports the parameters current values in the format:

#SSLCFG:
<SSId>,<cid>,<pktSz>,<maxTo>,<defTo>,<txTo>,<SSLSRingMode>,<noCarrierMode>,
<skipHostMismatch>,<equalizeTX>,0,0

AT#SSLCFG=?

Test command reports the ranges of all parameters values.

AT#SSLSECCFG - Configure Security Parameters of a SSL Socket

This command allows configuring SSL connection parameters.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECCFG=<SSId>,<CipherSuite>,<auth_mode>[,<cert_format>]

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier Value: 1÷10 : Secure Socket Identifier
<CipherSuite>	integer	-	identify the cipher suite. The value 0 means that the cipher suite is chosen by remote server, see Additional info section. Other values are expressed in hex format and identify the cipher suite though its RFC value. The supported values are listed in the Additional info section.
<auth_mode>	integer	0	authentication mode Values:

0	:	SSL Verify None
1	:	manage server authentication
2	:	manage server and client authentication if requested by the remote server

<cert_format>	integer	1	Select the format of the certificate to be stored via #SSLSECDATA command

Values:

0	:	DER format
1	:	PEM format

Additional info:

▶▶ With <CipherSuite> =0 cipher suite is chosen by remote server among the following:

TLS 1.2:

TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,

TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,

TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,

TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,

TLS_RSA_WITH_AES_128_CBC_SHA,

TLS_RSA_WITH_AES_256_CBC_SHA,

TLS_DHE_RSA_WITH_AES_128_CBC_SHA,

TLS_DHE_RSA_WITH_AES_256_CBC_SHA,

TLS_RSA_WITH_AES_128_CBC_SHA256,

TLS_DHE_RSA_WITH_AES_128_CBC_SHA256,

TLS_RSA_WITH_AES_256_CBC_SHA256,

TLS_DHE_RSA_WITH_AES_256_CBC_SHA256

▶▶ The first column shows the <CipherSuite> (RCF) values expressed in hex format, the second column indicates the related cipher suite name.

0x002F : TLS_RSA_WITH_AES_128_CBC_SHA
0x0033 : TLS_DHE_RSA_WITH_AES_128_CBC_SHA
0x0035 : TLS_RSA_WITH_AES_256_CBC_SHA
0x0039 : TLS_DHE_RSA_WITH_AES_256_CBC_SHA
0x003C : TLS_RSA_WITH_AES_128_CBC_SHA256
0x003D : TLS_RSA_WITH_AES_256_CBC_SHA256
0x0067 : TLS_DHE_RSA_WITH_AES_128_CBC_SHA256
0x006B : TLS_DHE_RSA_WITH_AES_256_CBC_SHA256
0x008A : TLS_PSK_WITH_RC4_128_SHA
0x008B : TLS_PSK_WITH_3DES_EDE_CBC_SHA
0x008C : TLS_PSK_WITH_AES_128_CBC_SHA
0x008D : TLS_PSK_WITH_AES_256_CBC_SHA
0x009C : TLS_RSA_WITH_AES_128_GCM_SHA256
0x009D : TLS_RSA_WITH_AES_256_GCM_SHA384
0x009E : TLS_DHE_RSA_WITH_AES_128_GCM_SHA256
0x009F : TLS_DHE_RSA_WITH_AES_256_GCM_SHA384
0x00A8 : TLS_PSK_WITH_AES_128_GCM_SHA256
0x00A9 : TLS_PSK_WITH_AES_256_GCM_SHA384
0x00AE : TLS_PSK_WITH_AES_128_CBC_SHA256
0x00AF : TLS_PSK_WITH_AES_256_CBC_SHA384
0xC004 : TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA
0xC005 : TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA
0xC009 : TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
0xC00A : TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
0xC00E : TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
0xC00F : TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
0xC013 : TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
0xC014 : TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
0xC023 : TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
0xC024 : TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
0xC025 : TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256
0xC026 : TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
0xC027 : TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
0xC028 : TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
0xC029 : TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
0xC02A : TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
0xC02B : TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

0xC02C : TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
0xC02D : TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256
0xC02E : TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384
0xC02F : TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
0xC030 : TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
0xC031 : TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256
0xC032 : TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384
0xC0A0 : TLS_RSA_WITH_AES_128_CCM_8
0xC0A1 : TLS_RSA_WITH_AES_256_CCM_8
0xC0A2 : TLS_DHE_RSA_WITH_AES_128_CCM_8
0xC0A3 : TLS_DHE_RSA_WITH_AES_256_CCM_8
0xCCA8 :
TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256
0xCCA9 :
TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256
0xCCAA : TLS_DHE_RSA_WITH_CHACHA20_POLY1305_SHA256

- Server CA certificate must be stored through #SSLSECDATA.
- Setting an obsolete cipher suite have the same effect as setting it to 0.

  If secure socket is not enabled using #SSLEN command, only test command can be used.
- If <auth_mode> is different from 0, the module uses its internal time and date to validate the certificate validity period.

  If time and date are incorrectly set, the certificate validation may fail. For additional information refer to AT+CCLK, AT#NITZ or AT#NTP.
- It is supposed that the module is just powered on and the AT#SSLSECCFG command is entered without <cert_format> parameter, the default format is PEM.

In this case the AT#SSLSECCFG? read command does not return the setting of the format in order to meet backward compatibility.

Now, let's assume that AT#SSLSECCFG command is entered again, but using the <cert_format> parameter for the first time: if the read command is entered, it reports the parameter value just used.

If subsequently the <cert_format> is omitted, the AT#SSLSECCFG? read command reports the parameter value entered the last time.

AT#SSLSECCFG?

Read command reports the currently selected parameters in the format:

#SSLSECCFG: <SSId>,<CipherSuite>,<auth_mode>,<cert_format>

Read command returns ERROR if secure socket has not been enabled using #SSLEN command.

AT#SSLSECCFG=?

Test command returns the range of supported values for all the parameters.

AT#SSLEN - Enable a SSL Socket

This command activates/deactivates a socket secured by SSL.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLEN=<SSId>,<Enable>

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<Enable>	integer	0	activate/deactivate secure socket
	Values:		
	0	:	deactivate
	1	:	activate

If the unique available secure socket is not activated, all the commands - belonging to the SSL set (example: #SSLSECDATA, #SSL..., etc.) and different from test commands - return an error message. #SSLS command is an exception, it can be issued also if the socket is deactivated.

If the unique available secure socket is connected, it cannot be deactivated issuing AT#SSLEN=1,0.

AT#SSLEN?

Read command reports the current status of secure socket in the format:

#SSLEN: <SSId>,<Enable>
OK

AT#SSLEN=?

Test command returns the range of supported values for all the parameters:

#SSLEN: (1-10),(0,1)

AT#SSLD - Open a SSL Socket to a Remote Server

This command opens a remote connection via socket secured through SSL.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#SSLD=<SSId>,<rPort>,<IPAddress>,<ClosureType>[,<connMode>[,<Timeout>]]

Execution command opens a remote connection via socket secured through SSL.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<rPort>	integer	1	Remote TCP port to contact
	Value:		
	1÷65535	:	TCP port number
<IPAddress>	string	-	address of SSL server.
<ClosureType>	integer	0	Closure type
	Value:		

0 : Until now only closure type 0 is supported. SSL session id and keys are free.

<connMode>	integer	1	connection mode

Values:

0 : online mode connection, see Additional info section.

1 : command mode connection, see Additional info section.

<Timeout>	integer	100	It represents the maximum allowed TCP inter-packet delay. It means that, when more data is expected during the handshake, the module awaits <Timeout> * 100 msec for the next packet. If no more data can be read, the module gives up the handshake and raises an ERROR response. <Timeout> is the total handshake timeout or, in other words, it is not the absolute maximum time between the #SSLD issue and the CONNECT/OK/ERROR response. Though by changing this parameter you can limit the handshake duration (for example in case of congested network or busy server), there is no way to be sure to get the command response within a certain amount of time, because it depends on the TCP connection time, the handshake time and the computation time (which depends

on the authentication mode and on the size of keys and certificates).

Value:

1÷5000	:	hundreds of ms

Additional info:

- <connMode>=0: online mode connection

  If <connMode> is set to online mode connection and the command is successful, the module enters into online data mode and returns the intermediate result code CONNECT. After the CONNECT, you can suspend the direct interface to the socket connection (the socket stays open) using the escape sequence (+++): the module moves back to command mode and returns the final result code OK after the suspension.

  After such a suspension, it is possible to resume it by using the #SSLO command with the corresponding <connId>.

- <connMode>=1: command mode connection

  If <connMode> is set to command mode connection and the command is successful, the socket is opened, the module remains in command mode and returns result code OK.

- If secure socket is not enabled using #SSLEN only test requests can be made.

- If timeout is not set for SSL connection the default timeout value, set by #SSLCFG, is used.

- In online mode the socket is closed after an inactivity period (configurable with #SSLCFG, with a default value of 90 seconds), and the NO CARRIER message is printed.

- In online mode data are transmitted as soon as the data packet size is reached or as after a transmission timeout. Both these parameters are configurable by using #SSLCFG.
- Before opening a SSL connection, the PDP context must have been activated by #SGACT=x,1
- Before opening a SSL connection, make sure to have stored the needed secure data (CA certificate), using #SSLSECDATA, for the security level set through AT#SSLSECCFG.
- The PDP context definition that will be used, is set by AT#SSLCFG command

AT#SSLD=?

Test command returns the range of supported values for all the parameters:

#SSLD: (1-10),(1-65535),,(0),(0,1),(1-5000)

See the following examples:

* Start online mode:

  AT#SSLD =1,8500,"84.94.194.21",0,1
  OK

  CONNECT

* Start command mode:

  AT#SSLD=1,8500,"84.94.194.21",1,1
  OK

* Configure correct PDP context with AT#SSLCFG command:

  AT#SGACT=3,1
  #SGACT: XX.XXX.XXX.XXX
  OK

Note the second parameter of #SSLCFG

```
AT#SSLCFG=1,3,300,90,100,50,0,0,0,0
OK
```

```
AT#SSLD=1,<port>,"IP or URL",0,1
OK
CONNECT
```

AT#SSLO - Restore a SSL Socket after a +++

This command restores a SSL connection (online mode) suspended by an escape sequence (+++).

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSLO=<SSId>

This command restores a SSL connection (online mode) suspended by an escape sequence (+++). After the connection restore, the CONNECT message is printed. Please note that this is possible even if the connection has been started in command mode (#SSLD with <connMode>=1).

Parameter:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier

- If secure socket has not be enabled through #SSLEN command, only test command can be used.
- Before opening a SSL connection, the PDP context must have been activated by AT#SGACT=X,1.
- If an error occurs during reconnection the socket cannot be reconnected, then a new connection has to be done.

AT#SSLO=?

Test command returns the range of supported values for all the parameters:

#SSLO: (1-10)

AT#SSLH - Close a SSL Socket

This command allows closing the SSL connection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLH=<SSId>[,<ClosureType>]

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<ClosureType>	integer	0	type of socket closure.
	Value:		
	0	:	only value 0 is supported

If secure socket has not be enabled through #SSLEN command, only test command can be used.

AT#SSLH=?

Test command returns the range of supported values for all the parameters:

#SSLH: (1-10),(0)

AT#SSLSEND - Send Data through a SSL Socket

This command allows sending data through a secure socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLSEND=<SSId>[,<Timeout>]

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<Timeout>	integer	100	socket send timeout
	Value:		
	1÷5000	:	timeout in 100 ms units

Additional info:

- After command line is terminated with <CR>, the command returns the following four-character sequence prompt:

  <CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

  and waits for the data to be send.

  to end the data editing and start the sending, enter Ctrl-Z char (0x1A hex). The maximum number of bytes to send is 1023, trying

to send more data will cause the data excess to be discarded and lost.

to exit without sending the message, enter ESC char (0x1B hex).

If data are successfully sent, the command returns OK. If data sending fails, an error code is reported.

If secure socket has not be enabled through #SSLEN command, only test command can be used.

If timeout is not set for SSL connection, is used the timeout value set by #SSLCFG.

Before sending data through the SSL connection, it must be established using #SSLD.

AT#SSLSEND=?

Test command returns the range of supported values for all the parameters.

AT#SSLRECV - Read Data from a SSL Socket

This command reads data from a SSL socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLRECV=<SSId>,<MaxNumByte>[,<Timeout>]

Set command allows to receive data, arrived through a connected secure socket. Data has been buffered and not read yet.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<MaxNumByte>	integer	N/A	maximum number of bytes to read
	Value:		
	1÷1000	:	maximum number of bytes to read
<Timeout>	integer	100	time-out in 100 ms units
	Value:		

1÷5000	:	hundreds of ms

Additional info:

- If data are received, the device responds:

  #SSLRECV: NumByteRead
  .. (Data read) ..
  OK

- If no data are received, the device responds:

  #SSLRECV: 0
  TIMEOUT
  OK

- If the remote host closes the connection, the device responds:

  #SSLRECV: 0
  DISCONNECTED
  OK

- If secure socket is not enabled using AT#SSLEN, only test requests can be made.
- If timeout is not set for SSL connection, the default timeout value, set through AT#SSLCFG, is used.

Before receiving data from the SSL connection, it has to be established using AT#SSLD.

AT#SSLRECV=?

The test command returns the ranges of the parameters values in the form:

#SSLRECV: (1-10),(1-1000),(10-5000)

AT#SSLS - Report the Status of a SSL Socket

This command reports the status of secure sockets.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLS=<SSId>

The command reports the status of secure sockets. The response message of the command can have the following formats:

if secure socket is connected, the format is:
#SSLS: <SSId>,<ConnectionStatus>,<CipherSuite>

otherwise:
#SSLS: <SSId>,<ConnectionStatus>

The response messages parameters are described in the Additional info section.

Parameter:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier

Value:

1÷10	:	Secure Socket Identifier

Additional info:

List of the meaning of the response message parameters.

Name	Type	Default	Description
<ConnectionStatus>	integer	N/A	Connection Status identifier Values: 0 : socket disabled 1 : connection closed 2 : connection open
<CipherSuite>	integer	-	identify the cipher suite. It is expressed in hex format and identified by RFC value. The supported values are listed in the #SSLSECCFG command

This command can be issued even if the <SSId> is not enabled.

In case of error when reading the <CipherSuite> 0xffff is returned

AT#SSLS=?

Test command returns the ranges of the parameter values in format:

#SSLS: (1-10)

```
* AT#SSLS=1
  #SSLS: 1,1
  OK

* AT#SSLS=1
  #SSLS: 1,2,0x0035
  OK
```

AT#SSLI - Secure Socket Info

This command is used to get information about secure socket data traffic.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLI=<SSId>

Execution command is used to get information about secure socket data traffic. The response is in the format:

#SSLI: <SSId>,<DataSent>,<DataRecv>,<PendingData>,<TCPConnWaitingAck>

Parameter:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier

Value:

1÷10	:	Secure Socket Identifier

Additional info:

Parameters returned by the response message and not described in the previous sections.

Name	Type	Default	Description

<DataSent>	integer	-	total amount (in bytes) of data sent to the TLS/SSL connection since the beginning of the connection itself (obviously: not yet encoded into TLS/SSL record)
<DataRecv>	integer	-	total number of bytes received from the TLS/SSL connection since the beginning of the connection itself (obviously: already decoded from TLS/SSL record)

<PendingData>	integer	-	number of bytes available to be read from the TLS/SSL record that is currently being processed (obviously: already decoded from TLS/SSL record) plus the number of bytes available to be read in the TCP/IP stack.
<TCPConnWaitingAck>	integer	N/A	indication of the underlying TCP socket condition, if there are TCP/IP packets sent but not yet acknowledged or not Values: 0 : no TCP/IP packets sent waiting for ack

1 : TCP/IP packets sent waiting for ack

AT#SSLI=?

Test command returns the range of supported values for all the parameters.

#SSLI: (1-10)

AT#SSLSECDATA - Manage the Security Data

The command stores, reads, and deletes security data (Certificate, CA certificate, private key) in / from NVM.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECDATA=<CertId>,<action>,<dataType>[,<size>[,<Password>[,<md5WhenReading>]]]

Parameters:

Name	Type	Default	Description
<CertId>	integer	1	Certificate Identifier.
	Values:		
	1 : Certificate identifier number 1		
<action>	integer	0	required action
	Values:		
	0 : delete data from NVM		
	1 : store data in NVM		
	2 : read data from NVM		
<dataType>	integer	0	security data type

Values:

0	:	certificate
1	:	CA certificate
2	:	RSA Private key. Private keys with password are not supported

<size>	integer	N/A	size of security data to be stored. The parameter is mandatory if write action is issued, can be omitted for delete or read actions.

Value:

1÷4095	:	size

<Password>	string	-	Dummy parameter, unused
<md5WhenReading>	integer	0	Dummy parameter (for backward compatibility)

Values:

0	:	default value

Additional info:

▶▶ Store security data in NVM

<action>=1 specifies the storing in NVM, <size> parameter is mandatory.

After command line is terminated with <CR>, the command returns the following four-character sequence prompt:

<CR><LF><greater_than><space> (see IRA 13, 10, 62, 32)

If data are successfully stored, the command returns OK, otherwise an error code is reported.

▶▶ Read security data from NVM

<action>=2 specifies the reading from NVM, the <size> parameter is not mandatory.

The command returns the following message:

#SSLSECDATA: <CertId>,<dataType>
<DATA>
OK

If the required security data has not been stored in NVM (or it has been deleted) the response has the following format:

#SSLSECDATA: <CertId>,<dataType>
No data stored
OK

▶▶ Read security data from NVM

<action>=2 is not allowed for RSA Private key.

Select certificate format

Secured data have to be in PEM or in DER format, depending on <cert_format> chosen with #SSLSECCFG.

If no <cert_format> has been specified with #SSLSECCFG, then PEM format is assumed.

PEM format (see #SSLSECCFG command): to complete the operation send Ctrl-Z char (0x1A hex); to exit without writing the message send ESC char (0x1B hex).

DER format (see #SSLSECCFG command): when <size> bytes are entered, the certificate is automatically stored; ESC or Ctrl-Z do not have any effect, because they are considered as possible octets contained in the certificate.

If secure socket has not be enabled through #SSLEN command, only test command can be used.

If socket is connected an error code is reported.

Only "rsa_sign" certificates are supported by the Telit Cinterion Modules in client authentication. The remote server must support this certificate type, otherwise the handshake will fail.

CA certificate stored in NVM

CA certificates stored in NVM are shared among all secure sockets <SSId>s

AT#SSLSECDATA?

Read command reports what security data are stored for each <CertId>. The returned message has the following format:

#SSLSECDATA: <CertId>,<CertIsSet>,<CACertIsSet>,<PrivKeyIsSet> [,<PasswordIsSet>,<md5WhenReading>]

<CertIsSet>, <CAcertIsSet>, <PrivKeIsset> are 1 if related data are stored into NVM otherwise 0.

Additional info:

▶▶ This Additional info section describes the parameters, if not described in the previous sections, returned by the AT#SSLSECDATA? read command.

Name	Type	Default	Description
<CertIsSet>	integer	0	identifies the certificate presence in the NVM Values: 0 : not present 1 : present
<CAcertIsSet>	integer	0	identifies the CA certificate presence in the NVM Values: 0 : not present 1 : present

<PrivKeyIsSet>	integer	0	identifies the RSA Private key presence in the NVM

Values:

0	:	not present
1	:	present

AT#SSLSECDATA=?

Test command returns the supported values for the parameters <CertId>, <action>, <dataType> and <size>.

AT#SSLSENDEXT - Send Data through a SSL Socket in Command Mode

This command sends data through a secure socket.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSLSENDEXT=<SSId>,<bytestosend>[,<Timeout>]

Execution command sends data through a secure socket.

After command line is terminated with <CR>, the command returns the following four-character sequence prompt:

<CR><LF><greater_than><space> (see IRA 13, 10, 62, 32)

and waits for the data to be send.

When <bytestosend> bytes have been sent, the sending is automatically completed. If data are successfully sent, the command returns OK. If data sending fails, an error code is reported.

Parameters:

Name	Type	Default	Description
<SSId>	integer	N/A	Secure Socket Identifier
	Value:		
	1÷10	:	Secure Socket Identifier
<bytestosend>	string	-	number of bytes to be sent. Refer to test command for range
<Timeout>	integer	100	time-out in 100 ms units.
	Value:		

1÷1500	:	hundreds of ms.

If secure socket has not be enabled through #SSLEN command, only test command can be used.

- If timeout is not set for SSL connection, is used the timeout value set by #SSLCFG.
- Before sending data through the SSL connection, it must be established using #SSLD.
- All special characters are sent like a generic byte. For instance: 0x08 BS (Backspace) is sent through the socket, it does not delete the previous character.

AT#SSLSENDEXT=?

Test command returns the range of supported values for parameters <SSId>, <bytestosend> and <Timeout>.

Open the socket in command mode:
AT#SSLD=1,443,<port>,"IP address",0,1
OK

Send data specifying total number of bytes:
AT#SSLSENDEXT=1,256,100
>...

AT#SSLSECCFG2 - Configure Additional Parameters of a SSL Socket

This command allows configuring additional SSL security parameters.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECCFG2=<SSId>,<version>[,<SNI>[,<CustomCA>[,<PreloadedCA>[,<MinVersion>]]]]

Parameters:

Name	Type	Default	Description
<SSId>	integer	N/A	Secure Socket Identifier
	Value:		
	1÷10	:	SSL socket Identifier
<version>	integer	3	select SSL/TLS protocol version
	Value:		
	0÷3	:	protocol version TLSv1.2
<SNI>	integer	0	enable/disable Service Name Indication
	Values:		

0	:	SNI disabled
1	:	SNI enabled

<CustomCA>	integer	0	Dummy (for backward compatibility)

Value:

0	:	No CA certificate

<PreloadedCA>	integer	0	Dummy (for backward compatibility)

Value:

0	:	No CA certificate

<MinVersion>	integer	1	Set the Minimum acceptable TLS version

Values:

1	:	protocol version TLSv1.0
2	:	protocol version TLSv1.1
3	:	protocol version TLSv1.2

Additional info:

▶▶ <MinVersion> must be minor or equal to <version>.

AT#SSLSECCFG2?

Read command reports the currently selected parameters in the format:

#SSLSECCFG2: <SSId>,<version>,<SNI>,
<CustomCA>,<PreloadedCA>,<MinVersion>

If <version> and <MinVersion> were set using a value inside a range, the read command will report the higher value of the range with the same meaning.

e.g. Setting <version> to 0 will result in the read value 3.

AT#SSLSECCFG2=?

Test command reports the range of supported values for all the parameters.

AT#SSLSECCS - Configure Cipher Suites List

This command configures a list of cipher suites, used for secure sockets

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECCS=<ListId>[,<Action>[,<CipherSuite>]]

Parameters:

Name	Type	Default	Description
<ListId>	integer	1	Cipher Suites list Identifier

Value:

1 : Cipher Suites list Identifier (currently limited to 1)

Name	Type	Default	Description
<Action>	integer	N/A	command action if this parameter is omitted, it prints the current content of the list.

Values:

0 : Remove <CipherSuite> from <ListId>

1 : Add <CipherSuite> to <ListId>

2 : Remove all cipher suites from <ListId>

3 : Add all supported cipher suites to <ListId>

4 : Reset <ListId> to the default set of cipher suites

<CipherSuite>	integer	-	Cipher Suite identifier The <CipherSuite> is identified by its RCF value and it is expressed in hex format. The range of supported values is available in the Additional info section

Additional info:

▶▶ The first column shows the <CipherSuite> (RCF) values expressed in hex format, the second column indicates the related cipher suite name.

0x002F : TLS_RSA_WITH_AES_128_CBC_SHA

0x0033 : TLS_DHE_RSA_WITH_AES_128_CBC_SHA

0x0035 : TLS_RSA_WITH_AES_256_CBC_SHA

0x0039 : TLS_DHE_RSA_WITH_AES_256_CBC_SHA

0x003C : TLS_RSA_WITH_AES_128_CBC_SHA256

0x003D : TLS_RSA_WITH_AES_256_CBC_SHA256

0x0067 : TLS_DHE_RSA_WITH_AES_128_CBC_SHA256

0x006B : TLS_DHE_RSA_WITH_AES_256_CBC_SHA256

0x008A : TLS_PSK_WITH_RC4_128_SHA

0x008B : TLS_PSK_WITH_3DES_EDE_CBC_SHA

0x008C : TLS_PSK_WITH_AES_128_CBC_SHA

0x008D : TLS_PSK_WITH_AES_256_CBC_SHA

0x009C : TLS_RSA_WITH_AES_128_GCM_SHA256

0x009D : TLS_RSA_WITH_AES_256_GCM_SHA384
0x009E : TLS_DHE_RSA_WITH_AES_128_GCM_SHA256
0x009F : TLS_DHE_RSA_WITH_AES_256_GCM_SHA384
0x00A8 : TLS_PSK_WITH_AES_128_GCM_SHA256
0x00A9 : TLS_PSK_WITH_AES_256_GCM_SHA384
0x00AE : TLS_PSK_WITH_AES_128_CBC_SHA256
0x00AF : TLS_PSK_WITH_AES_256_CBC_SHA384
0xC004 : TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA
0xC005 : TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA
0xC009 : TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
0xC00A : TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
0xC00E : TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
0xC00F : TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
0xC013 : TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
0xC014 : TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
0xC023 : TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
0xC024 : TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
0xC025 : TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256
0xC026 : TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
0xC027 : TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
0xC028 : TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
0xC029 : TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
0xC02A : TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
0xC02B : TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
0xC02C : TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
0xC02D : TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256
0xC02E : TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384
0xC02F : TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
0xC030 : TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
0xC031 : TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256

0xC032 : TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384

0xC0A0 : TLS_RSA_WITH_AES_128_CCM_8

0xC0A1 : TLS_RSA_WITH_AES_256_CCM_8

0xC0A2 : TLS_DHE_RSA_WITH_AES_128_CCM_8

0xC0A3 : TLS_DHE_RSA_WITH_AES_256_CCM_8

0xCCA8 : TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256

0xCCA9 : TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256

0xCCAA : TLS_DHE_RSA_WITH_CHACHA20_POLY1305_SHA256

- The Cipher Suites list (when configured) is used instead of the list of default ciphers, when parameter <CipherSuites> in AT#SSLSECCFG is set to 0 (cipher suite is chosen by remote server).
- The <CipherSuite>(s) are proposed to the server in the same order as they are added to the list.
- When <Action> is 3, all supported <CipherSuite>s are added to the list from strongest to weakest.

  PSK <CipherSuite>s are NOT included.
- If only <ListID> parameter is set, the command prints out the <CipherSuite>(s) currently configured on <ListID> in the following format:

AT#SSLSECCS=<ListId>

#SSLSECCS:

LIST<ListId>:

<01>: <CipherSuite_#1>

..

<n>: <CipherSuite_#n>

OK

AT#SSLSECCS=?

Test command returns the range of supported values for all the parameters

Add cipher suite 0x1302, 0x1301 and 0x003D to List 1 and display the new list content:

AT#SSLSECCS=1,1,0x1302

OK

AT#SSLSECCS=1,1,0x1301

OK

AT#SSLSECCS=1,1,0x003D

OK

AT#SSLSECCS=1

#SSLSECCS:

LIST1:

01: 0x1302

02: 0x1301

IoT Portal

AT#LWM2MSKIP - LwM2M Client startup mode

The set command enables and disables the LwM2M Client initialization mode at module startup. This command allows to the user the possibility to change the LwM2M client agents initialization.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#LWM2MSKIP=<enable>[,<agentMask>][,<reboot>]

Parameters:

Name	Type	Default	Description
<enable>	integer	0	enables or disables the "skipping client startup" modality Values: 0 : skip is disabled, thus the LwM2M clients selected in the <agentMask> parameter are configured to be started at module startup (Default) 1 : skip is enabled, thus the LwM2M clients selected in the <agentMask> parameter are configured to be not started at module startup
<agentMask>	hex	2	set of LwM2M clients affected by the <enable> parameter The command admits the parameter in hexadecimal format without the prefix 0x.

			The following values and combinations are accepted: 0x1: Telit client, bit mask 00000001 (not supported) 0x2: Verizon client, bit mask 00000010 0x4: AT&T client, bit mask 00000100 (not supported) 0x8: DOCOMO client, bit mask 00001000 (not supported) 0x10: reserved for internal use, bit mask 00010000 (not supported) 0x20: reserved for internal use, bit mask 00100000 (not supported) ERROR is returned if a bit of the mask has no LwM2M client mapped on. <agentMask>= 0 is not allowed since it is useless: it means that the skipping/not skipping required operation should not affect any clients. Giving the command without <agentMask> parameter will affect all LwM2M clients. Value: 1÷FF : bit mask length
<reboot>	integer	1	enables or disables the automatic reboot possibility after the command setting

Values:

0 : Reboot is disabled

1 : Reboot as part of issuing the command (Default)

If the command changes the current agents skip settings, in case of <reboot> parameter missing or set to 1, the module is rebooted. No reboots will be triggered if the command is not changing the current behavior.

The command affects only the clients indicated in the <agentMask>.

AT#LWM2MSKIP?

Read command reports the current values of parameters in the format:

#LWM2MSKIP: <enable>,<maskStatus>

AT#LWM2MSKIP=?

Test command reports the supported range of values for all the parameters.

Set the skip property for all agents available in the module: all the clients are stopped at startup (reboot is performed in particular cases, see additional notes) :

AT#LWM2MSKIP=1

OK

Set the skip property for all agents available in the module (reboot is not performed)

AT#LWM2MSKIP=1,,0

OK

Set the skip property for Verizon client. Reboot is not performed, it takes effect at next reboot

AT#LWM2MSKIP=1,2,0

OK

Reset the skip property for Verizon client. Reboot is not performed, it takes effect at next reboot:

AT#LWM2MSKIP=0,2,0

OK

AT#LWM2MREG - Registration to a LwM2M Server

This command allows the user to request a full registration, a deregistration or a registration update to a LwM2M server. It also allows to query the registration state of one or all the LwM2M servers related to a LwM2M agent

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#LWM2MREG=<agentId>,<actionID>[,<shortServerID>]

Parameters:

Name	Type	Default	Description
<agentId>	integer	N/A	identifier of the LwM2M agent related to the request

Values:

0 : Telit agent

1 : Verizon agent

2 : AT&T agent

3 : Docomo agent

4 : Reserved for internal use

5 : Reserved for internal use

<actionID>	integer	N/A	identifier of the required action

Values:

0	:	deregister
1	:	full registration
2	:	registration update
3	:	retrieve registration info

<shortServerID>	integer	N/A	identifier of the server related to the request. It is optional in case <actionID> is equal to 3

Value:

1÷65534	:	Short server ID of the addressed server

Expected result <actionID> 0 - deregister

The deregister operation will fail in the following conditions:

- The state of the <shortServerID> is not registered or the client is already deregistering for the addressed server

- The state of the <shortServerID> is disabled (a disable timeout has been requested for the addressed server)

Expected result <actionID> 1 - full registration

The register operation will fail in the following conditions:

- The state of the <shortServerID> is alredy registered or the client is already registering to the addressed server

- The state of the <shortServerID> is disabled (a disable timeout has been requested for the addressed server)

- Expected result <actionID> 2 - registration update

  The register operation will fail in the following conditions:

  - The state of the <shortServerID> is not registered

  - The state of the <shortServerID> is disabled (a disable timeout has been requested for the addressed server)

- Expected result <actionID> 3 - retrieve registration info

  The command will respond listing the registration details for the server <shortServerID>, or for all servers related to the <agentId> in case parameter <shortServerID> is not provided.

  The answer will be in the following form:

  #LWM2MREG: <shortServerID>, <regState>,<lastRegistration>,<nextUpdate>,<lastActivity>

Parameter	Description
<shortServerID>	Identifier of the server
<regState>	Identifier of the registration state: 0, not registered 1, disabled 2, registered
<lastRegistration>	Timestamp related to the last successful registration or registration updated. The timestamp is expressed in Unix local time. In case there is

	not a valid timestamp, 0 is reported
<nextUpdate>	Timestamp related to the next scheduled registration update.The timestamp is expressed in Unix local time. In case there is not a valid timestamp, 0 is reported
<lastActivity>	Timestamp related to the last succeeded data exchange between the LwM2M client and the server.The timestamp is expressed in Unix local time. In case there is not a valid timestamp, 0 is reported

AT#LWM2MREG?

Read command returns OK

AT#LWM2MREG=?

Test command reports the supported range of values for the parameters.

AT#LWM2MSTS - Select Bootstrap/Standard Server

This command configures the LwM2M server parameters which the client will connect to at the next module reboot.
It could be used to restore the default LwM2M server configuration or to set a new server configuration.
For security reasons, only DTLS encryption mode (CoAPS) is supported.
The command requires the existence of the corresponding LwM2M agent.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#LWM2MSTS=<agentId>[,<serverIndex>[,<URI>[,<isBootstrapServer>]]]

Parameters:

Name	Type	Default	Description
<agentId>	integer	0	identifier of the LwM2M agent related to the request. Issuing the command with only the <agentId> parameter will reset a formerly set server, restoring the condition to the factory default.

Values:

0 : Telit agent

1 : Verizon agent

2 : AT&T agent

		3 : Docomo agent 4 : Reserved for internal use 5 : Reserved for internal use	
<serverIndex>	integer	-	the server that will be used by the module at the next client reboot. Entries from: 0 to 9 reserved for Telit's agent 10 to 19 reserved for VZW's agent 20 to 29 reserved for AT&T's agent The correspondence table between <serverIndex> parameter and preconfigured server URI is shown in Additional info.
<URI>	string	-	URI
<isBootstrapServer>	integer	0	selects if the requested server is a bootstrap. Values:

0 : the specified server is not a bootstrap server

1 : the specified server is a bootstrap server

Additional info:

▶▶ Correspondence table between <serverIndex> parameter and server URI.

<serverIndex>	server URI	Note
0	reserved	for internal use
10	"coaps://ddocdpboot.do.motive.com:5684"	Bootstrap server
11	"coaps://ddocdp.do.motive.com:5684"	DM server
20	"coaps://xattmpct.xdev.motive.com:5684"	Bootstrap server
999	customized server URI entry	/

<serverIndex> special value '999' allows the user to insert a customized server string; this string should start with "coaps://" sub-string and may or may not contain the indicated port. In case the port is not indicated, the LwM2M client will use the proper default CoAPS port (5684).

- The servers list is not constrained in any way to any customizations; therefore, each entry is available to every agent.
- The command execution will invalidate the running <agentId> and performs the deregistration of <agentId> from all servers.
- After successful command execution a module reboot is required before AT#LWM2MREG set command for <agentId>, otherwise an ERROR or +CMEE: ERROR is returned
- If the command is given during BOOTSTRAP/REGISTRATION phase it will not stop them. New server will be used at the next module's reboot.

AT#LWM2MSTS?

Read command returns OK result code.

AT#LWM2MSTS=?

Test command returns OK result code.

It selects the "coaps://xvzwcdpiii.xdev.motive.com:5684" server for the Verizon agent.

AT#LWM2MSTS=1,10

It selects the "Leshan" test server, secure mode. Last parameter in this example could also be omitted.

AT#LWM2MSTS=0,999,"coaps://leshan.eclipse.org:5684",0

FOTA & OMA

FOTA Legacy

AT#FTPGETOTA - OTA FTP delta download

This command executes the OTA FTP delta download.

SIM Presence	Setting saved	Can be aborted
Required	No	Yes

AT#FTPGETOTA=<filename>[,<verboseLevel>[,<restart>[,<restartGranularity>]]]

Parameters:

Name	Type	Default	Description
<filename>	string	-	Delta-FW file name on ftp server
<verboseLevel>	integer	0	enable/disable verbose mode
	Value:		
	0 : disable		
	1 : enable		
<restart>	integer	0	enable/disable restart mode
	Value:		
	0 : disable		

		1 : enable	
<restartGranularity>	integer	4	write offset to file every X cycles
	Value:		
		1÷4 : write offset to file every X cycles	

AT#FTPGETOTA=?

Test command returns the OK result code.

AT#HTTPRCVOTA - Receive and Store HTTP Server Data in FOTA

This command read data from a HTTP server and save it into the FOTA partition.

SIM Presence	Setting saved	Can be aborted
Required	No	Yes

AT#HTTPRCVOTA=<prof_id>[,<verbose>]

Execution command allows to read data from a HTTP server, in response to a previous HTTP module request, and to save it into module FOTA partition.

The module is notified of these data by the #HTTPRING URC.

Parameters:

Name	Type	Default	Description
<prof_id>	integer	N/A	numeric parameter indicating the profile identifier Value: 0÷2 : profile identifier
<verbose>	integer	0	enable/disable verbose mode Value: 0 : disable 1 : enable

When verbose mode is enabled, i.e. <verbose> is set to 1, the "#" character is printed on the AT command port for every 4096 bytes written into the FOTA partition

AT#HTTPRCVOTA=?

Test command reports the range of supported values for parameters in the format:

HTTPRCVOTA:<prof_id> (list of supported <prof_id>s),<verbose> (list of supported <verbose>s)

AT#OTAUPW - OTA Delta Write

Execution command starts injection of a delta file into the device.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#OTAUPW=[<verbose_lev>[,<size>]]

Parameters:

Name	Type	Default	Description
<verbose_lev>	integer	0	Hash mark printing during injection Values: 0 : Hash mark printing disabled during injection 1 : Hash mark printing enabled during injection every 8192 bytes
<size>	integer	-	size in bytes of data to be injected

If the <size> parameter is added, #OTAUPW will finish with OK (or ERROR), otherwise must be closed with +++.

AT#OTAUPW<CR>

has the same behavior of

AT#OTAUPW=0

After AT#OTAUPW the FOTA URCs are available only on UART port even if the AT#OTAUPW is sent on USB port.

AT#OTAUPW=?

Test command returns OK result code.

AT#OTAUP - OTA Check and Start Update

Execution command performs validation of delta and firmware update process.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#OTAUP=<type>[,<verbose>]

Parameter:

Name	Type	Default	Description
<type>	integer	0	Type of FOTA action

Value:

0 : delta validation and, if delta valid, start of firmware update

<verbose>	integer	0	FOTA trace verbose level

Value:

0 : no output information during validation/update process

1 : same behavior as value 0 (not supported minimal output information during validation/ update process: validation percentage, update percentage, restart unsolicited message)

AT#OTAUP=?

Test command returns OK result code.

GNSS

GNSS Configuration

AT$GPSCFG - Set GNSS Configuration Parameters

This command sets the GNSS constellation.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GPSCFG=<parameter>,<value>

Parameters:

Name	Type	Default	Description
<parameter>	integer	N/A	selects the configuration parameter to be set
	Values:		
	0 : reserved		
	1 : reserved		
	2 : set constellation		
	3 : reserved		
	4 : reserved		
<value>	integer	-	value depends on the first parameter. See Additional info section.

Additional info:

- <parameter>=2, in this case the <value> assumes the meaning of <constellation>.

Name	Type	Default	Description
<constellation>	integer	10 or 12	selects the constellation

Values:

0 : reserved

1 : reserved

2 : reserved

3 : reserved

4 : GPS+QZSS

5 : reserved

6 : reserved

7 : GPS+GLONASS+QZSS

Not supported by ELS63 GNSS product variant

8 : GPS+GALILEO+QZSS

Not supported by ELS63 GNSS product variant

9 : GPS+BDS+QZSS

Not supported by ELS63 GNSS product variant

10 : GPS+GLONASS+GALILEO+BDS+QZSS

11 : NAVIC + QZSS

Not supported by LE910Q1 GNSS product variant

12 : GPS+GLONASS+GALILEO+BDS+NAVIC+QZSS

Not supported by LE910Q1 GNSS product variant

13 : GPS+GALILEO+NAVIC+QZSS

Not supported by LE910Q1 GNSS product variant

The current setting is stored through $GPSSAV command with $GPSSAV <GNSSParam> parameter set to 1.

AT$GPSCFG?

Read command returns current values of <reserved>, <reserved>, <constellation>, <reserved> and <reserved> in format:

$GPSCFG: <reserved>,<reserved>,<constellation>,<reserved>, <reserved>

AT$GPSCFG=?

Test command returns supported values range of parameters <reserved>, <reserved>, <constellation>, <reserved> and <reserved>

AT$GPSNHZ - GNSS Navigation Update Rate Configuration

This command set the GNSS Navigation Update Rate.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GPSNHZ=<update_rate>

Set command configures the GNSS Navigation Update Rate.

Parameter:

Name	Type	Default	Description
<update_rate>	integer	0	Navigation Update Rate

Values:

0 : 1Hz (1000 milliseconds)

1 : 2Hz (500 milliseconds)

2 : 5Hz (200 milliseconds)

3 : 10Hz (100 milliseconds)

The current setting is stored through $GPSSAV command with $GPSSAV <GNSSParam> parameter set to 1.

AT$GPSNHZ?

Read command returns the currently update rate setting , in the format

$GPSNHZ: <update_rate>

AT$GPSNHZ=?

Test command reports the range of supported values for parameter <update_rate>.

$GPSNHZ: (the supported range of <update_rate>)

AT#GNSSCMD - GNSS Command

This command allows to control the GNSS through direct GNSS proprietary commands.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT#GNSSCMD=<gnssCommandString>

Sends the GNSS proprietary command <gnssCommandString> directly to the GNSS and reports GNSS answer/response in the form:

#GNSSRESP: <gnssResponseString>

followed by

OK

See also Additional info section .

Parameter:

Name	Type	Default	Description
<gnssCommandString>	string	-	GNSS proprietary command string

Additional info:

Parameter meaning:

Name	Type	Default	Description

<gnssResponseString>	string	-	GNSS proprietary command answer/ response string

- AT#GNSSCMD is deprecated, please use equivalent AT$GNSSCMD.
- AT#GNSSCMD GNSS proprietary command string <gnssCommandString> is without checksum.
- If AT#GNSSCMD was used for any GNSS setting the current setting will be stored through $GPSSAV command with $GPSSAV <GNSSParam> parameter set to 1.

AT#GNSSCMD=?

Test command returns OK

AT$GNSSCMD - GNSS Command

This command allows to control the GNSS through direct GNSS proprietary commands.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GNSSCMD=<gnssCommandString>

Sends the GNSS proprietary command <gnssCommandString> directly to the GNSS and reports GNSS answer/response in the form:

$GNSSRESP: <gnssResponseString>

followed by

OK

See also Additional info section .

Parameter:

Name	Type	Default	Description
<gnssCommandString>	string	-	GNSS proprietary command string

Additional info:

Parameter meaning:

Name	Type	Default	Description

<gnssResponseString>	string	-	GNSS proprietary command answer/ response string

- AT$GNSSCMD GNSS proprietary command string <gnssCommandString> is without checksum.
- If AT$GNSSCMD was used for any GNSS setting the current setting will be stored through $GPSSAV command with $GPSSAV <GNSSParam> parameter set to 1.

AT$GNSSCMD=?

Test command returns OK

AT$GNSSMODE -GNSS Mode Configuration

This command sets the standalone or controlled mode of the GNSS.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$GNSSMODE=<mode>

Set command configures the standalone or controlled mode of the GNSS

Parameter:

Name	Type	Default	Description
<mode>	integer	0	standalone or controlled mode of the GNSS

Values:

0	:	GNSS is in standalone mode
1	:	GNSS is in controlled mode

For LE910Q1 GNSS product variant family:

AT$GNSSMODE=1 automatically sets GPIO7, GPIO4 and GPIO6 alternate functions

AT#GPIO=7,0,12

AT#GPIO=4,0,16

AT#GPIO=6,0,15

and reserves GPIO7, GPIO4 and GPIO6 for GNSS.

AT$GNSSMODE=0 automatically sets

AT#GPIO=7,0,0

AT#GPIO=4,0,0

AT#GPIO=6,0,0

and releases GPIO7, GPIO4 and GPIO6 to be used a normal GPIO.

For ELS63 GNSS product variant family:

AT$GNSSMODE=1 automatically sets GPIO7, GPIO22 and GPIO23 alternate functions

AT#GPIO=7,0,12

AT#GPIO=22,0,16

AT#GPIO=23,0,15

and reserves GPIO7, GPIO22 and GPIO23 for GNSS.

AT$GNSSMODE=0 automatically sets

AT#GPIO=7,0,0

AT#GPIO=22,0,0

AT#GPIO=23,0,0

and releases GPIO7, GPIO23 and GPIO22 to be used a normal GPIO.

AT$GNSSMODE?

Read command reports the current value of the <mode> parameter, in the format:

$GNSSMODE: <mode>

AT$GNSSMODE=?

Test command reports the supported range of values for parameter <mode>

AT$GNSSDWLD - GNSS Controller Download Mode

This command configures the module and the GNSS controller for download mode for GNSS controller's firmware update.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GNSSDWLD

The execution command configures the module and the GNSS controller for download mode for GNSS controller's firmware update.

The execution command:

- configures the UART or USB port where the command is executed in GNSS passthrough mode (all binary data on the module UART or USB where the command is executed are transferred as they are to GNSS UART and the same in the opposite direction)

- after 5 seconds triggers a GNSS reset

To exit GNSS passthrough mode use +++ escape sequence

Additional info:

GNSS firmware update procedure:

- select the modem UART or USB port

- execute AT$GNSSDWLD

- within 5 seconds, start GNSS firmware update according the specific GNSS firmware update procedure (the same used in GNSS standalone mode)

- 5 seconds after AT$GNSSDWLD execution it is expected that GNSS firmware proceeds successfully

- after GNSS firmware update procedure completed execute +++ escape sequence

Set AT$GNSSMODE=1 before using AT$GNSSDWLD command.

AT$GNSSDWLD returns ERROR if AT$GNSSMODE parameter <mode> is set to 0.

AT$GNSSDWLD command is not supported on USB in version 52.00.003.

AT$GNSSDWLD=?

Test command returns the OK result code.

AT$GPSSAV - Save GNSS Parameters Configuration

This command stores the current GNSS parameters in the NVM of the device.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSSAV=<GNSSParam>[,<GPSPParam>]

Set command stores the current GNSS parameters in the NVM of the device.

Parameters:

Name	Type	Default	Description
<GNSSParam>	integer	0	stores or does not store all the current GNSS parameters (except AT$GPSP <status>) in the NVM of the device

Values:

0 : do not store all the current GNSS parameters (except AT$GPSP <status>) in the NVM of the device

1 : store all the current GNSS parameters (except AT$GPSP <status>) in the NVM of the device

<GPSPParam>	integer	0	stores or does not store the current GNSS AT$GPSP <status> parameter in the NVM of the device

Values:

0 : do not store the current GNSS AT$GPSP <status> parameter in the NVM of the device

1 : store the current GNSS AT$GPSP <status> parameter in the NVM of the device

AT$GPSSAV=?

Test command command returns the supported range of values for parameters <GNSSParam>, <GPSPParam>..

AT$GPSSAV=1 or AT$GPSSAV=1,0

The saved parameters are those related to AT$GPSR, AT$GPSCFG, AT$GPSNHZ, AT$GPSNMUN, AT$GPSNMUNEX commands.

The saved parameters are also those configured by AT#GNSSCMD/AT$GNSSCMD commands.

AT$GPSSAV=0,1

The saved parameter is the AT$GPSP <status> parameter.

AT$GPSSAV=1,1

The saved parameters are both the one saved by AT$GPSSAV=1 and the one saved by AT$GPSSAV=0,1.

AT$GPSSAV=0,0

No parameters are saved.

```
AT$GPSSAV=1
OK

Or
AT$GPSSAV=0,1
OK

Or
AT$GPSSAV=1,1
OK
```

AT$AGNSSFINJ - AGNSS File Injection

This command handles Assisted GNSS file injection. AGNSS file could be downloaded through AT#AGNSSFDL.

GNSS receiver must be enabled with $GPSP=1

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$AGNSSFINJ=<provider>,<filename>

Parameter:

Name	Type	Default	Description
<provider>	integer	N/A	select the AGNSS provider to be set Value: 1 : use EPO AGNSS provider
<filename>	string	N/A	File to inject

AT$AGNSSFINJ=?

Test command returns OK

After the file is downloaded, enable GNSS receiver:

AT$GNSSMODE=1

OK

AT$GPSP=1

OK

Inject EPO file previously downloaded, and wait few seconds (it may depend on file length) until the OK reply appears:

AT$AGNSSFINJ=1,"EPO_GAL_3.DAT"

OK

Issue $AGNSSVAL and you should see the time validity of the injected data:

AT$AGNSSVAL?

$AGNSSVAL: 0,0,0,GPS

$AGNSSVAL: 0,0,0,GLO

$AGNSSVAL: 0,1750140000,1750528800,GAL

$AGNSSVAL: 0,0,0,BDS

AT$AGNSSTINJ - AGNSS Time Injection

This command handles Assisted GNSS time injection. GNSS receiver must be enabled before the injection.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$AGNSSTINJ=<provider>,<utc_time>

Parameters:

Name	Type	Default	Description
<provider>	integer	N/A	select the AGNSS provider to be set Value: 1 : use EPO AGNSS provider
<utc_time>	integer	-	provide UTC time. See Additional info section

Additional info:

- <utc_time> as Unix timestamp

Name	Type	Default	Description

<utc_time>	integer	-	provide time in Unix timestamp (seconds from midnight UTC on 1 January 1970)

AT$AGNSSTINJ=?

Test command returns OK.

Enable GNSS receiver:

AT$GNSSMODE=1

OK

AT$GPSP=1

OK

Time can now be injected into the GNSS engine:

AT$AGNSSTINJ=1,1747993320

OK

AT$AGNSSFDL - AGNSS File Download

This command handles Assisted GNSS file download. Currently, only HTTP protocol is supported. Before launching the command, AT#SGACT must be performed. After the download, the file can be injected using AT$AGNSSFINJ

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$AGNSSFDL=<provider>,<url>

Parameters:

Name	Type	Default	Description
<provider>	integer	N/A	select the AGNSS provider to be set Value: 0 : use EPO/ELPO AGNSS provider
<url>	string	-	Insert the HTTP url that points to the XTRA file. Port is not mandatory, if not present, the default value is 80 for HTTP and 443 for HTTPS.

Only HTTP protocol is currently supported. Secure (HTTPS) and unsecure (HTTP).

AT$AGNSSFDL=?

Test command returns OK.

Example:

AT$AGNSSFDL=0,"https://epo.telit.com/EPO_GPS_5.DAT"

OK

The structure of the URL field is as follow:

<protocol>://<username>:<password>@<hostname>[:<port>]/<URI>

Generic example:

https://user123:pass456@www.example.com:443/path/file.html

AT$AGNSSVAL - AGNSS Validity Data

This command shows information about the validity of AGNSS data. The command works only when the GNSS receiver is enabled.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$AGNSSVAL

The exec command is not supported and returns ERROR

AT$AGNSSVAL?

Read command reports information regarding the expiration date/time of AGNSS data, for each GNSS constellation.

Three comma separated values are printed:

the first field should be the current UNIX system timestamp (currently not supported, so 0 is printed),

the second field is the starting UNIX timestamp of AGNSS data validity,

and the third is the end UNIX timestamp of AGNSS data validity

AT$AGNSSVAL=?

Test command returns OK

Enable GNSS receiver:

AT$GNSSMODE=1

OK

AT$GPSP=1

OK

If GPS and GALILEO EPO data were previously injected, you should see something like this:

AT$AGNSSVAL?

$AGNSSVAL: 0,1750075200,1750939200,GPS

$AGNSSVAL: 0,0,0,GLO

$AGNSSVAL: 0,1750140000,1750528800,GAL

$AGNSSVAL: 0,0,0,BDS

OK

GNSS Control

AT$GPSP - GNSS Controller Power Management

This command powers on/off GNSS controller.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GPSP=<status>[,<powerOption>]

The set command manages the power-up/power-down of the GNSS controller.

Parameters:

Name	Type	Default	Description
<status>	integer	0	It is the power status that has to be set in GNSS power control
	Values:		
	0 : GNSS controller is powered off		
	1 : GNSS controller is powered on		
<powerOption>	integer	1	It is the power option that has to be set in GNSS power off control only
	Values:		
	0 : GNSS controller is fully powered off		
	1 : GNSS controller is partially powered off		

- Set AT$GNSSMODE=1 before using AT$GPSP command.

  AT$GPSP returns ERROR if AT$GNSSMODE parameter <mode> is set to 0.

- <powerOption> parameter is valid only if <status> parameter is 0.

  AT$GPSP=0 (default <powerOption> value 1) and AT$GPSP=0,1 partially power off GNSS (GNSS RTC still on) and GNSS will perform a warm/hot restart at next power on.

  AT$GPSP=0,0 fully power off GNSS (GNSS RTC off) and GNSS will perform a cold restart at next power on.

- GNSS data cleaning can be performed through AT$GPSR.

- The current setting of <status> parameter is stored through $GPSSAV command with $GPSSAV <GPSPParam> parameter set to 1.

AT$GPSP?

The read command reports the current value of the <status> parameter, in the format:

$GPSP: <status>

AT$GPSP=?

The test command reports the supported values range for parameter <status>.

GNSS controller is powered down

AT$GPSP=0

OK

AT$GPSSW - GNSS Software Version

This command provides the GNSS module software version.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSSW

Execution command returns the GNSS module software version in the format:

$GPSSW: <swVersion>

AT$GPSSW?

Read command has the same behavior as the execution command.

AT$GPSSW=?

Test command returns the OK result code

AT$GPSSW

$GPSSW: version string

OK

AT$GPSR - Reset the GNSS Controller

This command resets the GNSS controller.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSR=<resetType>

Execution command allows to reset the GNSS controller.

Parameter:

Name	Type	Default	Description
<resetType>	integer	0	set the type of GNSS controller reset.

Values:

0 : Factory Reset: this option clears all the GNSS memory including Clock Drift.

1 : Coldstart (No Almanac, No Ephemeris): this option clears all data that is currently stored in the internal memory of the GNSS receiver including Position, Almanac, Ephemeris and Time. The stored Clock Drift is retained.

2 : Warmstart (No ephemeris): this option clears all initialization data in the GNSS receiver and subsequently reloads the data that is currently displayed in the Receiver Initialization Setup screen. The Almanac is retained but the Ephemeris is cleared.

3 : Hotstart (with stored Almanac and Ephemeris): the GNSS receiver restarts by using all data that is currently stored in the internal memory of the GNSS receiver: validated Ephemeris and Almanac.

- AT$GPSR=0 changes and stores GNSS parameters values.
- AT$GPSR=1/2/3 triggers a cold/warm/hot restart of GNSS at AT command execution.

AT$GPSR?

Read command displays the current <resetType> value

AT$GPSR=?

Test command reports the range of supported values for parameter <resetType>.

Factory reset
AT$GPSR=0
OK

AT$GPSHWRST - GNSS Controller Hardware Reset

This command triggers the hardware reset of the GNSS controller.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSHWRST

The execution command manages the hardware reset of the GNSS controller.

Set AT$GNSSMODE=1 before using AT$GPSHWRST command.

AT$GPSHWRST returns ERROR if AT$GNSSMODE parameter <mode> is set to 0 .

AT$GPSHWRST=?

Test command returns the OK result code.

GNSS Position

AT$GPSACP - Get Acquired Position

This command returns information about the last GPS position.

[1] NMEA 0183 Standard

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSACP

Execution command returns information about the last GPS position in the format:

$GPSACP: <UTC>,<latitude>,<longitude>,<hdop>,<altitude>,<fix>,<cog>,

<spkm>,<spkn>, <date>,<nsat>,<hepe>,<vepe>

Additional info:

Meanings of the parameters returned by the command.

Name	Type	Default	Description
<UTC>	string	-	UTC time (hhmmss.sss) referred to GGA sentence
<latitude>	string	-	latitude in the format ddmm.mmmm N/S (referred to GGA sentence)

			where: dd: 00..90, degrees mm.mmmm: 00.0000..59.9999, minutes N/S: North/South
<longitude>	string	-	longitude in the format dddmm.mmmm E/W (referred to GGA sentence) where: ddd: 000..180, degrees mm.mmmm: 00.0000..59.9999, minutes E/W: East/West
<hdop>	string	-	Horizontal Dilution of Precision (referred to GGA sentence)
<altitude>	string	-	altitude - mean-sea-level (geoid) in meters (referred to GGA sentence)
<fix>	integer	N/A	fix type

Values:

0	:	invalid fix
1	:	invalid fix
2	:	2D fix
3	:	3D fix

<cog>	string	-	Course over Ground (degrees, True) (referred to RMC sentence)
<spkm>	string	-	speed over ground (Km/hr) (referred to VTG sentence)
<spkn>	string	-	speed over ground (knots) (referred to VTG sentence)
<date>	string	-	date of fix (referred to RMC sentence) in the format ddmmyy where: dd: 01..31, day mm: 01..12, month yy: 00..99, year 2000 to 2099
<nsat>	integer	N/A	total number of satellites in use (referred to GGA sentence)

Value:

0÷12	:	total number of satellites in use

<hepe>	float	-	Horizontal estimated position error (meters)
<vepe>	float	-	Vertical estimated position error (meters)

AT$GPSACP?

Read command has the same behavior as the Execution command.

AT$GPSACP=?

Test command returns the OK result code.

Execution and Read commands silently enable PTWSEPE (Estimated Position Error) in the GNSS.

```
AT$GPSP?
$GPSP: 0

OK
```

When the module is down there is no acquired position:

```
AT$GPSACP
$GPSACP: ,,,,,0,,,,,
OK

AT$GPSP=1
OK
```

Until first fix is received the command will display this:

```
AT$GPSACP
$GPSACP: ,,,,,1,,,,,

OK
```

Once fix has been received the command will display actual GPS time and position:

```
AT$GPSACP
$GPSACP:
091635.000,3913.6952N,00904.1505E,0.7,17.9,3,0.0,0.0,0.0,290920,10,3.79,2.50
OK
```

AT$GPSNMUN - Unsolicited NMEA Data Configuration

Unsolicited NMEA data configuration

NMEA 0183 Standard

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GPSNMUN=<enable>[,<GGA>,<GLL>,<GSA>,<GSV>,<RMC>,<VTG>]

Set command allows to activate an unsolicited GNSS data stream built with NMEA sentences on the standard serial port and defines which NMEA sentences will be available. Refer to document [1] to have information on the NMEA sentences contents and formats.

Parameters:

Name	Type	Default	Description
<enable>	integer	0	Enables unsolicited GNSS data stream and selects one of the available GNSS data stream format display. <enable> parameter is also used to disable the GNSS data stream. Here is the list of the <enable> values. See Additional info section to have information on GNSS data stream formats.

Values:

0 : disable GNSS data stream

1 : enable the first GNSS data stream format

2 : enable the second GNSS data stream format

3 : enable the second GNSS data stream format, and reserve the AT interface port only for the GNSS data stream

<GGA> integer 0 enables/disables the presence of the Global Positioning System Fix Data NMEA sentence (GGA) in the GNSS data stream.

Values:

0 : disable

1 : enable

<GLL> integer 0 enable/disable the presence of the Geographic Position - Latitude/ Longitude NMEA sentence (GLL) in the GNSS data stream.

Values:

0 : disable

1 : enable

<GSA> integer 0 enable/disable the presence of the GNSS DOP and Active Satellites NMEA sentence (GSA) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

<GSV>	integer	0	enable/disable the presence of the Satellites in View NMEA sentence (GSV) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

<RMC>	integer	0	enable/disable the presence of the Recommended Minimum Specific GNSS Data NMEA sentence (RMC) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

<VTG>	integer	0	enable/disable the presence of the GNSS Course Over Ground and Ground Speed NMEA sentence (VTG) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

Additional info:

- <enable>=1, GNSS data stream format:

  $GPSNMUN: <NMEA SENTENCE 1><CR><LF>
  ...
  $GPSNMUN: <NMEA SENTENCE N><CR><LF>
  ...

- <enable>=2, GNSS data stream format:

  <NMEA SENTENCE 1><CR><LF>
  ...
  <NMEA SENTENCE N><CR><LF>
  ...

- <enable>=3, in this case, the AT interface port is dedicated to NMEA sentences, it is not possible to send AT commands. Use the escape sequence "+++" to return in command mode. GNSS data stream format:

  <NMEA SENTENCE 1><CR><LF>
  ...
  <NMEA SENTENCE N><CR><LF>
  ...

- The unsolicited GNSS data stream built with NMEA sentences is enabled only for the AT instance where the AT$GPSNMUN was set.
- If the <enable> is 3, this <enable> option is not stored in NVM.
- The enable/disable filter for the presence of the NMEA sentence is independent from the first two letters (talker) of the NMEA sentence.

AT$GPSNMUN?

Read command returns whether the unsolicited GNSS data stream is currently enabled or not, along with the current NMEA mask configuration, in the format:

$GPSNMUN: <enable>,<GGA>,<GLL>,<GSA>,<GSV>,<RMC>,<VTG>

AT$GPSNMUN=?

Test command returns the supported range of values for parameters:

<enable>,<GGA>,<GLL>,<GSA>,<GSV>,<RMC>,<VTG>

The current setting is stored through $GPSSAV command with $GPSSAV either <GNSSParam> or <GPSPParam> parameters set to 1.

The setting is stored only for the AT instance where the AT$GPSNMUN was set .

Enable the unsolicited data stream and set the GSA as available sentence in the unsolicited message:

AT$GPSNMUN=1,0,0,1,0,0,0

OK

Read the current NMEA mask configuration:

AT$GPSNMUN?

$GPSNMUN: 1,0,0,1,0,0,0

OK

The unsolicited message will be:

$GPSNMUN: $GNGSA,A,3,28,32,,,,,,,,,,2.01,1.80,0.90,1*09

Turn-off the unsolicited data stream:

AT$GPSNMUN=0

OK

AT$GPSNMUNEX - Unsolicited NMEA Extended Data Configuration

Unsolicited NMEA extended data configuration.

NMEA 0183 Standard

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GPSNMUNEX=<GNGNS>[,<GNGSA>[,<GLGSV>[,<GPGRS>[,<BDGSA>[,<BDGSV>[,<GAGSA>[,<GAGSV>[,<GPDTM>[,<PTWSEPE>]]]]]]]]]

Set command defines which NMEA sentences will be available an unsolicited GNSS data stream built with extended NMEA sentences on the AT port. Refer to document [1] to have information on the NMEA sentences contents and formats.

Parameters:

Name	Type	Default	Description
<GNGNS>	integer	0	Fix data of GNSS receivers

Values:

0	:	disable
1	:	enable

<GNGSA>	integer	0	DOP and active satellites of GNSS

Values:

		0	: disable

<GLGSV>	integer	0	GLONASS satellites in view

Values:

0	:	disable

<GPGRS>	integer	0	GPS Range Residuals

Values:

0	:	disable
1	:	enable

<BDGSA>	integer	0	Beidou DOP and Active Satellites

Values:

0	:	disable

<BDGSV>	integer	0	Beidou Satellites in View

Values:

0	:	disable

<GAGSA>	integer	0	Galileo DOP and Active Satellites

Values:

		0	: disable

<GAGSV>	integer	0	Galileo Satellites in View

Values:

0	:	disable

<GPDTM>	integer	0	Datum reference information

Values:

0	:	disable
1	:	enable

<PTWSEPE>	integer	0	Estimated Position Error

Values:

0	:	disable
1	:	enable

<GNGNS>, <GPGRS>, <GPDTM> and <PTWSEPE> are the only parameters that support both values 0 and 1 for the ELS63 and LE910Q1 GNSS product variants.

The enable/disable filter for the presence of the NMEA sentence includes

- only GNGNS letters of the NMEA sentence for <GNGNS>

- GPGRS or GNGRS letters of the NMEA sentence for <GPGRS>

- GPDTM or GNDTM letters of the NMEA sentence for <GPDTM>

- only PTWSEPE letters of the NMEA sentence for <GNGNS>

The unsolicited GNSS extended data stream must be enabled/ disabled with AT$GPSNMUN command parameter <enable> and it is enabled only for the AT instance where the AT$GPSNMUN was set.

AT$GPSNMUNEX?

Read command returns the NMEA extended sentences availability status, in the format:

$GPSNMUNEX:
<GNGNS>,<GNGSA>,<GLGSV>,<GPGRS>,<BDGSA>,<BDGSV>,<GAGSA>,

<GAGSV>,<GPDTM>,<PWTSEPE>

AT$GPSNMUNEX=?

Test command returns the supported range of values for parameters

<GNGNS>, <GNGSA>, <GLGSV>, <GPGRS>, <BDGSA>, <BDGSV>, <GAGSA>, <GAGSV>, <GPDTM> and <PTWSEPE>.

The current setting is stored through $GPSSAV command with $GPSSAV <GNSSParam> parameter set to 1.

GNGNS sentence has field "mode indicator". The field is composed from the following order: gps, glonass, galileo and beidou.

- When gnss data is being received, there are four consecutive GNGSA sentences. The sentences are arranged at the following order: gps, glonass, galileo and beidou.

- All NMEA data stream is aligned with NMEA0183 v4.11.

  $GP--- NMEA data stream (for Gps)

  $GL--- NMEA data stream (for Glonass)

  $GQ--- NMEA data stream (for QZSS)

  $GN--- NMEA data stream (for multiconstellation)

  $GB--- NMEA data stream (for Beidou)

  $GA--- NMEA data stream (for Galileo)

- Galileo satellites are identified by their PRN numbers, ranging 1 to 36.

- Beidou satellites are identified by their PRN numbers, ranging 1 to 63.

- GPS satellites are identified by their PRN numbers, ranging 1 to 32.

- SBAS satellites are identified by their PRN numbers, ranging 33 to 64.

- Glonass satellites are identified by their PRN numbers, ranging 65 to 99.

- QZSS satellites are identified by their PRN numbers, ranging 1 to 7.

- GNSS Signal ID (the last field of $--GSV) are identified as GPSL1 C/A: 1, GLONASS L1:1 , GALILEO E1-BC:7, BEIDOU B1I:1, BEIDOU B1C:3, QZSS L1C/A:1.

```
AT$GPSNMUN=1
OK
```

```
AT$GPSNMUNEX=1,0,0,0,0,0,0,0,0,1
OK
```

These sets the GNGNS as available sentence in the unsolicited NMEA sentences.

```
AT$GPSNMUNEX?
$GPSNMUNEX: 1,0,0,0,0,0,0,0,0,1
OK
```

Give the current frame selected (GNGNS)

The unsolicited message will be:
$GNGNS,080558.0,3731.306144,N,12655.784429,E,AN,09,1.0,68.0,18.0,,*5B

$PTWSEPE,012101.000,3.54,2.50*64

AT$GNSSACP - Get Acquired Position

This command returns information about the last GNSS position.

NMEA 0183 Standard

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GNSSACP

Execution command returns information about the last GNSS position in the format:

$GNSSACP: <UTC>,<date>,<latitude>,<longitude>,<altitude>,<hdop>,<vdop>,<fix>,

<cog>, <spkm>,<spkn>,<mode>,<nsat>,<hepe>,<vepe>

Additional info:

Meaning of the parameters returned by the command

Name	Type	Default	Description
<UTC>	string	-	UTC time (hhmmss.sss) referred to GGA sentence
<date>	string	-	date of fix (referred to RMC sentence) in the format ddmmyy where:

			dd: 01..31, day mm: 01..12, month yy: 00..99, year 2000 to 2099
<latitude>	string	-	latitude in the format ddmm.mmmm S (referred to GGA sentence) where: dd: 00..90, degrees mm.mmmm: 00.0000..59.9999, minutes S: North/South
<longitude>	string	-	longitude in the format dddmm.mmmm W (referred to GGA sentence) where: ddd: 000..180, degrees mm.mmmm: 00.0000..59.9999, minutes W: East/West
<altitude>	string	-	altitude - mean-sea-level (geoid) in meters (referred to GGA sentence)
<hdop>	string	-	Horizontal Dilution of Precision (referred to GGA sentence)

<vdop>	string	-	Vertical Dilution of Precision (referred to GGA sentence)
<fix>	integer	N/A	fix type

Values:

0	:	invalid fix
1	:	invalid fix
2	:	2D fix
3	:	3D fix

<cog>	string	-	Course over Ground (degrees, True) (referred to RMC sentence)
<spkm>	string	-	speed over ground (km/h) (referred to VTG sentence)
<spkn>	string	-	speed over ground (knots) (referred to VTG sentence)
<mode>	string	-	Mode Indicator: five (LE910Q1) or six (ELS63) characters that indicate the operating mode of the five or six constellations GPS, GLONASS, Galileo, BDS, QZSS, NavIC (ELS63 only). The characters shall take one of the following values: A = Autonomous.

			D = Differential. E = Estimated (dead reckoning) Mode. F = Float RTK. M = Manual Input Mode. N = No fix. P = Precise. R = Real Time Kinematic. S = Simulator Mode. (referred to GNGNS sentence)
<nsat>	integer	-	total number of satellites in use (referred to GNS sentence)
<hepe>	float	-	Horizontal estimated position error (meters)
<vepe>	float	-	Vertical estimated position error (meters)

AT$GNSSACP?

Read command has the same behavior as the Execution command.

AT$GNSSACP=?

Test command returns the OK result code.

Execution and Read commands silently enable PTWSEPE (Estimated Position Error) in the GNSS.

AT$GPSP?

$GPSP: 0

OK

When the module is down there is no acquired position:

AT$GNSSACP

$GNSSACP: ,,,,,,,0,,,,,

OK

AT$GPSP=1

OK

Until first fix is received the command will display this:

AT$GNSSACP

$GNSSACP: ,,,,,,,1,,,,,

OK

Once fix has been received the command will display actual GNSS time and position:

AT$GNSSACP

$GNSSACP:
132807.000,130521,3913.6929N,00904.1545E,8.2,0.8,0.8,3,0.0,0.0,0.0,AANNN,14,3.79,2.50

OK

MQTT

AT#MQEN - Enable MQTT Feature

This command initializes a MQTT client and allocates the necessary resources.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQEN=<instanceNumber>,<enable>

Set command enables/disables the MQTT client for further configuration and usage.

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<enable>	integer	N/A	selects if client must be activated or deactivated Values: 0 : disable

1 : enable

AT#MQEN?

Read command returns the status of the MQTT stack in the format

#MQEN: <instanceNumber>,<enabled>

AT#MQEN=?

Test command reports the available range of values for parameters.

Read command
AT#MQEN?
#MQEN: 1,1
#MQEN: 2,0
#MQEN: 3,0
#MQEN: 4,0
OK

Test command
AT#MQEN=?
#MQEN: (1-maxClients),(0-1)
OK

AT#MQCFG - Configure MQTT Parameters

This sets the connection parameters for the selected MQTT client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQCFG=<instanceNumber>,<hostname>,<port>,<cid>[,<sslEn>[,<sslInstance>]]

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command Value: 1÷maxClients : client instance range. To know maxClients value use test command
<hostname>	string	-	URL of the MQTT broker. Default value: 0. Maximum length of <hostname> is 512 characters.
<port>	integer	0	TCP port of the MQTT broker Value:

			1÷65535 : TCP port range
<cid>	integer	N/A	PDP Context ID to be used for the connection. PDP context should be enabled before giving this command. Value: 1÷6 : <cid> range
<sslEn>	integer	0	enable/disable SSL Values: 0 : disable 1 : enable
<sslInstance>	integer	1	selects the ssl instance configuration to be used for this MQTT client. The list of available ssl instances is obtained with the read command. Value: 1÷maxClients : ssl instance range. To know ssl instance value use test command. If the value is

not provided it will select ssl instance 1 by default

AT#MQCFG?

Read command returns the configuration of all active MQTT clients in the format:

#MQCFG: <instanceNumber>,<hostname>,<port>,<cid>,<sslEn>

AT#MQCFG=?

Test command reports the available range of values for parameters.

</>

* SSL disabled (by default)

AT#MQCFG=1,api.mybroker.com,1883,3
OK

Read command
AT#MQCFG?
#MQCFG: 1,api.mybroker.com,1883,3,0
OK

Test command
AT#MQCFG=?
#MQCFG: (1-maxClients),512,(1-65535),(1-6),(0-1),(1-maxSSLInstance)
OK

* SSL enabled

AT#MQCFG=1,api.mybroker.com,8883,3,1
OK

Read command
AT#MQCFG?
#MQCFG: 1,api.mybroker.com,8883,3,1
OK

Test command
AT#MQCFG=?
#MQCFG: (1-maxClients),512,(1-65535),(1-6),(0-1)),(1-maxSSLInstance)
OK

AT#MQDISC - Log Out and Disconnect from the MQTT Broker

This command performs the logout and disconnection from to the MQTT broker.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQDISC=<instanceNumber>

Disconnects gracefully from the MQTT broker, then closes the network connection

Parameter:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate
	Value:		
	1÷maxClients	:	client instance range. To know maxClients value use test command

AT#MQDISC=?

Test command reports the available range of values for parameters.

```
AT#MQDISC=1
OK
```

Test command
AT#MQDISC=?
#MQDISC: (1-maxClients)
OK

AT#MQCONN - Connect and Log in the MQTT Broker

This command performs the connection and login to the MQTT broker.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQCONN=<instanceNumber>,<clientID>[,<userName>,<passWord>]

Set command performs network connection (using parameters set with #MQCFG) and sends the CONNECT packet to the MQTT broker.

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<clientID>	string	-	identifies each MQTT client that connects to the MQTT broker. Maximum length of <clientID> is 512.
<userName>	string	-	authentication and authorization.

			This is not mandatory, but this value must be pair with <passWord>. Maximum length of <userName> is 512.
<passWord>	string	-	authentication and authorization. Maximum length of <passWord> is 512.

If connection status is other than init state or connected, disconnect using #MQDISC before reconnecting using #MQCONN.

AT#MQCONN?

Read command reports the configuration of active MQTT connections in the format:

#MQCONN=<instanceNumber>,<state>

Additional info:

The following tables shows the <state> values and meanings:

Client status: <state>	Normal/ Failure Events Values
0	client is initialized but not connected
1	client performed MQTT authentication with broker

2	connection closed or reset by the server
3	the answer to the ping request packet was not received
4	the CONNACK packet was not received
8	client is disconnecting from the server

AT#MQCONN=?

Test command reports the available range of values for parameters.

```
AT#MQCONN=1,myClientID,myUser,myPassword
OK
```

Scenario (1)
Read command

```
AT#MQCONN?
#MQCONN: 1,1
#MQCONN: 2,0
OK
```

Test command

```
AT#MQCONN=?
#MQCONN: (1-maxClient),512,512,512
OK
```

Scenario (2)
Read command
AT#MQCONN?
#MQCONN: 1,2
OK

Set command
AT#MQDISC=1
OK

Read command
AT#MQCONN?
#MQCONN: 1,0
OK

AT#MQCFG2 - Configure Additional MQTT Parameters

This command sets the optional connection parameters for the selected MQTT client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQCFG2=<instanceNumber>,<keepAlive>,<cleanSession>

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<keepAlive>	integer	60	timeout of periodic packet to keep connection open Value: 60÷3600 : timeout expressed in sec

<cleanSession>	integer	1	Indicates whether a persistent connection is required. Without a persistent connection, when the client is offline all information and messages that are queued from a previous persistent session are lost.

Values:

1	:	clean session
0	:	persistent session

AT#MQCFG2?

Read command returns the configuration of all active MQTT clients in the format:

#MQCFG2: <instanceNumber>,<keepAlive>,<cleanSession>

AT#MQCFG2=?

Test command reports the available range of values for parameters.

```
AT# MQCFG2=1,60,1
OK
```

Read command

```
AT#MQCFG2?
#MQCFG2: 1,60,1
#MQCFG2: 2,70,1
OK
```

Test command

```
AT#MQCFG2=?
#MQCFG2: (1-maxClients),(60-3600),(0-1)
```

AT#MQWCFG - Configure MQTT Last Will and Testament

This command sets Last Will and Testament for the selected MQTT client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQWCFG=<instanceNumber>,<willFlag>
[,<willRetain>,<willQos>,<willTopic>,<willMsg>]

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command.
	Value:		
	1÷maxClients	:	client instance range. To know maxClients value use test command
<willFlag>	integer	0	selects whether the client needs to specify a Last Will and Testament. If set to 0, this is the last parameter to be set.
	Values:		

		0 : the client does not need to specify a Last Will and Testament 1 : the client needs to specify a Last Will and Testament	
<willRetain>	integer	0	selects whether the Last Will message needs to be retained by the server Values: 0 : the Last Will message does not need to be retained by the server 1 : the Last Will message needs to be retained by the server
<willQos>	integer	0	Quality of Service of the Last Will message Value: 0÷2 : Quality of Service range
<willTopic>	string	-	Topic to publish the Last Will message to (with quotes). Maximum length of <willTopic> is 256.
<willMsg>	string	-	Last Will message (with quotes). Maximum length of <willMsg> is 1024.

AT#MQWCFG?

Read command returns the Last Will and Testament (if any) of all active MQTT clients in the format:

#MQWCFG:<instanceNumber>,<willFlag> [,<willRetain>,<willQos>,<willTopic>,<willMsg>]

AT#MQWCFG=?

Test command reports the available range of values for parameters.

```
AT#MQWCFG=1,0
OK

AT#MQWCFG=2,1,1,2,"myLastWillTopic","myLastWillMessage"
OK

Read command
AT#MQWCFG?
#MQWCFG: 1,0
#MQWCFG: 2,1,1,2,myLastWillTopic,myLastWillMessage
OK

Test command
AT#MQWCFG=?
#MQWCFG: (1-maxClients),(0-1),(0-1),(0-2),256,1024
OK
```

AT#MQUNS - Unsubscribe from a Topic

This command revokes the subscription to a MQTT topic.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQUNS=<instanceNumber>,<topic>

Parameters:

Name	Type	Default	Description
<instanceNumber>	string	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic (with quotes). Maximum length of <topic> is 256.

AT#MQUNS=?

Test command reports the available range of values for parameters.

```
AT#MQUNS=1,"topicToUnsubscribe"
OK
```

Test command

```
AT#MQUNS=?
#MQUNS: (1-maxClients),256
OK
```

AT#MQTCFG - Configure Timeout Parameters for MQTT Transmission

This command writes the timeout options for the specified client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQTCFG=<instanceNumber>,<pktTimeout>

Parameters:

Name	Type	Default	Description
<instanceNumber>	string	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<pktTimeout>	integer	30	timeout of the packet delivery Value: 1÷60 : timeout range. Value expressed in seconds

AT#MQTCFG?

Read command returns the timeout configuration of all active MQTT clients in the format

#MQTCFG: <instanceNumber>,<pktTimeout>

AT#MQTCFG=?

Test command reports the available range of values for parameters

```
AT#MQTCFG=1,10
OK

Read command
AT#MQTCFG?
#MQTCFG: 1,10
#MQTCFG: 2,30
OK

Test command
AT#MQTCFG=?
#MQTCFG: (1-maxClients),(1-60)
OK
```

AT#MQSUB - Subscribe to a Topic

This command performs the subscription to a MQTT topic

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQSUB=<instanceNumber>,<topic>[,<qos>]

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic (with quotes). Maximum length of <topic> is 256.
<qos>	integer	1	Configures qos for the subscription. This is an optional parameter. If <qos> is not specified, <qos> 1 is taken as default.

Value:

0÷2	:	qos range

AT#MQSUB=?

Test command reports the available range of values for parameters.

```
AT#MQSUB=1,"topicToSubscribe"
OK

AT#MQSUB=1,"topicToSubscribe_2",0
OK

Test command
AT#MQSUB=?
#MQSUB: (1-maxClients),256, (0-2)
OK
```

AT#MQPUBS - Publish ASCII String

This command publishes an ASCII string to the specified MQTT topic.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQPUBS=<instanceNumber>,<topic>,<retain>,<qos>,<message>

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic. Maximum length of <topic> is 256.
<retain>	integer	N/A	specifies if the broker must retain this message or not Values: 0 : Not retained

	1	:	Retain

<qos>	integer	N/A	specifies the Quality of Service of this message
	Value:		
	0÷2	:	Quality of Service range
<message>	string	-	message to publish on the topic. Maximum length of <message> is 1024

AT#MQPUBS=?

Test command reports the available range of values for parameters.

```
AT#MQPUBS=1,"myTopic",0,0,"myMessage"
OK

Test command
AT#MQPUBS=?
#MQPUBS: (1-maxClients),256,(0-1),(0-2),1024
OK
```

AT#MQPUBSEXT - Publish Extension command

This command publishes data to the specified MQTT topic.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#MQPUBSEXT=<instanceNumber>,<topic>,<retain>,<qos>,<dataLen>

After entering AT#MQPUBSEXT=... command, and terminated the command line with <CR>, the module returns the following three characters sequence prompt, and waits for data to send:

<greater_than><greater_than><greater_than> (see IRA character set: 62, 62, 62)

Now, data can be entered from TE, sized <dataLen> bytes. If user does not enters any input for 10 seconds, command is terminated with error.

If sending ends successfully and acknowledgment is received for QOS > 0, the response will be OK, otherwise an error will be reported.

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate
	Value:		

			1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic. Maximum length of <topic> is 256
<retain>	integer	N/A	specifies if the broker must retain this message or not Values: 0 : Not retained 1 : Retain
<qos>	integer	N/A	specifies the Quality of Service of this message Value: 0÷2 : Quality of Service range
<dataLen>	integer	-	Length of message to publish on the topic. Maximum length of <dataLen> can go up to 65536

depending on system memory available.

AT#MQPUBSEXT=?

Test command reports the available range of values for parameters.

Publish 100 byte message

AT#MQPUBSEXT=1,myTopic,0,0,100

>>>

OK

Test command

AT#MQPUBSEXT=?

#MQPUBSEXT: (1-maxClients),256,(0-1),(0-2),(0-65536)

OK

AT#MQREAD - Read Messages Received from the MQTT Broker

This command reads the message payload from the queue slot provided.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQREAD=<instance_number>,<message_id>

After command line is terminated with <CR>, the module responds sending:
#MQREAD: <instanceNumber>,<topic>,<payload_len>,

then the module prompts the following characters sequence: <less_than><less_than><less_than><carriage return><line feed> (IRA 60, 60, 60, 13, 10)

followed by the data

Parameters:

Name	Type	Default	Description
<instance_number>	integer	N/A	Selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range; to know maxClients value use test command
<message_id>	integer	N/A	Message slot Id to be read. The read

operation will free the slot resource.

Value:

1÷30	:	message slot Id range

Additional info:

▶▶ UNSOLICITED MESSAGE:

#MQRING - Received data on subscribed topic

When a message is received on the subscribed topic, an URC message is sent to all AT commands interfaces. There are 30 messages slots available for incoming messages, and it is responsibility of the user to keep them empty by reading them with #MQREAD.

- If the message queue is full, and a new message arrives, the following URC #MQRING: 0 is received.

- If enough memory is not available in the system to store the received message, <len> field in URC message is -1. Example: #MQRING:1,0,my_topic,-1

- Otherwise, for normal messages, the URC format is:

#MQRING: <instance_number>,<message_id>,<topic>,<len>

In the Unsolicited fields section are described the URC message parameters not described in the previous sections.

Name	Type	Default	Description
<topic>	string	-	name of the topic from where the message was received
<len>	integer	-	length in bytes of the received payload

AT#MQREAD?

Read command returns the unread messages count for each instance number for all active MQTT clients in the format:

#MQREAD: <instanceNumber>,<unread>

AT#MQREAD=?

Test command reports the available range of values for parameters

#MQREAD: (1-maxClients),(1-30)

Set command

AT#MQREAD=1,2

#MQREAD: 1, "topic",10

<<<

0123456789

OK

Read command

AT#MQREAD?

#MQREAD: 1, 1

#MQREAD: 2, 10

OK

Test command

AT#MQREAD=?

#MQREAD: (1-4),(1-30)

OK

Mobile Broadband

Ethernet Control Mode (ECM)

AT#ECM - Ethernet Control Model Setup

This command sets up an Ethernet Control Model (ECM) session.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#ECM=<Cid>,<Did>

This command sets up an Ethernet Control Model (ECM) session

Parameters:

Name	Type	Default	Description
<Cid>	integer	-	context id that will be used by ECM. Refer to test command to know the supported range of values.
<Did>	integer	-	device id, currently limited to 0 (only one device)

- To activate an ECM session a compatible USB configuration has to be set; for details refer to the description of AT#USBCFG.
- AT#ECM takes immediate effect (no need for reboot) and at next power the ECM session is automatically setup.
- Change of AT#USBCFG set to different than ECM USB configuration shuts down current ECM session.

AT#ECM?

Read command returns the session state in the following format:

ECM: <Did>,<State>

OK

Additional info:

Here are the parameters meaning returned by the read command and not described in the previous sections.

Name	Type	Default	Description
<State>	integer	0	ECM session status.
	Values:		
	0	:	disabled
	1	:	enabled

AT#ECM=?

Test command returns the range of supported values for all the parameters.

If ECM session compatible USB configuration is not set by AT#USBCFG, AT#ECM test command will return ERROR.

AT#ECMD - ECM Shutdown

This command is used to shutdown an Ethernet Control Model (ECM) session.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#ECMD=<Did>

This command is used to shutdown an Ethernet Control Model (ECM) session.

Parameter:

Name	Type	Default	Description
<Did>	integer	N/A	Device id, currently limited to 0 (only one device)

Value:

0 : Device id (currently limited to only one device)

AT#ECMD takes immediate effect (no need for reboot) and at next power the ECM session is not automatically setup.

AT#ECMD?

Read command returns the session state in the following format:

#ECM: <Did>,<State>

OK

Additional info:

▶▶ Parameter returned by the read command and not described in the previous sections.

Name	Type	Default	Description
<State>	integer	0	State of Ethernet Control Model (ECM) session

Values:

0	:	disabled
1	:	enable

?

AT#ECMD=?

Test command returns the range of supported values for <Did>.

AT#RNDIS - Remote Network Driver Interface Specification setup

This command sets up Remote Network Driver Interface Specification (RNDIS) session.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#RNDIS=<Cid>,<Did>

This command sets up a Remote Network Driver Interface Specification (RNDIS) session.

Parameters:

Name	Type	Default	Description
<Cid>	integer	-	context id that will be used by RNDIS. Refer to test command to know the supported range of values
<Did>	integer	-	Device id, currently limited to 0 (only one device)

- To activate an RNDIS session a compatible USB configuration has to be set; for details refer to the description of AT#USBCFG.
- AT#RNDIS takes immediate effect (no need for reboot) and at next power the RNDIS session is automatically setup.
- Change of AT#USBCFG set to different than RNDIS USB configuration shuts down current RNDIS session.

AT#RNDIS?

Read command returns the session state in the following format:

#RNDIS: <Did>,<State>

OK

Additional info:

Here are the parameters meaning returned by the read command and not described in the previous sections

Name	Type	Default	Description
<State>	integer	0	RNDIS session status
	Values:		
	0	:	disabled
	1	:	enabled

AT#RNDIS=?

Test command returns the range of supported values for all the parameters.

If RNDIS session compatible USB configuration is not set by AT#USBCFG, AT#RNDIS test command will return ERROR.

AT#RNDISD - Remote Network Driver Interface Specification shutdown

This command is used to shutdown a Remote Network Driver Interface Specification (RNDIS) session.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#RNDISD=<Did>

This command is used to shutdown a Remote Network Driver Interface Specification (RNDIS) session.

Parameter:

Name	Type	Default	Description
<Did>	integer	-	Device id, currently limited to 0 (only one device)

AT#RNDISD takes immediate effect (no need for reboot) and at next power the RNDIS session is not automatically setup.

AT#RNDISD?

Read command returns the session state in the following format:

#RNDISD: <Did>,<State>

OK

Additional info:

▶▶ Parameter returned by the read command and not described in the previous sections

Name	Type	Default	Description
<State>	integer	0	RNDIS backhaul connection status

Values:

0	:	disabled
1	:	enabled

?⬚ AT#RNDISD=?

Test command returns the range of supported values for <Did>.

AT#IPPASSTH - ECM and RNDIS IP pass through

This command is used to enable/disable the IP pass through functionality for ECM and RNDIS.

Default, activating the ECM a small private local network is created between the module and the PC and a fixed IP address is used for the ECM interface (IP private mode); enabling the IP pass through the user can configure the ECM interface to use the IP address assigned to the module by the network operator avoiding the creation of the private local network (IP Public mode). The same for RNDIS.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#IPPASSTH=<enabled>[,<res_port>[,<res_range>]]

Set command enables/disables the IP pass through functionality.

Parameters:

Name	Type	Default	Description
<enabled>	integer	0	enable/disable the IP pass through functionality
	Values:		
	0 : Disabled		
	1 : Enabled		
<res_port>	integer	60000	the first reserved port; dummy value, not used
	Value:		

		5000÷65515	: TCP port
<res_range>	integer	20	the number of reserved ports starting from <res_port>; dummy value, not used
	Value:		
		20÷100	: Number of reserved ports

- The configuration will be applied after a reboot.
- This feature involves both the ECM and RNDIS behavior.
- In IP private mode the fixed IP address used for the ECM interface is 192.168.225.2.
- Enabling this feature could require configuring manually the network interface on the host.
- <res_port> + <res_range> cannot be greater than 65535 (dummy values, not used).

AT#IPPASSTH?

Read command presents the current configuration in the format:

#IPPASSTH: <status>,<res_port>,<res_range>

AT#IPPASSTH=?

Test command returns the supported values of the parameters.

PSM (Power Saving Mode)

AT+CPSMS - Power Saving Mode Setting

This command enables/disables Power Saving Mode (PSM) mode.

[1] 3GPP TS 27 007

[2] 3GPP TS 24.008

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CPSMS=[<mode>[,<ReqPeriodicRAU>[,<ReqGPRSreadyTimer>[,<ReqPeriodicTAU>[,<ReqActiveTime>]]]]]

The set command controls the setting of the UEs power saving mode (PSM) parameters. The command controls whether the UE wants to apply PSM or not, as well as the requested extended periodic RAU value and the requested GPRS READY timer value in GERAN, the requested extended periodic TAU value in E-UTRAN and the requested Active Time value.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	disables or enables the use of PSM in the UE.

Values:

0 : Disable

1 : Enable

2 : Disable the use of PSM and discard all parameters for PSM

<ReqPeriodicRAU>	string	-	one byte in an 8 bit format. Requested extended periodic RAU value (T3312) to be allocated to the UE in GERAN. The requested extended periodic RAU value is coded as one byte (octet 3) of the GPRS Timer 3 information element coded as bit format (e.g. "01000111" equals 70 hours). For the coding and the value range, see the GPRS Timer3 IE in 3GPP TS 24.008 Not supported. Reserved for future use
<ReqGPRSreadyTimer>	string	-	one byte in an 8 bit format. Requested GPRS READY timer value (T3314) to be allocated to the UE in GERAN. The requested GPRS READY timer value is coded as one byte (octet 2) of the GPRS Timer information element coded as bit format (e.g. "01000011" equals 3 decihours or 18 minutes). For the coding and the value range, see the GPRS Timer IE in 3GPP TS 24.008. Not supported. Reserved for future use
<ReqPeriodicTAU>	string	-	one byte in an 8 bit format. Requested extended periodic TAU value (T3412) to be allocated to the UE in

			E-UTRAN. The requested extended periodic TAU value is coded as one byte (octet 3) of the GPRS Timer 3 information element coded as bit format (e.g. "01000111" equals 70 hours). For the coding and the value range, see the GPRS Timer 3 IE in 3GPP TS 24.008.
<ReqActiveTime>	string	-	one byte in an 8 bit format. Requested Active Time value (T3324) to be allocated to the UE. The requested Active Time value is coded as one byte (octet 3) of the GPRS Timer 2 information element coded as bit format (e.g. "00100100" equals 4 minutes). For the coding and the value range, see the GPRS Timer 2 IE in 3GPP TS 24.008.

AT+CFUN=15 is not allowed in case of AT+CPSMS=1 configuration.

A special form of the command can be given as AT+CPSMS= (with all parameters omitted). In this form, the parameter <mode> will be set to 0, the use of PSM will be disabled and data for all parameters in command +CPSMS will be removed or, if available, set to the manufacturer specific default values.

AT+CPSMS?

Read command returns the current CPSMS configuration, in the format:

+CPSMS:<mode>,[<ReqPeriodicRAU>],[<ReqGPRSreadyTimer>], [<ReqPeriodicTAU>], [<ReqActiveTime>]

AT+CPSMS=?

Test command reports the range for the parameters in the format:

+CPSMS:(list of supported <mode>s),(list of supported <ReqPeriodicRAU>s),(list of supported <ReqGPRSreadyTimer>s),(list of supported <ReqPeriodicTAU>s),(list of supported <ReqActiveTime>s)

How to manage timer values octet.

T3412ext value:

Bits 5 to 1 represent the binary coded timer value.
Bits 6 to 8 defines the timer value unit as follows:

Bits

8 7 6
0 0 0 value is incremented in multiples of 10 minutes
0 0 1 value is incremented in multiples of 1 hour
0 1 0 value is incremented in multiples of 10 hours
0 1 1 value is incremented in multiples of 2 seconds
1 0 0 value is incremented in multiples of 30 seconds
1 0 1 value is incremented in multiples of 1 minute
1 1 0 value is incremented in multiples of 320 hours
1 1 1 value indicates that the timer is deactivated.

Example: 10101100 -> 101 means values is incremented in multiples of 1

minute, 01100 means 12 -> the obtained value is 12 minutes

T3324 value:

Bits 5 to 1 represent the binary coded timer value.
Bits 6 to 8 defines the timer value unit for the GPRS timer as follows:
Bits

8 7 6
0 0 0 value is incremented in multiples of 2 seconds
0 0 1 value is incremented in multiples of 1 minute
0 1 0 value is incremented in multiples of decihours
1 1 1 value indicates that the timer is deactivated.

How to exit from power saving mode (PSM) during hibernate state.

ON/OFF pin toggle exits from power saving mode (PSM) during hibernate state .

AT+CPSMS=1,,,"10101100","00100010"

OK

AT+CPSMS?

+CPSMS: 1,,,"10101100","00100010"

OK

It means that module requires to adopt the PSM for reducing its power consumption. If the network supports PSM and accepts that the UE uses PSM with requested timers value, module enters in PSM when the active timer expires (T3324=2 minutes) and stay in this mode for ten minutes (T3412=12 minutes).

AT+CPSMS=0

OK

It means that PSM is set to disable, the module does not go to Power Saving Mode in any case.

AT#PSMURC - Power Saving Mode URC

Set command enables/disables the URC that informs when modem entering in power saving mode.

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

AT#PSMURC=<en>

The URC format is:

#PSMURC=<ActiveTime>,<PSMTime>

Parameter:

Name	Type	Default	Description
<en>	integer	0	enable/disable URC message

Values:

0	:	disable URC message
1	:	enable URC message

Unsolicited fields:

Name	Type	Description

<ActiveTime>	integer	requested Active Time value, in seconds (T3324)
<PSMTime>	integer	low power phase duration in seconds (difference between T3412 and T3324 including boot time).

AT#PSMURC?

Read command reports the status (enable/disable):

#PSMURC: <en>

AT#PSMURC=?

Test command reports the supported range of value for parameter:

#PSMURC: (0,1)

The setting is saved using the &W command.

Debugging

AT#TRACE - Enable/Disable Trace

The command selects which trace outputs you want to display through the debugging tool.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#TRACE=[<mode>],[<configurationString>],[<baudrate>]

Parameters:

Name	Type	Default	Description
<mode>	integer	0	switches all trace outputs ON or OFF. If parameter <mode> is not entered in the command, the following <configurationString> will be used. Values: 0 : sets all trace outputs OFF, <configuration String> will be ignored 1 : sets all trace outputs ON, <configurationString> will be ignored
<configurationString>	string	-	enables/disables a set of trace outputs, its syntax is shown in the Additional info section.
<baudrate>	integer	921600	sets the trace baudrate. This setting is saved in NVM.

Values:

9600	:	bit/s
19200	:	bit/s
38400	:	bit/s
57600	:	bit/s
115200	:	bit/s
230400	:	bit/s
256000	:	bit/s
460800	:	bit/s
921600	:	bit/s
3000000	:	bit/s

Additional info:

▶▶ The syntax of the <configurationString> is:

["<unit>=<umode>[,<unit>=<umode>[,...]]"]

Here are the meanings and values of the string parameters.

Name	Type	Default	Description
<unit>	string	N/A	trace class name available to the user to select the trace output.

On the right side of each name, in lower case characters, there is the TC_XXX... string (Trace Class name) shown by the debugging tool.

Values:

generic	:	TC_GENERIC
socket	:	TC_SOCKET
clock	:	TC_CLOCK
pdp	:	TC_PDP
gnss	:	TC_GNSS
m2m_user	:	TC_M2M_USER
fota	:	TC_FOTA
fs	:	TC_FS
qmi	:	TC_QMI
sms	:	TC_SMS
info	:	TC_INFO
lwm2m	:	TC_LWM2M
net	:	TC_NET
sim	:	TC_SIM

spi	:	TC_SPI
usb	:	TC_USB,
nv	:	TC_NV
rtc	:	TC_RTC
m2m_uart	:	TC_UART
power	:	TC_POWER
ftpc	:	TC_FTPC
ati	:	TC_ATI
backup	:	TC_BACKUP
nipd	:	TC_NIPD
sys	:	TC_SYS
psm	:	TC_PSM
ssl	:	TC_SSL

<umode>	integer	0	enables/disables the trace output selected

Values:

0	:	disables
1	:	enables

AT#TRACE?

Read command reports the currently selected parameter values in the format:

#TRACE: "<unit>=<umode>,...,<unit>=<umode>",<baudrate>

AT#TRACE=?

Test command returns OK.

Here are some examples:

* Set all trace outputs OFF
  AT#TRACE=0
  OK

* Set all trace outputs ON
  AT#TRACE=1
  OK

* Enable/disable trace outputs selected
  AT#TRACE=,"generic=1,clock=0,lwm2m=1,pdp=0,gnss=0,ati=0",921600
  OK

Acronyms and Abbreviations

Table 8: Acronyms and Abbreviations

Acronym	Definition
ARFCN	Absolute Radio Frequency Channel Number
AT	Attention command
BA	BCCH Allocation
BCCH	Broadcast Control Channel
CA	Cell Allocation
CBM	Cell Broadcast Message
CBS	Cell Broadcast Service
CCM	Current Call Meter
CLIR	Calling Line Identification Restriction
CTS	Clear To Send
CUG	Closed User Group
DCD	Data Carrier Detect
DCE	Data Communication Equipment
DCS	Digital Cellular System
DGPS	Differential GPS, the use of GPS measurements, which are differentially corrected
DNS	Domain Name System
DSR	Data Set Ready
DTE	Data Terminal Equipment
DTMF	Dual Tone Multi Frequency

DTR	Data Terminal Ready
GGA	GPS Fix data
GLL	Geographic Position - Latitude/Longitude
GLONASS	Global positioning system maintained by the Russian Space Forces
GMT	Greenwich Mean Time
GNSS	Any single or combined satellite navigation system (GPS, GLONASS and combined GPS/GLONASS)
GPRS	Global Packet Radio Service
GPS	Global Positioning System
GSA	GPS DOP and Active satellites
GSM	Global System Mobile
GSV	GPS satellites in view
HDLC	High Level Data Link Control
HDOP	Horizontal Dilution of Precision
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
IP	Internet Protocol
IRA	International Reference Alphabet
IWF	Interworking Function
ME	Mobile Equipment
MO	Mobile Originated
MT	either Mobile Terminated or Mobile Terminal
NMEA	National Marine Electronics Association
NVM	Non-Volatile Memory

PCS	Personal Communication Service
PDP	Packet Data Protocol
PDU	Packet Data Unit
PIN	Personal Identification Number
PPP	Point to Point Protocol
PUK	Pin Unblocking Code
RLP	Radio Link Protocol
RMC	Recommended minimum Specific data
RTS	Request To Send
SAP	SIM Access Profile
SCA	Service Center Address
SMS	Short Message Service
SMSC	Short Message Service Center
SMTP	Simple Mail Transport Protocol
TA	Terminal Adapter
TCP	Transmission Control Protocol
TE	Terminal Equipment
UDP	User Datagram Protocol
USSD	Unstructured Supplementary Service Data
UTC	Coordinated Universal Time
VDOP	Vertical dilution of precision
VTG	Course over ground and ground speed
WAAS	Wide Area Augmentation System

Related Documents

Refer to

https://dz.telit.com/

for current documentation and downloads.

Document History

Table 9: Document History

Revision	Date	Changes
10	2025-12-12	Updated Commands: AT+CMUX
9	2025-12-05	Updated title adding LE310Q1 Updated Applicability Table Updated Commands: AT#FASTSHDN, AT#FTPCFG, AT#GPIO, AT#MQREAD, AT$GNSSACP, AT$GPSACP, AT+CFUN, AT+CGEREP, AT+CMUX, AT+CPSMS, AT+IPR New Commands: AT#SIMDET, AT#SIMINCFG
8	2025-08-28	Updated Commands: AT#FWSWITCH, AT#GPIO, AT#IPPASSTH, AT#OTAUPW, AT#RFSTS, AT#SSLSECCFG, AT#TESTMODE, AT#USBCFG, AT$GNSSACP, AT$GNSSDWLD, AT$GPSACP, AT$GPSNMUN, AT+CFUN, AT+CNMA, AT+CPLS, AT+CPSMS, AT+IPR New Commands: AT#JDR4GCFG, AT#JDRENH2, AT$AGNSSFDL, AT$AGNSSFINJ, AT$AGNSSTINJ, AT$AGNSSVAL, AT+CMUX
7	2025-05-25	Updated Commands: AT#SL, AT#SLUDP, AT#SSLSECDATA, AT#WIFISCAN, AT$GNSSACP, AT$GPSNMUN New Commands: AT#SCFGEXT4, AT$GNSSDWLD
6	2025-04-15	Updated Applicability Table Updated Commands: AT#GPIO, AT#TEMPCFG New Commands:

		AT#GNSSCMD, AT$GNSSACP, AT$GNSSCMD, AT$GNSSMODE, AT$GPSACP, AT$GPSCFG, AT$GPSHWRST, AT$GPSNHZ, AT$GPSNMUN, AT$GPSNMUNEX, AT$GPSP, AT$GPSR, AT$GPSSAV, AT$GPSSW
5	2025-02-25	Updated Applicability Table Removed Commands: AT+CMUX Updated Commands: AT#BND, AT#DNS, AT#ECM, AT#FWSWITCH, AT#GPIO, AT#I2CRD, AT#I2CWR, AT#NWDNS, AT#QDNS, AT#RNDIS, AT#TESTMODE, AT+CGCONTRDP, AT+CGDCONT, AT+CMGR, AT+CMGS, AT+CNMA, AT+CNMI, AT+CPLS, AT+CRES, AT+CSAS, AT+CSCA, AT+CSMP, AT+IPR New Commands: AT#CCMGR, AT#CCMGS, AT#CGSSN, AT#FPLMN, AT#IMEIW, AT#IPPASSTH, AT#LWM2MREG, AT#LWM2MSTS, AT#STGI, AT#STIA, AT#STSR
4	2024-12-06	Updated Commands: AT#ADC, AT#FRWL, AT#FTPCFG, AT#FWSWITCH , AT#GPIO, AT#NTP, AT#NTPCFG, AT#OTAUP, AT#PORTCFG, AT#PSMURC , AT#SD, AT#SECBTON, AT#SIMSELECT, AT#SL, AT#SSLCFG, AT#SSLSECDATA, AT#STIA, AT&C, AT+CEDRXRDP, AT+CEREG, AT+CFUN, AT+CGDCONT, AT+CGEREP, AT+COPS, AT+CPIN, AT+CPMS, AT+CREG, AT+CSMS, ATD, ATO, ATZ New Commands: AT#ECM, AT#ECMD, AT#ENHRST, AT#FRWL, AT#ICMP, AT#LWM2MSKIP, AT#NTPCFG, AT#RNDIS, AT#RNDISD, AT#SECBTON, AT#SIMSELECT, AT#STGI, AT#STIA, AT#STSR, AT#WIFISCAN
3	2024-06-27	Updated Commands: AT#CBCMON, AT#CFLO, AT#FASTSHDN, AT#FWSWITCH, AT#GPIO, AT#NITZ , AT#NTP, AT#SCFG, AT#SCFGEXT , AT#SCFGEXT3 , AT#SS, AT#TEMPCFG, AT#TEMPMON, AT&K, AT+CCLK, AT+CGEQOS, AT+CTZU, AT+ICF New Commands: AT#SLED, AT#SLEDSAV, AT+CGACT

2	2024-05-14	Changed Document title to "ELS63 and LE910Q1 AT Commands Reference Guide" Updated Applicability Table Updated chapters: Information Responses and Result Codes; Module Reboot Updated Commands: AT#FASTSHDN, AT#GPIO, AT#MQCFG, AT#SCFG, AT#SCFGEXT3, AT+CFUN, AT+CGDCONT, AT+CGEQOS, AT+CMGS, AT+CNMI, AT+CPMS, ATD, ATV New Commands: AT#CBC, AT#CBCMON, AT#DNS, AT#E2SLRI, AT#I2CRD, AT#I2CWR, AT#NWDNS, AT#QDNS, AT#OTAUPW, AT#SYSSTART, AT&S, AT+CMUX
1	2024-04-08	Updated chapter: Storage Updated Commands: AT#ADC, AT#BND, AT#CEERNETEXT, AT#CFLO, AT#FASTSHDN, AT#FTPCFG, AT#MQCFG, AT#MQCONN, AT#MQEN, AT#MQPUBS, AT#MQREAD, AT#MQSUB, AT#MQTCFG, AT#MQUNS, AT#MQWCFG, AT#NTP, AT#PORTCFG, AT#SCFG, AT#SCFG, AT#SGACT, AT#SL, AT#SS, AT#SSLCFG, AT#SSLS, AT#SSLSSECCFG, AT#TESTMODE, AT&K, AT+CEREG, AT+CFUN, AT+CREG, AT+ICCID, AT+IFC, AT+WS46, ATV New Commands: AT#FTPGETOTA, AT#HTTPRCVOTA, AT#MQPUBSEXT, AT#OTAUP, AT#SSLSECDATA, AT+CMSS, AT+GMI, AT+GMM, AT+GMR, ATH Removed Commands: AT#CMUXMODE, AT#E2SLRI, AT#I2CRD, AT#I2CWR, AT+CMUX, AT+SWITCHSIM
0	2024-01-16	First release

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Technical Documentation